

NASA TN D-6552

CH-1 FILE
COPY

GENENG — A PROGRAM FOR CALCULATING
DESIGN AND OFF-DESIGN PERFORMANCE
FOR TURBOJET AND TURBOFAN ENGINES

by Robert W. Koenig and Laurence H. Fishbach

*Lewis Research Center
Cleveland, Ohio 44135*

1. Report No. NASA TN D-6552		2. Government Accession No.		3. Recipient's Catalog No.	
4. Title and Subtitle GENENG - A PROGRAM FOR CALCULATING DESIGN AND OFF-DESIGN PERFORMANCE FOR TURBOJET AND TURBOFAN ENGINES				5. Report Date February 1972	
				6. Performing Organization Code	
7. Author(s) Robert W. Koenig and Laurence H. Fishbach				8. Performing Organization Report No. E-5867	
9. Performing Organization Name and Address Lewis Research Center National Aeronautics and Space Administration Cleveland, Ohio 44135				10. Work Unit No. 132-15	
				11. Contract or Grant No.	
12. Sponsoring Agency Name and Address National Aeronautics and Space Administration Washington, D. C. 20546				13. Type of Report and Period Covered Technical Note	
				14. Sponsoring Agency Code	
15. Supplementary Notes					
16. Abstract A computer program titled GENENG is described. The program uses component performance maps to enable the user to do analytical steady-state engine cycle calculations. Through a scaling procedure, each of the component maps can be used to represent a family of maps (different design values of pressure ratios, efficiency, weight flow, etc.). Either convergent or convergent-divergent nozzles may be used. Included is a complete FORTRAN IV listing of the program. Sample results and input explanations are shown for one-spool and two-spool turbojets and two-spool separate- and mixed-flow turbofans operating at design and off-design conditions. The computer program is available from the authors.					
17. Key Words (Suggested by Author(s)) Design Turbojet Off-design Computer code Matching Performance Turbofan				18. Distribution Statement Unclassified - unlimited	
19. Security Classif. (of this report) Unclassified		20. Security Classif. (of this page) Unclassified		21. No. of Pages 161	
				22. Price* \$3.00	

* For sale by the National Technical Information Service, Springfield, Virginia 22151

CONTENTS

	Page
SUMMARY	1
INTRODUCTION	1
THERMODYNAMIC ANALYSIS OF ENGINE TYPES	2
Two-Spool Afterburning Turbofan	3
Two-Spool Turbojet	6
One-Spool Turbojet	6
BALANCING TECHNIQUE	6
SUBROUTINES AND INPUTS FOR ENGINE PERFORMANCE CALCULATIONS	9
GENENG Subroutine Functions and Descriptions	9
Entering the Data	12
Choice of component maps - scaling laws	12
BLOCK DATA input	12
Inputs required at execution time	17
Inputs required for additional options to basic cycles	19
Means of Specifying Mode of Engine Operations	20
SAMPLE CALCULATIONS	21
Single-Spool Turbojet	21
Design point	21
Off-design	24
Two-Spool Turbojet	32
Design point	32
Off-design	33
Separate-Flow Turbofan	36
Design point	36
Off-design	38
Mixed-Flow Turbofan	40
Design point	40
Off-design	41
Design Case Without Pressure Ratio Rematch	43
CONCLUDING REMARKS	44

APPENDIXES	
A - GENENG PROGRAM LISTING	45
B - HUFF INPUT ROUTINE	137
C - SYMBOLS	141
REFERENCES	150

GENENG - A PROGRAM FOR CALCULATING DESIGN AND OFF-DESIGN

PERFORMANCE FOR TURBOJET AND TURBOFAN ENGINES

by Robert W. Koenig and Laurence H. Fishbach

Lewis Research Center

SUMMARY

A digital computer program titled GENENG is described. The original version of the computer program is titled SMOTE (SiMulation Of Turbofan Engine) and was developed by the Turbine Engine Division of the Air Force Aero Propulsion Laboratory, Wright-Patterson Air Force Base, Ohio. SMOTE is capable of calculating only turbofan design and off-design performance using specific component performance maps. GENENG calculates steady-state design and off-design turbofan and one- and two-spool turbojet engine performance. Discussed in this report are changes to SMOTE which enable the user to do jet engine calculations of a general nature, thereby allowing the mission analyst a greater freedom in selection of engine design characteristics. Typical of these changes is an automatic redesign of fan and compressor pressure ratios when the static pressures do not balance at the mix point for mixed-flow turbofans. A convergent-divergent nozzle performance map addition is shown. Other changes that permit greater flexibility for generalized cycle studies for GENENG are described.

Included as an appendix to the report is a complete FORTRAN IV listing of GENENG. Sample results and input explanations are included for two-spool turbojet, one-spool turbojet, and two-spool turbofan engines operating at design and off-design conditions.

INTRODUCTION

For preliminary as well as in-depth studies it is often necessary to study a broad range of engines operating at both design and off-design conditions in order to find an efficient airframe/engine combination. The spectrum of flight conditions through which an engine must operate will strongly affect the optimum design parameter for that engine.

The SMOTE code (SiMulation Of Turbofan Engines), discussed in references 1 and 2,

provided a computer program having off-design-point calculation capability for either existing engines or theoretical ones - a major advance. Theoretical engines are simulated by scaling component performances from existing engines to the design conditions of the theoretical engines.

GENENG (GENeralized ENGine), a computer code derived from SMOTE, was written to improve the versatility of SMOTE. Among the changes made are as follows:

- (1) One- and two-spool turbojets can be calculated, as well as turbofans.
- (2) Afterburner performance maps can be used.
- (3) Nozzle performance maps can be used.
- (4) Fan and compressor pressure ratios are automatically redesigned for mixed-flow turbofans if the static pressures at the mix point do not match.
- (5) Duct combustor pressure losses are calculated.
- (6) A new method of entering data into the program is used.

A derivative program from GENENG, called GENENG II, is reported in a companion report to this one (ref. 3). GENENG II calculates performance of two- or three-spool front- or aft-fan turbofan engines with as many as three nozzles (or airstreams).

These programs have proven to be very versatile, and minor changes to them can greatly increase the number of engine configurations that can be studied. As an example, GENENG was used to study nuclear-powered turbofan engines. This was accomplished by adding a heat-exchanger subroutine as a substitute for the combustor subroutine to simulate the use of nuclear power to raise the temperature of the air entering the first turbine.

In this report are included illustrative examples of the use of GENENG to study various one- and two-spool turbojets and separate- and mixed-flow turbofans.

GENENG is available from the authors upon request. This FORTRAN IV program can be used by computing centers having an IBM 7094 Mod 2 computer. With modifications, the program can be used on all machines that have a FORTRAN compiler.

THERMODYNAMIC ANALYSIS OF ENGINE TYPES

All thermodynamic properties of air and gas are calculated by considering variable specific heats and no dissociation. The air and gas property tables of reference 4 were curve fit and are used herein.

The following discussion presents the thermodynamic analysis of the engine cycles that can be studied using GENENG.

Two-Spool Afterburning Turbofan

The basic engine, a two-spool turbofan, is shown in figure 1. All other engine types are treated as variations of this basic engine. Free-stream conditions exist at station 1 and are determined by using the U. S. Standard Atmosphere Table of 1962 (ref. 5). The conditions at station 2 are determined by flight speed and inlet recovery.

GENENG compressor maps work with corrected values of airflow. At the entrance to the fan, the corrected airflow $WA_{F, c}$ is

$$WA_{F, c} = \frac{WA_F \sqrt{T_2/518.668}}{P_2/P_{SLS}} \quad (1)$$

where P_2 and P_{SLS} are in atmospheres and P_{SLS} equals 1.0. All symbols are defined in appendix C. Some symbols are formed as the combination of other symbols; thus WA is airflow, F is for fan, and c when following a compound symbol means corrected. Station numbers are defined on the appropriate figure.

All the fan air WA_F is compressed by the fan giving rise to conditions at station 21. The power required to do this is

$$\text{Fan power} = WA_F \times (H_{21} - H_2) \quad (2)$$

Some fan air may be lost to the cycle as fan bleed Bl_F , which is expressed as a fraction of the fan airflow

$$Bl_F = PC_{Bl, F} \times WA_F \quad (3)$$

The corrected airflow into the core compressor is

$$WA_{C, c} = \frac{WA_C \sqrt{T_{21}/518.668}}{P_{21}/1.0} \quad (4)$$

The remaining air goes through the fan duct where some leakage from the core air may also enter (see eq. (11)).

$$WA_D = WA_F - Bl_F - WA_C + Bl_{DU} \quad (5)$$

The air which may be heated by a duct burner to a temperature T_{24} undergoes a pressure drop

$$P_{25} = P_{24} \times \left[1 - \left(\frac{\Delta P}{P} \right)_{\text{DUCT}} \right] \quad (6)$$

The air would have been heated by the addition of fuel, which can be expressed as a fuel-air ratio so that

$$WG_{24} = WA_{23} \times \left[1 + (f/a)_{23} \right] \quad (7)$$

The gas is then either expanded through a nozzle (station 29) to produce thrust or is mixed with the core air as shown in figure 2 (mixed-flow turbofan). The bypass ratio of the engine is defined by

$$\text{BYPASS} = \frac{WA_D}{WA_C} \quad (8)$$

The air continuing into the core is compressed to conditions at station 3. The power required is

$$\text{Compressor power} = WA_C \times (H_3 - H_{21}) = WA_3 \times (H_3 - H_{21}) \quad (9)$$

Some core bleed air Bl_C may be used for turbine cooling. Some of the air is put back into the cycle into each of the two turbines, and some is lost to the cycle as overboard bleed or leakage into the fan duct.

$$Bl_C = PC_{Bl, C} \times WA_3 \quad (10)$$

$$Bl_{DU} = PC_{Bl, DU} \times Bl_C \quad (11)$$

$$Bl_{OB} = PC_{Bl, OB} \times Bl_C \quad (12)$$

$$Bl_{HP} = PC_{Bl, HP} \times Bl_C \quad (13)$$

$$Bl_{LP} = PC_{Bl, LP} \times Bl_C \quad (14)$$

Since $Bl_{DU} + Bl_{OB} + Bl_{HP} + Bl_{LP} = Bl_C$, the sum of $PC_{Bl, DU} + PC_{Bl, OB} + PC_{Bl, HP} + PC_{Bl, LP}$ must be equal to 1.0.

The remaining air is

$$WA_4 = WA_3 - Bl_C \quad (15)$$

and is heated to a turbine inlet temperature T_4 while undergoing a combustor pressure drop $(\Delta P/P)_{\text{COMB}}$. The fuel required to do this is expressed as a fuel-air ratio $(f/a)_4$ so that the weight of the gas entering the first (high pressure) turbine WG_4 can be expressed as

$$WG_4 = WA_4 \times [1 + (f/a)_4] \quad (16)$$

This gas is then expanded through the turbine to conditions at station 5. The enthalpy at station 5 is first calculated by making a power balance, since this turbine drives the compressor and supplies any work extracted (HPEXT). By using equation (9),

$$WG_4 \times (H_4 - H_5) = WA_3 \times (H_3 - H_{21}) + \text{HPEXT} \quad (17)$$

In addition, the physical speeds must match

$$N_{\text{HP, TURBINE}} = N_{\text{COMP}} \quad (18)$$

If high-pressure-turbine bleed air Bl_{HP} is added back into the cycle at this point, H_5 must be readjusted.

$$H_5 = \frac{(Bl_{\text{HP}} \times H_3) + WG_4 H_5}{WG_4 + Bl_{\text{HP}}} = \frac{(Bl_{\text{HP}} \times H_3) + WG_4 H_5}{WG_5} \quad (19)$$

Similarly,

$$WG_5 \times (H_5 - H_{55}) = WA_F \times (H_{21} - H_2) \quad (20)$$

$$N_{\text{LP, TURBINE}} = N_{\text{FAN}} \quad (21)$$

$$H_{55} = \frac{(Bl_{\text{LP}} \times H_3) + WG_5 H_{55}}{WG_5 + Bl_{\text{LP}}} = \frac{(Bl_{\text{LP}} \times H_3) + WG_5 H_{55}}{WG_{55}} \quad (22)$$

For non-mixed-flow turbfans, the gas flow at station 6, WG_6 , is identical with that at station 55, WG_{55} . For mixed-flow turbfans, the air in the fan duct is added.

$$WG_6 = WG_{55} + WA_D \quad (23)$$

Mixed-flow turbfans additionally require that the static pressures at station 25 and at station 55 (fig. 2) match.

$$PS_{55} = PS_{25} \quad (24)$$

The gas flow WG_6 then may be heated by an afterburner to a gas temperature T_7 and may undergo a pressure drop.

$$P_7 = P_6 \left[1 - (\Delta P/P)_{\text{AFTERBURNER}} \right] \quad (25)$$

And the gas flow rate would be increased by any fuel burned.

$$WG_7 = WG_{55} + WFA$$

The gas is then expanded through the nozzle (station 9) to produce the remainder of the engine thrust.

Two-Spool Turbojet

The two-spool turbojet is equivalent to a two-spool turbofan with a BYPASS of zero. This engine is shown in figure 3. In calculating this type of engine, there is no fan duct and the air entering the inner compressor is the same as the air entering the inlet less any bleed.

$$WA_C = WA_F - Bl_F \quad (26)$$

The thermodynamic calculations proceed identically to the previous case, the two-spool turbofan case, except that any equations referring to the fan duct are eliminated.

One-Spool Turbojet

The one-spool turbojet is shown in figure 4. As can be seen, to simulate this engine the inner compressor and its driving turbine are eliminated. That is, stations 21 and 3 become identical and stations 4 and 5 become identical.

The only calculation changes required therefore are (1) to eliminate any thermodynamic equations relating to the fan duct and the inner spool of the two-spool turbofan engine and (2) to add the horsepower extracted to the power requirements of the outer turbine.

BALANCING TECHNIQUE

An off-design engine cycle calculation requires satisfying various matching con-

straints (rotational speeds, airflows, compressor and turbine work functions, and nozzle flow functions) at each specified operating condition. GENENG internally searches for compressor and turbine operating points that will satisfy the constraints. It does this by generating differential errors caused by small changes in the independent variables. The program then uses a matrix that is loaded with the differential errors to solve for the zero-error condition. This procedure is known as the Newton-Raphson iteration technique.

For the two-spool turbfan or turbojet engines a solution for a set of six simultaneous linear algebraic equations is obtained; for the one-spool turbojet a set of three simultaneous linear equations is solved. The six independent variables selected are

ZF	Ratio of pressure ratios of outer compressor (fan) along a speed line, $ZF = \frac{(\text{Pressure ratio along speed line}) - (\text{Low pressure ratio on speed line})}{(\text{High pressure ratio on speed line}) - (\text{Low pressure ratio on speed line})}$
PCNF or T4	Percent fan speed or turbine inlet temperature
ZC	Ratio of pressure ratios of inner compressor along speed line; calculated same as ZF
PCNC or T4	Percent compressor speed or turbine inlet temperature
TFFHP	Inner (high pressure) turbine flow function, $WG_4 \sqrt{T_4/P_4}$
TFFLP	Outer (low pressure) turbine flow function, $WG_5 \sqrt{T_5/P_5}$

ZC, PCNC, and TFFHP are not used for the one-spool turbojet.

The program initially selects new (perturbed) values for the variables, based on the design values. It is then possible to proceed through the entire engine cycle, where six (or three) errors are generated. The initial values of the six (or three) variables and six (or three) errors are base values.

As per reference 1, the partial differential equations for $E = f(v)$ are

$$dE_i = \sum_{j=1}^{j_{\max}} \frac{\partial E_{ij}}{\partial V_j} dV_j \quad (27)$$

for i going from 1 to $j_{\max}$ where $j_{\max}$ is 6 for two-spool engines or 3 for one-spool turbojets; E is an error; V is a variable; and ∂E_{ij} is the change in E_i caused by a change in V_j .

The assumption of a small change in the variable results in the following approxima-

tions (B refers to a base value):

$$dE = E - EB \quad (28)$$

$$dV = V - VB \quad (29)$$

$$\frac{\partial E}{\partial V} = \frac{E}{V} \quad (30)$$

With these approximations and the knowledge that E should equal zero for the balanced engine, the set of partial differential equations (eq. (27)) reduces to

$$E_i - EB_i = \sum_{j=1}^{j_{\max}} \frac{\Delta E_{ij}}{\Delta V_j} dV_j = -EB_i \quad (31)$$

for i going from 1 to $j_{\max}$.

Thus the calculations made with the perturbed variables are used to compute $\Delta E/\Delta V$, and equation (31) is solved for dV_j . The variables V are then given new values from

$$V_j = V_j B + dV_j \quad (32)$$

If the engine cycle calculations were linear functions, the engine would balance (errors within some allowable limits) with the new values of the variables. However, this is not the case, and it is usually necessary to repeat the process of changing each variable by a small amount for each pass. A change in each error because of the small change in the variable is calculated for each pass, where the new values become base values. This process occurs several times before a balance is obtained.

A subroutine (MATRIX) to determine the solution of a matrix is used to solve the set of differential equations. After each pass through the engines, a matrix array is loaded with the appropriate values; after a number of passes equal to $1 + j_{\max}$ (base value plus perturbation on each of the independent variables), the matrix subroutine is used to solve the matrix. This solution (E within some allowable limit) yields the correct values of the independent variable and satisfies all the component matching constraints.

The most-often-used independent variable and the differential errors for four types of engines that can be run on GENENG are listed in table I.

SUBROUTINES AND INPUTS FOR ENGINE PERFORMANCE CALCULATIONS

Two forms of data are supplied to GENENG. Some data, such as all the constants and component map data, are in the form of a BLOCK DATA subprogram. The varying data are supplied at execution time by the use of input data cards.

The FORTRAN listings of GENENG are presented in appendix A. The function and description of the subroutines follows in the next section.

GENENG Subroutine Functions and Descriptions

A flow chart of the computer program with the subroutines is shown in figure 5. The functions of the subroutines are listed here and the purpose of each described.

GENENG	Dummy main program to initiate the calculations and cause the input of the controlled output variables. Because of the looping between subroutines, control is never transferred back to this routine.
ENGBAL	Main routine. Controls all engine balancing loops; checks tolerances and number of loops and loads matrix; calls input.
GUESS	Determines initial values of independent variable (see table I) at each point.
MATRIX	Solves error matrix.
PUTIN	Calls input subroutine package. Controls loop on static pressures for mixed-flow turbofan.
ZERO	Zeros nearly all of common and certain controls.
COINLT	Determines ram recovery and performs inlet calculations.
ATMOS	1962 U. S. Standard Atmosphere Table.
RAM	Calculates ram recovery defined by MIL-E-5008B specifications.
RAM2	Calculates special cases of input ram recovery as a function of flight Mach number.
COFAN	Uses BLOCK DATA to perform outer-compressor (fan) calculations.
COCOMP	Uses BLOCK DATA to perform inner-compressor calculations (two spools only).

COCOMB	Uses BLOCK DATA to perform combustor calculations. May use either T4 or WFB as the main parameter.
COHPTB	Uses BLOCK DATA to perform inner-turbine calculations (two spools only).
COLPTB	Uses BLOCK DATA to perform outer-turbine calculations.
CODUCT	Performs duct and duct-burning calculations for turbofans. May use either T24 or WFD as main parameters.
COMIX	Performs gas mixing calculations if in mixed-flow mode. At design points it calculates areas either from an input static pressure PS55 or from an input Mach number AM55 if PS55 = 0. At off-design points it calculates static pressures and Mach numbers from the design areas. Calculates ERR (5). Rescales pressure ratios for mixed-flow turbofans to match duct and core static pressures just prior to mixing. COMIX also calculates afterburner entrance area A6 as a function of afterburner entrance Mach number AM6.
COAFBN	Performs afterburning calculations. May use either T7 or WFA as the main parameters.
FRTOSD	Dummy routine to transfer values from common FRONT to common SIDE.
FASTBK	Dummy routine to transfer values from common SIDE to common BACK.
COMNOZ	Controls the main nozzle.
ERROR	Controls all printouts if an error occurs. Prints names of subroutine where error occurred and also prints the values of all variables in the main commons.
SYG	Controls printing from UNIT08. Throughout the program and particularly in ENGBAL, certain messages, variables, and matrix values are written on UNIT08 as an aid in determining why an error occurred or why a point did not balance. These values are printed out if subroutine ERROR is called and IDUMP is greater than zero, or after a good point if IDUMP = 2.
PERF	Calculates performance after the engine is balanced.
OUTPUT	Prints output except for controlled output. Prints the main commons after the design point.
CONOUT	Controls and prints the controlled output variables.

THCOMP	Performs isentropic calculations for compressors.
PROCOM	Calculates thermodynamic gas properties for either air or a fuel-air mixture based on JP-4 using curve fits of the tables of reference 4.
SEARCH	General table lookup and interpolation routine to obtain data from the BLOCK DATA subroutines.
MAPBAC	Used when calculations result in values not on the turbine maps. Changes the map value and an independent variable (PCNF, PCNC, or T4) in an attempt to rectify the situation.
CONVRG	Performs nozzle calculations for a convergent nozzle.
CONDIV	Performs nozzle calculations for a convergent-divergent (C-D) nozzle.
THTURB	Performs isentropic calculations for turbines.
THERMO	Provides thermodynamic conditions using PROCOM.
AFQUIR	General quadratic interpolation routine.
PARABO	Parabolic curve-fit routine.
BLKFAN	Performance data for outer compressor (fan) map (BLOCK DATA).
BLKCMP	Performance data for inner compressor map (BLOCK DATA; two-spool engines).
CMBDAT	BLOCK DATA for combustor.
HPTDAT	Performance data for inner turbine map (BLOCK DATA; two-spool engines).
LPTDAT	Performance data for outer turbine map (BLOCK DATA).
ETAAB	Generalized afterburner performance BLOCK DATA as a function of fuel-air ratio with correction factors for off-design afterburner entrance pressure and Mach number.
FRATIO	Convergent-divergent nozzle velocity coefficient (BLOCK DATA input as a function of nozzle pressure ratio and area expansion ratio).
INPUT	Package of Huff input subroutines. (The Huff Input Routine is a very versatile input mechanism further detailed in appendix B.)

The Huff Input Routine, used to enter input data into the program at execution time, is discussed in appendix B. Appendix C presents the individual symbols internal to the program, station numbers, etc., from which compound names such as WAFCDs (WA + F + C + DS) are formed. Table II and appendix C present the names of the variables, the values of which are supplied on data cards.

Choice of component maps - scaling laws. - Many of the engines that are studied using GENENG are theoretical. Therefore, actual component maps for these engines will be nonexistent. The program, however, does require component maps to do off-design calculations. To alleviate this problem, GENENG uses scaling laws to change data from one component map into a new component map. Hopefully, a component map can be found which could be expected to perform in a similar manner to the actual map for the engine type being studied. In fact, several maps that the authors have obtained are identified as to the range of pressure ratio and the engine component design type for which they are valid (i. e., pressure ratio range, 4 to 8; subsonic compressor or inner compressor). Thus, for example, a high-bypass-ratio, subsonic-flight-speed, low-pressure-ratio fan map for a CF6 engine would not properly simulate a low-bypass-ratio, high-pressure-ratio, supersonic multistage fan.

The scaling equations used for the compressor maps are

$$PR = \frac{PR_{\text{design}} - 1}{PR_{\text{map, design}} - 1} (PR_{\text{map}} - 1) + 1$$

$$WA = \frac{WA_{\text{design}}}{WA_{\text{map, design}}} \times WA_{\text{map}}$$

$$ETA = \frac{ETA_{\text{design}}}{ETA_{\text{map, design}}} \times ETA_{\text{map}}$$

Similar equations are used for combustor and turbine map scaling. These equations are found in the appropriate subroutines. In the output are printed the correction factors used in scaling the maps. The closer these values are to 1.0 (especially pressure ratio, a primary characteristic of a given compressor map), the more reasonable are the simulated maps of the engine. Conversely, however, not being close to 1.0 does not necessarily mean that the simulation is poor since many maps have been shown to be typical over quite a large range of variables.

BLOCK DATA input. - The two compressor performance maps are entered into the

code as the BLOCK DATA subprograms BLKFAN and BLKCMP. The subprograms supplied by the authors with the code and shown in appendix A are not to be taken as realistic maps. These maps are only of an illustrative nature and are the ones used to run the sample calculations.

Using subprogram BLKFAN as an example (the first nine cards of which are printed here) and referring to a typical compressor map (fig. 6), the data are programmed as follows:

```

$IBFTC BLKFAN  DECK
C      THIS IS A GENERALIZED FAN MAP FOR UNREALISTIC SUPERSONIC ENGINE      1
BLOCK DATA                                                                    2
COMMON / FAN/CN(15),PR(15,15),WAC(15,15),ETA(15,15),N,NP(15)                3
DATA N,NP/10,6,3*7,5*10,8,5*0/                                              4
DATA CV/0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,5*0./                      5
DATA (PR( 1,J),WAC( 1,J),ETA( 1,J),J=1, 6)/                                6
1  1.00000,  243.600,  0.75592,  1.01200,  229.800,  0.76120,                7
2  1.02800,  199.800,  0.76648,  1.03840,  166.800,  0.75592,                8
3  1.04480,  133.200,  0.72512,  1.04800,   86.400,  0.64152/                9

```

Card 1 reminds the reader that these maps are fictitious. Card 2 identifies subprogram as BLOCK DATA. Card 3 identifies common block FAN into which data are to be stored and dimensions the program variables. Card 4 indicates that there are 10 speed lines (N) and the number of points NF on each line - six on the lowest speed, seven on the next three lines, etc. Card 5 assigns the value of speed to each of the 10 lines (low to high). Cards 6 to 9 along speed line CN=0.3 set pressure ratio PR, corrected airflow WAC, and efficiency ETA in sets of three going from low pressure (PR = 1.0) to the surge line (PR = 1.048). Note there are six sets of three values (NP(1) = 6). The rest of the cards (appendix A) set the values for each speed line.

The combustor map is also a BLOCK DATA subprogram (CMBDAT). It is a plot of temperature rise across the combustor against efficiency for constant input pressure. Entry to the map is through temperature rise and input pressure with efficiency being output. The cards in the subprogram CMBDAT are reproduced here; a typical combustor map is shown in figure 7.

```

$IBFTC CMBDAT  DECK
BLOCK DATA                                                                    1
COMMON / COMB/PSI(15),DELT(15,15),ETA(15,15),N,NP(15)                      2
DATA N,NP / 15,15*15 /                                                       3
DATA PSI/4.9116,9.8232,14.735,19.646,24.558,29.470,34.381,                  4
139.293,44.207,73.674,100.,200.,300.,400.,500./                             5
DATA DELT/15*200.,15*300.,15*400.,15*500.,15*600.,15*700.,15*800.,        6
115*900.,15*1000.,15*1100.,15*1200.,15*1300.,15*1400.,15*1500.,          7
215*1500./                                                                    8
DATA ETA/                                                                      9
1.600,.726,.777,.806,.826,.843,.855,.865,7*.870,                          10
2.758,.825,.858,.875,.888,.898,.906,.912,.914,6*.915,                      11
3.868,.893,.911,.925,.935,.942,.947,.951,7*.953,                          12
4.925,.936,.946,.955,.963,.969,.974,.977,.978,6*.979,                      13
5.960,.966,.972,.977,.982,.985,.990,.992,.993,6*.995,                      14
6.988,.991,.992,.994,.995,.997,.998,8*.999,                                15
78*1.00,7*.999,120*1.00/                                                    16
END                                                                            17

```

Card 1 identifies subprogram as BLOCK DATA. Card 2 identifies common block COMB into which data are to be stored and the dimension of each variable. Card 3 indicates that there are 15 lines of constant PSI(P3) by the value of N, and that there are 15 values of DELT(DT) and ETA(ETAB) along each line of constant PSI(P3). Cards 4 and 5 assign values to each of the P3 lines from low to high pressure. Cards 6 to 8 assign values of ΔT to each of the P3 lines starting at low ΔT . The lowest value of ΔT on each of the P3 lines is given starting with the lowest value of ΔT on the lowest value of P3. Next comes the second lowest value of ΔT on each P3, etc. Again this map is unrealistic, being used for illustrative purposes only. Cards 9 to 16 assign the value of η_B in a one-to-one correspondence with the ΔT values just assigned. The order is the same. The combustor pressure loss is input as a design value (i. e., DPCODS = 0.05) for a combustor type being considered. During off-design operation, the pressure loss is related to the design value by the following equation: $DPCOM = DPCODS \times (WA3C/WA3CDS)$. This equation is found in the COCOMB routine.

Also entered as BLOCK DATA subprograms are the turbine maps (HPTDAT and LPTDAT). To illustrate the entering of turbine data, LPTDAT is used. A typical turbine map is shown in figure 8; the data are programmed as follows:

```

$IBFTC LPTDAT  DECK
BLOCK DATA
COMMON / LTURB/TFF(15),CN(15,15),DH(15,15),ETA(15,15),N,NP(15)
DATA N,NP/11,9*15,12,9,4*0/
DATA TFF / 88.470, 102.795, 116.835, 129.330, 141.045,
1 145.725, 150.000, 153.345, 156.405, 159.780, 163.170,4*0./
DATA (CN( 1,J),DH( 1,J),ETA( 1,J),J=1,15)/
1 0.3582, 0.0018, 0.7120, 0.5336, 0.0026, 0.7300,
2 0.7365, 0.0035, 0.7472, 0.9754, 0.0044, 0.7300,
3 1.2146, 0.0051, 0.7140, 1.4173, 0.0056, 0.7000,
4 1.5201, 0.0059, 0.6850, 1.7673, 0.0061, 0.6730,
5 2.3247, 0.0062, 0.6452, 2.2827, 0.0061, 0.6200,
6 2.4565, 0.0057, 0.6000, 2.6137, 0.0053, 0.5750,
7 2.3166, 0.0044, 0.5310, 2.9456, 0.0035, 0.5000,
8 3.3138, 0.0001, 0.3850/

```

Card 1 identifies the subprogram as BLOCK DATA. Card 2 identifies the common block into which data are to be loaded and dimensions the program variables. Card 3 indicates the number of constant turbine flow function lines TFF as 11(N) and the number of points on each line from low to high TFF. Cards 4 and 5 set values of TFF from low to high. Cards 6 to 14 set values of corrected speed CN, work function DH, and efficiency ETA along TFF(1) starting from low CN(0.3682) and ending at high CN(3.3138). The rest of the cards set values along higher TFF lines.

In many cases, turbine maps for high-performance engines operate at a choked condition (constant TFF). Thus, a turbine map to be represented could possibly have no lines representing constant TFF for a significant portion of the map. For complete map representation, lines of constant TFF may be estimated on the map up to the limit load-

ing line by inputting slight changes for the values of TFF (example: one line for TFF is 62.105, the next may be input as equal to 62.108). This will eliminate computational difficulties which would arise if constant values for TFF lines were input.

Afterburner performance has been programmed in a generalized form in subroutine COAFBN. The afterburner performance map included in the program is shown in figure 9(a). The performance map shows afterburner combustion efficiency ratio as a function of fuel-air ratio. The value of afterburner combustion efficiency correction factor $\Delta ETAA$ during off-design operation is shown against design afterburner inlet Mach number ratio (fig. 9(b)) and design afterburner inlet total-pressure ratio (fig. 9(c)). Other correction factors or performance maps could be added as desired. The afterburner efficiency, fuel-air ratio, inlet total pressure, and Mach number are generalized external to the program.

A specific afterburner performance is generalized by dividing the specific off-design values by the design values, as shown here. The symbols shown are the symbols used in the ABETTA subroutine, where the generalized and specific values are input. The generalized afterburner values are obtained as follows:

$$\text{Efficiency (ETABRT)} = \frac{ETAA}{ETAADS}$$

$$\text{Fuel-air ratio (FART)} = \frac{FART}{FARTDS}$$

$$\text{Entrance total pressure (P6T)} = \frac{P6}{P6DS}$$

$$\text{Entrance Mach number (EM6T)} = \frac{AM6}{AM6DS}$$

However, the correction factor for efficiency $\Delta ETAA$ is not a generalized value. The value of $\Delta ETAA$ is an efficiency degradation. These degradations are input in ABETTA as discrete points on a curve. The values of $\Delta ETAA$ are input as functions of the following:

(1) The change in efficiency as a function of EM6T is input as DELM6 (which is really $\Delta ETAA = f(AM6)$).

(2) The change in efficiency as a function of P6T is input as DELP6 (which is really $\Delta ETAA = f(P6)$).

At execution time for the design point, values of afterburner combustion efficiency ETAADS, exit total temperature T7DS, and entrance Mach number AM6DS are input. Design fuel-air ratio and entrance pressure ratio are calculated from the input values

and the other design engine characteristics.

The afterburner pressure loss is input as a design value (DPAFDS) during the design case. During afterburning the pressure loss due to combustion is calculated and the regular pressure drop is included for a total afterburner pressure loss. The equations for off-design and combustion pressure losses are contained in the COAFBN subroutines.

To achieve a reasonable accuracy in cycle calculations when using any generalized component map, the usage of the map should be limited within a certain range of the original design values and configuration changes. Therefore if, for example, an afterburner has a design task that differs significantly from an example used, a new generalized performance map should be used in order to simulate the component more accurately.

SMOTE, the original code, uses a single-point input for nozzle velocity coefficients (CVMNOZ and CVDNOZ) when calculating engine performance. GENENG, however, uses a convergent-divergent nozzle velocity coefficient which is input in map form in the FRATIO subroutine. The velocity coefficient is input as a function of nozzle total-pressure ratio (P_8/P_1 or P_{28}/P_1). The data for the nozzle are programmed as shown below for a performance map similar to that shown in figure 10.

```

$IBFTC FRATIO DECK
SUBROUTINE FRASHD (PRATIO,ARATIO,CF)
DIMENSION PR(30),AR(10),CFR(30,10),XX(3),YY(3),ZZ(2)
DATA PR/1.0,1.25,1.5,1.75,2.0,2.25,2.5,2.75,3.0,3.25,3.5,3.75,4.0,
14.25,4.5,4.75,5.0,5.25,5.5,5.75,6.0,6.25,6.5,6.75,7.0,7.25,7.5,7.75,8.0,8.25,8.5,8.75,9.0,9.25,9.5,9.75,10.0,10.25,10.5,10.75,11.0,11.25,11.5,11.75,12.0,12.25,12.5,12.75,13.0,13.25,13.5,13.75,14.0,14.25,14.5,14.75,15.0,15.25,15.5,15.75,16.0,16.25,16.5,16.75,17.0,17.25,17.5,17.75,18.0,18.25,18.5,18.75,19.0,19.25,19.5,19.75,20.0,20.25,20.5,20.75,21.0,21.25,21.5,21.75,22.0,22.25,22.5,22.75,23.0,23.25,23.5,23.75,24.0,24.25,24.5,24.75,25.0,25.25,25.5,25.75,26.0,26.25,26.5,26.75,27.0,27.25,27.5,27.75,28.0,28.25,28.5,28.75,29.0,29.25,29.5,29.75,30.0,30.25,30.5,30.75,31.0,31.25,31.5,31.75,32.0,32.25,32.5,32.75,33.0,33.25,33.5,33.75,34.0,34.25,34.5,34.75,35.0,35.25,35.5,35.75,36.0,36.25,36.5,36.75,37.0,37.25,37.5,37.75,38.0,38.25,38.5,38.75,39.0,39.25,39.5,39.75,40.0,40.25,40.5,40.75,41.0,41.25,41.5,41.75,42.0,42.25,42.5,42.75,43.0,43.25,43.5,43.75,44.0,44.25,44.5,44.75,45.0,45.25,45.5,45.75,46.0,46.25,46.5,46.75,47.0,47.25,47.5,47.75,48.0,48.25,48.5,48.75,49.0,49.25,49.5,49.75,50.0,50.25,50.5,50.75,51.0,51.25,51.5,51.75,52.0,52.25,52.5,52.75,53.0,53.25,53.5,53.75,54.0,54.25,54.5,54.75,55.0,55.25,55.5,55.75,56.0,56.25,56.5,56.75,57.0,57.25,57.5,57.75,58.0,58.25,58.5,58.75,59.0,59.25,59.5,59.75,60.0,60.25,60.5,60.75,61.0,61.25,61.5,61.75,62.0,62.25,62.5,62.75,63.0,63.25,63.5,63.75,64.0,64.25,64.5,64.75,65.0,65.25,65.5,65.75,66.0,66.25,66.5,66.75,67.0,67.25,67.5,67.75,68.0,68.25,68.5,68.75,69.0,69.25,69.5,69.75,70.0,70.25,70.5,70.75,71.0,71.25,71.5,71.75,72.0,72.25,72.5,72.75,73.0,73.25,73.5,73.75,74.0,74.25,74.5,74.75,75.0,75.25,75.5,75.75,76.0,76.25,76.5,76.75,77.0,77.25,77.5,77.75,78.0,78.25,78.5,78.75,79.0,79.25,79.5,79.75,80.0,80.25,80.5,80.75,81.0,81.25,81.5,81.75,82.0,82.25,82.5,82.75,83.0,83.25,83.5,83.75,84.0,84.25,84.5,84.75,85.0,85.25,85.5,85.75,86.0,86.25,86.5,86.75,87.0,87.25,87.5,87.75,88.0,88.25,88.5,88.75,89.0,89.25,89.5,89.75,90.0,90.25,90.5,90.75,91.0,91.25,91.5,91.75,92.0,92.25,92.5,92.75,93.0,93.25,93.5,93.75,94.0,94.25,94.5,94.75,95.0,95.25,95.5,95.75,96.0,96.25,96.5,96.75,97.0,97.25,97.5,97.75,98.0,98.25,98.5,98.75,99.0,99.25,99.5,99.75,100.0,100.25,100.5,100.75,101.0,101.25,101.5,101.75,102.0,102.25,102.5,102.75,103.0,103.25,103.5,103.75,104.0,104.25,104.5,104.75,105.0,105.25,105.5,105.75,106.0,106.25,106.5,106.75,107.0,107.25,107.5,107.75,108.0,108.25,108.5,108.75,109.0,109.25,109.5,109.75,110.0,110.25,110.5,110.75,111.0,111.25,111.5,111.75,112.0,112.25,112.5,112.75,113.0,113.25,113.5,113.75,114.0,114.25,114.5,114.75,115.0,115.25,115.5,115.75,116.0,116.25,116.5,116.75,117.0,117.25,117.5,117.75,118.0,118.25,118.5,118.75,119.0,119.25,119.5,119.75,120.0,120.25,120.5,120.75,121.0,121.25,121.5,121.75,122.0,122.25,122.5,122.75,123.0,123.25,123.5,123.75,124.0,124.25,124.5,124.75,125.0,125.25,125.5,125.75,126.0,126.25,126.5,126.75,127.0,127.25,127.5,127.75,128.0,128.25,128.5,128.75,129.0,129.25,129.5,129.75,130.0,130.25,130.5,130.75,131.0,131.25,131.5,131.75,132.0,132.25,132.5,132.75,133.0,133.25,133.5,133.75,134.0,134.25,134.5,134.75,135.0,135.25,135.5,135.75,136.0,136.25,136.5,136.75,137.0,137.25,137.5,137.75,138.0,138.25,138.5,138.75,139.0,139.25,139.5,139.75,140.0,140.25,140.5,140.75,141.0,141.25,141.5,141.75,142.0,142.25,142.5,142.75,143.0,143.25,143.5,143.75,144.0,144.25,144.5,144.75,145.0,145.25,145.5,145.75,146.0,146.25,146.5,146.75,147.0,147.25,147.5,147.75,148.0,148.25,148.5,148.75,149.0,149.25,149.5,149.75,150.0,150.25,150.5,150.75,151.0,151.25,151.5,151.75,152.0,152.25,152.5,152.75,153.0,153.25,153.5,153.75,154.0,154.25,154.5,154.75,155.0,155.25,155.5,155.75,156.0,156.25,156.5,156.75,157.0,157.25,157.5,157.75,158.0,158.25,158.5,158.75,159.0,159.25,159.5,159.75,160.0,160.25,160.5,160.75,161.0,161.25,161.5,161.75,162.0,162.25,162.5,162.75,163.0,163.25,163.5,163.75,164.0,164.25,164.5,164.75,165.0,165.25,165.5,165.75,166.0,166.25,166.5,166.75,167.0,167.25,167.5,167.75,168.0,168.25,168.5,168.75,169.0,169.25,169.5,169.75,170.0,170.25,170.5,170.75,171.0,171.25,171.5,171.75,172.0,172.25,172.5,172.75,173.0,173.25,173.5,173.75,174.0,174.25,174.5,174.75,175.0,175.25,175.5,175.75,176.0,176.25,176.5,176.75,177.0,177.25,177.5,177.75,178.0,178.25,178.5,178.75,179.0,179.25,179.5,179.75,180.0,180.25,180.5,180.75,181.0,181.25,181.5,181.75,182.0,182.25,182.5,182.75,183.0,183.25,183.5,183.75,184.0,184.25,184.5,184.75,185.0,185.25,185.5,185.75,186.0,186.25,186.5,186.75,187.0,187.25,187.5,187.75,188.0,188.25,188.5,188.75,189.0,189.25,189.5,189.75,190.0,190.25,190.5,190.75,191.0,191.25,191.5,191.75,192.0,192.25,192.5,192.75,193.0,193.25,193.5,193.75,194.0,194.25,194.5,194.75,195.0,195.25,195.5,195.75,196.0,196.25,196.5,196.75,197.0,197.25,197.5,197.75,198.0,198.25,198.5,198.75,199.0,199.25,199.5,199.75,200.0,200.25,200.5,200.75,201.0,201.25,201.5,201.75,202.0,202.25,202.5,202.75,203.0,203.25,203.5,203.75,204.0,204.25,204.5,204.75,205.0,205.25,205.5,205.75,206.0,206.25,206.5,206.75,207.0,207.25,207.5,207.75,208.0,208.25,208.5,208.75,209.0,209.25,209.5,209.75,210.0,210.25,210.5,210.75,211.0,211.25,211.5,211.75,212.0,212.25,212.5,212.75,213.0,213.25,213.5,213.75,214.0,214.25,214.5,214.75,215.0,215.25,215.5,215.75,216.0,216.25,216.5,216.75,217.0,217.25,217.5,217.75,218.0,218.25,218.5,218.75,219.0,219.25,219.5,219.75,220.0,220.25,220.5,220.75,221.0,221.25,221.5,221.75,222.0,222.25,222.5,222.75,223.0,223.25,223.5,223.75,224.0,224.25,224.5,224.75,225.0,225.25,225.5,225.75,226.0,226.25,226.5,226.75,227.0,227.25,227.5,227.75,228.0,228.25,228.5,228.75,229.0,229.25,229.5,229.75,230.0,230.25,230.5,230.75,231.0,231.25,231.5,231.75,232.0,232.25,232.5,232.75,233.0,233.25,233.5,233.75,234.0,234.25,234.5,234.75,235.0,235.25,235.5,235.75,236.0,236.25,236.5,236.75,237.0,237.25,237.5,237.75,238.0,238.25,238.5,238.75,239.0,239.25,239.5,239.75,240.0,240.25,240.5,240.75,241.0,241.25,241.5,241.75,242.0,242.25,242.5,242.75,243.0,243.25,243.5,243.75,244.0,244.25,244.5,244.75,245.0,245.25,245.5,245.75,246.0,246.25,246.5,246.75,247.0,247.25,247.5,247.75,248.0,248.25,248.5,248.75,249.0,249.25,249.5,249.75,250.0,250.25,250.5,250.75,251.0,251.25,251.5,251.75,252.0,252.25,252.5,252.75,253.0,253.25,253.5,253.75,254.0,254.25,254.5,254.75,255.0,255.25,255.5,255.75,256.0,256.25,256.5,256.75,257.0,257.25,257.5,257.75,258.0,258.25,258.5,258.75,259.0,259.25,259.5,259.75,260.0,260.25,260.5,260.75,261.0,261.25,261.5,261.75,262.0,262.25,262.5,262.75,263.0,263.25,263.5,263.75,264.0,264.25,264.5,264.75,265.0,265.25,265.5,265.75,266.0,266.25,266.5,266.75,267.0,267.25,267.5,267.75,268.0,268.25,268.5,268.75,269.0,269.25,269.5,269.75,270.0,270.25,270.5,270.75,271.0,271.25,271.5,271.75,272.0,272.25,272.5,272.75,273.0,273.25,273.5,273.75,274.0,274.25,274.5,274.75,275.0,275.25,275.5,275.75,276.0,276.25,276.5,276.75,277.0,277.25,277.5,277.75,278.0,278.25,278.5,278.75,279.0,279.25,279.5,279.75,280.0,280.25,280.5,280.75,281.0,281.25,281.5,281.75,282.0,282.25,282.5,282.75,283.0,283.25,283.5,283.75,284.0,284.25,284.5,284.75,285.0,285.25,285.5,285.75,286.0,286.25,286.5,286.75,287.0,287.25,287.5,287.75,288.0,288.25,288.5,288.75,289.0,289.25,289.5,289.75,290.0,290.25,290.5,290.75,291.0,291.25,291.5,291.75,292.0,292.25,292.5,292.75,293.0,293.25,293.5,293.75,294.0,294.25,294.5,294.75,295.0,295.25,295.5,295.75,296.0,296.25,296.5,296.75,297.0,297.25,297.5,297.75,298.0,298.25,298.5,298.75,299.0,299.25,299.5,299.75,300.0,300.25,300.5,300.75,301.0,301.25,301.5,301.75,302.0,302.25,302.5,302.75,303.0,303.25,303.5,303.75,304.0,304.25,304.5,304.75,305.0,305.25,305.5,305.75,306.0,306.25,306.5,306.75,307.0,307.25,307.5,307.75,308.0,308.25,308.5,308.75,309.0,309.25,309.5,309.75,310.0,310.25,310.5,310.75,311.0,311.25,311.5,311.75,312.0,312.25,312.5,312.75,313.0,313.25,313.5,313.75,314.0,314.25,314.5,314.75,315.0,315.25,315.5,315.75,316.0,316.25,316.5,316.75,317.0,317.25,317.5,317.75,318.0,318.25,318.5,318.75,319.0,319.25,319.5,319.75,320.0,320.25,320.5,320.75,321.0,321.25,321.5,321.75,322.0,322.25,322.5,322.75,323.0,323.25,323.5,323.75,324.0,324.25,324.5,324.75,325.0,325.25,325.5,325.75,326.0,326.25,326.5,326.75,327.0,327.25,327.5,327.75,328.0,328.25,328.5,328.75,329.0,329.25,329.5,329.75,330.0,330.25,330.5,330.75,331.0,331.25,331.5,331.75,332.0,332.25,332.5,332.75,333.0,333.25,333.5,333.75,334.0,334.25,334.5,334.75,335.0,335.25,335.5,335.75,336.0,336.25,336.5,336.75,337.0,337.25,337.5,337.75,338.0,338.25,338.5,338.75,339.0,339.25,339.5,339.75,340.0,340.25,340.5,340.75,341.0,341.25,341.5,341.75,342.0,342.25,342.5,342.75,343.0,343.25,343.5,343.75,344.0,344.25,344.5,344.75,345.0,345.25,345.5,345.75,346.0,346.25,346.5,346.75,347.0,347.25,347.5,347.75,348.0,348.25,348.5,348.75,349.0,349.25,349.5,349.75,350.0,350.25,350.5,350.75,351.0,351.25,351.5,351.75,352.0,352.25,352.5,352.75,353.0,353.25,353.5,353.75,354.0,354.25,354.5,354.75,355.0,355.25,355.5,355.75,356.0,356.25,356.5,356.75,357.0,357.25,357.5,357.75,358.0,358.25,358.5,358.75,359.0,359.25,359.5,359.75,360.0,360.25,360.5,360.75,361.0,361.25,361.5,361.75,362.0,362.25,362.5,362.75,363.0,363.25,363.5,363.75,364.0,364.25,364.5,364.75,365.0,365.25,365.5,365.75,366.0,366.25,366.5,366.75,367.0,367.25,367.5,367.75,368.0,368.25,368.5,368.75,369.0,369.25,369.5,369.75,370.0,370.25,370.5,370.75,371.0,371.25,371.5,371.75,372.0,372.25,372.5,372.75,373.0,373.25,373.5,373.75,374.0,374.25,374.5,374.75,375.0,375.25,375.5,375.75,376.0,376.25,376.5,376.75,377.0,377.25,377.5,377.75,378.0,378.25,378.5,378.75,379.0,379.25,379.5,379.75,380.0,380.25,380.5,380.75,381.0,381.25,381.5,381.75,382.0,382.25,382.5,382.75,383.0,383.25,383.5,383.75,384.0,384.25,384.5,384.75,385.0,385.25,385.5,385.75,386.0,386.25,386.5,386.75,387.0,387.25,387.5,387.75,388.0,388.25,388.5,388.75,389.0,389.25,389.5,389.75,390.0,390.25,390.5,390.75,391.0,391.25,391.5,391.75,392.0,392.25,392.5,392.75,393.0,393.25,393.5,393.75,394.0,394.25,394.5,394.75,395.0,395.25,395.5,395.75,396.0,396.25,396.5,396.75,397.0,397.25,397.5,397.75,398.0,398.25,398.5,398.75,399.0,399.25,399.5,399.75,400.0,400.25,400.5,400.75,401.0,401.25,401.5,401.75,402.0,402.25,402.5,402.75,403.0,403.25,403.5,403.75,404.0,404.25,404.5,404.75,405.0,405.25,405.5,405.75,406.0,406.25,406.5,406.75,407.0,407.25,407.5,407.75,408.0,408.25,408.5,408.75,409.0,409.25,409.5,409.75,410.0,410.25,410.5,410.75,411.0,411.25,411.5,411.75,412.0,412.25,412.5,412.75,413.0,413.25,413.5,413.75,414.0,414.25,414.5,414.75,415.0,415.25,415.5,415.75,416.0,416.25,416.5,416.75,417.0,417.25,417.5,417.75,418.0,418.25,418.5,418.75,419.0,419.25,419.5,419.75,420.0,420.25,420.5,420.75,421.0,421.25,421.5,421.75,422.0,422.25,422.5,422.75,423.0,423.25,423.5,423.75,424.0,424.25,424.5,424.75,425.0,425.25,425.5,425.75,426.0,426.25,426.5,426.75,427.0,427.25,427.5,427.75,428.0,428.25,428.5,428.75,429.0,429.25,429.5,429.75,430.0,430.25,430.5,430.75,431.0,431.25,431.5,431.75,432.0,432.25,432.5,432.75,433.0,433.25,433.5,433.75,434.0,434.25,434.5,434.75,435.0,435.25,435.5,435.75,436.0,436.25,436.5,436.75,437.0,437.25,437.5,437.75,438.0,438.25,438.5,438.75,439.0,439.25,439.5,439.75,440.0,440.25,440.5,440.75,441.0,441.25,441.5,441.75,442.0,442.25,442.5,442.75,443.0,443.25,443.5,443.75,444.0,444.25,444.5,444.75,445.0,445.25,445.5,445.75,446.0,446.25,446.5,446.75,447.0,447.25,447.5,447.75,448.0,448.25,448.5,448.75,449.0,449.25,449.5,449.75,450.0,450.25,450.5,450.75,451.0,451.25,451.5,451.75,452.0,452.25,452.5,452.75,453.0,453.25,453.5,453.75,454.0,454.25,454.5,454.75,455.0,455.25,455.5,455.75,456.0,456.25,456.5,456.75,457.0,457.25,457.5,457.75,458.0,458.25,458.5,458.75,459.0,459.25,459.5,459.75,460.0,460.25,460.5,460.75,461.0,461.25,461.5,461.75,462.0,462.25,462.5,462.75,463.0,463.25,463.5,463.75,464.0,464.25,464.5,464.75,465.0,465.25,465.5,465.75,466.0,466.25,466.5,466.75,467.0,467.25,467.5,467.75,468.0,468.25,468.5,468.75,469.0,469.25,469.5,469.75,470.0,470.25,470.5,470.75,471.0,471.25,471.5,471.75,472.0,472.25,472.5,472.75,473.0,473.25,473.5,473.75,474.0,474.2
```

ratios going from low to high.

After the data input, the equations required to interpolate between points of the input values are included in the FRATIO subroutine.

Usually, a turbojet or a mixed-flow turbofan has a single exhaust nozzle; therefore the velocity coefficient is determined as a function of $A9/A8$ and $P9/P1$. A separate-flow turbofan is considered to have two separate nozzles; thus in addition to using $A9/A8$ and $P9/P1$, it will use the duct $A29/A28$ and $P29/P28$ to determine a velocity coefficient to correct each stream separately. When desired, however, a different nozzle map (i. e., convergent nozzle) or single-point inputs can be readily incorporated.

Inputs required at execution time. - Basically, what must be supplied are a list of the desired output variables, design values of any component existing in the engine (compressors, combustors, turbines, etc.), and engine operational controls. The variables that are to be output are selected by the first section of data cards. Any variables that are in one of the main commons (DESIGN, FRONT, SIDE, BACK, or DUMMYS) may be selected for output by punching, in columns 1 to 6, the name of the variable as it appears in the common. Up to 150 variables (25 lines of six variables) may be chosen for a particular run. During the output phase, the name of the variable is printed out, with its value printed immediately below the name.

Another feature of the controlled output is the ability to change the name of the variable to be output; for example, it may be desired to change a station designation to one more common to a particular programmer. In this case, the variable name would be punched in columns 1 to 6 as previously described, but, in addition, the described name would be punched in columns 13 to 18. Special symbols, such as /, may be used in the new name. The last card of the controlled output must be a card with "THEEND" punched in columns 1 to 6.

Table II summarizes the design inputs for the four basic engines discussed in this report.

The following control variables should always be supplied at the design point. The value used is dependent on how the user wants the engine to operate. The symbols and their purpose are listed in subroutine PUTIN but are shown here for the reader's convenience. The superscripts (1) to (4) on the symbols have the following meanings: (1) means "automatically returned to zero after each point is calculated, must be re-input if option is again desired"; (2) means "option can be used for design or off-design", whereas the other two modes can only be used at off-design; (3) means "these input values remain as input unless changed by a new input"; (4) means "before using these options which are not equal to zero, a setup case must be run where all the components are first matched, and then the identical case may be repeated exercising these options."

IDES = 1 ⁽¹⁾	For calculating design points.
(2)MODE = 0 ⁽³⁾	Specify T4.
MODE = 1 ⁽³⁾	Specify PCNC.
(2)MODE = 2 ⁽³⁾	Specify WFB.
MODE = 3 ⁽³⁾	Specify PCNF.
INIT = 0 ⁽³⁾	Initializes point.
INIT = 1 ⁽³⁾	Will not initialize point.
IDUMP = 0 ⁽³⁾	Will not dump looping write-outs.
IDUMP = 1 ⁽³⁾	Will dump looping write-outs if error occurs.
IDUMP = 2 ⁽³⁾	Will dump looping write-outs after every point.
IAMTP = 0 ⁽³⁾	Will use input AM and military specification ETAR.
IAMTP = 1 ⁽³⁾	Will use input AM and input ETAR.
IAMTP = 2 ⁽³⁾	Will use input T2 to determine $T1 = 518.668 + T2$ and standard P1. (T2 value needs to be input at every point or an error will occur whenever used.)
IAMTP = 3 ⁽³⁾	Will use inputs P2 and standard T1.
IAMTP = 4 ⁽³⁾	Will use input T2 and input P2.
IAMTP = 5 ⁽³⁾	Will use specific schedule of ETAR located in subroutine RAMTWO.
IGASM ^X = -1 ⁽³⁾	Separate flow, input A6.
IGASM ^X = 0 ⁽³⁾	Separate flow, A6 = A55.
IGASM ^X = 1 ⁽³⁾	Will mix fan duct and main streams, A6 = A25 + A55.
IGASM ^X = 2 ⁽³⁾	Will mix fan duct and main streams, input A6.
IDBURN = 1 ⁽¹⁾	For duct burning (fan stream only), input T24. ⁽⁴⁾
IDBURN = 2 ⁽¹⁾	For duct burning (fan stream only), input WFD. ⁽⁴⁾

IAFTBN = 1 ⁽¹⁾	For afterburning (main stream or mixed stream of fan and main stream), input T7. ⁽⁴⁾
IAFTBN = 2 ⁽¹⁾	For afterburning (main stream or mixed stream of fan and main stream), input WFA. ⁽⁴⁾
IDCD = 1 ⁽³⁾	Fan duct nozzle will be convergent-divergent.
IMCD = 1 ⁽³⁾	Main nozzle will be convergent-divergent.
NOZFLT = 1 ⁽³⁾	For floating main nozzle exit area. ⁽⁴⁾
NOZFLT = 2 ⁽³⁾	For floating fan duct nozzle exit area. ⁽⁴⁾
NOZFLT = 3 ⁽³⁾	For floating fan duct and main nozzle exit area. ⁽⁴⁾
ITRYS = N	Number of passes through engine before quitting.
TOLALL = X	Tolerance which the errors must satisfy before engine is matched.

The following are other input variables for which some value, depending on the engine design, should be input at the discretion of the user:

DELFG, DELFN, DELSFC	Normally input as 1.0 unless correction is desired.
A6, AM55, AM23, AM6, HPEXT, AM, ALTP	See appendix C, Input Symbols.
PCBLF, PCBLC, PCBLDU, PCBLOB	Value for bleed out of the cycle; decimal equivalent of percent compressor flow.
PCBLHP, PCBLLP	Value of total bleed returned to turbines for cycle; fractional equivalent of flow. The sum of these variables plus PCBLDU and PCBLOB should equal 1.

Inputs required for additional options to basic cycles. - To run duct-burning (available only in fan stream duct) cases load ETAD, T24 or WFDDS, and DPDUDS. To run afterburning (mixed-flow fan or unmixed fan - available only for core and fan stream) cases, load T7DS or WFADS, ETAADS, and DPAFDS. Afterburner operation is the same as in reference 2 with the exception of a generalized afterburner performance map addition. For changing the generalized map to a specific map for a specific engine design, the preceding design values are needed at the design point.

Means of Specifying Mode of Engine Operations

Shown in the section SAMPLE CALCULATIONS (pp. 21 to 46) are the methods of specifying off-design operation points. The most common one and that used exclusively herein is to select a Mach number, altitude, and turbine inlet temperature other than design values. There are, however, several other possibilities which the user may employ. For example, changing the following controls:

MODE = 0 Specify a new turbine inlet temperature T4.

MODE = 1 Specify a compressor rotational speed PCNC.

MODE = 2 Specify a fuel flow rate WFB.

MODE = 3 Specify a fan rotational speed PCNF.

If the engine has all its nozzles fixed, an input such as turbine inlet temperature, fuel flow, or speed will set the thrust level. But other means of changing engine operation can be accomplished by varying such nozzle thrust areas as

A8 Main nozzle thrust area

A28 Fan nozzle thrust area

For example, an off-design condition may exist where, in an attempt to satisfy continuity of mass flow (one of the component matching requirements), the fan operating point may lie outside the limits of the data map that was input for the component map. A fan nozzle thrust area change could be used to return the fan operating condition on the map such that a match would occur. This would indicate a possibility exists that variable fan nozzle would be required on this engine for operation at the desired condition. The area is changed by inputting (example: $A28 = 1.2 \times A28$). Since the design areas are not known prior to running the design point, the Huff Input Routine provides the versatility in which A28 is increased by 20 percent, as was shown. It should be noted that any area changed remains changed until it is recalculated by a new design case or altered by a new input. The preceding example and statements would also apply if changes were made instead to A8.

The nozzle exit area (such as A9 or A29) may then be fully expanded (if A8 and A28 are sonic) after component matching is achieved by using the control variables NOZFLT = 1, 2, or 3. The significance of these values was explained in the previous section.

SAMPLE CALCULATIONS

Shown in this section are complete listings of sample calculations for a single-spool turbojet, a two-spool turbojet, a separate-flow turbofan, and a mixed-flow turbofan. Detailed instructions for operating the program are disclosed through means of these sample calculations plus comments in the program listing (appendix A) and the PUTIN routine (pp. 50 to 52). The input and output data associated with one complete design and off-design computer run are shown.

Single-Spool Turbojet

Design point. - This case is for design-point operation of a single-spool turbojet engine and is represented by three groups of data printout: (1) card images of input data, (2) results, and (3) the common variables printed out only at the design point.

For this point, the component maps were loaded as BLOCK DATA and scaled to maps for the desired engine design. The common variables appearing in the printout are defined by the relative locations of the common blocks of DESIGN, FRONT, SIDE, and BACK. The code for the common variables shown on pages 23 and 24 is as follows:

```

ZF, PCNF, ZC, PCNC, T4, MODE

COMMON /DESIGN/
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC,
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPFC,DHLPFC,T21DS ,
7T24DS ,WFDDDS ,DTDUODS,ETADDS,WA23DS,DPDUODS,DTDUFC,ETADCF,
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ,CVMNOZ,ABSAV ,A9SAV ,A28SAV,A29SAV
COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLJB,PCBLHP,PCBLLP
COMMON / SIDE/
1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS,DUMS3 ,
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29
COMMON / BACK/
1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,
2XWFB ,XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB ,
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC

```

It should be noted that the program uses fixed effective areas calculated at the design point. The design case which calculates the area uses a convergent nozzle. Thus, as seen on the listing (pp. 23 and 24), $A_8 = A_9$. Before an afterburning case or before the nozzle is allowed to float (become fully expanded to a convergent-divergent nozzle), the engine components must first be matched in a setup run. Then, the next case may be run with IAF TBN or NOZFLT control cards unequal to zero and may be used as shown in the off-design cases 4 and 5 or 6 and 7. Either a convergent or a convergent-divergent subroutine may be used, depending on the input controls. If afterburning has been selected, the nozzle area is allowed to float to satisfy continuity. However, the areas are returned to their original design values after the afterburning or nozzle float points are completed.

Because the first example is a single-spool turbojet, values for the inner compressor and turbine, as well as duct parameters, are either zero or 1. The actual or corrected values of compressor exit flow (WAC or WACC, respectively) are printed out prior to any bleed extraction. The value of WA3 is the combustor entrance airflow after bleed extraction.

The first 11 lines of the following example show the storage locations reserved for the input data. The locations are relative to the common block locations shown on page 21. Only locations for which values of data will be input are required. Following the end of the storage locations are the input data which are typical for running a single-spool turbojet engine.

```
$DATA(1) *1 INPUT DATA CARDS
$TABLE(.REAL,22=TOLALL,34=DELFG,35=DELFN,36=DELSFC,37=ZFDS,38=PCNFDS,39=PRFDS,
40=ETAFDS,41=WAFDS,45=ZCDS,46=PCNCDS,47=PRCDS,48=ETACDS,49=WACDS,53=T4DS,54=
WFBDS,55=DTCDSD,56=ETABDS,58=DPCDSD,61=TFHPDS,62=CNHPDS,63=ETHPDS,69=TFPLDS,
70=CNLPDS,71=ETLPDS,82=DPPDSD,85=T7DS,88=ETAADS,90=DPAFDS,95=A6,97=A8,101=PS55,
102=AM55,103=CVDNOZ,104=CVMNOZ,113=T2,114=P2,125=T4,144=WAFCDSD,152=WACCDSD,
172=HPEXT,173=AM,174=ALTP,175=ETAR,177=PCNF,179=PCNC,183=PCBLF,184=PCBLC,
185=PCBLDU,186=PCBLOB,187=PCBLHP,188=PCBLLP,205=T24,225=ETAD,257=T7,270=WFA,
273=ETAA,279=AM6,313=AM23,99=A28,180=WFB,222=WFD,.INTEGER.,2=IDES,5=MODE
,7=IDUMP,8=IAMTP,9=IGASMX,10=IDBURN,11=IAFTBN,12=IDCD,13=IMCD,16=NOZFLT,17=
ITRYS,314=ISPOOL,.LOGICAL.,319=FAN) * END OF TABLE GIVING NAMES AND LOCATIONS

*0 INPJT DATA TO RUN ONE SPOOL TURBOJET
FAN=.FALSE.,ISPOOL=1,ZFDS=.83333333,PRFDS=8,WAFCDSD=633,ETAFDS=.829,PCNFDS=100,
PCBLF=0,PCBLC=.097,PCBLHP=0,PCBLLP=1,PCBLOB=0,T4DS=3260,DPCDSD=.056,ETABDS=.985,
ETLPDS=.903,TFPLDS=130.0,CNLPDS=2.300,HPEXT=0,AM55=.238,AM6=.238,ETAADS=.90,
T7DS=3560,DPAFDS=.048,DELFG=1,DELFN=1,DELSFC=1,TOLALL=.005,ITRYS=210,IDCD=0,
IMCD=0,IDES=1,IAMTP=0,MODE=0,IDUMP=0,AM=0,ALTP=0,
```

The following is an example of the normal design-case printout. The fan, compressor, combustor, high-pressure- and low-pressure-turbine correction factors are printed out, as are the various designated design areas. Following these are the internal engine thermodynamic characteristics and thrust and specific fuel consumption.

*1 DESIGN POINT WITH NORMAL PRINTOUT WHEN IDES=1 (PRINTED OUT FOR DESIGN
* POINT ONLY) - - NEXT PAGE HAS THE CLOSE FORMAT (SEE PAGE OF KEY TO VARIABLES)

```

FAN DESIGN          PRFCF= 0.1750000E+02  ETACF= 0.9420454E+00  WACF= 0.1055000E+01  T2DS= 0.5186682E+03
COMPRESSOR DESIGN   PRCCF= 0.1000000E+01  ETACCF= 0.1000000E+01  WACCF= 0.1000000E+01  T2IDS= 0.1015159E+04
COMBUSTOR DESIGN    WACDS= 0.1552109E+03  ETABCF= 0.9850000E+00  DTCCF= 0.1000000E+01
L.P. TURBINE DESIGN CNLPCF= 0.1313217E+01  TFLPCF= 0.4257449E+00  ETLPCF= 0.1023534E+01  DHLPCF= 0.2318793E+01
TURBINE AREA DESIGN A55= 0.2093313E+02  AM55= 0.2382371E+00
AFTERBURNER DESIGN  WG6CDS= 0.8993930E+04
NOZZLE DESIGN       AB= 0.8585148E+01  AM8= 0.1000000E+01  A9= 0.8585148E+01  AM9= 0.1000000E+01
CUTPUT              AM= 0.  ALTP= 0.  T4= 3260.00  ETAR= 1.0000
PCNF  CNF  ZF  PRF  WAF  WAF
0.100000E+03  0.100000E+01  0.833333E+00  0.800000E+01  0.633000E+03  0.633000E+C3
PCNC  CNC  ZC  PRC  WACC  WAC
0.100000E+03  0.100000E+01  0.  0.100000E+01  0.110915E+03  0.633000E+03
T2  P2  T21  P21  T3  P3
0.518668E+03  0.100000E+01  0.101916E+04  0.800000E+01  0.101916E+04  0.800000E+C1
PCBLF  BLF  PCBL  BLC  PCBL0B  BLOB
0.  0.  0.970000E-01  0.614010E+02  0.  0.
PCBLHP  BLHP  PCBLLP  BLLP  T4  P4
0.  0.  0.100000E+01  0.614010E+02  0.326000E+04  0.755200E+C1
WA3  WFB  WG4  FAR4  ETAB  OPCOM
0.571599E+03  0.219359E+02  0.593535E+03  0.383764E-01  0.985000E+00  0.560000E-01
TFFHP  CNHP  DHTCHP  DHTC  T5  P5
0.  0.  0.  0.  0.326000E+04  0.755200E+01
TFFLP  CNLP  DHTCLP  DHTF  T55  P55
0.305347E+03  0.175142E+01  0.171750E-01  0.129830E+03  0.269721E+04  0.378187E+01
PCBLDU  RLDU  T24  P24  T25  P25
0.  0.  0.  0.  0.101916E+04  0.800000E+01
WAD  WFD  WG24  FAR24  ETAD  DPDU
0.  0.  0.633000E+03  0.  0.  0.
ETAF  ETAC  ETATHP  ETATLP  AM55  AM25
0.829000E+00  0.100000E+01  0.100000E+01  0.903000E+00  0.238237E+00  0.
T6  P6  PS6  AM5  V6  WG6
0.269721E+04  0.378187E+01  0.364695E+01  0.238237E+00  0.579433E+03  0.654936E+03
T7  WFA  WG7  FAR7  ETAA  DPAFT
0.269721E+04  0.  0.654936E+03  0.346539E-01  0.  0.480000E-01
PS8  AM8  V8  PS9  AM9  V9
0.198204E+01  0.100000E+01  0.228834E+04  0.198204E+01  0.100000E+01  0.228834E+04
PS28  AM28  V28  PS29  AM29  V29
0.  0.  0.  0.  0.  0.
BYPASS  HPEXT  WFT  WGT  VA  FRD
0.  0.  0.219359E+02  0.654936E+03  0.  0.
CVMNDZ  VJM  CVDNDZ  VJD  FGM  FGP
0.967632E+00  0.221427E+04  0.  0.  0.450738E+05  0.178418E+05
MAIN SCNIC CONVERGENT NOZZLE  FG= 62915.60  FN= 52915.60  SFC= 1.25516
DUCT SHOCK OUTSIDE C-D NOZZLE
CONVERGED AFTER 1 LOOPS

```

The following shows the common variables which can be read, if desired, by referring to the common designation (p. 21).

```

COMMON  0.833333E+00  0.100000E+03  -0.000000E-19  -0.000000E-19  0.  0.100000E+03  0.326000E+04  0
0.100000E+03  0.  0.326000E+04  0.100000E+03  0.  0.100000E+01  0.100000E+01  0.100000E+01
0.833333E+00  0.100000E+03  0.800000E+01  0.829000E+00  0.633000E+03  0.175000E+02  0.942045E+00  0.105500E+01
0.  0.  0.  0.  0.633000E+03  0.100000E+01  0.100000E+01  0.100000E+01
0.326000E+04  0.  0.224084E+04  0.985000E+00  0.155211E+03  0.560000E-01  0.100000E+01  0.985000E+00
0.  0.  0.  0.100000E+01  0.  0.100000E+01  0.100000E+01  0.518668E+03
0.130000E+03  0.230000E+01  0.903000E+00  0.425745E+00  0.131322E+01  0.102353E+01  0.231879E+01  0.101916E+04
0.  0.  0.  0.  0.  0.  0.  0.
0.358000E+04  0.  0.  0.900000E+00  0.899393E+04  0.480000E-01  0.  0.  0.
0.209331E+02  0.209331E+02  0.858515E+01  0.858515E+01  0.  0.  0.  0.
0.364595E+01  0.238237E+00  0.  0.967632E+00  0.  0.  0.  0.
0.518670E+03  0.100000E+01  0.123918E+03  0.159103E+01  0.518668E+03  0.100000E+01  0.123918E+03  0.159103E+01
0.101916E+04  0.800000E+01  0.245654E+03  0.161236E+01  0.101916E+04  0.800000E+01  0.245654E+03  0.161236E+01
0.326000E+04  0.755200E+01  0.916119E+03  0.196079E+01  0.326000E+04  0.755200E+01  0.916119E+03  0.196079E+01
0.269721E+04  0.378187E+01  0.735604E+03  0.194649E+01  0.  0.614010E+02  0.  0.
0.100000E+01  0.800000E+01  0.829000E+00  0.633000E+03  0.633000E+03  0.571599E+03  0.593535E+03  0.383764E-01
0.100000E+01  0.100000E+01  0.100000E+01  0.110915E+03  0.633000E+03  0.985000E+00  0.560000E-01  0.
0.100000E+01  0.100000E+01  0.  0.  0.  0.593535E+03  0.383764E-01  0.111699E+04
0.230000E+01  0.903000E+00  0.398252E-01  0.129830E+03  0.614010E+02  0.654936E+03  0.346539E-01  0.
0.  0.  0.100000E+01  0.833333E+00  0.100000E+03  0.  0.100000E+03  0.100000E+01
0.  0.130000E+03  0.  0.970000E-01  0.  0.  0.  0.100000E+01

```

G.1C0000E+01	0.633000E+03	0.633000E+03	0.	0.	0.245654E+03	0.	0.
G.101916E+04	C.800000E+01	0.245654E+03	0.161236E+01	0.	0.	0.	0.
0.	0.	0.	0.	0.101916E+04	0.800000E+01	0.245654E+03	0.161236E+01
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.633000E+03	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.269721E+04	0.378187E+01	0.735604E+03	0.194649E+01	0.101916E+04	0.800000E+01	0.245654E+03	0.161236E+01
0.219359E+02	0.654736E+03	0.346539E-01	0.	0.633000E+03	0.	0.100000E+01	0.
C.269721E+04	0.378187E+01	0.735604E+03	0.194649E+01	0.269721E+04	0.360034E+01	0.735601E+03	0.194987E+01
0.269721E+04	0.360034E+01	0.735601E+03	0.194987E+01	0.269721E+04	0.360034E+01	0.735601E+03	0.194987E+01
0.654936E+03	0.	0.654936E+03	0.346539E-01	0.	0.480000E-01	0.579433E+03	0.
0.364695E+01	0.579433E+03	0.238237E+00	0.267322E+04	0.345885E+01	0.610475E+03	0.251101E+00	0.
0.235513E+04	0.198204E+01	0.228834E+04	0.100000E+01	0.235513E+04	0.198204E+01	0.228834E+04	0.100000E+01
0.	0.	0.	0.	0.221427E+04	0.450738E+05	0.	0.178418E+05
0.450738E+05	0.178418E+05	0.219359E+02	0.654936E+03	0.346539E-01	0.629156E+05	0.629156E+05	0.125516E+01

Off-design. - In the following cases, only the input data that are changed are printed out as input. The other data (normal design-case printout, p. 23) are not affected. The common variables are not repeated because IDES was set to zero internally in the program. That printout is eliminated and the design printout converts to the normal off-design printout format.

Case 1: The engine is operating at a reduced turbine inlet temperature T4 at a sea-level-static condition. The only input change required is T4.

```

SD(1),T4=2660,+1 PART POWER ENGINE OPERATION -- OFF-DESIGN
# RESULTS NORMAL OFF-DESIGN OUTPUT

OUTPUT          AM= 0.          ALTP= 0.          T4= 2660.00          ETAR= 1.C000
PCNF            CNF            ZF            PRF            WAFc            WAF
0.843207E+02    0.843207E+00    0.766751E+00    0.584906E+01    0.525128E+03    0.525128E+03
PCNC            CNC            ZC            PRc            WACC            WAC
0.949398E+02    0.100000E+01    0.100000E+01    0.100000E+01    0.119482E+03    0.525128E+03
T2              P2              T21           P21              T3              P3
0.518668E+03    0.100000E+01    0.918624E+03    0.584906E+01    0.918615E+03    0.584906E+01
PCBLF           BLF            PCBLc         BLC              PCBL0B          BL0B
0.              0.              0.970000E-01    0.509374E+02    0.              0.
PCBLHP          BLHP           PCBLLP        BLLP            T4              P4
0.              0.              0.100000E+01    0.509374E+02    0.266000E+04    0.549621E+01
WA3             WFB            WG4           FAR4            ETAB            DPCOM
0.474191E+03    0.133727E+02    0.487563E+03    0.282011E-01    0.985000E+00    0.603254E-01
TFFHP           CVHP           DHTCHP        DHTC            T5              P5
0.              0.              0.              0.              0.266000E+04    0.549621E+01
TFFLP           CVLP           DHTCLP        DHTF            T55             P55
0.311322E+03    0.163491E+01    0.169092E-01    0.104294E+03    0.219476E+04    0.278698E+01
PCBLDU          BLDU           T24           P24             T25             P25
0.              0.              0.              0.              0.918624E+03    0.584906E+01
WAD             WFD            WG24          FAR24           ETAD            DPDUc
0.              0.              0.525128E+03    0.              0.              0.
ETAF            ETAC           ETATHP        ETATLP          AM55            AM25
0.843568E+00    0.100000E+01    0.100000E+01    0.907545E+00    0.238077E+00    0.
T6              P6             P56           AM6             V6              WG6
0.219476E+04    0.278698E+01    0.269024E+01    0.238077E+00    0.525868E+03    0.538501E+03
T7              WFA           WG7           FAR7            ETAA            DPAPT
0.219476E+04    0.              0.538501E+03    0.254656E-01    0.              0.483102E-01
PS8             AM8           V8            PS9             AM9             V9
0.145228E+01    0.100000E+01    0.207131E+04    0.145228E+01    0.100000E+01    0.207131E+04
PS28            AM28          V28           PS29            AM29            V29
0.              0.              0.              0.              0.              0.
RYPASS          HPExT         WFT           WGT             VA              FRD
0.              0.              0.133727E+02    0.538501E+03    0.              0.
CVMNOZ          VJM           CVDNOZ        VJD             FGM             FGP
0.977273E+00    0.202425E+04    0.              0.              0.338801E+05    0.821708E+04

MAIN SCNIC CONVERGENT NOZZLE          FG= 42097.19          FN= 42097.19          SFC= 1.143E
DUCT SHOCK OUTSIDE C-D NOZZLE

CONVERGED AFTER 14 LOOPS

```

Case 2: The engine is operating at design turbine inlet temperature at sea-level-static condition. The engine components must always be matched at the desired turbine inlet temperature prior to running the engine at afterburning conditions.

```

$D(1), T4=3260., * SET UP FOR AFTERBURNER OPERATION
OUTPUT      AM= 0.      ALTP= 0.      T4= 3260.00      ETAR= 1.0000
PCNF        CNF        ZF        PRF        WAF        WAF
0.100000E+03 0.100000E+01 0.833333E+00 0.800000E+01 0.633000E+03 0.633000E+03
PCNC        CNC        ZC        PRC        WACC        WAC
0.100000E+03 0.100000E+01 0.      0.100000E+01 0.110915E+03 0.633000E+03
T2          P2        T21       P21       T3        P3
0.518668E+03 0.100000E+01 0.101916E+04 0.800000E+01 0.101916E+04 0.800000E+01
PCBLF       BLF        PCBLC     BLC       PCBL0B     BLOB
0.      0.      0.970000E-01 0.614010E+02 0.      0.
PCBLHP      BLHP      PCBLLP    BLLP      T4        P4
0.      0.      0.100000E+01 0.614010E+02 0.326000E+04 0.755200E+01
WA3         WFB        WG4       FAR4      ETAB      DPCOM
0.571599E+03 0.219359E+02 0.593535E+03 0.383764E-01 0.985000E+00 0.560000E-01
TFFHP       CNHP      DHTCHP    DHTC      T5        P5
0.      0.      0.      0.      0.326000E+04 0.755200E+01
TFFLP       CNLP      DHTCLP    DHTF      T55       P55
0.305347E+03 0.175142E+01 0.171750E-01 0.129830E+03 0.269721E+04 0.378187E+01
PCBLDU      BLDU      T24       P24       T25       P25
0.      0.      0.      0.      0.101916E+04 0.800000E+01
WAD         WFD        WG24      FAR24     ETAD      DPDU
0.      0.      0.633000E+03 0.      0.      0.
ETAF        ETAC      ETATHP    ETATLP    AM55      AM25
0.829000E+00 0.100000E+01 0.100000E+01 0.903000E+00 0.238190E+00 0.
T6          P6        P56       AM6       V6        WG6
0.269721E+04 0.378187E+01 0.365225E+01 0.238190E+00 0.579408E+03 0.654936E+03
T7          WFA       WC7       FAR7      ETAA      DPAFT
0.269721E+04 0.      0.654936E+03 0.345539E-01 0.      0.480000E-01
PS8         AM8       V8        PS9       AM9       V9
0.198204E+01 0.100000E+01 0.228834E+04 0.198204E+01 0.100000E+01 0.228834E+04
PS28        AM28      V28       PS29      AM29      V29
0.      0.      0.      0.      0.      0.
BYPASS      HPXT      WFT       WGT       VA        FRD
0.      0.      0.219359E+02 0.654936E+03 0.      0.
CVMNOZ      VJM       CVDNOZ    VJD       FGM       FGP
0.967632E+00 0.221427E+04 0.      0.      0.450738E+05 0.178418E+05

MAIN SCALING CONVERGENT NOZZLE      FG= 62915.60      FN= 62915.60      SFC= 1.25516
DUCT SHOCK OUTSIDE C-D NOZZLE
CONVERGED AFTER 1 LOOPS

```

Case 3: The engine is operating at maximum design afterburner temperature at sea-level-static condition. Once the engine components are matched at the desired turbine inlet temperature, the engine performance for any afterburner temperature that is greater than the turbine exit temperature may be calculated.

A new value of A8 is computed when afterburning to account for the temperature, pressure, and mass flow changes. A9 is allowed to float to provide for complete expansion to PS9 = PS0 at station 9. The new values of A8 and A9 are printed at the top of the normal off-design output page. The value of IAFTBN is set to zero after each case; therefore, IAFTBN = 1 must be used for each afterburning case.

```

$D(1),T4=3260,T7=3560,IAFTBN=1, * AFTERBURNER OPERATION AT DESIGN TEMP
* RESULTS NORMAL OFF-DESIGN OUTPUT
NOZZLE DESIGN      A8= 0.10260896E+02      AM8= 0.10000000E+01      A9= 0.12334022E+02      AM9= 0.15117356E+01
OUTPUT      AM= 0.      ALTP= 0.      T4= 3260.00      T7= 3560.00      ETAR= 1.0000
PCNF        CNF        ZF        PRF        WAF        WAF
0.100000E+03 0.100000E+01 0.833333E+00 0.800000E+01 0.633000E+03 0.633000E+03
PCNC        CNC        ZC        PRC        WACC        WAC
0.100000E+03 0.100000E+01 0.      0.100000E+01 0.110915E+03 0.633000E+03
T2          P2        T21       P21       T3        P3
0.518668E+03 0.100000E+01 0.101916E+04 0.800000E+01 0.101916E+04 0.800000E+01
PCBLF       BLF        PCBLC     BLC       PCBL0B     BLOB
0.      0.      0.970000E-01 0.614010E+02 0.      0.
PCBLHP      BLHP      PCBLLP    BLLP      T4        P4
0.      0.      0.100000E+01 0.614010E+02 0.326000E+04 0.755200E+01
WA3         WFB        WG4       FAR4      ETAB      DPCOM
0.571599E+03 0.219359E+02 0.593535E+03 0.383764E-01 0.985000E+00 0.560000E-01
TFFHP       CNHP      DHTCHP    DHTC      T5        P5
0.      0.      0.      0.      0.326000E+04 0.755200E+01
TFFLP       CNLP      DHTCLP    DHTF      T55       P55
0.305347E+03 0.175142E+01 0.171750E-01 0.129830E+03 0.269721E+04 0.378187E+01
PCBLDU      BLDU      T24       P24       T25       P25
0.      0.      0.      0.      0.101916E+04 0.800000E+01
WAD         WFD        WG24      FAR24     ETAD      DPDU
0.      0.      0.633000E+03 0.      0.      0.

```

ETAF	ETAC	ETATHP	ETATLP	AM55	AM25
0.829000E+00	0.100000E+01	0.100000E+01	0.903000E+00	0.238190E+00	0.
T6	P6	PS6	AM6	V6	WG6
0.269721E+04	0.378187E+01	0.365225E+01	0.238190E+00	0.579408E+03	0.654936E+03
T7	WFA	WG7	FAR7	ETAA	DPAFT
0.356000E+04	0.106971E+02	0.665633E+03	0.515530E-01	0.900029E+00	0.480000E-01
PS8	AM8	V8	PS9	AM9	V9
0.196671E+01	0.100000E+01	0.261777E+04	0.100000E+01	0.151174E+01	0.369243E+04
PS28	AM28	V28	PS29	AM29	V29
0.	0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.	0.	0.326330E+02	0.665633E+03	0.	0.
CVMNOZ	VJM	CVDNOZ	VJD	FGM	FGP
0.977004E+00	0.360752E+04	0.	0.	0.746342E+05	0.

MAIN SHOCK OUTSIDE C-D NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CONVERGED AFTER 1 LOOPS

FG= 74634.20
 FN= 74634.20
 SFC= 1.57406

Case 4: The engine is operating at a subsonic Mach number (0.7) at 25 000 feet altitude with a turbine inlet temperature less than design. The performance shown is for A8 = A9 (i. e., the engine is using a convergent nozzle). If desired, a convergent-divergent nozzle could have been simulated by using this case as a setup and having an additional case with NOZFLT = 1 or IMCD = 1 for complete expansion to occur.

\$D(11),AM=.7,ALTP=25000,T4=2760, * PART POWER OPERATION AT CRUISE CONDITION
 * RESULTS NORMAL OFF-DESIGN OUTPUT

OUTPUT	AM= 0.700	ALTP= 25000.	T4= 2760.00	ETAR= 1.0000
PCNF	CNF	ZF	PRF	WAF
0.900352E+02	0.944024E+00	0.806890E+00	0.714555E+01	0.594428E+03
PCNC	CNC	ZC	PRC	WACC
0.936874E+02	0.100000E+01	0.100000E+01	0.100000E+01	0.114549E+03
T2	P2	T21	P21	T3
0.471787E+03	0.514853E+00	0.894556E+03	0.367891E+01	0.894555E+03
PCBLF	BLF	PCBLC	BLC	PCBLOB
0.	0.	0.970900E-01	0.311261E+02	0.
PCBLHP	BLHP	PCBLLP	BLLP	T4
0.	0.	0.100000E+01	0.311261E+02	0.276000E+04
WA3	WFB	WG4	FAR4	ETAR
0.289761E+03	0.882030E+01	0.298582E+03	0.304399E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.	0.	0.	0.	0.276000E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.307945E+03	0.171379E+01	0.171341E-01	0.109754E+03	0.227177E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.894556E+03
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.320887E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.836023E+00	0.100000E+01	0.100000E+01	0.905173E+00	0.238293E+00
T6	P6	PS6	AM6	V6
0.227177E+04	0.173692E+01	0.167666E+01	0.238293E+00	0.534812E+03
T7	WFA	WG7	FAR7	ETAA
0.227177E+04	0.	0.329708E+03	0.274872E-01	0.
PS8	AM8	V8	PS9	AM9
0.906790E+00	0.100000E+01	0.210597E+04	0.905790E+00	0.100000E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.882030E+01	0.329708E+03	0.711594E+03
CVMNOZ	VJM	CVDNOZ	VJD	FGM
0.957531E+00	0.201653E+04	0.	0.	0.206646E+05

MAIN SCNIC CONVERGENT NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CONVERGED AFTER 7 LOOPS

FG= 30397.21
 FN= 23300.14
 SFC= 1.36278

Case 5: The engine is operating at the same subsonic condition as in case 4, but at the design turbine inlet temperature. Afterburner operation may be added at this point. If a different turbine inlet temperature is desired, a new setup case, as in case 4, must be calculated for component matching purposes immediately prior to running the afterburner case.

OUTPUT	AM= 0.700	ALTP= 25000.	T4= 3260.00	ETAR= 1.0000
PCNF	CNF	ZF	PRF	WAF
0.984254E+02	0.103200E+01	0.856827E+00	0.869637E+01	0.660746E+03
PCNC	CNC	ZC	PRC	WACC
0.976137E+02	0.100000E+01	0.100000E+01	0.100000E+01	0.109007E+03
T2	P2	T21	P21	T3
0.471789E+03	0.514853E+00	0.971095E+03	0.447735E+01	0.971092E+03
PCBLF	BLF	PCBLC	BLC	PCBLOB
0.	0.	0.970000E-01	0.345987E+02	0.
PCBLHP	BLHP	PCBLLP	BLLP	T4
0.	0.	0.100000E+01	0.345987E+02	0.326000E+04
WA3	WFR	WG4	FAR4	ETAR
0.322089E+03	0.125982E+02	0.334688E+03	0.391140E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.	0.	0.	0.	0.326000E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.307337E+03	0.172385E+01	0.171404E-01	0.128965E+03	0.269690E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.971095E+03
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.356688E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.800114E+00	0.100000E+01	0.100000E+01	0.904580E+00	0.238236E+00
T6	P6	PS6	AM6	V6
0.269690E+04	0.213226E+01	0.205917E+01	0.238236E+00	0.579402E+03
T7	WFA	WG7	FAR7	ETAA
0.269690E+04	0.	0.369286E+03	0.353199E-01	0.
PS8	AM8	V8	PS9	AM9
0.111781E+01	0.100000E+01	0.228808E+04	0.111781E+01	0.100000E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.125982E+02	0.369286E+03	0.711594E+03
CVMNOZ	VJM	CVMNOZ	VJD	FGM
0.944453E+00	0.216190E+04	0.	0.	0.248138E+05

MAIN SCNIC CONVERGENT NOZZLE FG= 38380.11 FN= 30491.23 SFC= 1.4E743
DUCT SHOCK OUTSIDE C-U NOZZLE
CONVERGED AFTER 14 LOOPS

Case 6: The engine is now operating at a supersonic flight Mach number (2.5) at an altitude of 50 000 feet. The engine components are matched with an engine nozzle throat area A8 equal to the sea-level-static design value. This case is a setup case for matching the components only.

SD(1),AM=2.5,ALTP=50000,T4=3260, SET UP FOR MATCHING ONLY

OUTPUT	AM= 2.500	ALTP= 50000.	T4= 3260.00	ETAR= 0.8703
PCNF	CNF	ZF	PRF	WAF
0.832186E+02	0.640467E+00	0.665972E+00	0.361453E+01	0.385510E+03
PCNC	CNC	ZC	PRC	WACC
0.114102E+03	0.100000E+01	0.100000E+01	0.100000E+01	0.131289E+03
T2	P2	T21	P21	T3
0.875661E+03	0.170275E+01	0.132686E+04	0.615464E+01	0.132686E+04
PCBLF	BLF	PCBLC	BLC	PCBLOB
0.	0.	0.970000E-01	0.490043E+02	0.
PCBLHP	BLHP	PCBLLP	BLLP	T4
0.	0.	0.100000E+01	0.490043E+02	0.326000E+04
WA3	WFB	WG4	FAR4	ETAR
0.456195E+03	0.153057E+02	0.471501E+03	0.335508E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.	0.	0.	0.	0.326000E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.318769E+03	0.145751E+01	0.160734E-01	0.121704E+03	0.274010E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.132686E+04
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.505199E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.815437E+00	0.100000E+01	0.100000E+01	0.909909E+00	0.238010E+00
T6	P6	PS6	AM6	V6
0.274010E+04	0.302911E+01	0.292530E+01	0.238010E+00	0.583946E+03
T7	WFA	WG7	FAR7	ETAA
0.274010E+04	0.	0.520505E+03	0.302964E-01	0.
PS8	AM8	V8	PS9	AM9
0.158578E+01	0.100000E+01	0.230699E+04	0.158578E+01	0.100000E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.153057E+02	0.520505E+03	0.242123E+04
CVMNOZ	VJM	CVMNOZ	VJD	FGM
0.835769E+00	0.192812E+04	0.	0.	0.311927E+05

MAIN SCNIC CONVERGENT NOZZLE FG= 57923.74 FN= 19905.36 SFC= 2.76813
DUCT SHOCK OUTSIDE C-U NOZZLE
CONVERGED AFTER 8 LOOPS

Case 7: The performance data from this case reflect an expansion to the maximum attainable area ratio for A9/A8, thus simulating a convergent-divergent nozzle. Performance is calculated such that the nozzle static pressure PS9 is equal to the free-stream static pressure PS0. This is accomplished by setting NOZFLT equal to 1. The improvement in thrust and specific fuel consumption from case 6 should be noted.

Although engine performance is calculated as if complete expansion occurred ($A9/A8 = 3.449$), a performance penalty from the fully expanded case resulted because this particular nozzle has a physical expansion limit A9/A8 of 2.15. This means that the nozzle has a lower-than-optimum thrust coefficient, resulting in a performance penalty.

```

SD(1),NOZFLT=1, PERFORMANCE DATA WITH MAXIMUM ATTAINABLE NOZZLE RATIO
NOZZLE DESIGN          AB= 0.85821256E+01  AM8= 0.10000000E+01  A9= 0.29598886E+02  AM9= 0.27030654E+01
ARATIO= 3.449      OUT OF RANGE,USE DATA FOR 2.150
CUTPUT
AM= 2.500      ALTP= 50000.      T4= 3260.00      ETAR= 0.8703
PCNF          CNF          ZF          PRF          WAF          WAF
0.832186E+02  0.640467E+00  0.665972E+00  0.361453E+01  0.385510E+03  0.505199E+03
PCNC          CNC          LC          PRC          WACC          WAC
0.114102E+03  0.100000E+01  0.100000E+01  0.100000E+01  0.131289E+03  0.505199E+03
T2           P2          T21         P21          T3           P3
0.875661E+03  0.170275E+01  0.132686E+04  0.615464E+01  0.132686E+04  0.615464E+01
PCBLF         BLF          PCBLC        BLC          PCBL0B       RL0B
0.           0.          0.970000E-01  0.490043E+02  0.           0.
PCBLHP        BLHP        PCBLLP       BLLP         T4           P4
0.           0.          0.100000E+01  0.490043E+02  0.326000E+04  0.574667E+C1
WA3           WFB          WG4          FAR4         ETAB         DPCOM
0.456195E+03  0.153057E+02  0.471501E+03  0.335508E-01  0.985000E+00  0.662867E-01
TFFHP         CNHP         DHTCHP       DHTC         T5           P5
0.           0.          0.           0.           0.326000E+04  0.574667E+C1
TFFLP         CNLP         DHTCLP       DHTF         T55          P55
0.313767E+03  0.145751E+01  0.160734E-01  0.121704E+03  0.274010E+04  0.302911E+01
PCBLDU        BLDU         T24          P24          T25          P25
0.           0.          0.           0.           0.132686E+04  0.615464E+01
WAD           WFD          WG24         FAR24        ETAD         DPDU
0.           0.          0.505199E+03  0.           0.           0.
ETAF          ETAC          ETATHP       ETATLP       AM55         AM25
0.815437E+00  0.100000E+01  0.100000E+01  0.909909E+00  0.238010E+00  0.
T6           P6          PS6          AM5          V6          WG6
0.274010E+04  0.302911E+01  0.292530E+01  0.238010E+00  0.583946E+03  0.520505E+03
T7           WFA          WG7          FAR7         ETAA         DPAFT
0.274010E+04  0.           0.520505E+03  0.302964E-01  0.           0.480047E-01
PSH           AM8          V8           PS9          AM9          V9
0.158634E+01  0.100000E+01  0.230699E+04  0.114456E+00  0.270307E+01  0.462581E+04
PS28          AM28         V28          PS29         AM29         V29
0.           0.          0.           0.           0.           0.
BYPASS        HPEXT        WFT          WGT          VA           FRD
0.           0.          0.153057E+02  0.520505E+03  0.242123E+04  0.380184E+C5
CVMNOZ        VJM          CVDNOZ       VJD          FGM          FGP
0.958966E+00  0.443600E+04  0.           0.           0.717646E+05  0.
MAIN SHOCK DJTSIDE C-D NOZZLE      FG= 71764.62      FN= 33746.24      SFC= 1.62279
DUCT SHOCK DUTSIDE C-D NOZZLE
CONVERGED AFTER 1 LOOPS

```

Case 8: The engine is operating in the supersonic flight condition with partial afterburning. This case illustrates that for an afterburning condition (IAFTBN = 1), the program automatically expands the nozzle so PS9 is equal to PS0. There is no need to input NOZFLT = 1 as in the previous case.

\$D(1),T4=3260,T7=2900,IAFTBN=1, PARTIAL AFTERBURNING DURING SUPERSONIC OPER.

NOZZLE DESIGN A8= 0.88924085E+01 AM8= 0.10000000E+01 A9= 0.30845876E+02 AM9= 0.26977559E+01

ARATIO= 3.459 OUT OF RANGE,USE DATA FOR 2.150

OUTPUT

AM= 2.500	ALTP= 50000.	T4= 3260.00	T7= 2900.00	ETAR= 0.8703
PCNF	CNF	ZF	PRF	WAF
0.832186E+02	0.640467E+00	0.665972E+00	0.361453E+01	0.385510E+03
PCNC	CNC	ZC	PRC	WACC
0.114102E+03	0.100000E+01	0.100000E+01	0.100000E+01	0.131289E+03
T2	P2	T21	P21	T3
0.875661E+03	0.170275E+01	0.132686E+04	0.615464E+01	0.132686E+04
PCBLF	BLF	PCBLC	BLC	PCBL08
0.	0.	0.970000E-01	0.490043E+02	0.
PCRLHP	BLHP	PCRLLP	BLLP	T4
0.	0.	0.100000E+01	0.490043E+02	0.326000E+04
WA3	WFB	WG4	FAR4	ETA8
0.456193E+03	0.153057E+02	0.471501E+03	0.335508E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.	0.	0.	0.	0.326000E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.318769E+03	0.145751E+01	0.160734E-01	0.121704E+03	0.274010E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.132686E+04
WAD	WFD	WG24	FAR24	ETA0
0.	0.	0.505199E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.815437E+00	0.100000E+01	0.100000E+01	0.909909E+00	0.238010E+00
T6	P6	PS6	AM6	V6
0.274010E+04	0.302911E+01	0.292530E+01	0.238010E+00	0.583946E+03
T7	WFA	WG7	FAR7	ETAA
0.290000E+04	0.160194E+01	0.522107E+03	0.334673E-01	0.866827E+00
PS8	AM8	V8	PS9	AM9
0.158508E+01	0.100000E+01	0.237121E+04	0.114456E+00	0.269776E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.169076E+02	0.522107E+03	0.242123E+04
CVMNDZ	VJM	CVDNDZ	VJD	FGM
0.959036E+00	0.457282E+04	0.	0.	0.742059E+05

MAIN SHOCK OUTSIDE C-D NOZZLE FG= 74205.85 FN= 36187.47 SFC= 1.68201

DUCT SHOCK OUTSIDE C-D NOZZLE

CONVERGED AFTER 1 LOOPS

Case 9: The engine is operating in the supersonic condition at the maximum design afterburner temperature.

\$D(1),T4=3260,T7=3560,IAFTBN=1, MAX AFTERBURNING AT SUPERSONIC OPERATION

NOZZLE DESIGN A8= 0.10150120E+02 AM8= 0.10000000E+01 A9= 0.36261996E+02 AM9= 0.26857718E+01

ARATIO= 3.575 OUT OF RANGE,USE DATA FOR 2.150

OUTPUT

AM= 2.500	ALTP= 50000.	T4= 3260.00	T7= 3560.00	ETAR= 0.8703
PCNF	CNF	ZF	PRF	WAF
0.832186E+02	0.640467E+00	0.665972E+00	0.361453E+01	0.385510E+03
PCNC	CNC	ZC	PRC	WACC
0.114102E+03	0.100000E+01	0.100000E+01	0.100000E+01	0.131289E+03
T2	P2	T21	P21	T3
0.875661E+03	0.170275E+01	0.132686E+04	0.615464E+01	0.132686E+04
PCBLF	BLF	PCBLC	BLC	PCBL08
0.	0.	0.970000E-01	0.490043E+02	0.
PCRLHP	BLHP	PCRLLP	BLLP	T4
0.	0.	0.100000E+01	0.490043E+02	0.326000E+04
WA3	WFB	WG4	FAR4	ETA8
0.456193E+03	0.153057E+02	0.471501E+03	0.335508E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.	0.	0.	0.	0.326000E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.318769E+03	0.145751E+01	0.160734E-01	0.121704E+03	0.274010E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.132686E+04
WAD	WFD	WG24	FAR24	ETA0
0.	0.	0.505199E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.815437E+00	0.100000E+01	0.100000E+01	0.909909E+00	0.238010E+00
T6	P6	PS6	AM6	V6
0.274010E+04	0.302911E+01	0.292530E+01	0.238010E+00	0.583946E+03
T7	WFA	WG7	FAR7	ETAA
0.356000E+04	0.794422E+01	0.528449E+03	0.460213E-01	0.908862E+00
PS8	AM8	V8	PS9	AM9
0.157511E+01	0.100000E+01	0.261903E+04	0.114456E+00	0.268577E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.232499E+02	0.528449E+03	0.242123E+04
CVMNDZ	VJM	CVDNDZ	VJD	FGM
0.959380E+00	0.510720E+04	0.	0.	0.838843E+05

MAIN SHOCK OUTSIDE C-D NOZZLE FG= 83884.25 FN= 45865.87 SFC= 1.82498

DUCT SHOCK OUTSIDE C-D NOZZLE

CONVERGED AFTER 1 LOOPS

Case 10: Ability to fix the mechanical shaft speed PCNF is a revision to reference 2 which was supplied by its author. This engine operation method is accomplished by setting the control MODE equal to 3. MODE = 3 allows the engine to run at some desired value of shaft speed while the primary stream area A8 is varied. The value of turbine inlet temperature necessary to properly match the components is calculated in the program.

The supersonic flight condition is used for illustrative purposes. Any condition desired could have been used. In case 9, the outer compressor shaft speed was considerably less than 100 percent when the design turbine inlet temperature T4 and area A8 were used. By using MODE = 3 instead of MODE = 0, the design T4 was exceeded when a 100-percent shaft speed was specified. A8 was increased by 1 percent. An attempt was made to run the design value of A8; but the T4 exceeded the program limit value of 4000° R and the components did not match. The limit value of a 4000° R temperature is used because the equations for the gas properties do not account for the dissociation of the products of combustion which would become significant at higher temperatures.

```

SD(1),MODE=3,PCNF=100,A8=8.586*1.01,NOZFLT=0,*, SETUP FOR 100 PERCENT COMP. SPEED

OUTPUT      AM= 2.500      ALTP= 50000.      T4= 3962.97      ETAR= 0.8703
PCNF        CNF          ZF          PRF          WAF          WAF
0.100000E+03 0.769621E+00 0.736343E+00 0.502448E+01 0.473919E+03 0.621058E+03
PCNC        CNC          ZC          PRC          WAC          WAC
0.119204E+03 0.100000E+01 0.100000E+01 0.100000E+01 0.121299E+03 0.621058E+03
T2          P2          T21         P21         T3          P3
0.875661E+03 0.170275E+01 0.144819E+04 0.855543E+01 0.144819E+04 0.855543E+01
PCBLF       BLF         PCBLC      BLC         PCBL0B      BLOB
0.          0.          0.970000E-01 0.602426E+02 0.          0.
PCBLHP      BLHP       PCBLLP     BLLP        T4          P4
0.          0.          0.100000E+01 0.602426E+02 0.396297E+04 0.803147E+01
WA3         WFB        WG4        FAR4        ETAB        DPCOM
0.560815E+03 0.259670E+02 0.586782E+03 0.463022E-01 0.985000E+00 0.612431E-01
TFFHP      CNHP       DHTCHP     DHTC        T5          P5
0.          0.          0.          0.          0.396297E+04 0.803147E+01
TFFLP      CNLP       DHTCLP     DHTF        T55         P55
0.312963E+03 0.158851E+01 0.167439E-01 0.153692E+03 0.332634E+04 0.412729E+01
PC%LDU     BLDU       T24        P24         T25         P25
0.          0.          0.          0.          0.144819E+04 0.855543E+01
WAD        WFD        WG24       FAR24       ETAD        DPDU
0.          0.          0.621058E+03 0.          0.          0.
ETAF       ETAC       ETATHP     ETATLP      AM55        AM25
0.840396E+00 0.100000E+01 0.100000E+01 0.908362E+00 0.240982E+00 0.
T6         P6         PS6        AM6         V6          WG6
0.332634E+04 0.412729E+01 0.398387E+01 0.240982E+00 0.647423E+03 0.647025E+03
T7         WFA        WG7        FAR7        ETAA        DPAFT
0.332634E+04 0.          0.647025E+03 0.418109E-01 0.          0.482537E-01
PS8        AM8        V8         PS9         AM9         V9
0.217407E+01 0.100000E+01 0.253416E+04 0.217407E+01 0.100000E+01 0.253416E+04
PS28       AM28      V28        PS29        AM29        V29
0.          0.          0.          0.          0.          0.
BYPASS     HPEXT      WFT        WGT         VA          FRD
0.          0.          0.259670E+02 0.647025E+03 0.242123E+04 0.467372E+05
CVMNOZ     VJM        CVDNOZ     VJD         FGM         FGP
0.801550E+00 0.203126E+04 0.          0.          0.408489E+05 0.377970E+05

MAIN SCNIC CONVERGENT NOZZLE      FG= 78645.91      FN= 31908.72      SFC= 2.92964
DUCT SHOCK OUTSIDE C-D NOZZLE

CONVERGED AFTER 7 LOOPS

```

Case 11: This case indicates that if MODE = 3 and the area is changed, T4 will change. The nozzle throat area has been opened to 120 percent of its design value ($A8 = 8.586 \times 1.20$). The engine components match at a point where the calculated T4 is less than the design value and the compressor shaft speed is 100 percent of design. Area

variation A8 and compressor speed variations are the options that are available when MODE = 3. The engine components are matched in all setup cases with the engine operating with the throat area A8 equal to A9 as if the engine has a convergent nozzle. In order to return to a convergent nozzle, an input of the control variable NOZFLT = 0 is required when a new setup case is run because NOZFLT is not reset to zero internally. This case is a setup case and illustrates only one of the many options available for operation of the GENENG program.

```

$D(1),NOZFLT=1,*, PERFORMANCE WITH BEST NOZZLE AREA RATIO
NOZZLE DESIGN      A8= 0.86801134E+01  AM8= 0.10000000E+01  A9= 0.37183272E+02  AM9= 0.28765206E+01
ARATIO= 4.284      OUT OF RANGE,USE DATA FOR 2.150
OUTPUT
      AM= 2.500      ALTP= 50000.      T4= 3962.97      ETAR= 0.8703
      PCNF          CNF          ZF          PRF          WAF          WAF
0.100000E+03      0.769620E+00      0.736343E+00      0.502448E+01      0.473919E+03      0.621058E+03
      PCNC          CNC          ZC          PRC          WACC          WAC
0.119204E+03      0.100000E+01      0.100000E+01      0.100000E+01      0.121299E+03      0.621058E+03
      T2            P2            T21          P21          T3            P3
0.875661E+03      0.170275E+01      0.144819E+04      0.855543E+01      0.144819E+04      0.855543E+01
      PCBLF         BLF          PC8LC        BLC          PC8LOB        BLOB
0.            0.            0.970000E-01      0.602426E+02      0.            0.
      PCBLHP        BLHP        PC8LLP        BLLP        T4            P4
0.            0.            0.100000E+01      0.602426E+02      0.396297E+04      0.803147E+01
      WA3           WFB          WG4          FAR4        ETAB          DPCOM
0.560815E+03      0.259670E+02      0.586782E+03      0.463022E-01      0.985000E+00      0.612431E-01
      TFFHP         CNHP        DHTCHP       DHTC        T5            P5
0.            0.            0.            0.            0.396297E+04      0.803147E+01
      TFFLP         CNLP        DHTCLP       DHTF        T55           P55
0.312963E+03      0.159851E+01      0.167439E-01      0.153692E+03      0.332634E+04      0.412729E+01
      PCBLDU        BLDU        T24          P24          T25           P25
0.            0.            0.            0.            0.144819E+04      0.855543E+01
      WAD           WFD          WG24         FAR24       ETAD          OPDUC
0.            0.            0.621058E+03      0.            0.            0.
      ETAF          ETAC        ETATHP       ETATLP      AM55         AM25
0.840396E+00      0.100000E+01      0.100000E+01      0.908362E+00      0.240982E+00      0.
      T6            P6            PS6          AM6         V6           WG6
0.332634E+04      0.412729E+01      0.398386E+01      0.240982E+00      0.647423E+03      0.547025E+03
      T7            WFA          WG7          FAR7        ETAA         DPAFT
0.332634E+04      0.            0.647025E+03      0.418109E-01      0.            0.482538E-01
      PS8           AM8          V8           PS9         AM9         V9
0.217200E+01      0.100000E+01      0.253416E+04      0.114456E+00      0.287652E+01      0.529546E+04
      PS2B          AM2B        V28         PS29        AM29        V29
0.            0.            0.            0.            0.            0.
      BYPASS        HPEXT       WFT          WGT         VA           FRD
0.            0.            0.259670E+02      0.647025E+03      0.242123E+04      0.467372E+05
      CVMNDZ        VJM          CVDNUZ       VJD         FGM          FGP
0.948016E+00      0.502018E+04      0.            0.            0.100956E+06      0.
MAIN SHOCK OUTSIDE C-D NOZZLE      FG=100956.46      FN= 54219.26      SFC= 1.72413
DUCT SHOCK OUTSIDE C-D NOZZLE
CONVERGED AFTER 1 LOOPS

```

Case 12: The final example case is the performance for the previous case (11) when the maximum attainable area expansion is used (PS9 = PS0).

```

$D(1),MODE=3,PCNF=100,A8=8.586*1.20,NOZFLT=0,*, SET UP FOR 100 PERCENT
OUTPUT
      AM= 2.500      ALTP= 50000.      T4= 3157.72      ETAR= 0.8703
      PCNF          CNF          ZF          PRF          WAF          WAF
0.100000E+03      0.769620E+00      0.613211E+00      0.435150E+01      0.488534E+03      0.640210E+03
      PCNC          CNC          ZC          PRC          WACC          WAC
0.117487E+03      0.100000E+01      0.100000E+01      0.100000E+01      0.142298E+03      0.640210E+03
      T2            P2            T21          P21          T3            P3
0.875661E+03      0.170275E+01      0.140676E+04      0.740952E+01      0.140676E+04      0.740952E+01
      PCBLF         BLF          PC8LC        BLC          PC8LOB        BLOB
0.            0.            0.970000E-01      0.621004E+02      0.            0.
      PCBLHP        BLHP        PC8LLP        BLLP        T4            P4
0.            0.            0.100000E+01      0.621004E+02      0.315772E+04      0.687718E+01
      WA3           WFB          WG4          FAR4        ETAR          DPCOM
0.578110E+03      0.174823E+02      0.595592E+03      0.302405E-01      0.985000E+00      0.718451E-01
      TFFHP         CNHP        DHTCHP       DHTC        T5            P5
0.            0.            0.            0.            0.315772E+04      0.687718E+01
      TFFLP         CNLP        DHTCLP       DHTF        T55           P55
0.331151E+03      0.177956E+01      0.196409E-01      0.144425E+03      0.258009E+04      0.311714E+01

```

PCBLDU	BLDU	T24	P24	T25	P25
0.	0.	0.	0.	0.140676E+04	0.740952E+01
WAD	WFD	WG24	FAR24	ETAD	DPDUC
0.	0.	0.640210E+03	0.	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55	AM25
0.810314E+00	0.100000E+01	0.100000E+01	0.917729E+00	0.287188E+00	0.
T6	P6	PS6	AM6	V6	WG6
0.258009E+04	0.311714E+01	0.296110E+01	0.287188E+00	0.683798E+03	0.657693E+03
T7	WFA	WG7	FAR7	ETAA	DPAFT
0.258009E+04	0.	0.657693E+03	0.273072E-01	0.	0.571973E-01
PS8	AM8	V8	PS9	AM9	V9
0.161452E+01	0.100000E+01	0.224079E+04	0.161452E+01	0.100000E+01	0.224079E+04
PS28	AM28	V28	PS29	AM29	V29
0.	0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.	0.	0.174823E+02	0.657693E+03	0.242123E+04	0.481785E+C5
CVMNOZ	VJM	CVDNDZ	VJD	FGM	FGP
0.833962E+00	0.186873E+04	0.	0.	0.382001E+05	0.327071E+05

MAIN SONIC CONVERGENT NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CONVERGED AFTER 8 LUOPS

FG= 70907.18
 FN= 22728.66
 SFC= 2.769C3

Two-Spool Turbojet

Design point. - An example is given for design-point operation of a two-spool turbojet. For the design point the data are printed out in three groups: (1) card images of the input data that were changed from the original input data, (2) results, and (3) common variables at the design point. The card images show two cards having \$D(1), IDES = 1, imprinted. These two cards in sequence recall the original design inputs. This method can be used as many times as desired in any computer run to design a new engine. A typical use for this may be a study of several engines where design-point performance only is sufficient. Thus, only changes in engine design characteristics need be input and a new engine is run. Or, as in this case, the engine was changed from a single-spool turbojet to a two-spool turbojet by changing ISPOOL = 1 to ISPOOL = 2 and by inputting the design pressure ratio and efficiency of the inner compressor and the design efficiency speed and flow function of the inner turbine. Also a change to the front compressor pressure ratio was made to have an overall pressure ratio of 8:1. The MODE was reset to zero (MODE = 0) from the previous cases of the single-spool turbojet engine.

```

$D(1),IDES=1, * FIRST OF TWO IDES=1 CARDS NECESSARY TO CHANGE DESIGN POINT
$D(1),IDES=1, * SECOND OF TWO IDES=1 CARDS NECESSARY TO CHANGE DESIGN POINT
ISPOOL=2,* TWO SPOOL TURBOJET
PRFDS=2,PRCDS=4,ETACDS=.8>,PCNCDS=100,ZCDS=.81433225, * OVERALL P RATIO STILL 8
TFHPDS=50.00,CNHPDS=2.000,ETHPDS=.90,MODE=0, * ADDITIONAL INPUT WAS REQUIRED TO
* RUN TWO SPOOL TURBOJETS, MODE =0 RESETS TO USE OF T4 RATHER THAN SPEED

FAN DESIGN          PRFCF= 0.25000000E+01  ETAFCF= 0.94204545E+00  WAFCF= 0.10550000E+01  T2DS= 0.5186820E+03
COMPRESSOR DESIGN   PRCCF= 0.42857143E+00  ETACCF= 0.98837209E+00  WACCF= 0.35584073E+01  T21DS= 0.6556220E+03
COMBUSTOR DESIGN    WA3CDS= 0.15548823E+03  ETABCF= 0.98500000E+00  DTCCCF= 0.10000000E+01
H.P. TURBINE DESIGN CNHPCF= 0.11419282E+01  TFHPCF= 0.16375691E+00  ETHPCF= 0.10000000E+01  DHHPCF= 0.48959067E+0C
L.P. TURBINE DESIGN CNLPCF= 0.12507844E+01  TFLPCF= 0.27070545E+00  ETLPCF= 0.10235344E+01  DHLPCF= 0.68989028E+0C
TURBINE AREA DESIGN      A55= 0.21028949E+02  AM55= 0.23823737E+00
AFTERBURNER DESIGN  WG6CDS= 0.90353905E+04
NOZZLE DESIGN        A8= 0.86244577E+01  AM8= 0.10000000E+01  A9= 0.86244577E+01  AM9= 0.10000000E+01

```

OUTPUT	AM= 0.	CNF	ALTP= 0.	ZF	T4= 3260.00	PRF	WAF	ETAR= 1.0000
	PCNF	CNF	CNF	ZF	PRF	WAF	WAF	
	0.100000E+03	0.100000E+01	0.833333E+00	0.200000E+01	0.633000E+03	0.633000E+03	0.633000E+03	
	PCNC	CNC	ZC		PRC	WACC	WACC	
	0.100000E+03	0.100000E+01	0.814332E+00	0.400000E+01	0.355841E+03	0.633000E+03	0.633000E+03	
	T2	P2	T21		P21	T3	P3	
	0.518666E+03	0.100000E+01	0.655622E+03	0.200000E+01	0.102280E+04	0.800000E+01	0.800000E+01	
	PCHLF	BLF	PC8LC		BLC	PCBL0B	BL0B	
	0.	0.	0.970000E-01	0.614010E+02	0.	0.	0.	
	PCBLHP	BLHP	PC8LLP		BLLP	T4	P4	
	0.	0.	0.100000E+01	0.614010E+02	0.326000E+04	0.755200E+01	0.755200E+01	
	WA3	WFB	WG4		FA34	ETAB	DPCOM	
	0.571594E+03	0.219038E+02	0.593503E+03	0.383203E-01	0.985000E+00	0.560000E-01	0.560000E-01	
	TFFHP	CVHP	DHTCHP		DHTC	T5	P5	
	0.305331E+03	0.175142E+01	0.600000E-01	0.957639E+02	0.295739E+04	0.457332E+01	0.457332E+01	
	TFFLP	CNLP	DHTCLP		DHTF	T55	P55	
	0.480227E+03	0.183885E+01	0.171750E-01	0.350417E+02	0.269456E+04	0.376248E+01	0.376248E+01	
	PC8LOU	BLLU	T24		P24	T25	P25	
	0.	0.	0.	0.	0.655622E+03	0.200000E+01	0.200000E+01	
	WAD	WFD	WG24		FA24	ETAD	DPDUC	
	0.	0.	0.633000E+03	0.	0.	0.	0.	
	ETAF	ETAC	ETATHP		ETATLP	AM55	AM25	
	0.829000E+00	0.850000E+00	0.900000E+00	0.903000E+00	0.238237E+00	0.	0.	
	T6	P6	PS6		AMS	V6	WG6	
	0.269456E+04	0.376248E+01	0.362824E+01	0.238237E+00	0.579167E+03	0.654904E+03	0.654904E+03	
	T7	WFA	WG7		FA37	ETAA	DPAFT	
	0.269456E+04	0.	0.654904E+03	0.346032E-01	0.	0.	0.480000E-01	
	PS8	AM8	V8		PS9	AM9	V9	
	0.197183E+01	0.100000E+01	0.228725E+04	0.197183E+01	0.100000E+01	0.228725E+04	0.228725E+04	
	PS28	AM28	V28		PS29	AM29	V29	
	0.	0.	0.	0.	0.	0.	0.	
	BYPASS	HPEXT	WFT		WGT	VA	FRD	
	0.	0.	0.219038E+02	0.654904E+03	0.	0.	0.	
	CVMNOZ	VJM	CVDNOZ		VJD	FGM	FGP	
	0.967850E+00	0.221372E+04	0.	0.	0.450603E+05	0.177371E+05	0.177371E+05	
MAIN SCNIC CONVERGENT NOZZLE		FG= 62797.45		FN= 62797.45		SFC= 1.25568		
DUCT SHOCK OUTSIDE C-D NOZZLE								
CONVERGED AFTER 1 LOOPS								

COMMON	0.833333E+00	0.100000E+03	-0.000000E-19	-0.000000E-19	0.814332E+00	0.100000E+03	0.326000E+04	0
0.100000E+03	0.100000E+03	0.326000E+04	0.100000E+03	0.	0.100000E+01	0.100000E+01	0.100000E+01	
0.833333E+00	0.100000E+03	0.200000E+01	0.829000E+00	0.633000E+03	0.250000E+01	0.942045E+00	0.105500E+01	
0.814332E+00	0.100000E+03	0.400000E+01	0.850000E+00	0.633000E+03	0.428571E+00	0.988372E+00	0.355841E+01	
0.326000E+04	0.	0.223720E+04	0.985000E+00	0.155488E+03	0.560000E-01	0.100000E+01	0.985000E+00	
0.500000E+02	0.200000E+01	0.900000E+00	0.163757E+00	0.114193E+01	0.100000E+01	0.489591E+00	0.518666E+03	
0.130000E+03	0.230000E+01	0.903000E+00	0.270705E+00	0.125078E+01	0.102353E+01	0.689890E+00	0.655622E+03	
0.	0.	0.	0.	0.	0.	0.	0.	
0.356000E+04	0.	0.	0.900000E+00	0.903539E+04	0.480000E-01	0.	0.	
0.210289E+02	0.	0.210289E+02	0.210289E+02	0.862446E+01	0.862446E+01	0.	0.	
0.362824E+01	0.238237E+00	0.	0.967850E+00	0.	0.	0.	0.	
0.518670E+03	0.100000E+01	0.123918E+03	0.159103E+01	0.518668E+03	0.100000E+01	0.123918E+03	0.159103E+01	
0.655622E+03	0.200000E+01	0.156774E+03	0.159970E+01	0.102280E+04	0.800000E+01	0.246562E+03	0.161325E+01	
0.326000E+04	0.755200E+01	0.916050E+03	0.196075E+01	0.295739E+04	0.457332E+01	0.820286E+03	0.196436E+01	
0.269456E+04	0.376248E+01	0.734740E+03	0.194651E+01	0.	0.614010E+02	0.	0.	
0.100000E+01	0.200000E+01	0.829000E+00	0.633000E+03	0.633000E+03	0.571599E+03	0.593503E+03	0.383203E-01	
0.100000E+01	0.400000E+01	0.850000E+00	0.355841E+03	0.633000E+03	0.985000E+00	0.560000E-01	0.	
0.200000E+00	0.900000E+00	0.293754E-01	0.957639E+02	0.	0.593503E+03	0.383203E-01	0.111699E+04	
0.210000E+01	0.903000E+00	0.118488E-01	0.350417E+02	0.614010E+02	0.654904E+03	0.346032E-01	0.	
0.	0.	0.100000E+01	0.833333E+00	0.100000E+03	0.814332E+00	0.100000E+03	0.219038E+02	
0.500000E+02	0.130000E+03	0.	0.970000E-01	0.	0.	0.	0.100000E+01	
0.100000E+01	0.633000E+03	0.633000E+03	0.	0.	0.246562E+03	0.	0.	
0.655622E+03	0.200000E+01	0.156774E+03	0.159970E+01	0.	0.	0.	0.	
0.	0.	0.	0.	0.655622E+03	0.200000E+01	0.156774E+03	0.159970E+01	
0.	0.	0.	0.	0.	0.	0.	0.	
0.	0.	0.633000E+03	0.	0.	0.	0.	0.	
0.	0.	0.	0.	0.	0.	0.	0.	
0.269456E+04	0.376248E+01	0.734740E+03	0.194651E+01	0.655622E+03	0.200000E+01	0.156774E+03	0.159970E+01	
0.219038E+02	0.654904E+03	0.346032E-01	0.	0.633000E+03	0.	0.100000E+01	0.	
0.269456E+04	0.376248E+01	0.734740E+03	0.194651E+01	0.269456E+04	0.358188E+01	0.734737E+03	0.194989E+01	
0.269456E+04	0.376248E+01	0.734737E+03	0.194989E+01	0.269456E+04	0.358188E+01	0.734737E+03	0.194989E+01	
0.654904E+03	0.	0.654904E+03	0.346032E-01	0.	0.480000E-01	0.579167E+03	0.	
0.362824E+01	0.579167E+03	0.238237E+00	0.267059E+04	0.344111E+01	0.610194E+03	0.251101E+00	0.	
0.235272E+04	0.197183E+01	0.228725E+04	0.100000E+01	0.235272E+04	0.197183E+01	0.228725E+04	0.100000E+01	
0.	0.	0.	0.	0.221372E+04	0.450603E+05	0.	0.177371E+05	
0.450603E+05	0.177371E+05	0.219038E+02	0.654904E+03	0.346032E-01	0.627974E+05	0.627974E+05	0.125568E+01	

Off-design. - The following cases are off-design cases for the two-spool turbojet.

Case 1: The engine is operating at a reduced turbine inlet temperature at sea-level-static condition. The input change of T4 was required. The compressor shaft speed of both compressors is reduced as are the pressure ratios, airflows, etc., all of which are representative of part-power operation.

SD(1),T4=2660,PI PART POWER ENGINE OPERATION
RESULTS N O R M A L O F F - D E S I G N O U T P U T

OUTPUT	AM= 0.	ALTP= 0.	T4= 2660.00	ETAR= 1.0000
PCNF	CNF	ZF	PRF	WAF
0.836241E+02	0.836241E+00	0.849728E+00	0.175763E+01	0.503176E+03
PCNC	CNC	ZC	PRC	WACC
0.904097E+02	0.925609E+00	0.709173E+00	0.324591E+01	0.314397E+03
T2	P2	T21	P21	T3
0.518668E+03	0.100000E+01	0.625500E+03	0.175763E+01	0.930191E+03
PCBLF	BLF	PCRLC	BLC	PCBLOB
0.	0.	0.970000E-01	0.488098E+02	0.
PCRLHP	BLHP	PCBLLP	BLLP	T4
0.	0.	0.100000E+01	0.488098E+02	0.266000E+04
WA3	WFR	WG4	FAR4	ETAB
0.454384E+03	0.127365E+02	0.467120E+03	0.280302E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.305530E+03	0.175297E+01	0.613448E-01	0.797926E+02	0.239574E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.485755E+03	0.170849E+01	0.166673E-01	0.275883E+02	0.218600E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.625500E+03	0.175763E+01
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.503176E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.848567E+00	0.809756E+00	0.897395E+00	0.906332E+00	0.238214E+00
T6	P6	PS6	AM6	V6
0.218600E+04	0.265090E+01	0.255886E+01	0.238214E+00	0.525189E+03
T7	WFA	WG7	FAR7	ETAA
0.218600E+04	0.	0.515930E+03	0.253113E-01	0.
PS8	AM8	V8	PS9	AM9
0.138197E+01	0.100000E+01	0.206732E+04	0.138197E+01	0.100000E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.127365E+02	0.515913E+03	0.
CVMNOZ	VJM	CVDNOZ	VJD	FGM
0.978239E+00	0.202233E+04	0.	0.	0.324293E+05

MAIN SCNIC CONVERGENT NOZZLE FG= 39400.71 FN= 39400.71 SFC= 1.16372
DUCT SHOCK OUTSIDE C-D NOZZLE
CONVERGED AFTER 13 LOOPS

Case 2: The engine is now operating at a supersonic flight Mach number (2.5) at an altitude of 50 000 feet. The engine components are matched with an engine nozzle throat area A8 equal to the sea-level-static design value. This case is a setup case for matching components only.

SD(1),T4=3260,AM=2.5,ALTP=50000,PI SET UP FOR COMPONENT MATCHING ONLY
RESULTS N O R M A L O F F - D E S I G N O U T P U T

OUTPUT	AM= 2.500	ALTP= 50000.	T4= 3260.00	ETAR= 0.8703
PCNF	CNF	ZF	PRF	WAF
0.851964E+02	0.655689E+00	0.823071E+00	0.148416E+01	0.373082E+03
PCNC	CNC	ZC	PRC	WACC
0.102605E+03	0.832254E+00	0.556194E+00	0.245472E+01	0.268143E+03
T2	P2	T21	P21	T3
0.875661E+03	0.170275E+01	0.996510E+03	0.252716E+01	0.135728E+04
PCBLF	BLF	PCRLC	BLC	PCBLOB
0.	0.	0.970000E-01	0.474215E+02	0.
PCBLHP	BLHP	PCBLLP	BLLP	T4
0.	0.	0.100000E+01	0.474215E+02	0.326000E+04
WA3	WFR	WG4	FAR4	ETAB
0.441460E+03	0.145963E+02	0.456056E+03	0.330637E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.305235E+03	0.179705E+01	0.618535E-01	0.982400E+02	0.294654E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.486451E+03	0.156951E+01	0.157361E-01	0.319779E+02	0.271686E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.996510E+03	0.252716E+01
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.488914E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.840470E+00	0.758931E+00	0.898441E+00	0.905158E+00	0.238434E+00
T6	P6	PS6	AM6	V6
0.271686E+04	0.289862E+01	0.279887E+01	0.238434E+00	0.582650E+03
T7	WFA	WG7	FAR7	ETAA
0.271686E+04	0.	0.503478E+03	0.293565E-01	0.
PS8	AM8	V8	PS9	AM9
0.151968E+01	0.100000E+01	0.229751E+04	0.151968E+01	0.100000E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.145963E+02	0.503510E+03	0.242123E+04
CVMNOZ	VJM	CVDNOZ	VJD	FGM
0.839848E+00	0.192956E+04	0.	0.	0.301948E+05

MAIN SCNIC CONVERGENT NOZZLE FG= 55841.86 FN= 19049.05 SFC= 2.75849
DUCT SHOCK OUTSIDE C-D NOZZLE
CONVERGED AFTER 20 LOOPS

Case 3: The performance data for this case reflect an expansion to the maximum attainable area ratio for A9/A8, thus simulating a convergent-divergent nozzle. Performance is calculated such that the nozzle static pressure PS9 is equal to the free-stream static pressure PS0. This is accomplished by setting NOZFLT = 1.

```

SD(1),NOZFLT=1, * PERFORMANCE WITH MAX ATTAINABLE AREA RATIO
* RESULTS N O R M A L O F F - D E S I G N O U T P U T

NOZZLE DESIGN          A8= 0.86367638E+01    AM8= 0.10000000E+01    A9= 0.28983159E+02    AM9= 0.26768298E+01

ARATIO= 3.356    OUT OF RANGE,USE DATA FOR 2.150

OUTPUT
AM= 2.500    ALTP= 50000.    T4= 3260.00    ETAR= 0.8703
PCNF          CNF          ZF          PRF          WAF          WAF
0.851964E+02  0.655689E+00  0.823071E+00  0.148416E+01  0.373082E+03  0.488914E+03
PCNC          CNC          ZC          PRC          WACC          WAC
0.102605E+03  0.832254E+00  0.556194E+00  0.245472E+01  0.268143E+03  0.488881E+03
T2           P2          T21         P21         T3           P3
0.875661E+03  0.170275E+01  0.996510E+03  0.252716E+01  0.135728E+04  0.620347E+01
PCBLF         BLF          PCRLC        BLC         PCBLOB       BLOB
0.          0.          0.970000E-01  0.474215E+02  0.          0.
PCBLHP        BLHP        PCBLLP       BLLP        T4           P4
0.          0.          0.100000E+01  0.474215E+02  0.326000E+04  0.580489E+01
WAF          WFR          WG4          FAR4        ETAR         DPCDM
0.441460E+03  0.145963E+02  0.456056E+03  0.330637E-01  0.985000E+00  0.642514E-01
TFFHP        CNHP        DHTCHP       DHTC        T5           P5
0.305235E+03  0.179705E+01  0.618535E-01  0.982400E+02  0.294654E+04  0.346287E+01
TFFLP        CNLP        DHTCLP       DHTF        T55          P55
0.486451E+03  0.156951E+01  0.157361E-01  0.317779E+02  0.271686E+04  0.289862E+01
PCBLDU        BLDU        T24          P24         T25          P25
0.          0.          0.          0.          0.996510E+03  0.252716E+01
WAD           WFD          WG24         FAR24       ETAD         DPDUC
0.          0.          0.488914E+03  0.          0.          0.
ETAF          ETAC          ETATHP       ETATLP      AM55         AM25
0.840476E+00  0.758931E+00  0.898441E+00  0.906158E+00  0.238434E+00  0.
T6           P6          PS6          AM5         V6           WG6
0.271686E+04  0.289862E+01  0.279887E+01  0.238434E+00  0.582650E+03  0.503478E+03
T7           WFA          WG7          FAR7        ETAA         DPAPT
0.271686E+04  0.          0.503478E+03  0.298565E-01  0.          0.480968E-C1
PS8           AM8         V8           PS9         AM9          V9
0.151752E+01  0.100000E+01  0.229751E+04  0.114456E+00  0.267683E+01  0.458357E+04
PS28          AM28        V28          PS29        AM29         V29
0.          0.          0.          0.          0.          0.
BYPASS        HPEXT       WFT          WGT         VA           FRD
0.          0.          0.145963E+02  0.503510E+03  0.242123E+04  0.367928E+05
CVMNDZ        VJM          CVDNDZ       VJD         FGM          FGP
0.960271E+00  0.440147E+04  0.          0.          0.688767E+05  0.

MAIN SHOCK OUTSIDE C-D NOZZLE    FG= 68876.71    FN= 32083.89    SFC= 1.62779
DUCT SHOCK OUTSIDE C-D NOZZLE

CONVERGED AFTER 1 LOUPS

```

Case 4: The engine is operating in the supersonic flight condition with partial afterburning. No new setup case was required because the components are still matched from the previous two cases (2 and 3). IAFTBN = 1 automatically expands the nozzle so PS9 = PS0. It also adjusts A8 for the new temperature and pressure condition such that the nozzle throat is sonic. Both A8 and A9 are printed out at the top of the listing.

```

SD(1),T7=2900,IAFTBN=1, * PARTIAL AFTERBURNING AT SUPERSONIC OPERATION
* RESULTS N O R M A L O F F - D E S I G N O U T P U T

NOZZLE DESIGN          A8= 0.89961808E+01    AM8= 0.10000000E+01    A9= 0.30389738E+02    AM9= 0.26712209E+01

ARATIO= 3.378    OUT OF RANGE,USE DATA FOR 2.150

OUTPUT
AM= 2.500    ALTP= 50000.    T4= 3260.00    T7= 2900.00    ETAR= 0.8703
PCNF          CNF          ZF          PRF          WAF          WAF
0.851964E+02  0.655689E+00  0.823071E+00  0.148416E+01  0.373082E+03  0.488914E+03
PCNC          CNC          ZC          PRC          WACC          WAC
0.102605E+03  0.832254E+00  0.556194E+00  0.245472E+01  0.268143E+03  0.488881E+03
T2           P2          T21         P21         T3           P3
0.875661E+03  0.170275E+01  0.996510E+03  0.252716E+01  0.135728E+04  0.620347E+01
PCBLF         BLF          PCRLC        BLC         PCBLOB       BLOB
0.          0.          0.970000E-01  0.474215E+02  0.          0.
PCBLHP        BLHP        PCBLLP       BLLP        T4           P4
0.          0.          0.100000E+01  0.474215E+02  0.326000E+04  0.580489E+01

```

WA3	WFB	WG4	FAR4	ETAB	DPCOM
0.441460E+03	0.145963E+02	0.456056E+03	0.330637E-01	0.985000E+0C	0.642514E-01
TFFHP	CNHP	DHTCHP	DHTC	T5	P5
0.305235E+03	0.179705E+01	0.618535E-01	0.982400E+02	0.294654E+04	0.346287E+01
TFFLP	CNLP	DHTCLP	DHTF	T55	P55
0.486451E+03	0.156951E+01	0.157361E-01	0.319779E+02	0.271686E+04	0.289862E+01
PCBLDU	BLDU	T24	P24	T25	P25
0.	0.	0.	0.	0.996510E+03	0.252716E+0.
WAD	WFD	WG24	FAR24	ETAD	DPDUC
0.	0.	0.488914E+03	0.	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55	AM25
0.840470E+00	0.758931E+00	0.898441E+00	0.905158E+00	0.238435E+00	0.
T6	P6	PS6	AM6	V6	WG6
0.271686E+04	0.289862E+01	0.279887E+01	0.238435E+00	0.582650E+03	0.503477E+03
T7	WFA	WG7	FAR7	ETAA	DPAFT
0.290000E+04	0.171925E+01	0.505197E+03	0.333732E-01	0.893552E+0C	0.480968E-01
PS8	AM8	V8	PS9	AM9	V9
0.151594E+01	0.100000E+01	0.237123E+04	0.114456E+00	0.267122E+01	0.474598E+04
PS28	AM28	V28	PS29	AM29	V29
0.	0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.	0.	0.163155E+02	0.505229E+03	0.242123E+04	0.367928E+05
CVMNOZ	VJM	CVDNOZ	VJD	FGM	FGP
0.960350E+00	0.455780E+04	0.	0.	0.715666E+05	0.

MAIN SHOCK OUTSIDE C-D NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CGNVERGED AFTER 1 LOOPS

FG= 71566.56
 FN= 34773.74
 SFC= 1.66509

Case 5: The engine is operating in the supersonic flight condition at design after-burner temperature.

ID(1),T7=3500,IAFTBN=1, MAX AFTERBURNING AT SUPERSONIC OPERATION
 * RESULTS N D R M A L O F F - D E S I G N O U T P U T

NOZZLE DESIGN
 ARATIG= 3.470 OUT OF RANGE,USE DATA FOR 2.150

AM= 2.500	ALTP= 50000.	T4= 3260.00	T7= 3500.00	ETAR= C.8703
PCNV	CNF	ZF	WAF	WAF
0.851964E+07	0.655689E+00	0.823071E+00	0.373082E+03	0.488914E+03
PCNC	CNC	ZC	WAC	WAC
0.102605E+03	0.832254E+00	0.556194E+00	0.268143E+03	0.488881E+03
T2	P2	T21	T3	P3
0.875661E+03	0.170275E+01	0.996510E+03	0.135728E+04	0.620347E+01
PCBLF	BLF	PCRLC	RLC	PCBLOR
0.	0.	0.970000E-01	0.474215E+02	0.
PCBLHP	BLHP	PCRLLP	BLLP	T4
0.	0.	0.100000E+01	0.474215E+02	0.326000E+04
WA3	WFR	WG4	FAR4	ETAB
0.441460E+03	0.145963E+02	0.456056E+03	0.330637E-01	0.985000E+00
TFFHP	CNHP	DHTCHP	DHTC	T5
0.305235E+03	0.179705E+01	0.618535E-01	0.982400E+02	0.294654E+04
TFFLP	CNLP	DHTCLP	DHTF	T55
0.486451E+03	0.156951E+01	0.157361E-01	0.319779E+02	0.271686E+04
PCBLDU	BLDU	T24	P24	T25
0.	0.	0.	0.	0.996510E+03
WAD	WFD	WG24	FAR24	ETAD
0.	0.	0.488914E+03	0.	0.
ETAF	ETAC	ETATHP	ETATLP	AM55
0.840476E+00	0.758931E+00	0.898441E+00	0.906158E+00	0.238435E+0C
T6	P6	PS6	AM6	V6
0.271686E+04	0.289862E+01	0.279887E+01	0.238435E+00	0.582650E+03
T7	WFA	WG7	FAR7	ETAA
0.350000E+04	0.724094E+01	0.510718E+03	0.446677E-01	0.919029E+0C
PS8	AM8	V8	PS9	AM9
0.150725E+01	0.100000E+01	0.259758E+04	0.114456E+00	0.266076E+01
PS28	AM28	V28	PS29	AM29
0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA
0.	0.	0.218372E+02	0.510751E+03	0.242123E+04
CVMNOZ	VJM	CVDNOZ	VJD	FGM
0.960653E+00	0.504319E+04	0.	0.	0.800537E+05

MAIN SHOCK OUTSIDE C-D NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CONVERGED AFTER 1 LOOPS

FG= 80053.69
 FN= 43260.88
 SFC= 1.81721

Separate-Flow Turbofan

Design point. - This is the design case for the separate-flow turbofan. For the design point, the data that were changed are shown on the top of the following example.

The control card FAN = .TRUE. designates that there is now a bypass ratio. Thus the front compressor (fan) flow does not equal the inner compressor flow. Consequently, a value for the inner compressor flow must be input. This value determines the engine bypass ratio. Also, a duct Mach number AM23 must be input. If the design is to be a duct-burning turbofan engine, AM23 should remain low (i.e., similar to an afterburner inlet Mach number - between 0.15 and 0.25) to keep friction losses at a minimum in the burner. Also a computational problem could arise when the duct burner is operating if AM23 is too high. The large temperature increase due to fuel addition will cause choking at engine duct station 24, consequently a printout error would occur.

The values of AM and ALTP are input to zero which are the conditions designated for the engine design point. A duct pressure loss DPDUDS is also input.

In the output of the results, the duct parameters now have values associated with the engine design, which includes area (A28 and A29) and duct Mach numbers.

```

$D(1),IDES=1,
$D(1),IDES=1,
PRFDS=2.5,PRGDS=10,WAFCDS=500,WACCD=100,ALTP=0,AM=0,IGASMX=0,AM23=.15,
FAN=.TRUE.,ISPOOL=2,DPDUDS=.02,
* SEPARATE FLOW TURBOFAN ENGINE--DESIGN CASE

FAN DESIGN      PRFCF= 0.37500000E+01 ETAFCF= 0.94204545E+00  WAFCF= 0.83333334E+00  T20S= 0.51866820E+03
COMPRESSOR DESIGN  PRCCF= 0.12857143E+01 ETACCF= 0.98837209E+00  WACCF= 0.10000000E+01  T21DS= 0.70574329E+03
COMBUSTER DESIGN  WA3CDS= 0.19974841E+02 ETABCF= 0.98500000E+00  DTJOCF= 0.10000000E+01
H.P. TURBINE DESIGN  CNHPCF= 0.11419282E+01 TFHPCF= 0.15210490E+01  ETHPCF= 0.10000000E+01  DHHPCF= 0.10CE9C18E+01
L.P. TURBINE DESIGN  CNLPCF= 0.11782237E+01 TFLPCF= 0.14620719E+01  ETLPCF= 0.10235344E+01  DHLPCF= 0.24967145E+01
DUCT NOZZLE DESIGN  A28= 0.27472096E+01  AM28= 0.10000000E+01  A29= 0.27472096E+01  AM29= 0.10000000E+01
TURBINE AREA DESIGN  A55= 0.64546229E+01  AM55= 0.23825740E+00
AFTERBURNER DESIGN  WG6CDS= 0.27906742E+04
NOZZLE DESIGN      A8= 0.26471373E+01  AM8= 0.10000000E+01  A9= 0.26471373E+01  AM9= 0.10000000E+01

OUTPUT
AM= 0.      ALTP= 0.      T4= 3260.00      ETAR= 1.0000
PCNF      CWF      ZF      PRF      WAF      WAF
0.100000E+03  0.100000E+01  0.833333E+00  0.250000E+01  0.500000E+03  0.500000E+03
PCNC      CNC      ZC      PRC      WACC      WAC
0.100000E+03  0.100000E+01  0.814332E+00  0.100000E+02  0.100000E+03  0.214319E+03
T2      P2      T21      P21      T3      P3
0.518668E+03  0.100000E+01  0.705743E+03  0.250000E+01  0.143796E+04  0.250000E+02
PCBLF      BLF      PCRLC      RLC      PCBL08      BL08
0.      0.      0.970000E-01  0.207890E+02  0.      0.
PCBLHP      BLHP      PCBLLP      LLP      T4      P4
0.      0.      0.100000E+01  0.207890E+02  0.326000E+04  0.236000E+02
WA3      WFB      WG4      FAR4      ETAB      DPDCM
0.193530E+03  0.614714E+01  0.199678E+03  0.317632E-01  0.985000E+00  0.560000E-01
TFHHP      CNHP      DHTCHP      DHYC      T5      P5
0.328721E+02  0.175142E+01  0.600000E-01  0.197341E+03  0.262422E+04  0.782808E+01
TFFLP      CNLP      DHTCLP      DHYF      T55      P55
0.889143E+02  0.195209E+01  0.171750E-01  0.112529E+03  0.218109E+04  0.368952E+01
PCBLDU      BLDU      T24      P24      T25      P25
0.      0.      0.705743E+03  0.245000E+01  0.705743E+03  0.245000E+01
WAD      WFD      WG24      FAR24      ETAD      DPDU
0.285681E+03  0.      0.285681E+03  0.      0.      0.200000E-01
ETAF      ETAC      ETATHP      ETATLP      AM55      AM25
0.829000E+00  0.850000E+00  0.900000E+00  0.903000E+00  0.238257E+00  0.157446E+00
T6      P6      PS6      AM6      V6      WG6
0.216104E+04  0.368952E+01  0.355616E+01  0.238257E+00  0.524239E+03  0.270466E+03
T7      WFA      WG7      FAR7      ETAA      DPAPT
0.218109E+04  0.      0.220466E+03  0.285822E-01  0.      0.480000E-01
PS8      AM8      V8      PS9      AM9      V9
0.192429E+01  0.100000E+01  0.206441E+04  0.192429E+01  0.100000E+01  0.206441E+04
PS28      AM28      V28      PS29      AM29      V29
0.129756E+01  0.100000E+01  0.118926E+04  0.129756E+01  0.100000E+01  0.118926E+04
BYPAS      HPEXT      WFT      WGT      VA      FRD
0.133297E+01  0.      0.614714E+01  0.506147E+03  0.      0.
CVMQUZ      VJM      CVDVUZ      VJD      FGM      FGP
0.963650E+00  0.199971E+04  0.978768E+00  0.115401E+04  0.240381E+05  0.690771E+04

MAIN SCNIC CONVERGENT NOZZLE      FG= 30945.83      FN= 30945.83      SFC= 0.71511
DUCT SCNIC CONVERGENT NOZZLE
CONVERGED AFTER 1 LOOPS

```

COMMON	0.833333E+00	0.100000E+03	-0.000000E-19	-0.000000E-19	0.814332E+00	0.100000E+03	0.326000E+04	0
0.100000E+03	0.100000E+03	0.326000E+04	0.100000E+03	0.	0.100000E+01	0.100000E+01	0.100000E+01	0.100000E+01
0.833333E+00	0.100000E+03	0.250000E+01	0.829000E+00	0.500000E+03	0.375000E+01	0.942045E+00	0.833333E+00	0.833333E+00
0.814332E+00	0.100000E+03	0.100000E+03	0.850000E+00	0.214319E+03	0.128571E+01	0.988372E+00	0.100000E+01	0.100000E+01
0.326000E+04	0.	0.182204E+04	0.985000E+00	0.199748E+02	0.560000E-01	0.100000E+01	0.985000E+00	0.985000E+00
0.500000E+02	0.200000E+01	0.900000E+00	0.152105E+01	0.114193E+01	0.100000E+01	0.100890E+01	0.518668E+03	0.518668E+03
0.130000E+03	0.230000E+01	0.903000E+00	0.146207E+01	0.117822E+01	0.102353E+01	0.249671E+01	0.705743E+03	0.705743E+03
0.	0.	0.	0.	0.303574E+04	0.200000E-01	0.	0.	0.
0.356000E+04	0.	0.	0.900000E+00	0.279067E+04	0.480000E-01	0.	0.	0.
0.645402E+01	0.	0.645462E+01	0.645462E+01	0.264714E+01	0.264714E+01	0.274721E+01	0.274721E+01	0.274721E+01
0.355616E+01	0.238257E+00	0.978768E+00	0.968658E+00	0.	0.	0.	0.	0.
0.518670E+03	0.100000E+01	0.123918E+03	0.159103E+01	0.518668E+03	0.100000E+01	0.123918E+03	0.159103E+01	0.159103E+01
0.705743E+03	0.250000E+01	0.168857E+03	0.160216E+01	0.143796E+04	0.250000E+02	0.352717E+03	0.162209E+01	0.162209E+01
0.326000E+04	0.236000E+02	0.907906E+03	0.187798E+01	0.262422E+04	0.782808E+01	0.710565E+03	0.188645E+01	0.188645E+01
0.218109E+04	0.368952E+01	0.574903E+03	0.188086E+01	0.	0.207890E+02	0.	0.	0.
0.100000E+01	0.250000E+01	0.829000E+00	0.500000E+03	0.500000E+03	0.193530E+03	0.199678E+03	0.317632E-01	0.317632E-01
0.100000E+01	0.100000E+02	0.850000E+00	0.100000E+03	0.214319E+03	0.985000E+00	0.560000E-01	0.	0.
0.200000E+01	0.900000E+00	0.605341E-01	0.197341E+03	0.	0.199678E+03	0.317632E-01	0.111699E+04	0.111699E+04
0.230000E+01	0.903000E+00	0.428810E-01	0.112529E+03	0.207890E+02	0.220466E+03	0.286822E-01	0.	0.
0.	0.	0.100000E+01	0.833333E+00	0.100000E+03	0.814332E+00	0.100000E+03	0.614714E+01	0.614714E+01
0.500000E+02	0.	0.	0.970000E-01	0.	0.	0.	0.100000E+01	0.100000E+01
0.100000E+01	0.500000E+03	0.214319E+03	0.	0.	0.352717E+03	0.	0.	0.
0.705743E+03	0.250000E+01	0.168857E+03	0.160216E+01	0.705743E+03	0.250000E+01	0.168857E+03	0.160216E+01	0.160216E+01
0.705743E+03	0.245000E+01	0.168857E+03	0.160355E+01	0.705743E+03	0.245000E+01	0.168857E+03	0.160355E+01	0.160355E+01
0.705743E+03	0.245000E+01	0.168857E+03	0.160355E+01	0.705743E+03	0.245000E+01	0.168857E+03	0.160355E+01	0.160355E+01
0.285616E+03	0.	0.285681E+03	0.	0.	0.200000E-01	0.133297E+01	0.	0.
0.588488E+03	0.129756E+01	0.118926E+04	0.100000E+01	0.588488E+03	0.129756E+01	0.118926E+04	0.100000E+01	0.100000E+01
0.218109E+04	0.368952E+01	0.574903E+03	0.188086E+01	0.705743E+03	0.245000E+01	0.168857E+03	0.160355E+01	0.160355E+01
0.614714E+01	0.220466E+03	0.286822E-01	0.	0.285681E+03	0.	0.100000E+01	0.	0.
0.218109E+04	0.368952E+01	0.574903E+03	0.188086E+01	0.218109E+04	0.351242E+01	0.574905E+03	0.188423E+01	0.188423E+01
0.218109E+04	0.351242E+01	0.574905E+03	0.188423E+01	0.218109E+04	0.351242E+01	0.574905E+03	0.188423E+01	0.188423E+01
0.220466E+03	0.	0.220466E+03	0.286822E-01	0.	0.480000E-01	0.524239E+03	0.	0.
0.355616E+01	0.524239E+03	0.238257E+00	0.216058E+04	0.337259E+01	0.552311E+03	0.251121E+00	0.157446E+00	0.157446E+00
0.188900E+04	0.192429E+01	0.206441E+00	0.100000E+01	0.188900E+04	0.192429E+01	0.206441E+00	0.100000E+01	0.100000E+01
0.	0.	0.116401E+04	0.103355E+05	0.199971E+04	0.137026E+05	0.172992E+04	0.517779E+04	0.517779E+04
0.240316E+05	0.690771E+04	0.614714E+01	0.506147E+03	0.122943E-01	0.309458E+05	0.309458E+05	0.715112E+00	0.715112E+00

Off-design. - The following are the off-design cases for the separate-flow turbofan engine.

Case 1: The engine is operating at Mach 2.5 at an altitude of 50 000 feet. The engine components are matched, with the duct and primary nozzle throat areas (A28 and A8) equal to the sea-level-static design value. This is a setup case for matching components only. However, note the very low thrust of the turbofan engine at this flight condition because the ram drag FRD is nearly equal to the gross thrust of the two streams FG.

```

SD(1),AM=2.5,ALTP=50000,      #1
* OFF DESIGN OPERATION AT MAX T4 (SETUP CASE)

OUTPUT      AM= 2.500      ALTP= 50000.      T4= 3260.00      ETAR= C.8703
            PCNF          CNF          ZF          PRF          WAF          WAF
0.709714E+02 0.566211E+00 0.652696E+00 0.141563E+01 0.254159E+03 0.333068E+03
            PCNC          CNC          ZC          PRC          WACC          WAC
0.792542E+02 0.670088E+00 0.657516E+00 0.479978E+01 0.574781E+02 0.100423E+03
            T2           P2           T21          P21          T3           P3
0.875661E+03 0.170275E+01 0.987251E+03 0.241047E+01 0.173093E+04 0.115697E+02
            PCBLF        BLF          PCBLC        BLC          PCBLDB       BLDB
0.          0.          0.970000E-01 0.974107E+01 0.          0.
            PCBLHP       BLHP        PCBLLP       BLLP          T4           P4
0.          0.          0.100000E+01 0.974107E+01 0.326000E+04 0.108562E+02
            WA3          WFB          WG4          FAR4          ETAB        DPCDM
0.906823E+02 0.248834E+01 0.931707E+02 0.274401E-01 0.985000E+00 0.616665E-01
            TFFHP        CNHP        DHTCHP       DHTC          T5           P5
0.333433E+02 0.138808E+01 0.607172E-01 0.199384E+03 0.261240E+04 0.332216E+01
            TFFLP        CNLP        DHTCLP       DHTF          T55          P55
0.975392E+02 0.138856E+01 0.150943E-01 0.984124E+02 0.223271E+04 0.173915E+01
            PCBLDU       BLDU        T24          P24           T25          P25
0.          0.          0.987251E+03 0.236231E+01 0.987251E+03 0.236231E+01
            WAD          WFD          WG24         FAR24        ETAD        DPDUC
0.232645E+03 0.          0.232645E+03 0.          0.          0.199789E-01
            ETAF        ETAC        ETATHP       ETATLP       AM55       AM25
0.796334E+00 0.715568E+00 0.855728E+00 0.912016E+00 0.238583E+00 0.158118E+00
            T6           P6           PS6          AM6          V6          WG6
0.223271E+04 0.173915E+01 0.167855E+01 0.238583E+00 0.531377E+03 0.102912E+03
            T7           WFA         WG7          FAR7        ETAA       DPAFT
0.223271E+04 0.          0.102912E+03 0.247785E-01 0.          0.480925E-01
            PS8          AM8          V8           PS9          AM9          V9

```

0.903306E+00	0.100000E+01	0.208880E+04	0.908306E+00	0.100000E+01	0.208880E+04
PS28	AM28	V28	PS29	AM29	V29
0.125681E+01	0.100000E+01	0.140492E+04	0.125681E+01	0.100000E+01	0.140492E+04
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.231664E+01	0.	0.248834E+01	0.335557E+03	0.242123E+04	0.250648E+05
CVMNOZ	VJM	CVDNOZ	VJO	FGM	FGP
0.878344E+00	0.183469E+04	0.852815E+00	0.119813E+04	0.145319E+05	0.110884E+05

MAIN SCNIC CONVERGENT NOZZLE
 DUCT SCNIC CONVERGENT NOZZLE
 CONVERGED AFTER 34 LOOPS

FG= 25620.31
 FN= 555.53
 SFC=16.12516

Case 2: The same flight condition and engine operation is shown in this case as in case 1 because it is a continuation of the previous setup case. However, the engine is now duct burning. Therefore T24, IDBURN, and ETAD must be input. The duct burner does not have a performance map included in the subprogram as was the situation with the afterburner. Consequently, every duct-burning case requires the input of the burner efficiency (ETAD) and the control card (IDBURN = 1) in order to operate.

A new throat area (A28) is calculated reflecting the change in temperature, pressure, and mass flow of the gas in order that the nozzle is sonic. The value of A8 remains at the design value unless it is altered with a separate input. A permanent change in A8 would require a new setup case prior to running a duct-burning case.

\$D(11,T24=3460,ETAD=.90,IDBURN=1.
 * JFF DESIGN OPERATION WITH DUCTBURNING

DUCT NOZZLE DESIGN A28= 0.58485843E+01 AM28= 0.10000000E+01 A29= 0.18051276E+02 AM29= 0.25474720E+01

ARATIO= 3.086 OUT OF RANGE,USE DATA FOR 2.150

OUTPUT AM= 2.500 ALTP= 50000. T4= 3260.00 T24= 3460.00 ETAR= C.8703

PCNF	CNF	2F	PRF	WAF	WAF
0.709714E+02	0.546211E+00	0.652696E+00	0.141563E+01	0.254159E+03	0.333068E+03
PCNC	CNC	2C	PRC	WACC	WAC
0.792542E+02	0.670088E+00	0.657516E+00	0.479978E+01	0.574781E+02	0.100423E+03
T2	P2	T21	P21	T3	P3
0.875661E+03	0.170275E+01	0.987251E+03	0.241047E+01	0.170093E+04	0.115697E+02
PCBLF	RLF	PCBLC	BLC	PCBLOR	BLOR
0.	0.	0.970000E-01	0.974107E+01	0.	0.
PCBLHP	BLHP	PCBLLP	BLLP	T4	P4
0.	0.	0.100000E+01	0.974107E+01	0.326000E+04	0.108562E+02
WA3	WFR	WG4	FAR4	ETAR	DPCOM
0.906823E+02	0.248834E+01	0.931707E+02	0.274401E-01	0.985000E+00	0.616665E-01
TFFHP	CNHP	DHTCHP	DHTC	T5	P5
0.333433E+02	0.138808E+01	0.607172E-01	0.199384E+03	0.261240E+04	0.332216E+01
TFFLP	CNLP	DHTCLP	DHTF	T55	P55
0.975392E+02	0.138856E+01	0.150943E-01	0.984124E+02	0.223271E+04	0.173915E+01
PCBLDU	BLDU	T24	P24	T25	P25
0.	0.	0.346000E+04	0.224230E+01	0.346000E+04	0.224230E+01
WAD	WFD	WG24	FAR24	ETAD	DPOUC
0.232645E+03	0.109448E+02	0.243590E+03	0.470450E-01	0.900000E+00	0.199789E-01
ETAF	ETAC	ETATHP	ETATLP	AM55	AM25
0.796334E+00	0.715568E+00	0.855728E+00	0.912016E+00	0.238583E+00	0.268149E+07
T6	P6	AM6	V6	W6	WG6
0.223271E+04	0.173915E+01	0.167855E+01	0.238583E+00	0.531377E+03	0.102912E+03
T7	WFA	WG7	FAR7	ETAA	DPAFT
0.223271E+04	0.	0.102912E+03	0.247785E-01	0.	0.480924E-01
PS8	AM8	VR	PS9	AM9	V9
0.908306E+00	0.100000E+01	0.208880E+04	0.908306E+00	0.100000E+01	0.208880E+04
PS28	AM28	V28	PS29	AM29	V29
0.124170E+01	0.100000E+01	0.258242E+04	0.114456E+00	0.254747E+01	0.510859E+04
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.231664E+01	0.	0.134331E+02	0.346501E+03	0.242123E+04	0.250648E+05
CVMNOZ	VJM	CVDNOZ	VJO	FGM	FGP
0.878344E+00	0.183469E+04	0.965722E+00	0.493348E+04	0.432198E+05	0.444708E+04

MAIN SCNIC CONVERGENT NOZZLE
 DUCT SHOCK OUTSIDE C-D NOZZLE
 CONVERGED AFTER 1 LOOPS

FG= 47666.88
 FN= 22602.09
 SFC= 2.13959

Design point. - A mixed-flow turbofan engine is designed by using the input IGASMX = 1 and resetting the AM and ALTP to the desired value (sea-level-static in the example). This output listing shows the rebalancing of the fan and inner compressor pressure ratio to a value such that the static pressures at stations 25 and 55 will match. The overall compressor pressure ratio remains fixed. The values of the combination are not necessarily the optimum values; but it does give an indication, for the selected design values, as to what combination can be considered. Once the static-pressure values are satisfied, the program operates in a normal design mode. The other input cards are those required for the afterburner design. These values of design efficiency and temperature must be input for the design case in order to scale the generalized afterburner map to the proper design conditions. The afterburning takes place in the completely mixed streams - similar to an afterburning turbojet. The mixed-flow turbofan engine uses a single exhaust nozzle. The design areas are then printed out for this type of operation.

```

SD(1),IDES=1, *1
SD(1),IDES=1,
AM=0,ALTP=0,IGASMX=1, * MIXED FLOW TURBOFAN - DESIGN CASE
T2DS=3460,ETAADS=.89,AM6=.15,DPAFDS=.05, * AFTERBURNER DESIGN VALUES

FAN DESIGN          PRFCF= 0.37500000E+01 ETAFCF= 0.94204545E+00 WAFCF= 0.83333334E+00 T2DS= 0.51866820E+03
COMPRESSOR DESIGN   PRCCF= 0.12857143E+01 ETACCF= 0.98837209E+00 WACCF= 0.10000000E+01 T21DS= 0.70574329E+03
COMBUSTER DESIGN    WA3CDS= 0.19974841E+02 ETABCF= 0.98500000E+00 DTCCF= 0.10000000E+01
H.P. TURBINE DESIGN CNHPCF= 0.11419282E+01 TFHPCF= 0.15210490E+01 ETHPCF= 0.10000000E+01 DHMPCF= 0.10089018E+01
L.P. TURBINE DESIGN CVLPCF= 0.11782237E+01 TFLPCF= 0.14620719E+01 ETLPCF= 0.10235344E+01 DHLPCF= 0.24947145E+01
CHANGE PRFDS FROM 2.500 TO 3.701 AND PRCDs FROM 10.000 TO 6.754

FAN DESIGN          PRFCF= 0.67532611E+01 ETAFCF= 0.94204545E+00 WAFCF= 0.83333334E+00 T2DS= 0.51866820E+03
COMPRESSOR DESIGN   PRCCF= 0.82205365E+00 ETACCF= 0.98837209E+00 WACCF= 0.10000000E+01 T21DS= 0.80123660E+03
COMBUSTER DESIGN    WA3CDS= 0.27828602E+02 ETABCF= 0.98500000E+00 DTCCF= 0.10000000E+01
H.P. TURBINE DESIGN CNHPCF= 0.11419282E+01 TFHPCF= 0.10948068E+01 ETHPCF= 0.10000000E+01 DHMPCF= 0.89296639E+00
L.P. TURBINE DESIGN CVLPCF= 0.11947035E+01 TFLPCF= 0.11955020E+01 ETLPCF= 0.10235344E+01 DHLPCF= 0.26482349E+01
CHANGE PRFDS FROM 2.500 TO 4.057 AND PRCDs FROM 10.000 TO 6.163

FAN DESIGN          PRFCF= 0.76415902E+01 ETAFCF= 0.94204545E+00 WAFCF= 0.83333334E+00 T2DS= 0.51866820E+03
COMPRESSOR DESIGN   PRCCF= 0.73753452E+00 ETACCF= 0.98837209E+00 WACCF= 0.10000000E+01 T21DS= 0.82458885E+03
COMBUSTER DESIGN    WA3CDS= 0.30070055E+02 ETABCF= 0.98500000E+00 DTCCF= 0.10000000E+01
H.P. TURBINE DESIGN CNHPCF= 0.11419282E+01 TFHPCF= 0.10136263E+01 ETHPCF= 0.10000000E+01 DHMPCF= 0.86289610E+00
L.P. TURBINE DESIGN CVLPCF= 0.11789344E+01 TFLPCF= 0.11433495E+01 ETLPCF= 0.10235344E+01 DHLPCF= 0.26415626E+01
CHANGE PRFDS FROM 2.500 TO 4.215 AND PRCDs FROM 10.000 TO 5.931

FAN DESIGN          PRFCF= 0.80373937E+01 ETAFCF= 0.94204545E+00 WAFCF= 0.83333334E+00 T2DS= 0.51866820E+03
COMPRESSOR DESIGN   PRCCF= 0.70446542E+00 ETACCF= 0.98837209E+00 WACCF= 0.10000000E+01 T21DS= 0.83507622E+03
COMBUSTER DESIGN    WA3CDS= 0.31058852E+02 ETABCF= 0.98500000E+00 DTCCF= 0.10000000E+01
H.P. TURBINE DESIGN CNHPCF= 0.11419282E+01 TFHPCF= 0.98150524E+00 ETHPCF= 0.10000000E+01 DHMPCF= 0.84999365E+00
L.P. TURBINE DESIGN CVLPCF= 0.12007443E+01 TFLPCF= 0.11225456E+01 ETLPCF= 0.10235344E+01 DHLPCF= 0.26351416E+01
TURBINE/DUCT AREA DESIGN A55= 0.87603192E+01 AM55= 0.23825511E+00 A25= 0.10494306E+02 AM25= 0.57661657E+01
AFTERBURNER DESIGN WG6CDS= 0.51521512E+04
NOZZLE DESIGN       AG= 0.48660238E+01 AM8= 0.10000000E+01 A9= 0.48660238E+01 AM9= 0.10000000E+01

```

```

OUTPUT          AM= 0.          ALTP= 0.          T4= 3260.00          ETAR= 1.0000
PCNF           CNF           ZF           PRF           WAF           WAF
0.100000E+03  0.100000E+01  0.833333E+00  0.421496E+01  0.500000E+03  0.500000E+03
PCNC           CNL           ZC           PR1           WACC           WAC
0.100000E+03  0.100000E+01  0.814332E+00  0.593126E+01  0.100000E+03  0.332181E+03
T2            P2            T21           P21           T3            P3
0.514667E+03  0.100000E+01  0.835076E+03  0.421496E+01  0.144718E+04  0.250000E+03
PCBLF         BLF           PCALC        BL2           PCBLOR        BL0B
0.           0.           0.970000E-01  0.322216E+02  0.           0.
PCBLHP        BLHP        PCALLP       RL1P         T4            P4
0.           0.           0.100000E+01  0.322216E+02  0.326000E+04  0.236000E+02
WA3           W4           WG4          FAR4          ETAB          DPCOM
0.299959E+03  0.948284E+01  0.309442E+03  0.316137E-01  0.985000E+00  0.560000E-01
TFFHP        CNHP        DHTCHP       DHTC          T5            P5
0.509427E+02  0.175142E+01  0.600000E-01  0.165259E+03  0.272550E+04  0.949214E+01
TFFLP        CNLP        DHTCLP       DHTF          T55           P55
0.115803E+03  0.191548E+01  0.171750E-01  0.123352E+03  0.224356E+04  0.427559E+01
PCBLDI       RL0U        T24          P24           T25           P25
0.           0.           0.835076E+03  0.413066E+01  0.835076E+03  0.413066E+01
WAD           WFD         WG24         FAR24         ETAD          DPOUC
0.167819E+03  0.           0.167819E+03  0.           0.           0.200000E-01
ETAF         ETAC        ETATHP       ETATLP        AM55          AM25
0.829000E+00  0.850000E+00  0.900000E+00  0.903000E+00  0.238255E+00  0.578617E-01
T6            P6          PS6          AM6           V6            W66
0.181619E+04  0.421426E+01  0.415724E+01  0.143360E+00  0.290659E+03  0.509483E+03
T7            WFA        WG7          FAR7          ETAA          DPAFT
0.181619E+04  0.           0.509483E+03  0.189657E-01  0.           0.500000E-01
PS8           AM8        V8           PS9           AM9           V9
0.217454E+01  0.100000E+01  0.189038E+04  0.217954E+01  0.100000E+01  0.189038E+04
PS28          AM28       V28          PS29          AM29          V29
0.           0.           0.           0.           0.           0.
BYPASS        HPEXT       WFT          WGT           VA            FRD
0.505203E+00  0.           0.948284E+01  0.509483E+03  0.           0.
CVMNOZ        VJM        CVNOZ        VJD           FGM           FGP
0.963159E+00  0.182074E+04  0.           0.           0.288318E+05  0.121464E+05

MAIN SCNIC CONVERGENT NOZZLE          FG= 40978.13          FN= 40978.13          SFC= 0.83308
CONVERGED AFTER 1 LOOPS

```

```

COMMON          0.833333E+00  0.100000E+03  -0.000000E-19  -0.000000E-19  0.814332E+00  0.100000E+03  0.326000E+04  0
0.100000E+03  0.100000E+03  0.326000E+04  0.100000E+03  0.           0.100000E+01  0.100000E+01  0.100000E+01
0.833333E+00  0.100000E+03  0.421496E+01  0.829000E+00  0.500000E+03  0.803739E+01  0.942045E+00  0.833333E+00
0.814332E+00  0.100000E+03  0.593126E+01  0.850000E+00  0.332181E+03  0.704465E+00  0.988372E+00  0.100000E+01
0.326000E+04  0.           0.181282E+04  0.985000E+00  0.310589E+02  0.560000E-01  0.100000E+01  0.985000E+00
0.500000E+02  0.200000E+01  0.900000E+00  0.981505E+00  0.114193E+01  0.100000E+01  0.849994E+00  0.518668E+03
0.130000E+03  0.230000E+01  0.903000E+00  0.112255E+01  0.120074E+01  0.102353E+01  0.263514E+01  0.835076E+03
0.           0.           0.           0.           0.115056E+04  0.200000E-01  0.           0.
0.346000E+04  0.           0.           0.890000E+00  0.515215E+04  0.500000E-01  0.           0.
0.876032E+01  0.104943E+02  0.192546E+02  0.192546E+02  0.486602E+01  0.486602E+01  0.           0.
0.412124E+01  0.238255E+00  0.           0.963159E+00  0.           0.           0.           0.
0.518670E+03  0.100000E+01  0.123918E+03  0.159103E+01  0.518668E+03  0.100000E+01  0.123918E+03  0.159103E+01
0.835076E+03  0.421496E+01  0.200259E+03  0.160719E+01  0.144718E+04  0.250000E+02  0.355137E+03  0.162376E+01
0.326000E+04  0.236000E+02  0.907719E+03  0.187788E+01  0.272550E+04  0.949214E+01  0.741460E+03  0.188473E+01
0.224356E+04  0.427559E+01  0.593308E+03  0.187903E+01  0.           0.322216E+02  0.           0.
0.100000E+01  0.421496E+01  0.829000E+00  0.500000E+03  0.500000E+03  0.299959E+03  0.309442E+03  0.316137E-01
0.100000E+01  0.593126E+01  0.850000E+00  0.100000E+03  0.332181E+03  0.985000E+00  0.560000E-01  0.
0.200000E+01  0.900000E+00  0.509996E-01  0.166259E+03  0.           0.309442E+03  0.316137E-01  0.111659E+04
0.230000E+01  0.903000E+00  0.452585E-01  0.123352E+03  0.322216E+02  0.341664E+03  0.285472E-01  0.
0.           0.           0.100000E+01  0.833333E+00  0.100000E+03  0.814332E+00  0.100000E+03  0.948284E+01
0.500000E+02  0.130000E+03  0.           0.970000E-01  0.           0.           0.           0.100000E+01
0.100000E+01  0.500000E+03  0.332181E+03  0.           0.           0.355137E+03  0.           0.
0.835076E+03  0.421496E+01  0.200259E+03  0.160719E+01  0.835076E+03  0.421496E+01  0.200259E+03  0.160719E+01
0.835076E+03  0.413066E+01  0.200259E+03  0.160858E+01  0.835076E+03  0.413066E+01  0.200259E+03  0.160858E+01
0.           0.           0.           0.           0.           0.           0.           0.
0.167819E+03  0.           0.167819E+03  0.           0.           0.200000E-01  0.505203E+00  0.           0.
0.           0.           0.           0.           0.           0.           0.           0.
0.224356E+04  0.427559E+01  0.593308E+03  0.187903E+01  0.835076E+03  0.413066E+01  0.200259E+03  0.160858E+01
0.948284E+01  0.341664E+03  0.285472E-01  0.           0.167819E+03  0.           0.100000E+01  0.
0.181619E+04  0.421426E+01  0.463844E+03  0.181456E+01  0.181619E+04  0.400355E+01  0.463844E+03  0.181808E+01
0.181619E+04  0.400355E+01  0.463844E+03  0.181808E+01  0.181619E+04  0.400355E+01  0.463844E+03  0.181808E+01
0.509483E+03  0.           0.509483E+03  0.189657E-01  0.           0.500000E-01  0.531343E+03  0.816543E+02
0.415724E+01  0.290659E+03  0.143360E+00  0.181019E+04  0.394954E+01  0.308344E+03  0.151095E+00  0.578617E-01
0.155960E+04  0.217954E+01  0.189038E+04  0.100000E+01  0.155960E+04  0.217954E+01  0.189038E+04  0.100000E+01
0.           0.           0.           0.           0.182074E+04  0.288318E+05  0.           0.121464E+05
0.288318E+05  0.121464E+05  0.948284E+01  0.509483E+03  0.189657E-01  0.409781E+05  0.409781E+05  0.833084E+00

```

Off-design. - The following are off-design cases for the mixed-flow turbofan.

Case 1: The off-design case was selected to be supersonic Mach number (2.5) at 50 000 feet altitude. Case 1 illustrates a setup case for the afterburning case (case 2) which follows immediately after. Since the engine is operating at the design T4, the only input cards required are the AM and ALTP cards.

```

SD(1),AM=2.5,ALTP=50000, #1
* OFF DESIGN OPERATION AT MAX T4 (SETUP CASE)

CUTPUT      AM= 2.500      ALTP= 50000.      T4= 3260.00      ETAR= 0.8703
PCNF        CNF          ZF          PRF          WAF        WAF
0.563944E+02 0.434023E+00 0.435777E+00 0.140745E+01 0.222610E+03 0.291724E+C3
PCNC        CNC          ZC          PRC          WACC        WAC
0.856482E+02 0.786044E+00 0.696801E+00 0.370696E+01 0.683886E+02 0.118544E+03
T2          P2          T21         P21         T3          P3
0.875661E+03 0.170275E+01 0.991445E+03 0.239654E+01 0.153061E+04 0.888387E+C1
PCBLF       BLF         PCRLC      RLC         PCBL0B      BL0B
0.          0.          0.970000E-01 0.114987E+02 0.          0.
PCBLHP      BLHP       PCRLLP     LLLP        T4          P4
0.          0.          0.100000E+01 0.114987E+02 0.326000E+04 0.837006E+C1
WA3         WFB        WG4        FAR4        ETAR        DPCOM
0.107045E+03 0.323859E+01 0.110283E+03 0.302545E-01 0.985000E+00 0.578362E-C1
TFFHP      CNHP       DHTCHP     DHTC        T5          P5
0.511908E+02 0.150006E+01 0.538222E-01 0.148746E+03 0.278122E+04 0.369848E+01
TFFLP      CNLP       DHTCLP     DHTF        T55         P55
0.107005E+03 0.106935E+01 0.103248E-01 0.755756E+02 0.244645E+04 0.231426E+01
PCBLDU      RLDU       T24        P24         T25         P25
0.          0.          0.991445E+03 0.230175E+01 0.991445E+03 0.230175E+C1
WAD         WFD        WG24       FAR24       ETAD        DPDOC
0.173180E+03 0.          0.173180E+03 0.          0.          0.395519E-C1
ETAF        ETAC       ETATHP     ETATLP      AM55        AM25
0.753899E+00 0.773388E+00 0.889117E+00 0.889316E+00 0.161228E+00 0.117851E+C0
T6          P6          PS6        AM5         V6          WG6
0.163458E+04 0.230603E+01 0.227469E+01 0.143137E+00 0.276685E+03 0.294962E+C3
T7          WFA        WG7        FAR7        ETAA        DPAFT
0.163458E+04 0.          0.294962E+03 0.11016E-01 0.          0.501864E-01
PS8         AM8        V8         PS3         AM9         V9
0.118688E+01 0.100000E+01 0.179734E+04 0.118688E+01 0.100000E+01 0.179734E+04
PS28        AM28       V28        PS29        AM29        V29
0.          0.          0.          0.          0.          0.
BYPASS      HPEXT     WFT        WGT         VA          FRD
0.146090E+01 0.          0.323859E+01 0.294962E+03 0.242123E+04 0.219534E+05
CVMNOZ      VJM       CVDNOZ     VJD         FGM         FGP
0.858472E+00 0.154296E+04 0.          0.          0.141454E+05 0.110434E+05

MAIN SCNIC CONVERGENT NOZZLE      FG= 25188.82      FN= 3235.39      SFC= 3.66356
CONVERGED AFTER 21 LOOPS

```

Case 2: The previous setup case is used here for an afterburning example. Values of afterburner efficiency ETAA, pressure loss DPAFT, and Mach number AM6 indicate values selected from the afterburner performance map at off-design. The expanded nozzle areas are shown; a printout indicating that a nozzle having a greater expansion ratio than the convergent-divergent nozzle map used would have been more optimum (instead of 2.150). Consequently, a slight loss in velocity coefficient due to an under-expanded nozzle occurred.

```

SD(1), T7=3460,IAFTBN=1, # AFTERBURNING CASE

NOZZLE DESIGN      A8= 0.76637719E+01      AM8= 0.10000000E+01      A9= 0.23089376E+02      AM9= 0.25228365E+01
ARATIO= 1.013      OUT OF RANGE,USE DATA FOR 2.150

OUTPUT      AM= 2.500      ALTP= 50000.      T4= 3260.00      T7= 3460.00      ETAR= 0.8703
PCNF        CNF          ZF          PRF          WAF        WAF
0.563944E+02 0.434023E+00 0.435777E+00 0.140745E+01 0.222610E+03 0.291724E+C3
PCNC        CNC          ZC          PRC          WACC        WAC
0.856482E+02 0.786044E+00 0.696801E+00 0.370696E+01 0.683886E+02 0.118544E+03
T2          P2          T21         P21         T3          P3
0.875661E+03 0.170275E+01 0.991445E+03 0.239654E+01 0.153061E+04 0.888387E+C1
PCBLF       BLF         PCRLC      RLC         PCBL0B      BL0B
0.          0.          0.970000E-01 0.114987E+02 0.          0.
PCBLHP      BLHP       PCRLLP     LLLP        T4          P4
0.          0.          0.100000E+01 0.114987E+02 0.326000E+04 0.837006E+C1
WA3         WFB        WG4        FAR4        ETAR        DPCOM
0.107045E+03 0.323859E+01 0.110283E+03 0.302545E-01 0.985000E+00 0.578362E-C1
TFFHP      CNHP       DHTCHP     DHTC        T5          P5
0.511908E+02 0.150006E+01 0.538222E-01 0.148746E+03 0.278122E+04 0.369848E+01
TFFLP      CNLP       DHTCLP     DHTF        T55         P55
0.107005E+03 0.106935E+01 0.103248E-01 0.755756E+02 0.244645E+04 0.231426E+01
PCBLDU      RLDU       T24        P24         T25         P25
0.          0.          0.991445E+03 0.230175E+01 0.991445E+03 0.230175E+C1
WAD         WFD        WG24       FAR24       ETAD        DPDOC
0.173180E+03 0.          0.173180E+03 0.          0.          0.395519E-C1
ETAF        ETAC       ETATHP     ETATLP      AM55        AM25
0.753899E+00 0.773388E+00 0.889117E+00 0.889316E+00 0.161228E+00 0.117851E+C0

```

T6	P6	PS6	AMS	V6	WG6
0.163458E+04	0.230603E+01	0.227468E+01	0.143137E+00	0.276685E+03	0.294962E+C3
T7	WFA	WG7	FAR7	ETAA	UPAFT
0.346000E+04	0.109850E+02	0.305947E+03	0.487570E-01	0.774686E+00	0.501864E-01
PSR	AM8	V8	PS9	AM9	V9
0.119096E+01	0.100000E+01	0.258203E+04	0.114456E+00	0.252284E+01	0.508533E+04
PS28	AM28	V28	PS29	AM29	V29
0.	0.	0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT	VA	FRD
0.146090E+01	0.	0.142236E+02	0.305947E+03	0.242123E+04	0.219534E+C5
CVPMNOZ	VJM	CVDMOZ	VJD	FGM	FGP
0.966656E+00	0.491576E+04	0.	0.	0.467446E+05	0.

MAIN SHOCK OUTSIDE C-D NOZZLE FG= 46744.62 FN= 24791.19 SFC= 2.06544

CONVERGED AFTER 1 LOOPS

Design case without pressure ratio rematch. - The mixed-flow turbofan engine is shown again. However, the design fan and compressor pressure ratios were picked such that no rematch was required to balance the static pressures. This is another design case. As many design cases as desired can be run in succession. Care must always be taken to include the variables in input that need to be changed from the previous design case.

```

$DI(1),IDES=1,
$DI(1),IDES=1,
PRFDS=5.0,PRCDS=5.0,AM=0,ALTP=0,*, MIXED FLOW TURBOFAN WITHOUT REQUIRED P REMATCH

```

FAN DESIGN	PRFCF= 0.1000000E+02	ETACFC= 0.94204545E+00	WACFC= 0.8333333E+00	T2DS= 0.51866820E+03
COMPRESSOR DESIGN	PRCCF= 0.57142857E+00	ETACCF= 0.98837209E+00	WACCF= 0.1000000E+01	T2IDS= 0.88129484E+C3
COMBUSTER DESIGN	WA3CDS= 0.35881686E+02	ETABCF= 0.94500000E+00	DTCCF= 0.1000000E+01	
H.P. TURBINE DESIGN	CNHPCF= 0.11419282E+01	TFHPCF= 0.85001019E+00	ETHPCF= 0.1000000E+01	D+HPCF= 0.78792675E+00
L.P. TURBINE DESIGN	CNLPFC= 0.12091271E+01	TFLPCF= 0.10362098E+01	ETLPCF= 0.10235344E+01	DHLPFC= 0.2584124E+01
TURBINE/DUCT AREA DESIGN	A55= 0.93469959E+01	AM55= 0.23825344E+00	A25= 0.10994153E+01	AM25= 0.35644279E+00
AFTERRURNER DESIGN	WG6CDS= 0.48604900E+04			
NOZZLE DESIGN	A8= 0.46035268E+01	AM8= 0.10000000E+01	A9= 0.46035268E+01	AM9= 0.10000000E+01

OUTPUT

AM= 0.	ALTP= 0.	T4= 3260.00	ETAR= 1.0000
PCNF	CNF	ZF	PRF
0.100000E+03	0.100000E+01	0.833333E+00	0.500000E+01
PCNC	CNC	ZC	PRC
0.100000E+03	0.100000E+01	0.814332E+00	0.500000E+01
T2	P2	T21	P21
0.518668E+03	0.100000E+01	0.881300E+03	0.500000E+01
PCHLF	BLF	PCBLC	BLC
0.	0.	0.970000E-01	0.372070E+02
PCRLHP	RLHP	PC3LLP	BLLP
0.	0.	0.100000E+01	0.372070E+02
WA3	WFB	WG4	FAR4
0.346370E+03	0.109422E+02	0.357312E+03	0.315911E-01
TFHHP	CNHP	DHTCHP	DHTC
0.588227E+02	0.175142E+01	0.600000E-01	0.154510E+03
TFFLP	CNLP	DHTCLP	DHTF
0.125457E+03	0.190220E+01	0.171750E-01	0.122659E+03
PCPLDU	BLDU	T24	P24
0.	0.	0.881300E+03	0.490000E+01
WAD	WFD	WG24	FAR24
0.115423E+03	0.	0.116423E+03	0.
ETAF	ETAC	ETATHP	ETATLP
0.929000E+00	0.850000E+00	0.900000E+00	0.903000E+00
T6	PA	PS6	AM6
0.199038E+04	0.468986E+01	0.449153E+01	0.256683E+00
T7	WFA	WG7	FAR7
0.199038E+04	0.	0.510942E+03	0.218844E-01
PS8	AM8	V8	PS9
0.243244E+01	0.100000E+01	0.197586E+04	0.243248E+01
PS28	AM28	V28	PS29
0.	0.	0.	0.
BYPASS	HPEXT	WFT	WGT
0.303514E+00	0.	0.109422E+02	0.510942E+03
CVPMNOZ	VJM	CVDMOZ	VJD
0.957521E+00	0.189192E+04	0.	0.
			FGM
			0.300448E+05

MAIN SENIC CONVERGENT NOZZLE FG= 44000.11 FN= 44000.11 SFC= 0.85527

CONVERGED AFTER 1 LOOPS

COMMON	0.833333E+00	0.100000E+03	-0.000000E-19	-0.000000E-19	0.814332E+00	0.100000E+03	0.326000E+04	0
0.100000E+03	0.100000E+03	0.326000E+04	0.100000E+03	0.	0.100000E+01	0.100000E+01	0.100000E+01	0.100000E+01
0.833333E+00	0.100000E+03	0.500000E+01	0.829000E+00	0.500000E+03	0.100000E+02	0.942045E+00	0.833333E+00	0.833333E+00
0.814332E+00	0.100000E+03	0.500000E+01	0.850000E+00	0.383577E+03	0.571429E+00	0.988372E+00	0.100000E+01	0.100000E+01
0.326000E+04	0.	0.181142E+04	0.985000E+00	0.358817E+02	0.560000E-01	0.100000E+01	0.985000E+00	0.985000E+00
0.500000E+02	0.200000E+01	0.900000E+00	0.850010E+00	0.114193E+01	0.100000E+01	0.789927E+00	0.518668E+03	0.518668E+03
0.130030E+03	0.230000E+01	0.903000E+00	0.103621E+01	0.120913E+01	0.102353E+01	0.258413E+01	0.881300E+03	0.881300E+03
0.	0.	0.	0.	0.691241E+03	0.200000E-01	0.	0.	0.
0.346030E+04	0.	0.	0.	0.890000E+00	0.486049E+04	0.500000E-01	0.	0.
0.934700E+01	0.109942E+01	0.104464E+02	0.104464E+02	0.460353E+01	0.460353E+01	0.	0.	0.
0.449964E+01	0.238253E+00	0.	0.957521E+00	0.	0.	0.	0.	0.
0.518670E+03	0.100000E+01	0.123918E+03	0.159103E+01	0.518668E+03	0.100000E+01	0.123918E+03	0.159103E+01	0.159103E+01
0.841300E+03	0.500000E+01	0.211573E+03	0.160867E+01	0.144858E+04	0.250000E+02	0.355503E+03	0.162402E+01	0.162402E+01
0.326000E+04	0.236000E+02	0.907691E+03	0.187786E+01	0.276368E+04	0.101882E+02	0.753181E+03	0.188413E+01	0.188413E+01
0.228157E+04	0.466804E+01	0.604585E+03	0.187798E+01	0.	0.372070E+02	0.	0.	0.
0.100000E+01	0.500000E+01	0.829000E+00	0.500000E+03	0.500000E+03	0.346370E+03	0.357312E+03	0.315911E-01	0.315911E-01
0.100000E+01	0.500000E+01	0.850000E+00	0.100000E+03	0.383577E+03	0.985000E+00	0.560000E-01	0.	0.
0.200000E+01	0.900000E+00	0.473956E-01	0.154510E+03	0.	0.357312E+03	0.315911E-01	0.111699E+04	0.111699E+04
0.230000E+01	0.903000E+00	0.443824E-01	0.122659E+03	0.372070E+02	0.394519E+03	0.285267E-01	0.	0.
0.	0.	0.100000E+01	0.833333E+00	0.100000E+03	0.814332E+00	0.100000E+03	0.109422E+02	0.109422E+02
0.500000E+02	0.130000E+03	0.	0.970000E-01	0.	0.	0.	0.100000E+01	0.100000E+01
0.100000E+01	0.500000E+03	0.383577E+03	0.	0.	0.355503E+03	0.	0.	0.
0.881300E+03	0.500000E+01	0.211573E+03	0.160867E+01	0.881300E+03	0.500000E+01	0.211573E+03	0.160867E+01	0.160867E+01
0.881300E+03	0.490000E+01	0.211573E+03	0.161005E+01	0.881300E+03	0.490000E+01	0.211573E+03	0.161005E+01	0.161005E+01
0.	0.	0.	0.	0.	0.	0.	0.	0.
0.116423E+03	0.	0.116423E+03	0.	0.	0.200000E-01	0.303518E+00	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.
0.228157E+04	0.466804E+01	0.604585E+03	0.187798E+01	0.881300E+03	0.490000E+01	0.211573E+03	0.161005E+01	0.161005E+01
0.109422E+02	0.394519E+03	0.285267E-01	0.	0.116423E+03	0.	0.100000E+01	0.	0.
0.199038E+04	0.468986E+01	0.515033E+03	0.183454E+01	0.199038E+04	0.445537E+01	0.515033E+03	0.183806E+01	0.183806E+01
0.199038E+04	0.445537E+01	0.515033E+03	0.183806E+01	0.199038E+04	0.445537E+01	0.515033E+03	0.183806E+01	0.183806E+01
0.510942E+03	0.	0.510942E+03	0.218844E-01	0.	0.500000E-01	0.535611E+03	0.510401E+03	0.510401E+03
0.449153E+01	0.541234E+03	0.256683E+00	0.196783E+04	0.424783E+01	0.571843E+03	0.271337E+00	0.356444E+00	0.356444E+00
0.171607E+04	0.243248E+01	0.197586E+04	0.100000E+01	0.171607E+04	0.243248E+01	0.197586E+04	0.100000E+01	0.100000E+01
0.	0.	0.	0.	0.189192E+04	0.300448E+05	0.	0.139553E+05	0.139553E+05
0.300448E+05	0.139553E+05	0.109422E+02	0.510942E+03	0.218844E-01	0.440001E+05	0.440001E+05	0.895270E+00	0.895270E+00

CONCLUDING REMARKS

The computer code (GENENG) presented herein has proven to be an indispensable tool for steady-state cycle analysis of various types of jet engines, namely two-spool mixed- or unmixed-flow turbofans and one- or two-spool turbojets.

This program has been found to be valuable for many applications because it has the capability of studying a broad range of engine types having different design characteristics, while it also has low-execution-time requirements. The laborious task of matching components manually at off-design operating has been eliminated.

The program has proven itself to be easy to use, especially in terms of input requirements. It is felt that with a minimum of effort the reader can become proficient in using the computer code.

The code is available to be reproduced on the requestor's tape upon application to the authors.

Lewis Research Center,

National Aeronautics and Space Administration,

Cleveland, Ohio, October 1, 1971,

132-15.

APPENDIX A

GENENG PROGRAM LISTING

```

$IBFTC GENENG  DECK
COMMON /POINT/IDATPT
COMMON/LOOPPR/KKGO,PRFNEW,PRCNEW
DATAIII/O/
DIMENSION XLO(5),XHI(5)
EQUIVALENCE(XLO,DHLOSV),(XHI,DHHISV)
COMMON/ALL/X(28)/DESIGN/Y(80)/FRONT/Z(80)/SIDE/W(48)/BACK/V(72)
COMMON/DUMMYS/DUMMY(100)
EQUIVALENCE (P6DSAV,DUMMY(7)),(AM6DSV,DUMMY(8)),(ETAASV,DUMMY(9)),
1(FAR7SV,DUMMY(10))
LOGICAL ERRER,CLEAR
DATA CLEAR/.TRUE./
COMMON/ERER/ERRER
ERRER=.FALSE.
IF (.NOT.CLEAR) CALL ENGBAL
CLEAR=.FALSE.
DO 1 J=1,408
1 X(J)=0.
IF (III.EQ.0) KKGO=0
III=1
IDATPT=0
CALL CONOUT (1)
P6DSAV=1.
AM6DSV=1.
ETAASV=1.
FAR7SV=1.
DO 2 I=1,5
2 XLO(I)=100.
XHI(I)=100.
CALL ENGBAL
STOP
END

```

```

$IBFTC ENGBAL  DECK
SUBROJTINE ENGBAL
COMMON / ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX,IDBURN,IAFTBN,IDCD ,IMCD ,IDSHJC,IMSHOC,NOZFLT,
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)
COMMON /DESIGN/
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELF6 ,DELFN ,DELSFC,
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFFDS ,PRFCF ,ETAFCF,WAFCF ,
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,
5TFHPDS,CNHPS,ETHPS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPFC,DHLPFC,T21DS ,
7T24DS ,WFDDSD,DTODSD,ETADSD,WA23DS,DPDUDSD,DTDUFC,ETADCF,
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ,CVMNJZ,A8SAV ,A9SAV ,A28SAV,A29SAV

```

	COMMON / FRONT/	17
	1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,	18
	2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,	19
	3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,	20
	4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLOU ,BLOB ,	21
	5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,	22
	6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,	23
	7CNHP ,ETATHP ,DHTCHP ,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,	24
	8CNLP ,ETATLP ,DHTCLP ,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,	25
	9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,	26
	\$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP	27
	COMMON / SIDE/	28
	1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,	29
	2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,	30
	3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,	31
	4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,	32
	5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,JPDUC ,BYPASS ,DUMS3 ,	33
	6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29	34
	COMMON / BACK/	35
	1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,	36
	2XWFB ,XWG55 ,XFAR55 ,XWFD ,XWG24 ,XFAR24 ,XXP1 ,DUMB ,	37
	3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,	38
	4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,	39
	5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,	40
	6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,	41
	7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,	42
	8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,	43
	9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC	44
	COMMON/DUMMYS/DUMMY(100)	45
	LOGICAL ERRER,FAN	46
	EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))	47
	COMMON/ERER/ERRER	48
	DIMENSION VAR(6),DEL(6),ERRB(6),DELVAR(6),EMAT(6,6),VMAT(6),	49
	1AMAT(5)	50
	DATA AWORD/6HENGBAL/	51
	DATA VDELTA,VLIM,VCHNGE,NOMISX/	52
	1 0.001,0.100,0.850,4/	53
	CALL PUTIN	54
	IF (INIT.EQ.1) GO TO 1	55
	TFFHP=TFHPDS	56
	TFFLP=TFLPDS	57
1	LOOPER=0	58
	NUMMAP=0	59
	NOMISS=0	60
2	LOOP=0	61
	MISMAT=0	62
	NOMAP=0	63
	IGO=2	64
	DO 3 I=1,6	65
	VMAT(I)=0.	66
	AMAT(I)=0.	67
	DELVAR(I)=0.	68
	DO 3 L=1,6	69
3	EMAT(I,L)=0.	70
4	LOOPER=LOOPER+1	71
	CALL CDFAN	72
	WORD=AWORD	73
	IF (LJOPER.GT.ITRYS) ERRER=.TRUE.	74
	IF (LJOPER.GT.ITRYS) GO TO 24	75
	IF (NOMAP.GT.0) GO TO 2	76
	NUMMAP=0	77

5	VAR(1)=ZF*100.	78
	IF (MODE.NE.3) VAR(2)=PCNF	79
	IF (MODE.EQ.3) VAR(2)=T4/10.	80
	VAR(3)=ZC*100.	81
	IF (MODE.NE.1) VAR(4)=PCNC	82
	IF (MODE.EQ.1) VAR(4)=T4/10.	83
	VAR(5)=TFFHP	84
	VAR(6)=TFFLP	85
	NMAX=5	86
	IF (FAN.OR.ISPOOL.EQ.2) GO TO 6	87
	NMAX=3	88
	VAR(3)=TFFLP	89
6	DO 7 I=1,NMAX	90
7	IF (ABS(ERR(I)).GT.TOLALL) GO TO 8	91
	CALL PERF	92
	CALL ERROR	93
8	IF (LJJP.GT.0) GO TO 10	94
	MAPEDG=0	95
	MAPSET=0	96
	DO 9 I=1,NMAX	97
	ERRB(I)=ERR(I)	98
9	DEL(I)=VDELTA*VAR(I)	99
	GO TO 13	100
10	IF (MISMAT.GT.0) GO TO 27	101
	IF (MAPEDG.EQ.0) GO TO 11	102
	MAPEDG=0	103
	MAPSET=1	104
	VAR(LJJP)=VAR(LOOP)+2.*DEL(LOOP)	105
	GO TO 14	106
11	IF (MAPSET.EQ.0) VAR(LOOP)=VAR(LOOP)+DEL(LOOP)	107
	IF (MAPSET.EQ.1) VAR(LOOP)=VAR(LOOP)-DEL(LOOP)	108
	MAPSET=0	109
	DO 12 I=1,NMAX	110
12	EMAT(I,LOOP)=(ERRB(I)-ERR(I))/DEL(LOOP)	111
13	LOOP=LOOP+1	112
	IF (LJJP.GT.NMAX) GO TO 15	113
	VAR(LJJP)=VAR(LOOP)-DEL(LOOP)	114
14	ZF=VAR(1)/100.	115
	IF (MODE.NE.3) PCNF=VAR(2)	116
	IF (MODE.EQ.3) T4=VAR(2)*10.	117
	ZC=VAR(3)/100.	118
	IF (MODE.NE.1) PCNC=VAR(4)	119
	IF (MODE.EQ.1) T4=VAR(4)*10.	120
	TFFHP=VAR(5)	121
	TFFLP=VAR(6)	122
	IF (ISPOOL.EQ.1) TFFLP=VAR(3)	123
	IF (ZF.LT.0.) ZF=0.05	124
	IF (ZC.LT.0.) ZC=0.05	125
	GO TO (2,4),IGO	126
15	DO 16 I=1,NMAX	127
16	AMAT(I)=-ERRB(I)	128
	DO 18 I=1,NMAX	129
	IZERO=0	130
	DO 17 LOOP=1,NMAX	131
17	IF (EMAT(I,LOOP).EQ.0.) IZERO=IZERO+1	132
	IF (IZERO.LT.NMAX) GO TO 18	133
	WRITE (6,30) I	134
	LOOPER=ITRYS+100	135
	GO TO 24	136
18	CONTINUE	137
	DO 20 LOOP=1,NMAX	138
	IZERO=0	139
	DO 19 I=1,NMAX	140

19	IF (EMAT(I,LOOP).EQ.0.) IZERO=IZERO+1	141
	IF (IZERO.LT.NMAX) GO TO 20	142
	WRITE (6,31) LOOP	143
	LOOPER=ITRYS+100	144
	GO TO 24	145
20	CONTINUE	146
21	CALL MATRIX (EMAT,VMAT,AMAT,NMAX)	147
	LBIG=0	148
	VARBIG=0.0	149
	DO 22 L=1,NMAX	150
	ABSVAR=ABS(VMAT(L))	151
	IF (ABSVAR.LE.VLIM*VAR(L)) GO TO 22	152
	IF (ABSVAR.LE.VARBIG) GO TO 22	153
	LBIG=L	154
	VARBIG=ABSVAR	155
22	CONTINUE	156
	VRATIO=1.0	157
	IF (LBIG.GT.0) VRATIO=VLIM*VAR(LBIG)/VARBIG	158
	ERRAVE=0.0	159
	VMTAVE=0.0	160
	DELAVE=0.0	161
	DO 23 L=1,NMAX	162
	DELVAR(L)=VRATIO*VMAT(L)	163
	ERRAVE=ERRAVE+ABS(AMAT(L))	164
	VMTAVE=VMTAVE+ABS(VMAT(L))	165
	DELAVE=DELAVE+ABS(DELVAR(L))	166
23	VAR(L)=VAR(L)+DELVAR(L)	167
	ERRAVE=ERRAVE/FLOAT(NMAX)	168
	VMTAVE=VMTAVE/FLOAT(NMAX)	169
	DELAVE=DELAVE/FLOAT(NMAX)	170
	IF (MISMAT.GT.0) GO TO 29	171
	IF (NOMISS.EQ.0) MISMAT=1	172
	IF (MISMAT.EQ.0) IGD=1	173
24	WRITE (8,32) LOOPER	174
	DO 25 I=1,NMAX	175
25	WRITE (8,33) AMAT(I),(EMAT(I,L),L=1,6),VMAT(I),DELVAR(I),VAR(I)	176
	WRITE (8,34) ERRAVE,VMTAVE,DELAVE	177
26	IF (LJOPER.LT.ITRYS) GO TO 14	178
	CALL ERROR	179
	RETURN	180
27	VMTAVX=VMTAVE	181
	DO 28 I=1,NMAX	182
28	AMAT(I)=-ERR(I)	183
	GO TO 21	184
29	WRITE (8,35) AMAT,ERRAVE,DELVAR,DELAVE,VMAT,VMTAVE,VAR	185
	MISMAT=MISMAT+1	186
	IF (VMTAVE.LT.VCHNGE*VMTAVX) GO TO 26	187
	WRITE (8,36)	188
	IF (MISMAT.LT.NOMISX) NOMISS=1	189
	MISMAT=0	190
	LOOP=0	191
	IGD=2	192
	GO TO 5	193
C		194
30	FORMAT (4HROW,12,16H IS ZERO IN EMAT)	195
31	FORMAT (7HCOLUMN,12,16H IS ZERO IN EMAT)	196
32	FORMAT (8H8 ERR8,28X23HERROR MATRIX AFTER LOOP,I4,29X4HVMAT,6X5H	197
	1DELVAR,7X14HVARIALE\$\$\$\$\$)	198
33	FORMAT (1H0,F8.4,8X6F10.4,10XF10.4,F11.4,4XF11.4,6H\$\$\$\$\$)	199
34	FORMAT (1H0,F8.4,32X14HAVERAGE VALUES,31X,2F11.4,6H\$\$\$\$\$)	200

35	FORMAT (12H0----- AMAT,7F16.6,6H\$\$\$\$\$,/,12H -----DELVAR,7F16.6,6	201
	1H\$\$\$\$\$,/,12H ----- VMAT,7F16.6,6H\$\$\$\$\$,/,12H ----- VAR,6F16.6	202
	2,6H\$\$\$\$\$)	203
36	FORMAT (1H0,50X22HCHANGE TOO SMALL\$\$\$\$\$)	204
	END	205

\$IBFTC GUESSS DECK

FUNCTION GUESS(M,T,TD,P,PD,W,WD,D,DD,VD)	1
IF (M.EQ.0) GUESS=VD*((T/TD)**1.60)*((DD/D)**0.50)	2
IF (M.EQ.1) GUESS=VD*((P/PD)**1.80)*((DD/D)**0.33)	3
IF (M.EQ.2) GUESS=VD*((W/WD)**0.33)*((DD/D)**1.00)	4
IF (M.EQ.3) GUESS=VD*((W/WD)**0.00)*((P/PD)**0.50)	5
IF (M.EQ.4) GUESS=VD*((W/WD)**0.00)*((P/PD)**0.50)	6
IF (M.EQ.5) GUESS=VD*((T/TD)**1.10)*((DD/D)**0.60)	7
IF (M.EQ.6) GUESS=VD*((P/PD)**1.00)*((D/DD)**0.25)	8
IF (M.EQ.7) GUESS=VD*((P/PD)**0.62)*((D/DD)**0.31)	9
RETURN	10
END	11

\$IBFTC MATRIX DECK

SUBROUTINE MATRIX (E,V,A,N)	1
DIMENSION E(6,6),V(6),A(6),PIV(7),T(6,7)	2
NN=N+1	3
NM=N-1	4
DO 1 I=1,N	5
T(I,NN)=A(I)	6
DO 1 J=1,N	7
1 T(I,J)=E(I,J)	8
DO 7 I=1,N	9
TEMP=0.	10
DO 2 J=I,N	11
IF (TEMP.GT.ABS(T(J,I))) GO TO 2	12
TEMP=ABS(T(J,I))	13
IPIV=J	14
2 CONTINUE	15
IPI=I+1	16
DO 3 J=IPI,NN	17
3' PIV(J)=T(IPIV,J)/T(IPIV,I)	18
IFROM=N	19
ITO=N	20
4 IF (IFROM.EQ.IPIV) GO TO 6	21
RM=-T(IFROM,I)	22
DO 5 J=IPI,NN	23
5 T(ITO,J)=T(IFROM,J)+RM*PIV(J)	24
ITO=ITO-1	25
6 IFROM=IFROM-1	26
IF (IFROM.GE.I) GO TO 4	27
DO 7 J=IPI,NN	28
7 T(I,J)=PIV(J)	29
DO 8 I=1,NM	30
J=NN-I	31
K=N-I	32
DO 8 L=J,N	33

8	T(K,NN)=T(K,NN)-T(K,L)*T(L,NN)	34
	DO 9 I=1,N	35
9	V(I)=T(I,NN)	36
	RETURN	37
	END	38

SIBFTC PUTIN DECK		
SUBROJTINE PUTIN		1
COMMON /POINT/IDATPT		2
COMMON / ALL/		3
1WORD ,	IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	4
2IGASMX,	IDBURN,IAFTBN, IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	5
3ITRYS ,	LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)	6
COMMON /DESIGN/		7
1PCNFGJ,PCNCGU,T4GU ,	DUMD1 ,DUMD2 ,DELF6 ,DELFN ,DELSFC,	8
2ZFDS ,	PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,	9
3ZCDS ,	PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,	10
4T4DS ,	WFBDS ,DTCODS,ETABDS,WA3CDS,DPCCDS,DTCOCF,ETABCF,	11
5TFHPDS,	CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,	12
6TFLPDS,	CNLPDS,ETLPDS,TFLPCF,CNLPDF,ETLPDF,DHLPDF,T21DS ,	13
7T24DS ,	WFDDDS ,DTODDS,ETADDS,WA23DS,DPDUDS,DTODCF,ETADCF,	14
8T7DS ,	WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,	15
9A55 ,	A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	16
\$PS55 ,	AM55 ,CVDNOZ,CVMNJZ,A8SAV ,A9SAV ,A28SAV,A29SAV	17
COMMON / FRONT/		18
1T1 ,	P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,	19
2T21 ,	P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,	20
3T4 ,	P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,	21
4T55 ,	P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,	22
5CNF ,	PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,	23
6CNC ,	PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,	24
7CNHP ,	ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,	25
8CNLP ,	ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,	26
9AM ,	ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,	27
\$TFFHP ,	TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP	28
COMMON / SIDE/		29
1XP1 ,	XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,	30
2XT21 ,	XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,	31
3T24 ,	P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,	32
4T28 ,	P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,	33
5WAD ,	WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS,DUMS3 ,	34
6TS28 ,	PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29	35
COMMON / BACK/		36
1XT55 ,	XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,	37
2XWFB ,	XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB ,	38
3T6 ,	P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,	39
4T8 ,	P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,	40
5WG6 ,	WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,	41
6PS6 ,	V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,	42
7TS8 ,	PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,	43
8VA ,	FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,	44
9FGM ,	FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC	45
COMMON/DUMMYS/DUMMY(100)		46
COMMON/LOOPPR/KKGO,PRFNEW,PRCNEW		47
DIMENSION XSAVE(308),XFILL(1)		48
EQUIVALENCE (XFILL,WORD)		49
LOGICAL ERROR		50
COMMON/ERER/ERRER		51

```

C *** ITITLE=1 WILL READ IN TITLE 52
C *** MODE =0 FOR CONSTANT T4 53
C *** MODE =1 FOR CONSTANT PCNC 54
C *** MODE =2 FOR CONSTANT WFB 55
C *** MODE=3 FOR CONSTANT PCNF 56
C *** INIT =1 WILL NOT INITIALIZE POINT 57
C *** IDUMP =1 WILL DUMP LOOPING WRITE-OUTS IF ERROR OCCURS 58
C *** IDUMP =2 WILL DUMP LOOPING WRITE-OUTS AFTER EVERY POINT 59
C *** IAMTP =0 WILL USE INPUT AM AND MIL SPEC ETAR 60
C *** IAMTP =1 WILL USE INPUT AM AND INPUT ETAR 61
C *** IAMTP =2 WILL USE T2 AS T1=T1+T2 AND STANDARD P1 62
C *** IAMTP =3 WILL USE P2 AND STANDARD T1 63
C *** IAMTP =4 WILL USE T2 AND P2 64
C *** IAMTP =5 WILL USE RAM2 FOR SPECIAL RECOVERY 65
C *** IGASMX=-1 SEPERATE FLOW, INPUT AM6 66
C *** IGASMX=0 SEPARATE FLOW, A6=A55 67
C *** IGASMX=1 WILL MIX DUCT AND MAIN STREAMS, A6=A25+A55 68
C *** IGASMX=2 WILL MIX DUCT AND MAIN STREAMS, INPUT AM6 69
C *** IDBURN=1 FOR DUCT BURNING, INPUT T24 70
C *** IDBURN=2 FOR DUCT BURNING, INPUT WFD 71
C *** IAFTBN=1 FOR AFTERBURNING, INPUT T7 72
C *** IAFTBN=2 FOR AFTERBURNING, INPUT WFA 73
C *** IDCD =1 DUCT NOZZLE WILL BE C-D 74
C *** IMCD =1 MAIN NOZZLE WILL BE C-D 75
C *** NOZFLT=1 FOR FLOATING MAIN NOZZLE 76
C *** NOZFLT=2 FOR FLOATING DUCT NOZZLE 77
C *** NOZFLT=3 FOR FLOATING MAIN AND DUCT NOZZLES 78
C *** ITRY5 =N NUMBER OF PASSES THRU ENGINE BEFORE QUITTING 79
      DIMENSION ITABLE(400) 80
      DATA (ITABLE(I),I=1,3)/0,400,0/ 81
1      CALL ZERO 82
      IF (K<GO.EQ.1) GO TO 5 83
      IDES=0 84
      CALL INPUT (5,6,1,WORD,ITABLE) 85
      IF (ERRER.AND.IAFTBN.GT.0) GO TO 1 86
      ERRER=.FALSE. 87
C      TABLE IS REFERENCED TO COMMON/ALL/FIRST ENTRY 88
      IF (IDES.EQ.0) GO TO 7 89
      IF (K<GO.NE.2) GO TO 3 90
      DO 2 I=1,308 91
2      XFILL(I)=XSAVE(I) 92
      CALL INPUT (5,6,1,WORD,ITABLE) 93
3      CONTINUE 94
C      SAVE INPUT IN CASE JF LOOP ON PRESSURE RATIOS 95
      DO 4 I=1,308 96
4      XSAVE(I)=XFILL(I) 97
      GO TO 7 98
5      DO 6 I=1,308 99
6      XFILL(I)=XSAVE(I) 100
      WRITE (6,8) PRFDS,PRFNEW,PRCDS,PRCNEW 101
      PRCDS=PRCNEW 102
      PRFDS=PRFNEW 103
7      CONTINUE 104
      KKGO=2 105
      IF (IAFTBN.GT.0.OR.IDBURN.GT.0) INIT=1 106
      IF (NOZFLT.GT.0) INIT=1 107
      IF (MODE.EQ.0) WRITE (8,9) IDES,AM,ALTP,T4,T24,T7 108
      IF (MODE.EQ.1) WRITE (8,10) IDES,AM,ALTP,PCNC,T24,T7 109
      IF (MODE.EQ.2) WRITE (8,11) IDES,AM,ALTP,WFB,T24,T7 110
      CALL COINLT 111
      RETURN 112
C 113

```

C		114
C		115
8	FORMAT (18HOCHANGE PRFDS FROM,F9.3,4H TO,F9.3,16H AND PRCD\$ FROM	116
	1,F10.3,4H TO,F10.3)	117
9	FORMAT (1H0,7H IDE\$=,I3,10X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H	118
	1 T4=,F8.2,5X7H T24=,F8.2,5X7H T7=,F8.2,6H\$\$\$\$\$)	119
10	FORMAT (1H0,7H IDE\$=,I3,10X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H	120
	1PCNC=,F8.3,5X7H T24=,F8.2,5X7H T7=,F8.2,6H\$\$\$\$\$)	121
11	FORMAT (1H0,7H IDE\$=,I3,10X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H	122
	1 WFB=,F8.4,5X7H T24=,F8.2,5X7H T7=,F8.2,6H\$\$\$\$\$)	123
	END	124

\$IBFTC ZERO	DECK	
SUBROUTINE ZERO		1
COMMON / ALL/		2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,		3
2IGASMX,IDBURN,IAFTBV,IDCD ,IMCD ,IDSHOC,IMSHOC,VOZFLT,		4
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPELG,TOLALL,ERR(6)		5
COMMON/DESIGN/QXQ(80)		6
COMMON / FRONT/		7
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,		8
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,		9
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,		10
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,		11
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,		12
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,		13
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,		14
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,		15
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,		16
\$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP		17
COMMON / SIDE/		18
1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,		19
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,		20
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,		21
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,		22
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS,DUMS3 ,		23
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29		24
COMMON / BACK/		25
1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,		26
2XWFB ,XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB ,		27
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,		28
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,		29
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,		30
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,		31
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,		32
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,		33
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC		34
COMMON/DUMMYS/DUMMY(100)		35
DIMENSION Z1(63),Z2(48),Z3(72)		36
EQUIVALENCE (Z1,T1),(Z2,XP1),(Z3,XT55)		37
IDES=0		38
JDES=0		39
INIT=0		40
IDBURN=0		41
IAFTBV=0		42
IDSHOC=3		43
IMSHOC=3		44
T2Q=T2		45

	P2Q=P2	45
	T4Q=T4	47
	DO 1 I=1,63	48
1	Z1(I)=0.	49
	DO 2 I=1,48	50
2	Z2(I)=0.	51
	DO 3 I=1,72	52
3	Z3(I)=0.	53
	T2=T2Q	54
	P2=P2Q	55
	T4=T4Q	56
	CALL SYG (1)	57
	RETURN	58
	END	59

\$IBFTC COINLT DECK		
SUBROJTIME COINLT		1
COMMON / ALL/		2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,		3
2IGASMX ,IDBURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,		4
3ITRYS ,LOOPER ,NOMAP ,NUMMAP ,MAPEDG ,TOLALL ,ERR (6)		5
COMMON / DESIGN/		6
1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELFQ ,DELFN ,DELSFC ,		7
2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,		8
3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,		9
4T4DS ,WFBDS ,DTCODS ,ETABDS ,WA3CDS ,DPCODS ,DTCOCF ,ETABCF ,		10
5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHHPCF ,T2DS ,		11
6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,		12
7T24DS ,WFDDDS ,DTDUDS ,ETADDS ,WA23DS ,DPDUDS ,DTDUCF ,ETADCF ,		13
8T7DS ,WFADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,		14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,		15
\$PS55 ,AM55 ,CVDNOZ ,CVMNOZ ,A8SAV ,A9SAV ,A28SAV ,A29SAV		16
COMMON / FRONT/		17
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,		18
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,		19
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,		20
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,		21
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,		22
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,		23
7CNHP ,ETATHP ,DHTCHP ,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,		24
8CNLP ,ETATLP ,DHTCLP ,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,		25
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,		26
\$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP		27
COMMON / SIDE / ZYX (48) / BACK / YZX (72)		28
COMMON / DUMMYS / DUMMY (100)		29
DATA AWORD / 6HCOINLT /		30
WORD = AWORD		31
AJ = 773.26		32
G = 32.174049		33
ALT = ALTP * 2.0855531E+07 / (2.0855531E+07 - ALTP)		34
CALL ATMOS (ALT , T1 , XX1 , XX2 , XX3 , P1 , CS , XX4 , IIER)		35
IF (IAMTP .EQ. 2) T1 = T1 + T2		36
IF (IAMTP .EQ. 5) CALL RAM2 (AM , ETAR)		37
IF (IAMTP .NE. 1 .AND. IAMTP .VE. 5) CALL RAM (AM , ETAR)		38
FAR = 0.0		39
CALL PROCOM (FAR , T1 , CS , XX2 , XX3 , R1 , PHI1 , H1)		40
S1 = PHI1 - R1 * ALOG (P1)		41
H2 = H1 + (AM * CS) ** 2 / (2. * AJ * G)		42

```

P2T=1.
DO 1 I=1,10
CALL THERMO (P2T,H2,T2T,S2T,AW,0,0.0,1)
IF (ABS(S2T-S1).LE.0.0001*S1) GO TO 2
1 P2T=P1*EXP((AW/1.986375)*((S2T-S1)+(1.986375/AW)*ALOG(P2T/P1)))
CALL ERROR
RETURN
2 IF (IAMTP.EQ.3.OR.IAMTP.EQ.4) ETAR=P2/P2T
P2=ETAR*P2T
IF (IAMTP.NE.4) CALL THERMO (P2,H2,T2,S2,XX5,0,0.0,1)
IF (IAMTP.EQ.4) CALL THERMO (P2,H2,T2,S2,XX5,0,0.0,0)
IF (INIT.EQ.1) GO TO 5
IF (IDES.EQ.1) GO TO 3
IF (MODE.EQ.3) GO TO 4
PCNF=3JESS(MODE,T4,T4DS,PCNC,PCNCDS,WFB,WFBDS,T2,T2DS,PCNFDS)
PCNFGJ=PCNF
GO TO 4
3 PCNF=PCNFDS
PCNFGJ=PCNF
T2DS=T2
4 ZF=ZFDS
5 RETURN
END

```

```

$IBFTC ATMOS DECK
SUBROJTIME ATMOS (ZFT,TM,SIGMA,RHO,THETA,DELTA,CA,AMU,K)
C THIS IS A SUBROUTINE TO COMPUTE CERTAIN ELEMENTS OF THE 1962
C U.S. STANDARD ATMOSPHERE UP TO 90 KILOMETERS.
C CALLING SEQUENCE...
C
C CALL ATMOS (ZFT, TM, SIGMA, RHO, THETA, DELTA, CA, AMU, K)
C ZFT = GEOMETRIC ALTITUDE (FEET)
C TM = MOLECULAR SCALE TEMPERATURE (DEGREES RANKINE)
C SIGMA = RATIO OF DENSITY TO THAT AT SEA LEVEL
C RHO = DENSITY (LB-SEC**2-FT**(-4) OR SLUGS-FT**3)
C THETA = RATIO OF TEMPERATURE TO THAT AT SEA LEVEL
C DELTA = RATIO OF PRESSURE TO THAT AT SEA LEVEL
C CA = SPEED OF SOUND (FT/SEC)
C AMU = VISCOSITY COEFFICIENT (LB-SEC/FT**2)
C
C K = 1 NORMAL
C = 2 ALTITUDE LESS THAN -5000 METERS OR GREATER THAN 90 KM
C = 3 FLOATING POINT OVERFLOW
C
C ALL DATA AND FUNDAMENTAL CONSTANTS ARE IN THE METRIC SYSTEM AS
C THESE QJANTITIES ARE DEFINED AS EXACT IN THIS SYSTEM.
C
C THE RADIUS OF THE EARTH (REFT59) IS THE VALUE ASSOCIATED WITH THE
C 1959 ARJC ATMOSPHERE SO THAT PROGRAMS CURRENTLY USING THE LIBRARY
C ROUTINE WILL NOT REQUIRE ALTERATION TO USE THIS ROUTINE.
C
C DIMENSION HB(10),TMB(10),DELTAB(10),ALM(10)
C DATA(HB(I), TMB(I), DELTAB(I), ALM(I),I=1,10)/
1 -5.0, 320.65, 1.75363E 00, -6.5,
2 0.0, 288.15, 1.00000E 00, -6.5,
3 11.0, 216.65, 2.23361E-01, 0.0,
4 20.0, 216.65, 5.40328E-02, 1.0,
5 32.0, 228.65, 8.56663E-03, 2.8,
6 47.0, 270.65, 1.09455E-03, 0.0,

```

7	52.0,	270.65,	5.82289E-04,	-2.0,	34
8	61.0,	252.65,	1.79718E-04,	-4.0,	35
9	79.0,	180.65,	1.0241 E-05,	0.0,	36
\$	88.743,	180.65,	1.6223 E-06,	0.0/	37
	DATA REFT59/2.0855531E 07/, GZ /9.80665/,				38
1	AMZ	/28.9644	/, RSTAR	/8.31432/,	39
2	FTTOKM	/3.048E-04	/, S	/110.4 /,	40
3	AMUZ	/1.2024E-05	/, CAZ	/1116.45/,	41
4	RHOZ	/0.076474	/, GZENG	/32.1741/	42
C	CONVERT GEOMETRIC ALTITUDE TO GEOPOTENTIAL ALTITUDE				43
	HFT=(REFT59/(REFT59+ZFT))*ZFT				44
C	CONVERT HFT AND ZFT TO KILOMETERS				45
	Z=FTTOKM*ZFT				46
	H=FTTOKM*HFT				47
	K=1				48
	TMZ=TMZ(2)				49
	IF (H.LT.-5.0.OR.Z.GT.90.0) GO TO 7				50
	DO 1 M=1,10				51
	IF (H-HB(M)) 2,3,1				52
1	CONTINUE				53
	GO TO 7				54
2	M=M-1				55
3	DELH=1-HB(M)				56
	IF (ALM(M).EQ.0.0) GO TO 4				57
	TMK=TMZ(M)+ALM(M)*DELH				58
C	GRADIENT IS NON ZERO, PAGE 10, EQUATION 1.2.10-(3)				59
	DELTA=DELTAB(M)*((TMZ(M)/TMK)**(GZ*AMZ/(RSTAR*ALM(M))))				60
	GO TO 5				61
4	TMK=TMZ(M)				62
C	GRADIENT IS ZERO, PAGE 10, EQUATION 1.2.10-(4)				63
	DELTA=DELTAB(M)*EXP(-GZ*AMZ*DELH/(RSTAR*TMZ(M)))				64
5	THETA=TMK/TMZ				65
	SIGMA=DELTA/THETA				66
	ALPHA=SQRT(THETA**3)*((TMZ+S)/(TMK+S))				67
C	CONVERSION TO ENGLISH UNITS				68
	TM=1.8*TMK				69
	RHO=RHOZ*SIGMA/GZENG				70
	CA=CAZ*SQRT(THETA)				71
	AMU=AMUZ*ALPHA/GZENG				72
	CALL JVERFL (J)				73
	GO TO (6,8),J				74
6	K=K+2				75
	GO TO 8				76
7	K=2				77
8	RETURN				78
	END				79

\$IBFTC RAMS DECK		
	SUBROUTINE RAM (AM,ETAR)	1
	IF (AM.GT.1.) GO TO 2	2
	ETAR=1.	3
1	RETURN	4
2	IF (AM.GT.5.) GO TO 3	5
	ETAR=1.-0.075*((AM-1.)*1.35)	6
	GO TO 1	7
3	ETAR=900./((AM**4)+935.)	8
	GO TO 1	9
	END	10

```

$IBFTC RAMT#0 DECK
SUBROJTINE RAM2 (AM,ETAR)
DIMENSION PRINLT(15),FMN(15)
DIMENSION Y(3),X(3)
DATA FMN/0.,.1,.2,.3,.4,.5,.8,1.1,1.2,1.4,1.6,1.8,2.2,2.4,2.7/
DATA PRINLT/.9,.932,.95,.961,.968,.97,.9701,.97,.9681,.958,.94,
1.9181,.858,.8201,.75/
M=0
DO 1 J=1,15
1 IF (AM.GE.FMN(J)) M=J-1
IF (M.EQ.0) M=1
IF (M.GE.14) M=13
DO 2 I=1,3
MM=M-1+I
X(I)=FMN(MM)
2 Y(I)=PRINLT(MM)
CALL PARABO (X,Y,AM,ETAR)
RETURN
END

```

```

$IBFTC COFAN DECK
SUBROJTINE COFAN
COMMON / ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX,IDBURN,IAFTBN,IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)
COMMON /DESIGN/
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELEF ,DELFN ,DELSFC,
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCDS,DTCOCF,ETABCF,
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPCF,DHLPFC,T21DS ,
7T24DS ,WFDDDS ,DTDDDS,ETADDS,WA23DS,DPDDDS,DTDDCF,ETADCF,
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV
COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP
COMMON/SIDE/ZYX(48)/BACK/YZX(72)
COMMON/DUMMYS/DUMMY(100)
COMMON / FAN/CNX(15),PRX(15,15),WACX(15,15),ETAX(15,15),
1NCN,NPT(15)
DIMENSION WLH(2)
DATA A#ORD,WLH/6H COFAN,6H (LO) ,6H (HI) /
WORD=A#ORD
THETA=SQRT(T2/518.668)
IF (IDES.EQ.1) THETAD=THETA

```

	CNF=PCNF*THETAD/(100.*THETA)	37
	IF (IDES.EQ.1) WAFDS=W AFC*P2/THETA	38
	IF (ZF.LT.0.) ZF=0.	39
	IF (ZF.GT.1.) ZF=1.	40
	CNFS=CNF	41
	CALL SEARCH (ZF,CNF,PRF,W AFC,ETAF,CNX(1),NCN,PRX(1,1),WACX(1,1),ET	42
	1AX(1,1),NPT(1),15,15,IGD)	43
	IF ((CNF-CNFS).GT.0.0005*CNF) MAPEDG=1	44
	IF (IGD.EQ.1.OR.IGD.EQ.2) WRITE (8,11) CNFS,WLH(IGD)	45
	WAF=W AFC*P2/THETA	46
	IF (IDES.NE.1) GO TO 1	47
	PRFCF=(PRFDS-1.)/(PRF-1.)	48
	ETAFCF=ETAFDS/ETAF	49
	W AFCF=WAFDS/WAF	50
	WRITE (6,12) PRFCF,ETAFCF,W AFCF,T2DS	51
1	PRF=PRFCF*(PRF-1.)+1.	52
	ETAF=ETAFCF*ETAF	53
	WAF=W AFCF*WAF	54
	W AFC=W AFCF*W AFC	55
	PCNF=100.*THETA*CNF/THETAD	56
	DUMD1=PCNF	57
	CALL THCOMP (PRF,ETAF,T2,H2,S2,P2,T21,H21,S21,P21)	58
	IF (PCBLF.GT.0.) BLF=PCBLF*WAF	59
	IF (JDES.EQ.1) GO TO 8	60
	JDES=1	61
	IF (INIT.EQ.1) GO TO 7	62
	IF (IDES.EQ.1) GO TO 5	63
	IF (MJDE.NE.2) GO TO 2	64
	T4=GUESS(3,Y1,Y2,PCNF,PCNFDS,WFB,WFBDS,Y7,Y8,T4DS)	65
	PCNC=GJESS(4,Y1,Y2,PCNF,PCNFDS,WFB,WFBDS,Y7,Y8,PCNCDS)	66
	GO TO 6	67
2	IF (MJDE.EQ.1) GO TO 4	68
	IF (MJDE.EQ.0) GO TO 3	69
	T4=GUESS(7,Y1,Y2,PCNF,PCNFDS,Y5,Y6,T2,T2DS,T4DS)	70
3	PCNC=GUESS(5,T4,T4DS,Y3,Y4,Y5,Y6,T21,T21DS,PCNCDS)	71
	GO TO 6	72
4	T4=GUESS(6,Y1,Y2,PCNC,PCNCDS,Y5,Y6,T21,T21DS,T4DS)	73
	GO TO 5	74
5	PCNC=PCNCDS	75
	T4=T4DS	76
	WFB=WFBDS	77
	T21DS=T21	78
6	ZC=ZCDS	79
	PCNCGU=PCNC	80
	T4GU=T4	81
7	INIT=0	82
8	IF (MJDE.NE.3) GO TO 9	83
	IF (ABS(CNF-CNFS).LE.0.001*CNFS) GO TO 10	84
	WRITE (8,13) CNFS,CNF	85
	CALL ERROR	86
9	PCNF=100.*THETA*CNF/THETAD	87
10	CALL CCOMP	88
	RETURN	89
C		90
C		91
11	FORMAT (19H0* * * CNF OFF MAP,F10.4,2XA6,11H* * *\$\$\$\$\$)	92
12	FORMAT (11HOFAN DESIGN,13X8H PRFCF=,E15.8,8H ETAFCF=,E15.8,8H WA	93
	1FCF=,E15.8,8H T2DS=,E15.8)	94
13	FORMAT (10HOCNF WAS= ,E15.8,11H AND NOW= ,E15.8,24H CHECK PCNF I	95
	1INPUT\$\$\$\$\$)	96
	END	97

```

$IBFTC COCOMP  DECK
SUBROJTIME COCOMP
COMMON /  ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASM ,IDBURN,IAFTBN,IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)
COMMON /DESIGN/
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC,
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCCOF,ETABCF,
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPDF,ETLPDF,DHLPDF,T21DS ,
7T24DS ,WFDD ,DTDUDS,ETADDS,WA23DS,DPDUDS,DTDUCF,ETADCF,
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV
COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
$TFFHP ,TFFLP ,PCBLF ,PCBL ,PCBLDU,PCBLJB,PCBLHP,PCBLLP
COMMON/SIDE/ZYX(48)/BACK/YZX(72)
COMMON/DUMMYS/DUMMY(100)
LOGICAL ERRER,FAN
EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))
COMMON / COMP/CNX(15),PRX(15,15),WACX(15,15),ETAX(15,15),
1NCN,NPT(15)
DIMENSION WLH(2)
DATA AWORD,WLH/6HCOCOMP,6H (LO) ,6H (HI) /
WORD=AWORD
THETA=SQRT(T21/518.668)
IF (IDES.EQ.1) THETAD=THETA
CNC=PCNC*THETAD/(100.*THETA)
IF (.NOT.FAN) WACC=(WAF-BLF)*THETA/P21
IF (IDES.EQ.1) WACDS=WACC*P21/THETA
IF (ZC.LT.0.) ZC=0.
IF (ZC.GT.1.) ZC=1.
CNCS=CNC
IF (ISPOOL.EQ.1) GO TO 1
CALL SEARCH (ZC,CNC,PRC,WACC,ETAC,CNX(1),NCN,PRX(1,1),WACX(1,1),ET
1AX(1,1),NPT(1),15,15,IGO)
GO TO 2
1 PRC=1.
ETAC=1.
CNC=1.
PRCCF=1.
2 CONTINUE
IF (MODE.EQ.1) GO TO 3
IF ((CNC-CNCS).GT.0.0005*CNC) MAPEDG=1
3 IF (IGO.EQ.1.OR.IGO.EQ.2) WRITE (8,7) CNCS,WLH(IGO)
WAC=WACC*P21/THETA
IF (IDES.NE.1) GO TO 4
IF (ISPOOL.EQ.2) PRCCF=(PRCDS-1.)/(PRC-1.)
ETACCF=ETACDS/ETAC
IF (ISPOOL.EQ.1) ETACCF=1.0

```

	WACCF=WACDS/WAC	62
	WRITE (6,8) PRCCF,ETACCF,WACCF,T21DS	63
4	PRC=PRCCF*(PRC-1.)*1.	64
	ETAC=ETACCF*ETAC	65
	WAC=WACCF*WAC	66
	WACC=WACCF*WACC	67
	IF (.NOT.FAN) ERR(5)=(WAF-WAC-BLF)/WAC	68
	CALL TCOMP (PRC,ETAC,T21,H21,S21,P21,T3,H3,S3,P3)	69
	IF (PCBLC.GT.0.) BLC=PCBLC*WAC	70
	WA3=WAC-BLC	71
	BLDU=PCBLDU*BLC	72
	BLOB=PCBLOB*BLC	73
	BLHP=PCBLHP*BLC	74
	BLLP=PCBLLP*BLC	75
	IF (MODE.NE.1) GO TO 5	76
	IF (ABS(CNC-CNCS).LE.0.001*CNCS) GO TO 6	77
	WRITE (8,9) CNCS,CNC	78
	CALL ERROR	79
5	PCNC=100.*THETA*CNC/THETAD	80
6	CALL CCOMB	81
	RETURN	82
C		83
C		84
7	FORMAT (19H0* * * CNC OFF MAP,F10.4,2XA6,11H* * *\$\$\$\$\$)	85
8	FORMAT (18HCOMPRESSOR DESIGN,6X8H PRCCF=,E15.8,8H ETACCF=,E15.8,	86
	18H WACCF=,E15.8,8H T21DS=,E15.8)	87
9	FORMAT (10H0CNC WAS= ,E15.8,11H AND NOW= ,E15.8,24H CHECK PCNC I	88
	INPUT\$\$\$\$\$)	89
	END	90

\$IBFTC COCOMB DECK

	SUBROUTINE COCOMB	1
	COMMON / ALL/	2
	1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
	2IGASM ,IDBURN,IAFTBN,IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	4
	3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)	5
	COMMON /DESIGN/	6
	1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELF ,DELFN ,DELSFC,	7
	2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,	8
	3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,	9
	4T4DS ,WFBDS ,DTCQDS,ETABDS,WA3CDS,DPCQDS,DTCQCF,ETABCF,	10
	5TFHPDS,CNHPS,ETHPS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,	11
	6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPFC,DHLPFC,T21DS ,	12
	7T24DS ,WFDD ,DTDD ,ETADD ,WA23DS,DPOUDS,DTDDCF,ETADD ,	13
	8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,	14
	9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	15
	\$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV	16
	COMMON / FRONT/	17
	1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,	18
	2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,	19
	3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,	20
	4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,	21
	5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,	22
	6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,	23
	7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,	24
	8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,APEXT ,	25
	9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,	26

	\$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP	27
	COMMON/SIDE/ZYX(48)/BACK/YZX(72)	28
	COMMON/DUMMYS/DIMMY(100)	29
	LOGICAL ERROR,FAN	30
	EQUIVALENCE (FAN,DIMMY(11)),(ISPOOL,DIMMY(6))	31
	COMMON / COMB/PSI(15),DELT(15,15),ETA(15,15),NPS,NPT(15)	32
	DIMENSION Q(9),DUMBO(15,15)	33
	DATA AWORD/6HCOCOMB/	34
	WORD=AWORD	35
	Q(2)=0.	36
	Q(3)=0.	37
	P3PSI=14.696*P3	38
	WA3C=WA3*SQRT(T3)/P3PSI	39
	IF (IDES.EQ.1) WA3CDS=WA3C	40
	DPCOM=DPCODS*(WA3C/WA3CDS)	41
	IF (DPCOM.GT.1.) DPCOM=1.	42
	P4=P3*(1.-DPCOM)	43
1	IF (T4.GT.4000.) T4=4000.	44
	IF (T4.GE.1000.) GO TO 2	45
	T4=1000.	46
	IF (MJDE.EQ.1) MAPEDG=1	47
2	DTCO=T4-T3	48
	IF (IDES.NE.1) GO TO 3	49
	DTCODS=DTCO	50
	DTCCOF=DTCCODS/DTCO	51
3	DTCO=DTCCOF*DTCO	52
	P3PSIN=P3PSI	53
	CALL SEARCH (-1.,P3PSIN,DTCO,ETAB,DUMMY,PSI(1),NPS,DELT(1,1),ETA(1	54
	1,1),DJMBO(1,1),NPT(1),15,15,IGO)	55
	IF (IGO.EQ.7) CALL ERROR	56
	IF (IDES.NE.1) GO TO 4	57
	ETABCF=ETABDS/ETAB	58
4	ETAB=ETABCF*ETAB	59
	HV=((((-0.4594317E-19*T4)-.2034116E-15)*T4+.2783643E-11)*T4+.2051	60
	1501E-07)*T4-.2453116E-03)*T4-.9433296E-01)*T4+.1845537E+05	61
	CALL THERMO (P4,HA,T4,XX1,XX2,0,0,0,0)	62
	FAR4=(HA-H3)/(HV*ETAB)	63
	IF (FAR4.LT.0.) FAR4=0.	64
	WFBX=FAR4*WA3	65
	IF (MJDE.NE.2) GO TO 7	66
	ERRW=(WFB-WFBX)/WFB	67
	DIR=SQRT(WFB/WFBX)	68
	CALL AFQUIR (Q(1),T4,ERRW,0.,20.,0.0001,DIR,T4T,IGO)	69
	GO TO (5,8,6),IGO	70
5	T4=T4T	71
	GO TO 1	72
6	CALL ERROR	73
7	WFB=WFBX	74
8	CALL THERMO (P4,H4,T4,S4,XX2,1,FAR4,0)	75
	WG4=WFB+WA3	76
	IF (IDES.EQ.1) WRITE (6,10) WA3CDS,ETABCF,DTCCOF	77
	IF (ISPOOL.EQ.1) GO TO 9	78
	CALL COMPTB	79
	RETURN	80
9	FAR5=FAR4	81
	T5=T4	82
	WG5=WG4	83
	S5=S4	84
	H5=H4	85
	P5=P4	86
	ETATH=1.	87
	CNHP=1.	88

	DHHPCF=1.	89
	ETHPCF=1.	90
	TFHPCF=1.	91
	THHCAL=0.	92
	DHTCHP=0.	93
	DHTC=3.	94
	CALL C3LPTB	95
	RETURN	96
C		97
10	FORMAT (17H0COMBUSTER DESIGN,7X8H WA3CDS=,E15.8,8H ETABCF=,E15.8,8	98
	1H DTCJCF=,E15.8)	99
	END	100

```

$IBFTC COHPTB DECK
  SUBROJTIME COHPTB
  COMMON / ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX ,IDURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,
3ITRYS ,LODPER ,NOMAP ,NUMMAP ,MAPEDG ,TOLALL ,ERR(6)
  COMMON / DESIGN/
1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC ,
2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,
3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,
4T4DS ,WFBDS ,DTCODS ,ETABDS ,WA3CDS ,DPCODS ,DTCOCF ,ETABCF ,
5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHHPCF ,T2DS ,
6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,
7T24DS ,WFDDDS ,DTDUDS ,ETADDS ,WA23DS ,DPUUDS ,DTDUCF ,ETADCF ,
8T7DS ,WFAADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ ,CVMNOZ ,A8SAV ,A9SAV ,A28SAV ,A29SAV
  COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP ,DHTCHP ,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP ,DHTCLP ,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP
  COMMON/SIDE/QXQ(48)/BACK/QWQ(72)
  COMMON/DUMMYS/DUMMY(100)
  EQUIVALENCE(TFFACT,DUMMY(12)),(CNACT,DUMMY(13)),(DHCACT,DUMMY(14))
  COMMON/TERBHI/DHHISV,TFHISV,CNHISV,ETHISV,DHHPDS
  COMMON /HTURB/TFFX(15),CNX(15,15),D4TCX(15,15),ETATX(15,15),
1NTFFS,NPTTFF(15)
  DATA AWORD,WLO,WHI/6HCOHPTB,6H (LO) ,6H (HI) /
  WORD=AWORD
  IF (IDES.EQ.0) GO TO 1
  CNHPCF=CNHPDS*SQRT(T4)/PCNC
  CNHP=CNHPCF*PCNC/SQRT(T4)
  CNHPS=CNHP
  TFFHPS=TFFHP
  CALL SEARCH (-1.,TFFHP,CNHP,DHTCHP,ETATHP,TFFX(1),NTFFS,CNX(1,1),D
1HTCX(1,1),ETATX(1,1),NPTTFF(1),15,15,IGO)
  IF (IGO.EQ.1.OR.IGO.EQ.11.OR.IGO.EQ.21) WRITE (8,8) TFFHPS,WLO
  IF (IGO.EQ.2.OR.IGO.EQ.12.OR.IGO.EQ.22) WRITE (8,8) TFFHPS,WHI

```

	IF (IGJ.EQ.10.OR.IGJ.EQ.11.OR.IGJ.EQ.12) WRITE (8,9) CNHPS,WLD	45
	IF (IGJ.EQ.20.OR.IGJ.EQ.21.OR.IGJ.EQ.22) WRITE (8,9) CNHPS,WHI	46
	IF (IGJ.NE.7) GO TO 2	47
	CALL ERROR	48
	RETURN	49
2	MAPGO=0	50
	IF (ABS(TFFHPS-TFFHP).LE.0.001*TFFHPS) GO TO 3	51
	MAPGO=1	52
	IF (ABS(CNHPS-CNHP).GT.0.001*CNHPS) MAPGO=3	53
	GO TO 4	54
3	IF (ABS(CNHPS-CNHP).GT.0.001*CNHPS) MAPGO=2	55
4	IF (MAPGO.GT.0) CALL MAPBAC (1,MAPGO,TFFHPS,TFFHP,CNHPS,CNHP,PCNC,	56
	1T4,MODE,NOMAP,NUMMAP)	57
	IF (NJMAP.GT.0) RETURN	58
	TFHCAL=WG4*SQRT(T4)/(14.696*P4)	59
	BTUEXT=0.706705*HPEXT	60
	DHTCC=(BTUEXT+WAC*(H3-H21))/(WG4*T4)	61
	IF (IDES.EQ.0) GO TO 5	62
	TFHPCF=TFHPDS/TFHCAL	63
	DHHPCF=DHTCC/DHTCHP	64
	ETHPCF=ETHPDS/ETATHP	65
	WRITE (6,10) CNHPCF,TFHPCF,ETHPCF,DHHPCF	66
5	TFHCAL=TFHPCF*TFHCAL	67
	DHTCHP=DHHPCF*DHTCHP	68
	ETATHP=ETHPCF*ETATHP	69
	DHTC=DHTCC*T4	70
	TFFACT=TFHCAL/TFHPCF	71
	CNACT=CNHP/CNHPCF	72
	DHCACF=DHTCHP/DHHPCF	73
	ERR(1)=(TFHCAL-TFFHP)/TFHCAL	74
	ERR(2)=(DHTCC-DHTCHP)/DHTCC	75
	CALL THTURB (DHTC,ETATHP,FAR4,H4,S4,P4,T5,H5,S5,P5)	76
	IF (BLHP.LE.0.) GO TO 6	77
	FAR5=HFB/(WA3+BLHP)	78
	WG5=WG4+BLHP	79
	H5=(BLHP*H3+WG4*H5)/WG5	80
	CALL THERMO (P5,H5,T5,S5,XX2,1,FAR5,1)	81
	GO TO 7	82
6	FAR5=FAR4	83
	WG5=WG4	84
7	CALL COLPTB	85
	RETURN	86
C		87
C		88
8	FORMAT (19H0*****TFFHP OFF MAP,F10.4,2XA6,11H*****\$\$\$\$\$\$)	89
9	FORMAT (19H0*****CNHP OFF MAP,F10.4,2XA6,11H*****\$\$\$\$\$\$)	90
10	FORMAT (20H0H.P. TURBINE DESIGN,5X7HCNHPCF=,E15.8,8H TFHPCF=,E15.8	91
	1,8H ETHPCF=,E15.8,8H DHHPCF=,E15.8)	92
	END	93

\$IBFTC COLPTB DECK

SUBROUTINE COLPTB	1
COMMON / ALL/	2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
2IGASM, IDBURN,IAFTBN, IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	4
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)	5
COMMON /DESIGN/	6
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC,	7

```

2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,      8
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,      9
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,     10
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,     11
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLP CF,ETLP CF,DHLP CF,T21DS ,   12
7T24DS ,WFDDSD ,DTODSD,ETADSD,WA23DS,DPDUDS,DTDUCF,ETADCF,   13
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,     14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,                       15
$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV        16
COMMON / FRONT/                                              17
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,                          18
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,                      19
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,                          20
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,                21
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,                22
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,            23
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,           24
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,      25
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,                 26
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP    27
COMMON/SIDE/QXQ(48)/BACK/QWQ(72)                            28
COMMON/DUMMYS/DUMMY(100)                                     29
EQUIVALENCE(TFFACT,DUMMY(15)),(CNACT,DUMMY(16)),(DHCACT,DUMMY(17)) 30
LOGICAL ERROR,FAN                                           31
EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))               32
COMMON/TERBLO/DHLOSV,TFLOSV,CNLOSV,ETLOSV,DHLPDS           33
COMMON /LTURB/TFFX(15),CNX(15,15),DHTCX(15,15),ETATX(15,15), 34
1NTFFS,NPTTFF(15)                                           35
DATA AWORD,WLO,WHI/6HCOLPTB,6H (LO) ,6H (HI) /              36
WORD=AWORD                                                    37
IF (IDES.EQ.0) GO TO 1                                       38
CNLP CF=CNLPDS*SQRT(T5)/PCNF                                  39
CNLP=CNLP CF*PCNF/SQRT(T5)                                    40
CNLPS=CNLP                                                    41
TFFLPS=TFFLP                                                  42
CALL SEARCH (-1.,TFFLP,CNLP,DHTCLP,ETATLP,TFFX(1),NTFFS,CNX(1,1),D 43
1HTCX(1,1),ETATX(1,1),NPTTFF(1),15,15,IGO)                 44
IF (IGO.EQ.1.OR.IGO.EQ.11.OR.IGO.EQ.21) WRITE (8,10) TFFLPS,WLO 45
IF (IGO.EQ.2.OR.IGO.EQ.12.OR.IGO.EQ.22) WRITE (8,10) TFFLPS,WHI 46
IF (IGO.EQ.10.OR.IGO.EQ.11.OR.IGO.EQ.12) WRITE (8,11) CNLPS,WLO 47
IF (IGO.EQ.20.OR.IGO.EQ.21.OR.IGO.EQ.22) WRITE (8,11) CNLPS,WHI 48
IF (IGO.NE.7) GO TO 2                                         49
CALL ERROR                                                    50
RETURN                                                         51
2 MAPGO=0                                                      52
IF (ABS(TFFLPS-TFFLP).LE.0.001*TFFLPS) GO TO 3              53
MAPGO=1                                                         54
IF (ABS(CNLPS-CNLP).GT.0.001*CNLPS) MAPGO=3                  55
GO TO 4                                                         56
3 IF (ABS(CNLPS-CNLP).GT.0.001*CNLPS) MAPGO=2                57
4 IF (MAPGO.GT.0) CALL MAPBAC (2,MAPGO,TFFLPS,TFFLP,CNLPS,CNLP,PCNF, 58
1T4,MODE,NOMAP,NUMMAP)                                         59
IF (NOMAP.GT.0) RETJRN                                         60
TFLCAL=WG5*SQRT(T5)/(14.696*P5)                                61
DHTCF=WAF*(H21-H2)/(WG5*T5)                                    62
IF (ISPOOL.EQ.2) GO TO 5                                       63
BTUEXT=0.706705*HPEXT                                         64
DHTCF=(BTUEXT+WAF*(H21-H2))/(WG5*T5)                          65
CONTINUE                                                       66
IF (IDES.EQ.0) GO TO 6                                         67
TFLPCF=TFLPDS/TFLCAL                                           68
DHLPCF=DHTCF/DHTCLP                                           69

```

	ETLPCF=ETLPDS/ETATLP	70
	WRITE (6,12) CNLPCF,TFLPCF,ETLPCF,DHLPCF	71
6	TFLCAL=TFLPCF*TFLCAL	72
	DHTCLP=DHLPCF*DHTCLP	73
	ETATLP=ETLPCF*ETATLP	74
	DHTF=DHTCF*T5	75
	TFFACT=TFLCAL/TFLPCF	76
	CNACT=CNLP/CNLPCF	77
	DHCACT=DHTCLP/DHLPCF	78
	I1=3	79
	I2=4	80
	IF (ISPOOL.NE.1) GO TO 7	81
	I1=1	82
	I2=2	83
7	ERR(I1)=(TFLCAL-TFFLP)/TFLCAL	84
	ERR(I2)=(DHTCF-DHTCLP)/DHTCF	85
	CALL THTURB (DHTF,ETATLP,FAR5,H5,S5,P5,T55,H55,S55,P55)	86
	IF (BLLP.LE.0.) GO TO 8	87
	FAR55=WFB/(WA3+BLHP+BLLP)	88
	WG55=WG5+BLLP	89
	H55=(BLLP*H3+WG5*H55)/WG55	90
	CALL THERMO (P55,H55,T55,S55,XX2,1,FAR55,1)	91
	GO TO 9	92
8	FAR55=FAR5	93
	WG55=WG5	94
9	CALL FRTOSD	95
	RETURN	96
C		97
C		98
10	FORMAT (19H0*****TFFLP OFF MAP,F10.4,2XA6,11H*****\$\$\$\$\$)	99
11	FORMAT (19H0*****CNLP OFF MAP,F10.4,2XA6,11H*****\$\$\$\$\$)	100
12	FORMAT (20H0L.P. TURBINE DESIGN,5X7HCNLPCF=,E15.8,8H TFLPCF=,E15.8	101
	1,8H ETLPCF=,E15.8,8H DHLPCF=,E15.8)	102
	END	103

\$IBFTC CODUCT DECK

SUBROJTIME CODUCT	1
COMMON / ALL/	2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
2IGASMX ,IDBURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,	4
3ITRYS ,LOOPER ,NOMAP ,NUMMAP ,MAPEDG ,TOLALL ,ERR(6)	5
COMMON /DESIGN/	6
1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELF6 ,DELFN ,DELSFC ,	7
2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,	8
3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,	9
4T4DS ,WFBDS ,DTCODS ,ETABDS ,WA3CDS ,DPCODS ,DTCOCF ,ETABCF ,	10
5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHHPCF ,T2DS ,	11
6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,	12
7T24DS ,WFDDDS ,DTODDS ,ETADDS ,WA23DS ,DPDUDS ,DTDUCF ,ETADCF ,	13
8T7DS ,WFADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,	14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	15
\$PS55 ,AM55 ,CVDNOZ ,CVMNJZ ,A8SAV ,A9SAV ,A28SAV ,A29SAV	16
COMMON /FRONT/XX(80)	17
COMMON / SIDE/	18
1P1 ,WAF ,WAC ,BLF ,BLDU ,H3 ,DUMS1 ,DUMS2 ,	19
2T21 ,P21 ,H21 ,S21 ,T23 ,P23 ,H23 ,S23 ,	20
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,	21

4	T28	,P28	,H28	,S28	,T29	,P29	,H29	,S29	,22
	5	WAD	,WFD	,WG24	,FAR24	,ETAD	,DPDUC	,BYPASS,DUMS3	,23
	6	TS28	,PS28	,V28	,AM28	,TS29	,PS29	,V29	,AM29
		COMMON/BACK/ZZ(72)							24
		COMMON/DUMMYS/DUMMY(100)							25
		EQUIVALENCE(A24,DUMMY(4)),(AM23,DUMMY(5))							26
		EQUIVALENCE(AM25,ZZ(48))							27
		DIMENSION Q(9)							28
		DATA AWORD1,AWORD2/6HCONDUCT,6HDNOZZL/							29
		WORD=AWORD1							30
		Q(2)=3.							31
		Q(3)=3.							32
		AJ=778.26							33
		CAPSF=2116.2170							34
		GOGO=3.0							35
		G=32.174049							36
		WAX=WAF-WAC-BLF							37
		WAD=WAX+BLDU							38
		P23=P21							39
C***		DRY LJSS							40
		H23=(BLDU*H3+WAX*H21)/WAD							41
		CALL THERMD(P23,H23,T23,S23,XX2,1,0.0,1)							42
		WA23C=WAD*SQRT(T23)/P23							43
		IF(IDES.EQ.1) WA23DS=WA23C							44
		BYPASS=(WAF-WAC)/WAC							45
		DPDUC=DPDUCS*(WA23C/WA23DS)							46
		IF(DPDUC.GT.1.) DPDUC=1.0							47
		P24=P23*(1.-DPDUC)							48
		CALL PROCOM(0.,T23,XX1,XX2,XX3,XX4,PHI23,XX6)							49
		IF(IGASM.GT.0) IDBURN=0							50
		AM24=AM23							51
		TS24=T23*0.875							52
1		DO 2 I=1,15							53
		CALL PROCOM(0.,TS24,CS24,AK24,CP24,REX24,PHIS24,HS24)							54
		V24=AM24*CS24							55
		HSCAL=H23-V24**2/(2.*G*AJ)							56
		DELHS=HSCAL-HS24							57
		IF(ABS(DELHS).LE.0.001*HSCAL) GO TO 3							58
2		TS24=TS24+DELHS/CP24							59
		GO TO 10							60
3		C1=P24*SQRT(G/(T23*AJ))*CAPSF							61
		IF(IDES.NE.1) GO TO 4							62
		IF(GOGO.GT.0.) GO TO 4							63
		ASTOA=((AK24+1.)/2.)*((AK24+1.)/(2.*(AK24-1.)))*AM24*(1.+(((AK24-							64
		11.)/2.)*AM24**2))*((AK24+1.)/(2.*(AK24-1.)))							65
		EQWCR=SQRT(G*AK24/REX24/AJ)/(SQRT(518.69)/2116.2)*(2.0/(AK24+1.))*							66
		1*((AK24+1.)/2.)/(AK24-1.))							67
		WA23CC=WA23C/SQRT(518.69)							68
		A24=1./ASTOA*WA23CC/EQWCR							69
		GOGO=1.0							70
4		WQA=WAD/A24							71
		WQAT=C1*SQRT(AK24/REX24)*AM24/(1.+(AK24-1.)*AM24**2/2.)*((AK24+1.							72
		1)/(2.*(AK24-1.)))							73
		DIR=WQA/WQAT							74
		EW=(WQA-WQAT)/WQA							75
		CALL AFQUIR(Q(1),AM24,EW,0.,30.,0.001,DIR,AM24T,IGO)							76
		GO TO (5,6,10),IGO							77
5		AM24=AM24T							78
		IF(AM24.GT.1.0) AM24=0.5							79
		GO TO 1							80
6		PS24=P24/EXP((PHI23-PHIS24)/REX24)							81
		IF(IDBURN.GT.0) GO TO 7							82
									83

C***	NON-DJCT BURNING	84
	T24=T23	85
	WFD=0.	86
	FAR24=0	87
	GO TO 16	88
7	IF (IJBURN.EQ.2) T24=T23+2000.	89
8	IF (T24.GT.4000.) T24=4000.	90
	IF (T24.LT.T23) T24=T23	91
C***	DUCT 3 JURNING	92
	RHO42=CAPSF*PS24/(AJ*REX24*TS24)	93
	PS42=PS24	94
	V42=V24	95
	Q(2)=0.	96
	Q(3)=0.	97
C ***	IF DESIRED, ENTER CALCULATIONS FOR ETAD HERE	98
	HV=((((-0.4594317E-19*T24)-0.2034116E-15)*T24+.2783643E-11)*T24+.2	99
	1051501E-07)*T24-.2453116E-03)*T24-.9433296E-01)*T24+.1845537E+05	100
	CALL THERMO (P24,HA,T24,XX1,XX2,0,0,0,0)	101
	FAR24=(HA-H23)/(HV*ETAD)	102
	IF (FAR24.LT.0.) FAR24=0.	103
	WFDX=FAR24*WAD	104
	IF (IJBURN.NE.2) GO TO 11	105
	ERRW=(WFD-WFDX)/WFD	106
	DIR=SQR(T(WFD/WFDX))	107
	CALL AFQUIR (Q(1),T24,ERRW,0.,20.,0.0001,DIR,T24T,IGO)	108
	GO TO (9,12,10),IGO	109
9	T24=T24T	110
	GO TO 8	111
10	CALL ERROR	112
11	WFD=WFDX	113
12	CONTINUE	114
C***	MOMENT JM LOSS	115
	WG24=WFD+WAD	116
	CALL PROCOM (FAR24,T24,XX1,XX2,XX3,REX24,PHI24,H24)	117
	RHO24=CAPSF*P24/(AJ*REX24*T24)	118
	V24=WG24/(RHO24*A24)	119
	Q(2)=0.	120
	Q(3)=0.	121
	PS24=PS42-0.01	122
13	RHO24=WG24/(V24*A24)	123
	HS24=H24-V24**2/(2.*G*AJ)	124
	CALL THERMO (1.0,HS24,TS24,PHI24,XX2,1,FAR24,1)	125
	IF (TS24.GE.301.) GO TO 14	126
	CALL THERMO (1.0,HS24,400.,PHI24,XX2,1,FAR24,1)	127
	V24=SQR(2.*G*AJ*(H24-HS24))	128
	GO TO 13	129
14	PS24=RHO24*AJ*REX24*TS24/CAPSF	130
	PS24A=PS42+(RHO42*V42**2-RHO24*V24**2)/(G*CAPSF)	131
	DIR=SQR(T(ABS(PS24/PS24A)))	132
	EP=(PS24-PS24A)/PS24	133
	CALL AFQUIR (Q(1),V24,EP,0.,50.,0.001,DIR,V24T,IGO)	134
	V24=V24T	135
	IF (V24.LT.25.) V24=25.	136
	GO TO (13,15,10),IGO	137
15	P24=PS24*EXP((PHI24-PHI24)/REX24)	138
	CALL PROCOM (FAR24,TS24,CS24,XX2,XX3,XX4,XX5,XX6)	139
	AM24=V24*CS24	140
16	CALL THERMO (P24,H24,T24,S24,XX1,1,FAR24,0)	141
	WG24=WFD+WAD	142
	T25=T24	143
	P25=P24	144
	H25=H24	145

	S25=S24	146
	AM25=AM24	147
	IF (IGASM.GT.0) GO TO 20	148
	WORD=WORD2	149
	A28SAV=A28	150
	A29SAV=A29	151
	NOZD=0	152
	IDNOZ=0	153
	IF (NOZFLT.EQ.2.OR.NOZFLT.EQ.3) NOZD=1	154
	IF (IDES.EQ.1.OR.IDBURN.GT.0.OR.NOZD.EQ.1) IDNOZ=1	155
	IF (IDBURN.EQ.1.OR.NOZFLT.EQ.2.OR.NOZFLT.EQ.3) IDCD=1	156
	IF (IDBURN.EQ.0.AND.NOZFLT.EQ.0) IDCD=0	157
	IF (IDCD.EQ.1) GO TO 17	158
	CALL CONVRG (T25,H25,P25,S25,FAR24,WG24,P1,IDNOZ,A28,P25R,T28,H28,	159
	1P28,S29,TS28,PS28,V28,AM28,ICON)	160
	GO TO (18,18,18,10),ICON	161
17	CALL CONDIV (T25,H25,P25,S25,FAR24,WG24,P1,IDNOZ,A28,A29,P25R,T28,	162
	1H28,P28,S28,T29,H29,P29,S29,TS28,TS29,PS28,PS29,V28,V29,AM28,AM29,	163
	2ICON)	164
	IDSHOC=ICON	165
	GO TO (19,19,19,10),ICON	166
18	T29=T28	167
	H29=H28	168
	P29=P28	169
	S29=S28	170
	TS29=TS28	171
	PS29=PS28	172
	V29=V28	173
	AM29=AM28	174
	A29=A28	175
	IDSHOC=ICON+3	176
19	ERR(5)=(P25R-P25)/P25R	177
	IF (IDNOZ.EQ.1) WRITE (6,21) A28,AM28,A29,AM29	178
20	CALL FASTBK	179
	RETURN	180
C		181
C		182
21	FORMAT (19HODUCT NOZZLE DESIGN,5X8H A28=,E15.8,8H AM28=,E15.8	183
	1,8H A29=,E15.8,8H AM29=,E15.8)	184
	END	185

\$IBFTC COMIX DECK

SUBROUTINE COMIX	1
COMMON / ALL/	2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
2IGASM, IDBURN, IAFTBN, IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	4
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)	5
COMMON /DESIGN/	5
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC,	7
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAFCF,WAFCF ,	8
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,	9
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,	10
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPDF,T2DS ,	11
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPDF,ETLPDF,DHLPDF,T21DS ,	12
7T24DS ,WFDDDS ,DTDDDS,ETADDS,WA23DS,DPOUDS,DTDUCF,ETADCF,	13
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,	14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	15

\$PS55	,AM55	,CVDNOZ	,CVMNOZ	,A8SAV	,A9SAV	,A28SAV	,A29SAV	16
COMMON/FRONT/QZQ(80)/SIDE/QWQ(48)								17
COMMON / BACK/								18
1T55	,P55	,H55	,S55	,T25	,P25	,H25	,S25	19
2WFB	,WG55	,FAR55	,WFD	,WG24	,FAR24	,P1	,DUMB	20
3T6	,P6	,H6	,S6	,T7	,P7	,H7	,S7	21
4T8	,P8	,H8	,S8	,T9	,P9	,H9	,S9	22
5WG6	,WFA	,WG7	,FAR7	,ETAA	,DPAFT	,V55	,V25	23
6PS6	,V6	,AM6	,TS7	,PS7	,V7	,AM7	,AM25	24
7TS8	,PS8	,V8	,AM8	,TS9	,PS9	,V9	,AM9	25
8VA	,FRD	,VJD	,FGMD	,VJM	,FGMM	,FGPD	,FGPM	26
9FGM	,FGP	,WFT	,WGT	,FART	,FG	,FN	,SFC	27
EQUIVALENCE(ZF,QZQ(68)),(PCNF,QZQ(69))								28
COMMON/DUMMYS/DUMMY(100)								29
COMMON/LOOPPR/KKGO,PRFNEW,PRCNEW								30
DATA AWORD/6H COMIX/								31
DIMENSION QQ(9)								32
WORD=AWORD								33
AJ=778.26								34
CAPSF=2116.2170								35
G=32.174049								36
CALL PROCOM (FAR55,T55,XX1,XX2,XX3,XX4,PHI55,XX5)								37
CALL PROCOM (FAR24,T25,XX1,XX2,XX3,XX4,PHI25,XX5)								38
IF (IDES.EQ.0) GO TO 12								39
C ***	CALCULATE A55 AND A25 WITH PS25=PS55							40
IF (PS55.EQ.0.) GO TO 3								41
TS55=T55*(PS55/P55)**0.286								42
DO 1 I=1,15								43
CALL PROCOM (FAR55,TS55,CS55,AK55,CP55,REX55,PHI55,HS55)								44
PHIS=PHI55-REX55*ALOG(P55/PS55)								45
DELPHI=PHIS-PHIS55								46
IF (ABS(DELPHI).LE.0.0001*PHIS) GO TO 6								47
1	TS55=TS55*EXP(4.0*DELPHI)							48
2	CALL ERROR							49
RETURN								50
3	TS55=0.875*T55							51
DO 4 I=1,15								52
CALL PROCOM (FAR55,TS55,CS55,AK55,CP55,REX55,PHI55,HS55)								53
V55=AM55*CS55								54
HSCAL=155-V55**2/(2.*G*AJ)								55
DELHS=HSCAL-HS55								56
IF (ABS(DELHS).LE.0.0005*HSCAL) GO TO 5								57
4	TS55=TS55+DELHS/CP55							58
GO TO 2								59
5	PS55=P55/EXP((PHI55-PHIS55)/REX55)							60
IF (PS55.GT.P25.AND.IDES.EQ.1.AND.IGASM.X.GT.0) GO TO 47								61
6	IF (H55.GT.HS55) GO TO 7							62
WRITE (8,48) P55,PS55,T55,TS55,H55,HS55								63
CALL ERROR								64
7	V55=SQRT(2.*G*AJ*(H55-HS55))							65
RHO=CAPSF*PS55/(AJ*REX55*TS55)								66
A55=W355/(RHO*V55)								67
AM55=V55/CS55								68
IF (IGASM.X.GT.0) GO TO 8								69
WRITE (6,49) A55,AM55								70
IF (IGASM.X.EQ.-1) GO TO 35								71
IF (IGASM.X.EQ.0) GO TO 43								72
8	PS25=PS55							73
TS25=T25*(PS25/P25)**0.286								74
DO 9 I=1,15								75
CALL PROCOM (FAR24,TS25,CS25,AK25,CP25,REX25,PHI25,HS25)								76
PHIS=P4I25-REX25*ALOG(P25/PS25)								77

	QQ(2)=0.	141
	QQ(3)=0.	142
	AM25=0.25	143
	TS25=0.875*T25	144
23	DO 24 I=1,15	145
	CALL PROCOM (FAR24,TS25,CS25,AK25,CP25,REX25,PHIS25,HS25)	146
	V25=AM25*CS25	147
	HSCAL=H25-V25**2/(2.*G*AJ)	148
	DELHS=HSCAL-HS25	149
	IF (ABS(DELHS).LE.0.0005*HSCAL) GO TO 25	150
24	TS25=TS25+DELHS/CP25	151
	GO TO 2	152
25	WQAT=C1*SQRT(AK25/REX25)*AM25/(1.+(AK25-1.)*AM25**2/2.)*((AK25+1.1)/(2.*(AK25-1.)))	153
	AMX=AM25	154
	IGOGO=1	155
	GO TO 16	156
26	PS25=P25/EXP((PHI25-PHIS25)/REX25)	157
27	WG6=WG24+WG55	158
	ERR(5)=(PS25-PS55)/PS25	159
	WF6=WFD+WFB	160
	FAR6=WF6/(WG6-WF6)	161
	H6=(WG24*H25+WG55*H55)/WG6	162
	CALL THERMO (1.,H6,T6,PHI6,AMX,1,FAR6,1)	163
	C1=PS55*A55*(1.+AK55*AM55**2)+PS25*A25*(1.+AK25*AM25**2)	164
	TS6=0.833*T6	165
	DO 32 I=1,15	166
	CALL PROCOM (FAR6,TS6,CS6,AK6,CP6,REX6,PHIS6,HS6)	167
	C2=WG5*SQRT(AJ*REX6*T6/(AK6*G))	168
	C3=C2/(CAPSF*C1)	169
	C4=(AK6-1.)/2.-(C3*AK6)**2	170
	C5=1.-2.*AK6*C3**2	171
	C6=C5**2+4.*C4*C3**2	172
	IF (C5) 28,29,30	173
28	CALL ERROR	174
	RETURN	175
29	AM62G=-C5/(2.*C4)	176
	GO TO 31	177
30	AM62G=(SQRT(C6)-C5)/(2.*C4)	178
31	IF (AM62G.LE.0.) GO TO 28	179
	AM6G=SQRT(AM62G)	180
	V6=AM5G*CS6	181
	HSCAL=H6-V6**2/(2.*G*AJ)	182
	DELHS=HSCAL-HS6	183
	IF (ABS(DELHS).LE.0.0005*HSCAL) GO TO 33	184
32	TS6=TS5+DELHS/CP6	185
	GO TO 28	186
33	IF (IGASMX.GT.0) GO TO 34	187
34	A6G=A25+A55	188
	C7=SQRT(1.+(AK6-1.)*AM62G/2.)	189
	PS6=C2/(CAPSF*A6G*AM6G*C7)	190
	P6=PS6*EXP((PHI6-PHIS6)/REX6)	191
	CALL THERMO (P6,H6,T6,S6,XX1,1,FAR6,0)	192
	S6AVE=(WG24*S25+WG55*S55)/WG6	193
	IF (S6.GE.S6AVE) GO TO 35	194
	S6=S6AVE	195
	P6=EXP(AMX*(PHI6-S6)/1.986375)	196
35	IF (IGASMX.EQ.1) GO TO 45	197
	IF (IGASMX.EQ.-1) GO TO 36	198
	IF (IGASMX.EQ.2) GO TO 37	199
36	T6=T55	200
	P6=P55	201
	H6=H55	202
		203

	S6=S55	204
	WG6=WG55	205
	PS6=PS55	206
	FAR6=FAR55	207
	AK6=AK55	208
37	IF (IDES.EQ.0) GO TO 40	209
C***	CALCULATES A6 AS A FUNCTION OF INPUT AM6	210
	TS6=TS/(1.0+(((AK6-1.0)/2.0)*AM6**2))	211
	DO 38 JJ=1,15	212
	AK6P=AK6	213
	CALL PROCOM (FAR6,TS6,CS6,AK6,CP6,REX6,PHIS6,HS6)	214
	V6=AM6*CS6	215
	DELA6=AK6P-AK6	216
	IF (ABS(DELA6).LE.0.0005*AK6) GO TO 39	217
38	TS6=TS/(1.0+(((AK6-1.0)/2.0)*AM6**2))	218
	GO TO 28	219
39	PS6=PS/((1.0+(((AK6-1.0)/2.0)*AM6**2))*((AK6/(AK6-1.0)))	220
	AM6ABD=AM6	221
	RHO=CAPSF*PS6/(AJ*REX6*TS6)	222
	A6=WG6/(RHO*V6)	223
	WRITE (6,54) A6	224
	GO TO 46	225
C	CALCULATES M6=F(A6DESIGN)	226
40	TS6P=TS/(1.0+(((AK6-1.0)/2.0)*AM6ABD**2))	227
	DO 41 I=1,15	228
	CALL PROCOM (FAR6,TS6P,CS6,AK6,CP6,REX6,PHIS6,HS6)	229
	PS6P=PS6*(TS6P/TS6)**(AK6/(AK6-1.0))	230
	RHO6=CAPSF*PS6P/(AJ*REX6*TS6P)	231
	V6=SQRT(2.*G*AJ*(H6-HS6))	232
	IF ((H6-HS6).LT.0.0) GO TO 44	233
	A6P=WG6/(RHO6*V6)	234
	DELA6=A6P-A6	235
	V6=WG6/(RHO6*A6)	236
	AM6=V6/CS6	237
	AM62=AM6**2	238
	IF (ABS(DELA6).LE.00.002*A6) GO TO 42	239
41	TS6P=TS/(1.0+(((AK6-1.0)/2.0)*AM62))	240
	GO TO 28	241
42	TS6=TS6P	242
	PS6=PS6P	243
	GO TO 46	244
43	T6=T55	245
	P6=P55	246
	H6=H55	247
	S6=S55	248
	WG6=WG55	249
	PS6=PS55	250
	V6=V55	251
	AM6=A455	252
	IF (IGASM.X.EQ.0) A6=A55	253
	GO TO 46	254
44	WRITE (6,55) H6,HS6	255
	GO TO 28	256
45	AM62=AM62G	257
	AM6=A46G	258
	A6=A25+A55	259
46	CALL CJAFBN	260
	RETURN	261
47	KKGO=1	262
	OPRDS=PRFDS*PRCDS	263
	PRFNEW=PRFDS*PS55/P25*1.02	264
	PRCNEW=OPRDS/PRFNEW	265

CALL ENGBAL	265
RETURN	267
C	268
C	269
48 FORMAT (22HOSQRT OF H55-HS55 NEG ,6E15.6,6H\$\$\$\$\$)	270
49 FORMAT (20HOTURBINE AREA DESIGN,6X6H A55=,E15.8,8H AM55=,E15.8)	271
50 FORMAT (22HOSQRT OF H25-HS25 NEG ,6E15.6,6H\$\$\$\$\$)	272
51 FORMAT (25HOTURBINE/DUCT AREA DESIGN,7H A55=,E15.8,8H AM55=,E15.8,8H A25=,E15.8,8H AM25=,E15.8)	273
52 FORMAT (12HOCOMIX PCNF=,F7.4,4H AM=,F8.6,5H P55=,F9.5,6H PS55=,F9.5,5H P25=,F9.5,6H PS25=,F9.5,6H\$\$\$\$\$)	275
53 FORMAT (10HOCOMIX ZF=,F8.5,4H AM=,F8.6,5H P55=,F9.5,6H PS55=,F9.5,5H P25=,F9.5,6H PS25=,F9.5,6H\$\$\$\$\$)	277
54 FORMAT (3X,27HAFTERBURNER DESIGN AREA A6 F8.3)	279
55 FORMAT (3X,18HNEG.HS6 FACTOR H6 F9.4,3X,4HHS6 F9.4)	280
END	281

\$IBFTC COAFBN DECK	
SUBROUTINE COAFBN	1
COMMON / ALL/	2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
2IGASM, IDBURN, IAFTBN, IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	4
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)	5
COMMON /DESIGN/	6
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELF6 ,DELFN ,DELSFC,	7
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFDS ,PRFCF ,ETAF6F,WAF6F ,	8
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,	9
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,	10
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,	11
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLP6F,ETLP6F,DHLP6F,T21DS ,	12
7T24DS ,WFDDDS ,DTDUDS,ETADDSD,WA23DS,DPDUDS,DTDUCF,ETADCF,	13
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAF6F,ETAACF,	14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	15
\$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV	16
COMMON/FRONT/QXQ(80)/SIDE/QYQ(48)	17
COMMON / BACK/	18
1T55 ,P55 ,H55 ,S55 ,T25 ,P25 ,H25 ,S25 ,	19
2WFB ,WG55 ,FAR55 ,WFD ,WG24 ,FAR24 ,P1 ,DUMB ,	20
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,	21
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,	22
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,	23
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,	24
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,	25
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,	26
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC	27
COMMON/DUMMYS/DUMMY(100)	28
EQUIVALENCE (P6DSAV,DUMMY(7)),(AM6DSV,DUMMY(8)),(ETAASV,DUMMY(9)),	29
1(FAR7SV,DUMMY(10))	30
DIMENSION Q(9)	31
DATA AWORD/6HCDAFBN/	32
WORD=AWORD	33
Q(2)=0.	34
Q(3)=0.	35
AJ=778.26	36
CAPSF=2116.2170	37
G=32.174049	38
C*** P6DS AND AM6DS ARE SET FOR GENERALIZATION OF AFTERBURNER	39

C***	EFFICIENCY MAP GENERALIZATION	40
	IF (IDES.EQ.1) P6DS=P6*14.696	41
	IF (IDES.EQ.1) AM6DS=AM6	42
	WF6=WFB	43
	IF (I3ASM.XGT.0) WF6=WF6+WFD	44
	WA6=W36-WF6	45
C ***	DRY L3SS	46
	WG6C=N36*SQRT(T6)/P6	47
	IF (IDES.EQ.1) WG6CDS=WG6C	48
	DPAFT=DPAFDS*(WG6C/WG6CDS)	49
	IF (DPAFT.GT.1.) DPAFT=1.	50
	P7=P6*(1.-DPAFT)	51
	A7=A6	52
	FAR6=N6/WA6	53
	CALL PROCOM (FAR6,T6,XX1,XX2,XX3,XX4,PHI6,XX6)	54
	WQA=W35/A7	55
	C1=P7*SQRT(G/(T6*AJ))*CAPSF	56
	AM7=A46	57
	TS7=0.875*T6	58
1	DO 2 I=1,15	59
	CALL PROCOM (FAR6,TS7,CS7,AK7,CP7,REX7,PHIS7,HS7)	60
	V7=AM7*CS7	61
	HSCAL=46-V7**2/(2.*G*AJ)	62
	DELHS=HSCAL-HS7	63
	IF (ABS(DELHS).LE.0.0005*HSCAL) GO TO 3	64
2	TS7=TS7+DELHS/CP7	65
	GO TO 14	66
3	WQAT=C1*SQRT(AK7/REX7)*AM7/(1.+(AK7-1.)*AM7**2/2.)*((AK7+1.)/(2.*	67
	1(AK7-1.)))	68
	DIR=WQA/WQAT	69
	EW=(WQA-WQAT)/WQA	70
	CALL AFQUIR (Q(1),AM7,EW,0.,40.,.001,DIR,AM7T,IGO)	71
	GO TO (4,5,14),IGO	72
4	AM7=AM7T	73
	IF (AM7.GE.1.0) AM7=0.9	74
	GO TO 1	75
5	PS7=P7/EXP((PHI6-PHIS7)/REX7)	76
	IF (IAFTBN.GT.0) GO TO 7	77
C ***	NON-AFTERBURNING	78
6	T7=T6	79
	WFA=0.0	80
	FAR7=FAR6	81
	WG7=W35	82
	IF (IDES.EQ.1.AND.T7DS.NE.0.) GO TO 7	83
	GO TO 20	84
C ***	AFTERBJRNING	85
7	IF (IAFTBN.EQ.2) T7=T6+2000.	86
	IF (IDES.EQ.1) T7=T7DS	87
	IF (T7.LE.T6) GO TO 6	88
	RHO65=CAPSF*PS7/(AJ*REX7*TS7)	89
	PS65=PS7	90
	V65=V7	91
	Q(2)=0.	92
	Q(3)=0.	93
8	IF (T7.GT.4000.) T7=4000.	94
	HV=((((((-.4594317E-19*T7)-.2034116E-15)*T7+.2783643E-11)*T7+.2051	95
	1501E-07)*T4-.2453116E-03)*T4-.9433296E-01)*T4+.1845537E+05	96
	CALL THERMO (P7,HA,T7,XX1,XX2,1,FAR6,0)	97
C***	TO ALTER DESIGN ABETAA MAP FROM GENERAL TO SPECIFIC MAP	98
	IF (IDES.NE.1) GO TO 9	99
	FAR7DS=(HA-H6)/(HV*ETAADS)	100
	CALL ETAAB (0.,0.,0.,0.,ETAADS,ETAASV,P6DS,P6DSAV,AM6DS,AM6DSV,IDE	101
	1S,FAR7DS,FAR7SV)	102

	T7=T6	103
	GO TO 20	104
9	P6GS=25*14.696	105
	FAR7GS=(HA-H6)/(HV*ETAADS)	106
	DO 10 II=1,15	107
	CALL ETAAB (FAR7GS,AM6,P6GS,ETAA,ETAADS,ETAASV,P6DS,P6DSAV,AM6DS,A	108
	IM6DSV,IDES,FAR7DS,FAR7SV)	109
	FAR7=(HA-H6)/(HV*ETAA)	110
	DELFAT=ABS(FAR7-FAR7GS)	111
	IF (DELFAT.LE.0.01*FAR7) GO TO 11	112
10	FAR7GS=FAR7	113
11	CONTINUE	114
	IF (FAR7.GT.0.) GO TO 12	115
	CALL ERROR	116
12	WFA=FAR7*WG6	117
	IF (IAFTBN.EQ.1) GO TO 15	118
	ERRW=(WFA-WFAX)/WFA	119
	DIR=SQRT(WFA/WFAX)	120
	CALL AFQUIR (Q(1),T7,ERRW,0.,30.,.0005,DIR,T7T,IGD)	121
	GO TO (13,16,14),IGD	122
13	T7=T7T	123
	GO TO 8	124
14	CALL ERROR	125
15	WFA=WFAX	126
16	FAR7=(WF6+WFA)/WA6	127
	WG7=WG6+WFA	128
C ***	MOMENTUM LOSS	129
	CALL PROCOM (FAR7,T7,XX1,XX2,XX3,REX7,PHI7,H7)	130
	RHO7=CAPSF*P7/(AJ*REX7*T7)	131
	V7=WG7/(RHO7*A7)	132
	Q(2)=0.	133
	Q(3)=0.	134
	PS7=PS65-0.01	135
17	RHO7=WG7/(V7*A7)	136
	HS7=H7-V7**2/(2.*G*AJ)	137
	CALL THERMO (1.0,HS7,TS7,PHIS7,XX2,1,FAR7,1)	138
	IF (TS7.GE.301.) GO TO 18	139
	CALL THERMO (1.0,HS7,400.,PHIS7,XX2,1,FAR7,0)	140
	V7=SQRT(2.*G*AJ*(H7-HS7))	141
	GO TO 17	142
18	PS7=RHO7*AJ*REX7*TS7/CAPSF	143
	PS7A=PS65+(RHO65*V65**2-RHO7*V7**2)/(G*CAPSF)	144
	DIR=SQRT(ABS(PS7/PS7A))	145
	EP=(PS7-PS7A)/PS7	146
	CALL AFQUIR (Q(1),V7,EP,0.,50.,.001,DIR,V7T,IGD)	147
	V7=V7T	148
	IF (V7.LT.100.) V7=100.	149
	GO TO (17,19,14),IGD	150
19	P7=PS7*EXP((PHI7-PHIS7)/REX7)	151
	CALL PROCOM (FAR7,TS7,CS7,XX2,XX3,XX4,XX5,XX6)	152
	AM7=V7/CS7	153
20	CALL THERMO (P7,H7,T7,S7,XX2,1,FAR7,0)	154
	IF (IDES.EQ.1) WRITE (6,21) WG6CDS	155
	CALL COMNOZ	156
	RETURN	157
C		158
C		159
C		160
21	FORMAT (19H0AFTERBURNER DESIGN,5X8H WG6CDS=,E15.8)	161
	END	162

```

$IBFTC FRTOSD  DECK
SUBROUTINE FRTOSD
COMMON/ALL/XX(28)/DESIGN/YY(80)
COMMON / FRONT/
1T1    ,P1    ,H1    ,S1    ,T2    ,P2    ,H2    ,S2    ,
2T21   ,P21   ,H21   ,S21   ,T3    ,P3    ,H3    ,S3    ,
3T4    ,P4    ,H4    ,S4    ,T5    ,P5    ,H5    ,S5    ,
4T55   ,P55   ,H55   ,S55   ,BLF    ,BLC    ,BLDU    ,BLOB    ,
5CNF   ,PRF    ,ETAF   ,WAF    ,WAF    ,WA3    ,WG4    ,FAR4    ,
6CNC   ,PRC    ,ETAC   ,WACC   ,WAC    ,ETAB   ,DPCOM   ,DUMF    ,
7CNHP  ,ETATHP ,DHTCHP ,DHTC   ,BLHP   ,WG5    ,FAR5    ,CS     ,
8CNLP  ,ETATLP ,DHTCLP ,DHTF   ,BLLP   ,WG55   ,FAR55   ,HPEXT   ,
9AM    ,ALTP   ,ETAR   ,ZF     ,PCNF   ,ZC     ,PCNC    ,WFB     ,
$TFFHP ,TFFLP  ,PCBLF  ,PCBLC  ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP ,
COMMON / SIDE/
1XP1   ,XWAF   ,XWAC   ,XBLF   ,XBLDU ,XH3    ,DUMS1   ,DUMS2   ,
2XT21  ,XP21   ,XH21   ,XS21   ,T23   ,P23    ,H23    ,S23    ,
3T24   ,P24    ,H24    ,S24    ,T25   ,P25    ,H25    ,S25    ,
4T28   ,P28    ,H28    ,S28    ,T29   ,P29    ,H29    ,S29    ,
5WAD   ,WFD    ,WG24   ,FAR24  ,ETAD   ,DPDUC   ,BYPASS  ,DUMS3   ,
6TS28  ,PS28   ,V28    ,AM28   ,TS29   ,PS29   ,V29    ,AM29   ,
COMMON/BACK/ZZ(72)
COMMON/DUMMYS/DUMMY(100)
LOGICAL ERRER,FAN
EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))
XP1=P1
XWAF=WAF
XWAC=WAC
XBLF=BLF
XBLDU=BLDU
XH3=H3
XT21=T21
XP21=P21
XH21=H21
XS21=S21
IF (FAN) CALL CODUCT
IF (FAN) RETURN
CALL FASTBK
RETURN
END

```

```

$IBFTC FASTBC  DECK
SUBROUTINE FASTBK
COMMON/ALL/XX(28)/DESIGN/YY(80)
COMMON / FRONT/
1T1    ,P1    ,H1    ,S1    ,T2    ,P2    ,H2    ,S2    ,
2T21   ,P21   ,H21   ,S21   ,T3    ,P3    ,H3    ,S3    ,
3T4    ,P4    ,H4    ,S4    ,T5    ,P5    ,H5    ,S5    ,
4T55   ,P55   ,H55   ,S55   ,BLF    ,BLC    ,BLDU    ,BLOB    ,
5CNF   ,PRF    ,ETAF   ,WAF    ,WAF    ,WA3    ,WG4    ,FAR4    ,
6CNC   ,PRC    ,ETAC   ,WACC   ,WAC    ,ETAB   ,DPCOM   ,DUMF    ,
7CNHP  ,ETATHP ,DHTCHP ,DHTC   ,BLHP   ,WG5    ,FAR5    ,CS     ,
8CNLP  ,ETATLP ,DHTCLP ,DHTF   ,BLLP   ,WG55   ,FAR55   ,HPEXT   ,
9AM    ,ALTP   ,ETAR   ,ZF     ,PCNF   ,ZC     ,PCNC    ,WFB     ,
$TFFHP ,TFFLP  ,PCBLF  ,PCBLC  ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP ,
COMMON / SIDE/
1XP1   ,XWAF   ,XWAC   ,XBLF   ,XBLDU ,XH3    ,DUMS1   ,DUMS2   ,
2XT21  ,XP21   ,XH21   ,XS21   ,T23   ,P23    ,H23    ,S23    ,

```

3T24	,P24	,H24	,S24	,T25	,P25	,H25	,S25	,	17
4T28	,P28	,H28	,S28	,T29	,P29	,H29	,S29	,	18
5WAD	,WFD	,WG24	,FAR24	,ETAD	,DPDUC	,BYPASS	,DUMS3	,	19
6TS28	,PS28	,V28	,AM28	,TS29	,PS29	,V29	,AM29	,	20
COMMON / BACK/									21
1XT55	,XP55	,XH55	,XS55	,XT25	,XP25	,XH25	,XS25	,	22
2XWFB	,XWG55	,XFAR55	,XWFD	,XWG24	,XFAR24	,XXP1	,DUMB	,	23
3T6	,P6	,H6	,S6	,T7	,P7	,H7	,S7	,	24
4T8	,P8	,H8	,S8	,T9	,P9	,H9	,S9	,	25
5WG6	,WFA	,WG7	,FAR7	,ETAA	,DPAFT	,V55	,V25	,	25
6PS6	,V6	,AM6	,TS7	,PS7	,V7	,AM7	,AM25	,	27
7TS8	,PS8	,V8	,AM8	,TS9	,PS9	,V9	,AM9	,	28
8VA	,FRD	,VJD	,FGMD	,VJM	,FGMM	,FGPD	,FGPM	,	29
9FGM	,FGP	,WFT	,WGT	,FART	,FG	,FN	,SFC	,	30
COMMON/DUMMYS/DUMMY(100)									31
LOGICAL ERROR,FAN									32
EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))									33
XT55=T55									34
XP55=P55									35
XH55=H55									36
XS55=S55									37
IF (FAN) GO TO 1									38
T25=T21									39
P25=P21									40
H25=H21									41
S25=S21									42
WG24=HAF-BLF									43
1	XT25=T25								44
XP25=P25									45
XH25=H25									46
XS25=S25									47
XWFB=WFB									48
XWG55=HGG55									49
XFAR55=FAR55									50
XWFD=HFD									51
XWG24=WG24									52
XFAR24=FAR24									53
XXP1=P1									54
CALL COMIX									55
RETURN									56
END									57

\$18FTC COMN02 DECK

SUBROUTINE COMNOZ

COMMON / ALL/

1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,

2IGASMX ,IDBURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,

3ITRYS ,LOOPER ,NOMAP ,NUMMAP ,MAPEDG ,TOLALL ,ERR(6)

COMMON /DESIGN/

1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC ,

2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,

3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,

4T4DS ,WFBDS ,DTCODS ,ETABDS ,WA3CDS ,DPCJDS ,DTCOCF ,ETABCF ,

5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHHPCF ,T2DS ,

6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,

7T24DS ,WFDDDS ,DTDUDS ,ETADDS ,WA23DS ,DPDUDS ,DTDUCF ,ETADCF ,

8T7DS ,WFADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,

9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,

	\$PS55	,AM55	,CVDNOZ	,CVMNOZ	,A8SAV	,A9SAV	,A28SAV	,A29SAV	15
	COMMON/FRONT/QXQ(80)/SIDE/QYQ(48)								17
	EQUIVALENCE(AM,QXQ(65))								18
	COMMON / BACK/								19
	1T55	,P55	,H55	,S55	,T25	,P25	,H25	,S25	20
	2WFB	,WG55	,FAR55	,WFD	,WG24	,FAR24	,P1	,DUMB	21
	3T6	,P6	,H6	,S6	,T7	,P7	,H7	,S7	22
	4T8	,P8	,H8	,S8	,T9	,P9	,H9	,S9	23
	5WG6	,WFA	,WG7	,FAR7	,ETAA	,DPAFT	,V55	,V25	24
	6PS6	,V6	,AM6	,TS7	,PS7	,V7	,AM7	,AM25	25
	7TS8	,PS8	,V8	,AM8	,TS9	,PS9	,V9	,AM9	26
	8VA	,FRD	,VJD	,FGMD	,VJM	,FGMM	,FGPD	,FGPM	27
	9FGM	,FGP	,WFT	,WGT	,FART	,FG	,FN	,SFC	28
	COMMON/DUMMYS/DUMMY(100)								29
	LOGICAL ERROR,FAN								30
	EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))								31
	DATA A#ORD/6HMNOZZL/								32
	WORD=4#ORD								33
	A8SAV=A8								34
	A9SAV=A9								35
	NOZM=0								36
	IMNOZ=0								37
	IF (NOZFLT.EQ.1.OR.NOZFLT.EQ.3) NOZM=1								38
	IF (IDES.EQ.1.OR.IAFTBN.GT.0.OR.NOZM.EQ.1) IMNOZ=1								39
	IF (IAFTBN.EQ.0.AND.NOZFLT.EQ.0) IMCD=0								40
	IF (IAFTBN.EQ.1.OR.NOZFLT.EQ.1.OR.NOZFLT.EQ.3) IMCD=1								41
	IF (IMCD.EQ.1) GO TO 1								42
	CALL CONVRG (T7,H7,P7,S7,FAR7,WG7,P1,IMNOZ,A8,P7R,T8,H8,P8,S8,TS8,								43
	1PS8,V8,AM8,ICON)								44
	GO TO (3,3,3,2),ICON								45
1	CALL CONDIV (T7,H7,P7,S7,FAR7,WG7,P1,IMNOZ,A8,A9,P7R,T8,H8,P8,S8,T								46
	19,H9,P9,S9,TS8,TS9,PS8,PS9,V8,V9,AM8,AM9,ICON)								47
	IMSHOC=ICON								48
	GO TO (4,4,4,2),ICON								49
2	CALL ERROR								50
3	T9=T8								51
	H9=H8								52
	P9=P8								53
	S9=S8								54
	TS9=TS8								55
	PS9=PS8								56
	V9=V8								57
	AM9=AM8								58
	A9=A8								59
	IMSHOC=ICON+3								60
4	ERR(6)=(P7R-P7)/P7R								61
	IF (ISPOOL.EQ.1) ERR(3)=ERR(6)								62
	IF (IMNOZ.EQ.1) WRITE (6,5) A8,AM8,A9,AM9								63
	RETURN								64
C									65
C									66
5	FORMAT (14HONNOZZLE DESIGN,10X8H A8=,E15.8,8H AM8=,E15.8,8H								67
1	A9=,E15.8,8H AM9=,E15.8)								68
	END								69

```

$IBFTC ERROR   DECK
SUBROUTINE ERROR
COMMON /  ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX,IOBURN,IAFTBN,IDCD ,IMCD ,IDSHJC,IMSHOC,NOZFLT,
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ERR(6)
COMMON /DESIGN/
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFQ ,DELFN ,DELSFC,
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFFS ,PRFCF ,ETAFCF,WAFCF ,
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPFC,DHLPFC,T21DS ,
7T24DS ,WFDDDS ,DTODDS,ETADDS,WA23DS,DPDUDS,DTDUCF,ETADCF,
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV
COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WF8 ,
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLDB,PCBLHP,PCBLLP
COMMON / SIDE/
1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS,DUMS3 ,
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29
COMMON / BACK/
1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,
2XWFB ,XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB ,
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,JPAFT ,V55 ,V25 ,
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC
COMMON/DUMMYS/DUMMY(100)
LOGICAL ERROR
COMMON/ERER/ERRER
DIMENSION TRASH1(80),TRASH2(80),TRASH3(48),TRASH4(72)
EQUIVALENCE (TRASH1,PCNFGU),(TRASH2,T1),(TRASH3,XP1),(TRASH4,XT55)
DATA AWORD/6HCOMMON/
ERRER=.TRUE.
WRITE (6,2) WORD
WORD=AWORD
WRITE (6,3) WORD,ZF,PCNF,ZC,PCNC,T4,MODE
WRITE (6,4)
WRITE (6,5) (TRASH1(I),I=1,80)
WRITE (6,6)
WRITE (6,5) (TRASH2(I),I=1,80)
WRITE (6,4)
WRITE (6,5) (TRASH3(I),I=1,48)
WRITE (6,4)

```

	WRITE (6,5) (TRASH4(I),I=1,72)	62
	WRITE (6,4)	63
	WRITE (6,7) LOOPER	64
	IF (IDJMP.EQ.0) GO TO 1	65
	WRITE (6,6)	66
	CALL SYG (2)	67
1	CALL ENGBAL	68
	RETURN	69
C		70
C		71
2	FORMAT (28H0AN ERROR HAS BEEN FOUND IN ,A6)	72
3	FORMAT (1H0,A6,9X,5E15.6,I4)	73
4	FORMAT (2H0)	74
5	FORMAT (1H0,8E15.6)	75
6	FORMAT (1H1)	76
7	FORMAT (25H0FAILED TO CONVERGE AFTER,I4,6H LOOPS)	77
	END	78

\$IBFTC	SYGS	DECK	
	SUBROUTINE SYG (ICON)		1
	DIMENSION WORD(132)		2
	DATA ONEDOL/6H\$ /		3
	GO TO (1,2),ICON		4
1	END FILE 8		5
	REWIND 8		6
	RETURN		7
C	TERMINATE THE FILE		8
2	WRITE (8,10)		9
	END FILE 8		10
	REWIND 8		11
C	READ RECORD		12
3	READ (8,11) (WORD(I),I=1,132)		13
C	CHECK FOR 12 LEADING DOLLAR SIGNS		14
	DO 4 I=1,12		15
	IF (WORD(I)-ONEDOL) 5,4,5		16
4	CONTINUE		17
	RETURN		18
C	CHECK FOR 6 TRAILING DOLLAR SIGNS		19
5	DO 8 I=1,132		20
	I=I		21
	IF (WORD(I)-ONEDOL) 8,6,8		22
6	K=I+5		23
	DO 7 J=I,K		24
	IF (WORD(J)-ONEDOL) 8,7,8		25
7	CONTINUE		26
	GO TO 9		27
8	CONTINUE		28
	WRITE (6,12)		29
	RETURN		30
C	PRINT LINE		31
9	I=I-1		32
	WRITE (6,11) (WORD(M),M=1,I)		33
	GO TO 3		34
C			35
C			36
10	FORMAT (12H\$\$\$\$\$\$\$\$\$\$\$)		37
11	FORMAT (132A1)		38
12	FORMAT (1H0,12HERROR IN SYG)		39
	END		40

```

$IBFTC PERFOR DECK
SUBROJTIME PERF
COMMON / ALL/
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX ,IDBURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,
3ITRYS ,LOOPER ,NOMAP ,NUMMAP ,MAPEDG ,TOLALL ,ERR(6)
COMMON /DESIGN/
1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC ,
2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,
3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,
4T4DS ,WF8DS ,DTCODS ,ETABDS ,WA3CDS ,DPCODS ,DTCOCF ,ETABCF ,
5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHHPCF ,T2DS ,
6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,
7T24DS ,WFDDDS ,DTDUDS ,ETADDS ,WA23DS ,DPDUDS ,DTDUCF ,ETADCF ,
8T7DS ,WFAADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
$PS55 ,AM55 ,CVDNOZ ,CVMNOZ ,A8SAV ,A9SAV ,A28SAV ,A29SAV
COMMON / FRONT/
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHP ,ETATHP ,DHTCHP ,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLP ,ETATLP ,DHTCLP ,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP
COMMON / SIDE/
1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS ,DUMS3 ,
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29
COMMON / BACK/
1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,
2XWFB ,XWG55 ,XFAR55 ,XWFD ,XWG24 ,XFAR24 ,XXP1 ,DUMB ,
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC
COMMON/DUMMYS/DUMMY(100)
LOGICAL ERRER,FAN
EQUIVALENCE (FAN,DUMMY(11)),(ISPOOL,DUMMY(6))
DATA AWORD/6H PERF/
WORD=AWORD
G=32.174049
CAPSF=2116.2170
WFT=WFB+WFD+WFA
WAT=WAF-BLOB
WGT=WAT+WFT
FART=WFT/WAT
VA=AM*CS
FRD=VA*WAF/G
C TO REMOVE CD NOZ COEFF, CHANGE CALL FRASHJ,CUMNOZ EQ.
C ALSO SUB CONDUCT REMOVE IF (IDBURN.EQ.1) IDCD=1
C ALSO SUB COMNOZ REMOVE IF (IAFTBN.EQ.1) IMCD=1
ARATIO=A9/A8

```

PTRATIO=P8/P1	62
PRATIO=PS9/P1	63
CALL PROCOM (FAR7,TS9,CS9,AK9,CP9,REX9,PHIS9,HS9)	64
CALL FRASHO (PTRATIO,ARATIO,CUMNOZ)	65
CVMNOZ=CUMNOZ	66
VJM=CVMNOZ*V9	67
FGMM=VJM*WG7/G	68
FGPM=CAPSF*(PS9-P1)*A9	69
IF (IGASM*GT.0.OR..NOT.FAN) GO TO 1	70
PRATIO=PS29/P1	71
ARATIO=A29/A28	72
PTRATIO=P28/P1	73
CALL PROCOM (FAR24,TS29,CS9,AK29,CP9,REX9,PHIS9,HS9)	74
CALL FRASHO (PTRATIO,ARATIO,CUDNOZ)	75
CVDNOZ=CUDNOZ	76
VJD=CVDNOZ*V29	77
FGMD=VJD*WG24/G	78
FGPD=CAPSF*(PS29-P1)*A29	79
1 FGM=FGMM+FGMD	80
FGP=F3PM+FGPD	81
FG=FGM+FGP	82
FN=FG-FRD	83
SFC=3500.*WFT/FN	84
FG=DELFG*FG	85
FN=DELFN*FN	86
SFC=DELSFC*SFC	87
CALL JJTPUT	88
CALL ERROR	89
RETURN	90
END	91

\$IBFTC OUTPUT DECK

SUBROUTINE OUTPUT	1
COMMON / ALL/	2
1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,	3
2IGASM, IDURN, IAFTBN, IDCD ,IMCD ,IDSHOC,IMSHOC,NOZFLT,	4
3ITRYS ,LOOPER,NOMAP ,NUMMAP,MAPEDG,TOLALL,ARR(6)	5
COMMON /DESIGN/	6
1PCNFGJ,PCNCGU,T4GU ,DUMD1 ,DUMD2 ,DELFG ,DELFN ,DELSFC,	7
2ZFDS ,PCNFDS,PRFDS ,ETAFDS,WAFFDS ,PRFCF ,ETAFCF,WAFCF ,	8
3ZCDS ,PCNCDS,PRCDS ,ETACDS,WACDS ,PRCCF ,ETACCF,WACCF ,	9
4T4DS ,WFBDS ,DTCODS,ETABDS,WA3CDS,DPCODS,DTCOCF,ETABCF,	10
5TFHPDS,CNHPDS,ETHPDS,TFHPCF,CNHPCF,ETHPCF,DHHPCF,T2DS ,	11
6TFLPDS,CNLPDS,ETLPDS,TFLPCF,CNLPFC,ETLPCF,DHLPFC,T21DS ,	12
7T24DS ,WFDDDS ,DTDUDS,ETADDS,WA23DS,DPDUDS,DTDUCF,ETADCF,	13
8T7DS ,WFADS ,DTAFDS,ETAADS,WG6CDS,DPAFDS,DTAFCF,ETAACF,	14
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,	15
\$PS55 ,AM55 ,CVDNOZ,CVMNOZ,A8SAV ,A9SAV ,A28SAV,A29SAV	16
COMMON / FRONT/	17
1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,	18
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,	19
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,	20
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,	21
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,	22
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,	23
7CNHP ,ETATHP,DHTCHP,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,	24
8CNLP ,ETATLP,DHTCLP,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,	25
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,	26

\$TFFHP ,TFFLP ,PCBLF ,PCBLC ,PCBLDU,PCBLOB,PCBLHP,PCBLLP	27
COMMON / SIDE/	28
1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,	29
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,	30
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,	31
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,	32
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS,DUMS3 ,	33
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29	34
COMMON / BACK/	35
1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,	36
2XWFB ,XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB ,	37
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 ,	38
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 ,	39
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 ,	40
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 ,	41
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 ,	42
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM ,	43
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC	44
COMMON / DUMMYS / DUMMY(100)	45
EQUIVALENCE (T4PBL,DUMMY(2)),(T41,DUMMY(3))	46
DIMENSION W(5,4),ANS1(80),ANS2(80),ANS3(48),ANS4(72)	47
EQUIVALENCE (ANS1,PCNFGU),(ANS2,T1),(ANS3,XP1),(ANS4,XT55)	48
DATA AWORD1,AWORD2/6HOUTPUT,6HCOMMON/	49
DATA (W(1,I),I=1,4)/6HSUBSON,6HIC C-D,6H NOZZL,6HE /	50
DATA (W(2,I),I=1,4)/6HSHOCK ,6HINSIDE,6H C-D N,6HNOZZLE /	51
DATA (W(3,I),I=1,4)/6HSHOCK ,6HOUTSID,6HE C-D ,6HNOZZLE/	52
DATA (W(4,I),I=1,4)/6HSUBSON,6HIC CON,6HVERG. ,6HNOZZLE/	53
DATA (W(5,I),I=1,4)/6HSONIC ,6HCONVER,6HGENT N,6HNOZZLE /	54
WORD=AWORD1	55
IF (ICBURN.GT.0) GO TO 2	56
IF (IAFTBN.GT.0) GO TO 1	57
WRITE (6,7) WORD,AM,ALTP,T4,ETAR	58
GO TO 3	59
1 WRITE (6,8) WORD,AM,ALTP,T4,T7,ETAR	60
GO TO 3	61
2 WRITE (6,9) WORD,AM,ALTP,T4,T24,ETAR	62
3 CALL CONOUT (2)	63
WRITE (6,10) (W(IMSHOC,I),I=1,4),FG,FN,SFC	64
IF (IGASMX.GT.0) GO TO 4	65
WRITE (6,11) (W(IDSHOC,I),I=1,4)	66
4 WRITE (6,12) LOOPER	67
IF (IDES.NE.1) GO TO 5	68
WORD=AWORD2	69
WRITE (6,13) WORD,ZF,PCNF,ZI,PCNI,ZC,PCNC,T4,MODE	70
WRITE (6,14)	71
WRITE (6,15) (ANS1(I),I=1,80)	72
WRITE (6,14)	73
WRITE (6,15) (ANS2(I),I=1,80)	74
WRITE (6,14)	75
WRITE (6,15) (ANS3(I),I=1,48)	76
WRITE (6,14)	77
WRITE (6,15) (ANS4(I),I=1,72)	78
WRITE (6,16)	79
IF (IDES.EQ.1) GO TO 6	80
5 CONTINUE	81
A8=A8SAV	82
A9=A9SAV	83
A28=A28SAV	84
A29=A29SAV	85
IF (IJUMP.NE.2) GO TO 6	86
WRITE (6,16)	87
CALL SYG (2)	88

6	CALL ENGBAL	89
	RETURN	90
C		91
C		92
C		93
7	FORMAT (1HB,A6,14X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H T4=,F8.2	94
	1,25X7H ETAR=,F7.4)	95
8	FORMAT (1HB,A6,14X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H T4=,F8.2	96
	1,5X7H T7=,F8.2,5X7H ETAR=,F7.4)	97
9	FORMAT (1HB,A6,14X7H AM=,F7.3,6X7H ALTP=,F7.0,6X7H T4=,F8.2	98
	1,5X7H T24=,F8.2,5X7H ETAR=,F7.4)	99
10	FORMAT (6HOMAIN ,4A5,9X3HFG=,F9.2,18X3HFN=,F9.2,18X4HSFC=,F8.5)	100
11	FORMAT (6H DUCT ,4A5)	101
12	FORMAT (16HOCONVERGED AFTER,I4,6H LOOPS,/,1H1)	102
13	FORMAT (1H ,A6,9X,7E15.6,I4)	103
14	FORMAT (1H)	104
15	FORMAT (1H ,8E15.6)	105
16	FORMAT (1H1)	106
	END	107

\$IBFTC CONOJIT DECK

SUBROJTIME CONOUT (ICON)

COMMON / ALL/

1WORD ,IDES ,JDES ,KDES ,MODE ,INIT ,IDUMP ,IAMTP ,
2IGASMX ,IDBURN ,IAFTBN ,IDCD ,IMCD ,IDSHOC ,IMSHOC ,NOZFLT ,
3ITRYS ,LOOPER ,NDMAP ,NUMMAP ,MAPELG ,TOLALL ,ERR(6)

COMMON /DESIGN/

1PCNFGJ ,PCNCGU ,T4GU ,DUMD1 ,DUMD2 ,DELF6 ,DELFN ,DELSFC ,
2ZFDS ,PCNFDS ,PRFDS ,ETAFDS ,WAFDS ,PRFCF ,ETAFCF ,WAFCF ,
3ZCDS ,PCNCDS ,PRCDS ,ETACDS ,WACDS ,PRCCF ,ETACCF ,WACCF ,
4T4DS ,WFBDS ,DTCODS ,ETABDS ,WA3CDS ,DPCODS ,DTCOCF ,ETABCF ,
5TFHPDS ,CNHPDS ,ETHPDS ,TFHPCF ,CNHPCF ,ETHPCF ,DHPCF ,T2DS ,
6TFLPDS ,CNLPDS ,ETLPDS ,TFLPCF ,CNLPCF ,ETLPCF ,DHLPCF ,T21DS ,
7T24DS ,WFDDDS ,DTDUDS ,ETADDS ,WA23DS ,DPDUDS ,DTDUCF ,ETADCF ,
8T7DS ,WFAADS ,DTAFDS ,ETAADS ,WG6CDS ,DPAFDS ,DTAFCF ,ETAACF ,
9A55 ,A25 ,A6 ,A7 ,A8 ,A9 ,A28 ,A29 ,
\$PS55 ,AM55 ,CVDNOZ ,CVMNOZ ,A8SAV ,A9SAV ,A28SAV ,A29SAV

COMMON / FRONT/

1T1 ,P1 ,H1 ,S1 ,T2 ,P2 ,H2 ,S2 ,
2T21 ,P21 ,H21 ,S21 ,T3 ,P3 ,H3 ,S3 ,
3T4 ,P4 ,H4 ,S4 ,T5 ,P5 ,H5 ,S5 ,
4T55 ,P55 ,H55 ,S55 ,BLF ,BLC ,BLDU ,BLOB ,
5CNF ,PRF ,ETAF ,WAF ,WAF ,WA3 ,WG4 ,FAR4 ,
6CNC ,PRC ,ETAC ,WACC ,WAC ,ETAB ,DPCOM ,DUMF ,
7CNHPM ,ETATHP ,DHTCHM ,DHTC ,BLHP ,WG5 ,FAR5 ,CS ,
8CNLPM ,ETATLP ,DHTCLM ,DHTF ,BLLP ,WG55 ,FAR55 ,HPEXT ,
9AM ,ALTP ,ETAR ,ZF ,PCNF ,ZC ,PCNC ,WFB ,
\$TFFHPM ,TFFLPM ,PCBLF ,PCBLC ,PCBLDU ,PCBLOB ,PCBLHP ,PCBLLP

COMMON / SIDE/

1XP1 ,XWAF ,XWAC ,XBLF ,XBLDU ,XH3 ,DUMS1 ,DUMS2 ,
2XT21 ,XP21 ,XH21 ,XS21 ,T23 ,P23 ,H23 ,S23 ,
3T24 ,P24 ,H24 ,S24 ,T25 ,P25 ,H25 ,S25 ,
4T28 ,P28 ,H28 ,S28 ,T29 ,P29 ,H29 ,S29 ,
5WAD ,WFD ,WG24 ,FAR24 ,ETAD ,DPDUC ,BYPASS ,DUMS3 ,
6TS28 ,PS28 ,V28 ,AM28 ,TS29 ,PS29 ,V29 ,AM29

COMMON / BACK/

1XT55 ,XP55 ,XH55 ,XS55 ,XT25 ,XP25 ,XH25 ,XS25 ,

```

2XWFB ,XWG55 ,XFAR55,XWFD ,XWG24 ,XFAR24,XXP1 ,DUMB , 37
3T6 ,P6 ,H6 ,S6 ,T7 ,P7 ,H7 ,S7 , 38
4T8 ,P8 ,H8 ,S8 ,T9 ,P9 ,H9 ,S9 , 39
5WG6 ,WFA ,WG7 ,FAR7 ,ETAA ,DPAFT ,V55 ,V25 , 40
6PS6 ,V6 ,AM6 ,TS7 ,PS7 ,V7 ,AM7 ,AM25 , 41
7TS8 ,PS8 ,V8 ,AM8 ,TS9 ,PS9 ,V9 ,AM9 , 42
8VA ,FRD ,VJD ,FGMD ,VJM ,FGMM ,FGPD ,FGPM , 43
9FGM ,FGP ,WFT ,WGT ,FART ,FG ,FN ,SFC , 44
COMMON/DUMMYS/DUMMY(100) 45
EQUIVALENCE (TFFHP,DUMMY(12)),(CNHP,DUMMY(13)),(DHTCHP,DUMMY(14)) 46
EQUIVALENCE (TFFLP,DUMMY(15)),(CNLP,DUMMY(16)),(DHTCLP,DUMMY(17)) 47
DIMENSION PARAM(380),WORDY(380),IDUT(103),AOUT(6),WOUT(6) 48
EQUIVALENCE (PARAM,PCNFGU) 49
DATA (WORDY(I),I=1,98)/ 50
16HPCNFGU,6HPCNCGU,6HT4GU ,6HDUMD1 ,6HDUMD2 ,6HDELF6 ,6HDELFN , 51
26HDELSFC,6HZFDS ,6HPCNFDS,6HPRFDS ,6HETAFDS,6HWAFFDS ,6HPRFCF , 52
36HETAFCF,6HWAFCF ,6HZCDS ,6HPCNCDS,6HPRCDS ,6HETACDS,6HWACDS , 53
46HPRCCF ,6HETACCF,6HWACCF ,6HT4DS ,6HWFBDs ,6HDTCODS,6HETABDS , 54
56HWA3CDS,6HDPCDSD,6HDTCCCF,6HETABCF,6HTFHPDS,6HCNHPDS,6HETHPDS , 55
66HTFHPCF,6HCNHPCF,6HETHPCF,6HDHHPCF,6HT2DS ,6HTFLPDS,6HCNLPDS , 56
76HETLPDS,6HTFLPCF,6HCNLPDF,6HETLPCF,6HDHLPDF,6HT21DS ,6HT24DS , 57
86HWFODS ,6HDTDUOS,6HETADDS,6HWA23DS,6HDPDUOS,6HDTDUOF,6HETADCF , 58
96HT7DS ,6HWFADS ,6HDTAFDS,6HETAADS,6HWG6CDS,6HDPAFDS,6HDTAFCF , 59
$6HETAACF,6HA55 ,6HA25 ,6HA6 ,6HA7 ,6HA8 ,6HA9 , 60
$6HA28 ,6HA29 ,6HPS55 ,6HAM55 ,6HCVDNOZ,6HCVMOZ,6HABSAV , 61
$6HA9SAV ,6HA28SAV,6HA29SAV,6HT1 ,6HP1 ,6H41 ,6HS1 , 62
$6HT2 ,6HP2 ,6HH2 ,6HS2 ,6HT21 ,6HP21 ,6HH21 , 63
$6HS21 ,6HT3 ,6HP3 ,6HH3 ,6HS3 ,6HT4 ,6HP4 / 64
DATA (WORDY(I),I=99,189)/ 65
16HH4 ,6HS4 ,6HT5 ,6HP5 ,6HH5 ,6HS5 ,6HT55 , 66
26HP55 ,6HH55 ,6HS55 ,6HBLF ,6HBLC ,6HBLDU ,6HBLOB , 67
36HCNF ,6HPRF ,6HETAF ,6HWAFC ,6HWAFF ,6HWA3 ,6HWG4 , 68
46HFAR4 ,6HCNC ,6HPRC ,6HETAC ,6HWACC ,6HWAC ,6HETAB , 69
56HDPCJM ,6HDUMF ,6HCNHPM ,6HETATHP,6HDHTCHM,6HDHTC ,6HBLHP , 70
66HWG5 ,6HFAR5 ,6HCS ,6HCNLP ,6HETATLP,6HDHTCLM,6HDHTF , 71
76HBLLP ,6HWG55 ,6HFAR55 ,6HHPEXT ,6HAM ,6HALTP ,6HETAR , 72
86HZF ,6HPCNF ,6HZC ,6HPCNC ,6HWF6 ,6HTFFHPM,6HTFFLPM , 73
96HPCBLF ,6HPCBLC ,6HPCBLDU,6HPCBLOB,6HPCBLHP,6HPCBLLP,6HXP1 , 74
$6HXWAF ,6HXWAC ,6HXBLF ,6HXBLDU ,6HXH3 ,6HDUMS1 ,6HDUMS2 , 75
$6HXT21 ,6HXP21 ,6HXH21 ,6HXS21 ,6HT23 ,6HP23 ,6HH23 , 76
$6HS23 ,6HT24 ,6HP24 ,6HH24 ,6HS24 ,6HT25 ,6HP25 , 77
$6HH25 ,6HS25 ,6HT28 ,6HP28 ,6HH28 ,6HS28 ,6HT29 / 78
DATA (WORDY(I),I=190,280)/ 79
16HP29 ,6HH29 ,6HS29 ,6HWAD ,6HWFD ,6HWG24 ,6HFAR24 , 80
26HETAD ,6HDPDUC ,6HBYPASS,6HDUMS3 ,6HTS28 ,6HPS28 ,6HV28 , 81
36HAM23 ,6HTS29 ,6HPS29 ,6HV29 ,6HAM29 ,6HXT55 ,6HXP55 , 82
46HXH55 ,6HXS55 ,6HXT25 ,6HXP25 ,6HXH25 ,6HXS25 ,6HXWFB , 83
56HXWG55 ,6HXFAR55,6HXXWFD ,6HXXG24 ,6HXXFAR24,6HXXP1 ,6HDUMB , 84
66HT6 ,6HP6 ,6HH6 ,6HS6 ,6HT7 ,6HP7 ,6HH7 , 85
76HS7 ,6HT8 ,6HP8 ,6HH8 ,6HS8 ,6HT9 ,6HP9 , 86
86HH9 ,6HS9 ,6HWG6 ,6HWF4 ,6HWG7 ,6HFAR7 ,6HETAA , 87
96HDPAFT ,6HV55 ,6HV25 ,6HPS6 ,6HV6 ,6HAM6 ,6HTS7 , 88
$6HPS7 ,6HV7 ,6HAM7 ,6HAM25 ,6HTS8 ,6HPS8 ,6HV8 , 89
$6HAM8 ,6HTS9 ,6HPS9 ,6HV9 ,6HAM9 ,6HVA ,6HFRD , 90
$6HVJD ,6HFGMD ,6HVJM ,6HFGMM ,6HFGPD ,6HFGPM ,6HFGM , 91
$6HFGP ,6HWFT ,6HWGT ,6HFART ,6HFG ,6HFN ,6HSFC / 92
DATA (WORDY(I),I=281,297)/ 93
111*0. ,6HTFFHP ,6HCNHP ,6HDHTCHP,6HTFFLP ,6HCNLP ,6HDHTCLP/ 94
DATA THEEND,BLANK,LIMIT/ 6HTHEEND,6H ,297/ 95
GO TO (1,6),ICON 96

```

C *** INPUT SECTION	97
1 DO 4 N=1,103	98
NUM=N	99
READ (5,11) AIN,CHANGE	100
IF (AIN.EQ.THEEND) GO TO 5	101
DO 2 J=1,LIMIT	102
JJ=J	103
IF (AIN.EQ.WORDY(J)) GO TO 3	104
2 CONTINUE	105
WRITE (6,12) AIN	106
GO TO 4	107
3 IOUT(NUM)=JJ	108
IF (CHANGE.NE.BLANK) WORDY(JJ)=CHANGE	109
4 CONTINUE	110
WRITE (6,13)	111
5 NUM=NUM-1	112
RETURN	113
C *** OUTPUT SECTION	114
6 IF (NUM.EQ.1) GO TO 10	115
N=NUM	116
J=6	117
DO 9 I=1,NUM,6	118
IF (N.GT.6) GO TO 7	119
J=N	120
7 N=N-6	121
DO 8 K=1,J	122
L=I+K-1	123
M=IOUT(L)	124
WOUT(K)=WORDY(M)	125
8 AOUT(K)=PARAM(M)	126
WRITE (6,14) (WOUT(K),K=1,J)	127
WRITE (6,15) (AOUT(K),K=1,J)	128
IF (N.LE.0) GO TO 10	129
9 CONTINUE	130
10 RETURN	131
C	132
C	133
11 FORMAT (A6,6X,A6)	134
12 FORMAT (10H0THE WORD ,A6,26H NOT FOUND IN COMMON ARRAY)	135
13 FORMAT (22H0ERROR IN CONOUT INPUT)	136
14 FORMAT (1H ,25XA6,5(9XA6))	137
15 FORMAT (1H ,20X6E15.6)	138
END	139

\$IBFTC THCOMP DECK	
SUBROUTINE THCOMP (PR,ETA,T,H,S,P,TO,HO,SO,PO)	1
PO=P*PR	2
TP=T*PR**0.28572	3
DO 1 I=1,25	4
CALL THERMO (PO,HP,TP,SP,X1,0,X2,0)	5
DELS=SP-S	6
IF (ABS(DELS).LE.0.00005*S) GO TO 2	7
1 TP=TP/EXP(4.*DELS)	8
CALL ERROR	9
2 HO=H+((HP-H)/ETA)	10
CALL THERMO (PO,HO,TO,SO,X1,0,X2,1)	11
RETURN	12
END	13

\$IBFTC	PROCJM	DECK		
	SUBROUTINE	PROCJM (FARX,TEX,CSEX,AKEX,CPEX,REX,PHI,HEX)		1
	IF (FARX.LE.0.067623)	GO TO 1		2
	FARX=	0.067623		3
1	IF (TEX.GE.300.)	GO TO 2		4
	TEX=	300.		5
2	IF (TEX.LE.4000.)	GO TO 3		6
	TEX=	4000.		7
3	IF (FARX.GE.0.0)	GO TO 4		8
	FARX=	0.0		9
C	AIR PATH			10
4	CPA=(((1.0115540E-25*TEX-1.4526770E-21)*TEX+7.6215767E-18)*TEX-			11
	11.5128259E-14)*TEX-6.7178376E-12)*TEX+6.5519486E-08)*TEX-5.1536879			12
	2E-05)*TEX+2.5020051E-01			13
	HEA=(((1.2544425E-26*TEX-2.0752522E-22)*TEX+1.2702630E-18)*TEX			14
	1-3.0255518E-15)*TEX-1.6794594E-12)*TEX+2.1839826E-08)*TEX-2.576844			15
	20E-05)*TEX+2.5020051E-01)*TEX-1.7558886E+00			16
	SEA=+2.5020051E-01*ALOG(TEX)+(((1.4450767E-26*TEX-2.4211288E-22			17
	1)*TEX+1.5243153E-18)*TEX-3.7820648E-15)*TEX-2.2392790E-12)*TEX+3.2			18
	2759743E-08)*TEX-5.1576879E-05)*TEX+4.5432300E-02			19
	IF (FARX.LE.0.0)	GO TO 5		20
C	FUEL/AIR PATH			21
	CPF=(((1.7.2678710E-25*TEX-1.3335668E-20)*TEX+1.0212913E-16)*TEX-			22
	14.2051104E-13)*TEX+9.9686793E-10)*TEX-1.3771901E-06)*TEX+1.2258630			23
	2E-03)*TEX+7.3816638E-02			24
	HEF=(((1.9.0848388E-26*TEX-1.9050949E-21)*TEX+1.7021525E-17)*TEX			25
	1-8.4102208E-14)*TEX+2.4921698E-10)*TEX-4.5906332E-07)*TEX+6.129315			26
	20E-04)*TEX+7.3816638E-02)*TEX+3.0581530E+01			27
	SEF=+7.3816638E-02*ALOG(TEX)+(((1.0382670E-25*TEX-2.2226118E-21			28
	1)*TEX+2.0425826E-17)*TEX-1.0512776E-13)*TEX+3.3228928E-10)*TEX-6.8			29
	2859505E-07)*TEX+1.2258630E-03)*TEX+6.483398E-01			30
5	CPEX=(CPA+FARX*CPF)/(1.+FARX)			31
	HEX=(HEA+FARX*HEF)/(1.+FARX)			32
	PHI=(SEA+FARX*SEF)/(1.+FARX)			33
	AMW=29.97-.946186*FARX			34
	REX=1.986375/AMW			35
	AKEX=CPEX/(CPEX-REX)			36
	CSEX=SQRT(AKEX*REX*TEX*25031.37)			37
	RETURN			38
	END			39

\$IBFTC	SERCH	DECK		
	SUBROUTINE	SEARCH (P,A,B,C,D,AX,NA,BX,CX,DX,NO,NAM,NOM,NCODE)		1
	DIMENSION	AX(NAM),BX(NAM,NOM),CX(NAM,NOM),DX(NAM,NOM),NO(NAM),Q(9)		2
C ***	NEEDS	SUBROUTINE AFQUIR		3
C ***	AX AND BX	MUST BE STORED LO TO HI		4
C ***	P=INPJT	PROPORTION BETWEEN 0.0 AND 1.0		5
C	IF NOT	INPUT, P MUST EQUAL -1.		6
C ***	NCODE=00	OK		7
C	NCODE=01	A LO		8
C	NCODE=02	A HI		9
C	NCODE=07	ERROR		10
C	NCODE=10	B LO		11
C	NCODE=20	B HI		12
	NCODE=0			13
	C=0.			14
	D=0.			15

C ***	FIND A	16
	DO 1 I=1,NA	17
	IH=I	18
	IF (A.LT.AX(I)) GO TO 2	19
1	CONTINUE	20
	IF (A.GT.AX(IH)) NCODE=2	21
	A=AX(IH)	22
	GO TO 3	23
2	IF (IH.GT.1) GO TO 3	24
	NCODE=1	25
	IH=2	26
	A=AX(1)	27
3	IL=IH-1	28
	LIMH=ND(IH)	29
	LIML=ND(IL)	30
C ***	FIND B	31
	PRM=(A-AX(IL))/(AX(IH)-AX(IL))	32
	PP=P	33
	IF (P.GE.0.) GO TO 6	34
	BL=BX(IL,1)+PRM*(BX(IH,1)-BX(IL,1))	35
	BH=BX(IL,LIML)+PRM*(BX(IH,LIMH)-BX(IL,LIML))	36
	IF (B.GE.BL) GO TO 4	37
	NCODE=NCODE+10	38
	B=BL	39
	GO TO 5	40
4	IF (B.LE.BH) GO TO 5	41
	NCODE=NCODE+20	42
	B=BH	43
5	PP=0.5	44
	Q(2)=0.	45
	Q(3)=0.	46
6	BH=PP*(BX(IH,LIMH)-BX(IH,1))+BX(IH,1)	47
	BL=PP*(BX(IL,LIML)-BX(IL,1))+BX(IL,1)	48
	DO 7 J=2,LIMH	49
	JH=J	50
	IF (BH.LT.BX(IH,J)) GO TO 8	51
7	CONTINUE	52
8	JL=JH-1	53
	DO 9 K=2,LIML	54
	KH=K	55
	IF (BL.LT.BX(IL,K)) GO TO 10	56
9	CONTINUE	57
10	KL=KH-1	58
	PR=(BX(IH,JL)-BH)/(BX(IH,JH)-BX(IH,JL))	59
	CH=CX(IH,JL)-PR*(CX(IH,JH)-CX(IH,JL))	60
	DH=DX(IH,JL)-PR*(DX(IH,JH)-DX(IH,JL))	61
	PR=(BX(IL,KL)-BL)/(BX(IL,KH)-BX(IL,KL))	62
	CL=CX(IL,KL)-PR*(CX(IL,KH)-CX(IL,KL))	63
	DL=DX(IL,KL)-PR*(DX(IL,KH)-DX(IL,KL))	64
	BT=BL+PRM*(BH-BL)	65
	CT=CL+PRM*(CH-CL)	66
	DT=DL+PRM*(DH-DL)	67
	IF (P.GE.0.) GO TO 13	68
	DIR=SQRT(B/BT)	69
	ERR=(B-BT)/B	70
	CALL AFQUIR (Q(1),PP,ERR,0.,25.,0.001,DIR,PT,ICON)	71
	GO TO (11,13,12),ICON	72
11	PP=PT	73
	IF (P>.LT.0.) PP=0.	74
	IF (P>.GT.1.) PP=1.	75
	GO TO 6	76
12	NCODE=7	77

13	B=BT	78
	C=CT	79
	D=DT	80
	RETURN	81
	END	82

```

$IBFTC MAPBAK  DECK
SUBROJTIME MAPBAC (MAP,MAPGO,TFFS,TFF,CNS,CN,PCN,T,MODE,IGO,NUM)
DATA W1,WL,WT,WS/6H H.P. ,6H L.P. ,6H TFF ,6HSPEED /
IF (NJMH.GT.0) GO TO 1
NUMH=0
NUML=0
1   IGO=MAPGO+3*(MAP-1)
GO TO (2,3,5,6,7,9),IGO
C *** HIGH PRESSURE TURBINE
2   TFF=TFF+0.1*(TFF-TFFS)
WRITE (8,10) WH,WT,TFFS,TFF
RETURN
3   CN=CN+0.05*(CN-CNS)
IF (MODE.NE.1) PCN=PCN*(CN/CNS)
IF (MODE.EQ.1) T=T*(CNS/CN)**2
WRITE (8,10) WH,WS,CNS,CN
IF (NJMH.GT.2) GO TO 4
NUM=1
NUMH=NUMH+1
RETURN
4   DELCN=CN-CNS
IF (DELCN.GE.0.) RETURN
TFF=TFF*(1.+DELCN/CN)
WRITE (8,11) WH,WT,TFFS,TFF
RETURN
5   TFF=TFF+0.1*(TFF-TFFS)
WRITE (8,10) WH,WT,TFFS,TFF
GO TO 3
C *** LOW PRESSURE TURBINE
6   TFF=TFF+0.1*(TFF-TFFS)
WRITE (8,10) WL,WT,TFFS,TFF
RETURN
7   CN=CN+0.05*(CN-CNS)
IF (MODE.NE.3) PCN=PCN*(CN/CNS)
IF (MODE.EQ.3) T=T*(CNS/CN)
WRITE (8,10) WL,WS,CNS,CN
IF (NJML.GT.2) GO TO 8
NUM=1
NUML=NUML+1
RETURN
8   DELCN=CN-CNS
IF (DELCN.GE.0.) RETURN
TFF=TFF*(1.+DELCN/CN)
WRITE (8,11) WL,WT,TFFS,TFF
RETURN
9   TFF=TFF+0.1*(TFF-TFFS)
WRITE (8,10) WL,WT,TFFS,TFF
GO TO 7
C
C
10  FORMAT (1H0,A6,12HTURBINE MAP ,A6,4HWAS=,E13.6,10H  AND NOW=,E13.6
1,6H$$$$$)

```

11	FORMAT (1H0,A6,A6,22H) WAS ALSO CHANGED FROM ,E13.6,5H TO ,E13.6,6H	52
	1\$\$\$\$\$)	53
	END	54
\$IBFTC	CONVRG DECK	
	SUBROUTINE CONVRG (TI,HI,PI,SI,FAR,WG,PA,IDES,AO,PR,TO,HO,PO,SO,TS	1
	10,PSO,VO,AMO,ICON)	2
C	ICON=1 SUBSONIC, COMPARE PI WITH PR	3
C	ICON=2 SONIC, COMPARE PI WITH PR	4
C	ICON=4 ERROR	5
	AJ=778.26	6
	CAPSF=2116.217	7
	G=32.174049	8
	CALL PROCOM (FAR,TI,XX1,XX2,XX3,XX4,PHII,XX6)	9
C ***	SONIC CALCULATIONS	10
	J=0	11
	TSS=0.833*TI	12
1	J=J+1	13
	CALL PROCOM (FAR,TSS,CSS,AKS,CP,REXS,PHISS,HSS)	14
	HSCAL=TI-CSS**2/(2.*G*AJ)	15
	DELHS=HSCAL-HSS	16
	IF (ABS(DELHS)-0.0005*HSCAL) 4,4,2	17
2	TSS=TSS+DELHS/CP	18
	IF (J-15) 1,1,3	19
3	ICON=4	20
	RETURN	21
4	IF (IDES) 12,12,5	22
C ***	ISENTROPIC EXPANSION CALCULATIONS	23
5	J=0	24
	TSI=TI*(PA/PI)**0.286	25
6	J=J+1	26
	CALL THERMO (PA,HSI,TSI,SSI,XX1,1,FAR,0)	27
	IF (ABS(SSI-SI)-0.0001*SI) 8,8,7	28
7	TSI=TSI/EXP(4.*(SSI-SI))	29
	IF (J-30) 6,6,3	30
8	VIS=SQRT(2.*G*AJ*(HI-HSI))	31
	IF (VIS-CSS) 9,11,11	32
C ***	SUBSONIC DESIGN, CALCULATE AO	33
9	VO=VIS	34
	TSO=TSI	35
	PSO=PA	36
	CALL PROCOM (FAR,TSO,CSO,XX2,XX3,REX,PHISO,HSO)	37
	RHO=CAPSF*PSO/(AJ*REX*TSO)	38
	AO=WG/(RHO*VO)	39
	AMO=VO/CSO	40
	PR=PI	41
	ICON=1	42
10	TO=TI	43
	HO=HI	44
	PO=PI	45
	SO=SI	46
	RETURN	47
C ***	SONIC DESIGN, CALCULATE AO	48
11	VO=CSS	49
	TSO=TSS	50
	PSO=PI*(TSO/TI)**(AKS/(AKS-1.))	51
	RHO=CAPSF*PSO/(AJ*REXS*TSO)	52
	AO=WG/(RHO*VO)	53

AMD=1.0	54
PR=PI	55
ICON=2	56
GO TO 10	57
C *** NON-DESIGN, CALCULATE CRITICAL CONDITIONS	58
12 VO=CSS	59
TSO=TSS	60
PSO=PA	61
RHO=CAPSF*PSO/(AJ*REXS*TSO)	62
AOCRIT=WG/(RHO*VO)	63
AMD=1.0	64
PR=PSJ*(TI/TSO)**(AKS/(AKS-1.))	65
IF (AJ-AOCRIT) 13,13,14	66
C *** NON-DESIGN, CRITICAL AND SUPERCRITICAL CONDITIONS	67
13 PSO=PSJ*AOCRIT/AO	68
PR=PR*AOCRIT/AO	69
ICON=2	70
GO TO 10	71
C *** NON-DESIGN, SUBSONIC CALCULATIONS	72
14 PSO=PA	73
J=0	74
TSO=0.833*TSO	75
15 J=J+1	76
CALL PROCOM (FAR,TSO,CSO,AKO,CP,REX,PHISO,HSS)	77
RHO=CAPSF*PSO/(AJ*REX*TSO)	78
VO=WG/(RHO*AO)	79
HSCAL=HI-VO**2/(2.*G*AJ)	80
DELHS=HSCAL-HSO	81
IF (ABS(DELHS)-0.0005*HSCAL) 17,17,16	82
16 TSO=TSO+DELHS/CP	83
IF (J-15) 15,15,3	84
17 AMO=VO/CSO	85
PR=PSJ*(TI/TSO)**(AKO/(AKO-1.))	86
ICON=1	87
GO TO 10	88
END	89

\$IBFTC CONDIV DECK	
SUBROJTIME CONDIV (TI,HI,PI,SI,FAR,WG,PA,IDES,AT,AO,PIR,TT,HT,PT,S	1
IT,TO,IO,PO,SO,TST,TSO,PST,PSO,VT,VO,AMT,AMO,ICON)	2
C ICON=1 SUBSONIC, COMPARE PIR WITH PI	3
C ICON=2 SONIC, SHOCK INSIDE NOZZLE, COMPARE PIR WITH PI	4
C ICON=3 SONIC, SHOCK OUTSIDE NOZZLE, COMPARE PIR WITH PI	5
C ICON=4 ERROR	6
DIMENSION Q(9)	7
Q(2)=0.	8
Q(3)=0.	9
AJ=779.26	10
CAPSF=2116.2170	11
G=32.174049	12
CALL PROCOM (FAR,TI,XX1,XX2,XX3,XX4,PHII,XX6)	13
C *** SONIC CALCULATIONS	14
J=0	15
TSS=0.833*TI	16
1 J=J+1	17
CALL PROCOM (FAR,TSS,CSS,AK,CP,REXS,PHISS,HSS)	18
HSCAL=HI-CSS**2/(2.*G*AJ)	19

	DELHS=HSCAL-HSS	20
	IF (ABS(DELHS)-0.0005*HSCAL) 4,4,2	21
2	TSS=TSS+DELHS/CP	22
	IF (J-15) 1,1,3	23
3	ICON=4	24
	RETURN	25
4	IF (IDES) 11,11,5	26
C ***	SONIC DESIGN, CALCULATE AT	27
5	VT=CSS	28
	TST=TSS	29
	PST=PI*(TST/TI)**(AK/(AK-1.))	30
	RHO=CAPSF*PST/(AJ*REXS*TST)	31
	AT=WG/(RHO*VT)	32
	AMT=1.0	33
C ***	IDEAL EXPANSION DESIGN, CALCULATE AJ	34
	PSO=PA	35
	J=0	36
	TSO=TI*(PSO/PI)**.286	37
6	J=J+1	38
	CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSO)	39
	PHICAL=PHII-REX*ALOG(PI/PSO)	40
	DELPHI=PHICAL-PHISO	41
	IF (ABS(DELPHI)-0.0001*PHICAL) 8,8,7	42
7	TSO=TSO*EXP(4.*DELPHI)	43
	IF (J-15) 6,6,3	44
8	VO=SQRT(2.*G*AJ*(HI-HSO))	45
	AMO=VO/CSO	46
	AO=(AT/AMO)*(2.*(1.+(AK-1.)*AMO**2/2.)/(AK+1.))**((AK+1.)/(2.*(AK-	47
	11.)))	48
	PIR=PI	49
	ICON=3	50
9	TO=TI	51
	HO=HI	52
	PO=PI	53
	SO=SI	54
10	TT=TI	55
	HT=HI	56
	PT=PI	57
	ST=SI	58
	RETURN	59
C ***	ASSUME SONIC THROAT AND ISENTROPIC EXPANSION TO AO	60
11	VT=CSS	61
	AMT=1.0	62
	TST=TSS	63
	RHO=WG/(AT*VT)	64
	PST=RHO*AJ*REXS*TST/CAPSF	65
	PIR=PST*(TI/TST)**(AK/(AK-1.))	66
	IF (PST-PA) 12,27,27	67
12	TSO=0.95*TI	68
	MAM=0	69
13	CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSO)	70
	AMO=SQRT(2.*((TI/TSO)-1.)/(AK-1.))	71
	AOCAL=(AT/AMO)*(2.*(1.+(AK-1.)*AMO**2/2.)/(AK+1.))**((AK+1.)/(2.*(	72
	1AK-1.)))	73
	EA=(AJ-AOCAL)/AO	74
	DIR=SQRT(AO/AOCAL)	75
	CALL AFQUIR (Q(1),TSO,EA,0.,100.,0.0001,DIR,TSOT,JCON)	76
	GO TO (14,18,3),JCON	77
14	TSO=TSOT	78
	IF (TSO-TI) 15,13,16	79
15	TSC=2.*TI/(AK+1.)	80
	IF (TSO.GT.TSC) GO TO 17	81

16	TSO=0.98*TI	82
	GO TO 13	83
17	IF (Q(2).LT.30.0.OR.AMO.LT.0.95.OR.MAM.EQ.1) GO TO 13	84
	TSO=2.*TI/(2.+0.98*(AK-1.))	85
	MAM=1	85
	GO TO 13	87
18	PSO=PIR*(TSO/TI)**(AK/(AK-1.))	88
	IF (PSO-PA) 20,19,27	89
C ***	CRITICAL FLOW, ISENTROPIC EXPANSION TO PA	90
19	VO=AMJ*CSO	91
	ICON=1	92
	GO TO 9	93
C ***	SUBSONIC FLOW	94
20	PSO=PA	95
	Q(2)=0.	96
	Q(3)=0.	97
	J=0	98
	TSO=0.833*TI	99
21	J=J+1	100
	CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSD)	101
	RHO=CAPSF*PSO/(AJ*REX*TSO)	102
	VO=WG/(RHO*AO)	103
	HSCAL=HI-VO**2/(2.*G*AJ)	104
	DELHS=HSCAL-HSD	105
	IF (ABS(DELHS)-0.0005*HSCAL) 23,23,22	105
22	TSO=TSO+DELHS/CP	107
	IF (J-15) 21,21,3	108
23	AMO=VO/CSO	109
	PIR=PSO*(TI/TSO)**(AK/(AK-1.))	110
	TST=TSO	111
24	CALL PROCOM (FAR,TST,CST,AK,CP,REX,PHIST,HST)	112
	PST=PIR*(TST/TI)**(AK/(AK-1.))	113
	RHO=PST*CAPSF/(AJ*REX*TST)	114
	VT=WG/(RHO*AT)	115
	HSCAL=HI-VT**2/(2.*G*AJ)	116
	EH=(HSCAL-HST)/HSCAL	117
	DIR=1.+(HSCAL-HST)/(CP*TST)	118
	CALL AFQUIR (Q(1),TST,EH,0.,20.,0.0005,DIR,TSTT,JCON)	119
	GO TO (25,26,3),JCON	120
25	TST=TSTT	121
	GO TO 24	122
26	AMT=VT/CST	123
	ICON=1	124
	GO TO 9	125
C ***	SUPERCRITICAL FLOW, ISENTROPIC EXPANSION TO PA	126
27	PSO=PA	127
	J=0	128
	TSO=TI*(PSO/PIR)**.286	129
28	J=J+1	130
	CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSD)	131
	PHICAL=PHII-REX*ALOG(PIR/PSO)	132
	DELPHI=PHICAL-PHISO	133
	IF (ABS(DELPHI)-0.0001*PHICAL) 30,30,29	134
29	TSO=TSO*EXP(4.0*DELPHI)	135
	IF (J-15) 28,28,3	136
30	VO=SQRT(2.*G*AJ*(HI-HSD))	137
	AMO=VO/CSO	138
	AOID=(AT/AMO)*(2.*(1.+(AK-1.)*AMO**2/2.)/(AK+1.))**((AK+1.)/(2.*(AK-1.)))	139
	ICON=3	141
	N=0	142
	IF (AO-AOID) 31,9,32	143

C *** SUPERCRITICAL FLOW, ISENTROPIC EXPANSION TO AO	144
31 N=1	145
32 TSO=0.833*TI	146
J=0	147
33 J=J+1	148
CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSO)	149
AMO=SQRT(2.*((TI/TSO)-1.)/(AK-1.))	150
AOCAL=(AT/AMO)*(2.*(1.+(AK-1.)*AMO**2/2.)/(AK+1.))*((AK+1.)/(2.*(AK-1.)))	151
DELA=AO-AOCAL	152
IF (ABS(DELA)-0.0001*AO) 35,35,34	153
34 TSO=TSO*SQRT(AOCAL/AO)	154
IF (J-50) 33,33,3	155
35 IF (N) 37,37,36	156
C *** UNDEREXPANDED, SHOCK OUTSIDE NOZZLE	157
36 PSO=PIR*(TSO/TI)**(AK/(AK-1.))	158
VO=AMJ*CSO	159
GO TO 9	160
C *** OVEREXPANDED, FIND SHOCK POSITION	161
37 PSX=PIR*(TSO/TI)**(AK/(AK-1.))	162
PSY=PSX*(2.*AK*AMO**2/(AK+1.)-(AK-1.)/(AK+1.))	163
IF (PA-PSY) 38,39,39	164
C *** OVEREXPANDED, SHOCK OUTSIDE NOZZLE	165
38 PSO=PSX	166
VO=AMJ*CSO	167
GO TO 9	168
C *** OVEREXPANDED, SHOCK INSIDE NOZZLE	169
39 PSO=PA	170
J=0	171
TSO=0.833*TI	172
40 J=J+1	173
CALL PROCOM (FAR,TSO,CSO,AK,CP,REX,PHISO,HSO)	174
RHO=CAPSF*PSO/(AJ*REX*TSO)	175
VO=WG/(RHO*AO)	176
HSCAL=4I-VO**2/(2.*G*AJ)	177
DELHS=HSCAL-HSO	178
IF (ABS(DELHS)-0.0005*HSCAL) 42,42,41	179
41 TSO=TSO+DELHS/CP	180
IF (J-15) 40,40,3	181
42 AMO=VO/CSO	182
TO=TI	183
HO=HI	184
PO=PSJ*(TO/TSO)**(AK/(AK-1.))	185
SO=PHI I-REX*ALOG(PO)	186
ICON=2	187
GO TO 10	188
END	189
	190

\$IBFTC THTERB DECK

SUBROUTINE THTURB (DH,ETA,FAR,H,S,P,TO,HQ,SD,PO)	1
HO=H-DH	2
HOP=H-DH/ETA	3
PT=P/2.	4
DO 1 I=1,25	5
CALL THERMO (PT,HOP,TT,ST,AMWT,1,FAR,1)	6
DELS=ST-S	7
IF (ABS(DELS).LE.0.00005*S) GO TO 2	8

1	PT=P*EXP(DELS*AMWT/1.986375+ALOG(PT/P))	9
	CALL ERROR	10
2	PO=PT	11
	CALL THERMO (PO,HO,TO,SO,X1,1,FAR,1)	12
	RETURN	13
	END	14

\$IBFTC	THERMO	DECK	
	SUBROUTINE	THERMO (PX,HX,TX,SX,AMX,L,FAR,K)	1
		FX=0.	2
		IF (L.EQ.1) FX=FAR	3
		IF (K.EQ.1) GO TO 1	4
		CALL PROCOM (FX,TX,CS,AK,CP,R,PHI,HX)	5
		GO TO 3	6
1		TX=4.*HX	7
		DO 2 I=1,15	8
		CALL PROCOM (FX,TX,CS,AK,CP,R,PHI,H)	9
		DELH=HX-H	10
		IF (ABS(DELH).LE.0.00001*HX) GO TO 3	11
2		TX=TX+4.*DELH	12
		WRITE (8,4)	13
3		SX=PHI-R*ALOG(PX)	14
		AMX=1.986375/R	15
		RETURN	16
C			17
C			18
4		FORMAT (31HONO CONVERGENCE IN THERMO\$\$\$\$\$)	19
		END	20

\$IBFTC	AFQUER	DECK	
	SUBROUTINE	AFQUIR (X,AIND,DEPEND,ANS,AJ,TOL,DIR,ANEW,ICON)	1
		DIMENSION X(9)	2
C	X(1)=NAME OF ARRAY TO USE		3
C	AIND=INDEPENDANT VARIABLE		4
C	DEPEND= DEPENDANT VARIABLE		5
C	ANS=ANSWER UPON WHICH TO CONVERGE		6
C	AJ=MAX NUMBER OF TRYs		7
C	TOL=PERCENT TOLERANCE FOR CONVERGENCE		8
C	DIR=DIRECTION AND PERCENTAGE FOR FIRST GUESS		9
C	ANEW=CALCULATED VALUE OF NEXT TRY AT INDEPENDANT VARIABLE		10
C	ICON=CONTROL =1 GO THRU LOOP AGAIN		11
C	=2 YOU HAVE REACHED THE ANSWER		12
C	=3 COUNTER HAS HIT LIMITS		13
C	X(2)=COUNTER STORAGE		14
C	X(3)=CHOOSES METHOD OF CONVERGENCE		15
C	X(4)=THIRD DEPEND VAR		16
C	X(5)=THIRD IND VAR		17
C	X(6)=SECOND DEPEND VAR		18
C	X(7)=SECOND IND VAR		19
C	X(8)=FIRST DEPEND VAR		20
C	X(9)=FIRST IND VAR		21
C	X(3) MUST BE ZERO UPON FIRST ENTRY TO ROUTINE		22
	Y=0.		23
	IF (ANS) 1,2,1		24

1	DEP=DEPEND-ANS	25
	TOLANS=TOL*ANS	26
	GO TO 3	27
2	DEP=DEPEND	28
	TOLANS=TOL	29
3	IF (ABS(DEP)-TOLANS) 5,5,4	30
4	IF (X(2)-AJ) 8,8,7	31
5	ANEW=AIND	32
	X(2)=0.	33
	ICON=2	34
	RETURN	35
6	ANEW=Y	36
	X(2)=X(2)+1.	37
	ICON=1	38
	RETURN	39
7	ANEW=Y	40
	X(2)=0.	41
	ICON=3	42
	RETURN	43
8	IF (X(3)) 9,9,12	44
C ***	FIRST GUESS USING DIR	45
9	X(3)=1.	46
	X(8)=DEP	47
	X(9)=AIND	48
	IF (AIND) 10,11,10	49
10	Y=DIR*AIND	50
	GO TO 5	51
11	Y=DIR	52
	GO TO 6	53
12	IF (X(3)-1.) 13,13,16	54
C ***	LINEAR GUESS	55
13	X(3)=2.	56
	X(6)=DEP	57
	X(7)=AIND	58
	IF (X(8)-X(6)) 14,9,14	59
14	IF (X(9)-X(7)) 15,9,15	60
15	A=(X(9)-X(7))/(X(8)-X(6))	61
	Y=X(9)-A*X(8)	62
	IF (ABS(10.*X(9))-ABS(Y)) 9,9,6	63
C ***	QUADRATIC GUESS	64
16	X(4)=DEP	65
	X(5)=AIND	66
	IF (X(7)-X(5)) 18,17,18	67
17	IF (X(6)-X(4)) 13,9,13	68
18	IF (X(6)-X(4)) 19,13,19	69
19	IF (X(9)-X(5)) 23,20,23	70
20	IF (X(8)-X(4)) 21,22,21	71
21	X(9)=X(7)	72
	X(8)=X(6)	73
	GO TO 13	74
22	X(9)=X(7)	75
	X(8)=X(6)	76
	X(3)=1.	77
	IF (X(9)) 10,11,10	78
23	IF (X(8)-X(4)) 24,21,24	79
24	F=(X(5)-X(4))/(X(7)-X(5))	80
	A=(X(3)-X(4)-F*(X(9)-X(5)))/((X(9)-X(7))*(X(9)-X(5)))	81
	B=F-A*(X(5)+X(7))	82
	C=X(4)+X(5)*(A*X(7)-F)	83
	IF (A) 27,25,27	84
25	IF (B) 26,7,26	85

26	Y=-C/B	86
	GO TO 47	87
27	IF (B) 32,28,32	88
28	IF (C) 30,29,30	89
29	Y=0.	90
	GO TO 47	91
30	G=-C/A	92
	IF (G) 7,7,31	93
31	Y=SQRT(G)	94
	YY=-SQRT(G)	95
	GO TO 37	96
32	IF (C) 34,33,34	97
33	Y=-B/A	98
	YY=0.	99
	GO TO 37	100
34	D=4.*A*C/B**2	101
	IF (1.-D) 13,35,36	102
35	Y=-B/(2.*A)	103
	GO TO 47	104
36	E=SQRT(1.-D)	105
	Y=(-B/(2.*A))*(1.+E)	106
	YY=(-B/(2.*A))*(1.-E)	107
37	J=4	108
	DEPMIN=ABS(X(4))	109
	DO 39 I=6,8,2	110
	IF (DEPMIN-ABS(X(I))) 39,39,38	111
38	J=I	112
	DEPMIN=ABS(X(I))	113
39	CONTINUE	114
	K=J+1	115
	IF ((X(K)-Y)*(X(K)-YY)) 42,42,40	116
40	IF (ABS(X(K)-Y)-ABS(X(K)-YY)) 47,47,41	117
41	Y=YY	118
	GO TO 47	119
42	IF (J-6) 43,44,44	120
43	JJ=J+2	121
	KK=K+2	122
	GO TO 45	123
44	JJ=J-2	124
	KK=K-2	125
45	SLOPE=(X(KK)-X(K))/(X(JJ)-X(J))	126
	IF (SLOPE*X(J)*(X(K)-Y)) 46,46,47	127
46	Y=YY	128
47	X(9)=X(7)	129
	X(8)=X(6)	130
	X(7)=X(5)	131
	X(6)=X(4)	132
	GO TO 6	133
	END	134

\$IBFTC PARABO DECK

	SUBROUTINE PARABO (X,Y,XD,YANS)	1
	DIMENSION X(3),Y(3)	2
	A=((X(1)-X(2))*(Y(1)-Y(3))-(X(1)-X(3))*(Y(1)-Y(2)))/((X(1)-X(2))*	3
	1X(1)-X(3))*(X(3)-X(2)))	4
	B=((X(1)**2-X(2)**2)*(Y(1)-Y(3))-(X(1)**2-X(3)**2)*(Y(1)-Y(2)))/((	5
	1X(1)-X(2))*(X(1)-X(3))*(X(2)-X(3)))	6

```

D=(Y(1)*X(2)**2-Y(2)*X(1)**2-B*X(2)*X(1)*(X(2)-X(1)))/(X(2)**2-X(1)
1)**2)
YANS=(A*XD+B)*XD+D
RETURN
END

```

7
8
9
10
11

\$IBFTC BLKFAN DECK

C THIS IS A GENERALIZED FAN MAP FOR UNREALISTIC SUPERSONIC ENGINE

1

BLOCK DATA

2

COMMON / FAN/CN(15),PR(15,15),WAC(15,15),ETA(15,15),N,NP(15)

3

DATA V,NP/10,6,3*7,5*10,8,5*0/

4

DATA CN/0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,5*0./

5

DATA (PR(1,J),WAC(1,J),ETA(1,J),J=1, 6)/

6

1 1.00000, 243.600, 0.75592, 1.01200, 229.800, 0.76120,

7

2 1.02800, 199.800, 0.76648, 1.03840, 166.800, 0.75592,

8

3 1.04480, 133.200, 0.72512, 1.04800, 86.400, 0.64152/

9

DATA (PR(2,J),WAC(2,J),ETA(2,J),J=1, 7)/

10

1 1.00000, 286.800, 0.75592, 1.02000, 270.000, 0.77616,

11

2 1.04000, 253.200, 0.79200, 1.05840, 233.400, 0.79728,

12

3 1.07520, 209.400, 0.80256, 1.09200, 183.600, 0.77616,

13

4 1.10000, 156.600, 0.74008/

14

DATA (PR(3,J),WAC(3,J),ETA(3,J),J=1, 7)/

15

1 1.00000, 333.600, 0.75064, 1.02560, 322.800, 0.77616,

16

2 1.05120, 310.200, 0.80256, 1.08000, 291.600, 0.82808,

17

3 1.11600, 259.800, 0.84392, 1.13200, 240.000, 0.82808,

18

4 1.14800, 213.600, 0.77616/

19

DATA (PR(4,J),WAC(4,J),ETA(4,J),J=1, 7)/

20

1 1.00000, 383.400, 0.74536, 1.03680, 376.200, 0.77616,

21

2 1.09800, 358.200, 0.82808, 1.12400, 340.200, 0.85448,

22

3 1.15000, 313.200, 0.88000, 1.18960, 276.600, 0.82808,

23

4 1.19520, 266.400, 0.80784/

24

DATA (PR(5,J),WAC(5,J),ETA(5,J),J=1,10)/

25

1 1.00000, 439.800, 0.72512, 1.06400, 436.800, 0.77616,

26

2 1.11840, 428.400, 0.82808, 1.14800, 420.600, 0.85448,

27

3 1.13400, 406.800, 0.88000, 1.20960, 393.600, 0.90112,

28

4 1.21760, 388.200, 0.90376, 1.22400, 383.400, 0.90112,

29

5 1.24400, 368.400, 0.88000, 1.26720, 342.600, 0.82808/

30

DATA (PR(6,J),WAC(6,J),ETA(6,J),J=1,10)/

31

1 1.00000, 499.800, 0.68816, 1.10000, 499.800, 0.77616,

32

2 1.15000, 493.200, 0.82808, 1.20000, 485.400, 0.85448,

33

3 1.22800, 476.400, 0.88000, 1.25520, 466.800, 0.90112,

34

4 1.27200, 456.600, 0.91080, 1.28640, 448.200, 0.90112,

35

5 1.30240, 433.200, 0.88000, 1.33200, 406.800, 0.82720/

36

DATA (PR(7,J),WAC(7,J),ETA(7,J),J=1,10)/

37

1 1.00000, 566.400, 0.64152, 1.07600, 566.400, 0.72512,

38

2 1.15200, 566.400, 0.77616, 1.21920, 559.800, 0.82808,

39

3 1.25000, 553.200, 0.85888, 1.28960, 544.800, 0.88000,

40

4 1.33120, 528.600, 0.90112, 1.36160, 509.400, 0.88000,

41

5 1.39120, 483.600, 0.82808, 1.40000, 474.000, 0.81752/

42

DATA (PR(8,J),WAC(8,J),ETA(8,J),J=1,10)/

43

1 1.00000, 633.600, 0.60016, 1.04400, 633.600, 0.64152,

44

2 1.13520, 633.600, 0.72512, 1.22080, 633.000, 0.77616,

45

3 1.23440, 625.800, 0.82808, 1.34000, 616.800, 0.85888,

46

4 1.40000, 600.000, 0.88000, 1.42800, 586.800, 0.85888,

47

5 1.44800, 576.600, 0.82808, 1.48000, 553.200, 0.78672/

48

DATA (PR(9,J),WAC(9,J),ETA(9,J),J=1,10)/

49

1 1.00000, 700.200, 0.56936, 1.10400, 700.200, 0.64152,

50

2 1.22000, 700.200, 0.72512, 1.32400, 700.200, 0.77616,

51

3	1.43000,	700.200,	0.80256,	1.44800,	698.400,	0.80784,	52
4	1.50000,	693.600,	0.80256,	1.53360,	683.400,	0.77616,	53
5	1.55800,	666.600,	0.74536,	1.58400,	656.400,	0.72512/	54
	DATA (PR(10,J),WAC(10,J),ETA(10,J),J=1, 8)/						55
1	1.00000,	750.000,	0.51744,	1.16320,	750.000,	0.64152,	56
2	1.31200,	750.000,	0.72512,	1.40000,	750.000,	0.75592,	57
3	1.43000,	750.000,	0.76120,	1.54000,	750.000,	0.75064,	58
4	1.58000,	749.400,	0.72512,	1.66000,	736.800,	0.64152/	59
	END						60

\$IBFTC BLKCMF DECK

C	THIS IS GENERALIZED COMP. MAP FOR UNREALISTIC SUPERSONIC ENGINE						1
	BLOCK DATA						2
	COMMON / COMP/CN(15),PR(15,15),WAC(15,15),ETA(15,15),N,NP(15)						3
	DATA N,NP/10,2*6,2*8,4*10,2*8,5*0/						4
	DATA CV/.562,.674,.787,.899,1.,1.034,1.067,1.124,1.236,1.292,5*0./						5
	DATA (PR(1,J),WAC(1,J),ETA(1,J),J=1, 6)/						6
1	1.00000,	51.000,	0.59082,	1.84000,	50.200,	0.62178,	7
2	2.42800,	49.500,	0.64242,	2.86900,	48.800,	0.65274,	8
3	3.83500,	46.700,	0.67338,	4.54900,	44.500,	0.64242/	9
	DATA (PR(2,J),WAC(2,J),ETA(2,J),J=1, 6)/						10
1	1.00000,	59.300,	0.59082,	1.96600,	59.300,	0.64242,	11
2	3.09300,	58.800,	0.69402,	3.93300,	57.900,	0.72498,	12
3	4.68900,	56.700,	0.74562,	5.52900,	55.000,	0.72498/	13
	DATA (PR(3,J),WAC(3,J),ETA(3,J),J=1, 8)/						14
1	1.00000,	70.000,	0.58566,	1.84000,	70.000,	0.64242,	15
2	2.63000,	70.000,	0.68370,	3.40800,	69.500,	0.72498,	16
3	4.52100,	68.800,	0.77744,	5.44500,	67.900,	0.79292,	17
4	6.31300,	66.400,	0.77744,	6.52300,	65.700,	0.76970/	18
	DATA (PR(4,J),WAC(4,J),ETA(4,J),J=1, 8)/						19
1	1.00000,	84.800,	0.58050,	2.00800,	84.800,	0.64242,	20
2	3.42900,	84.800,	0.72498,	4.60500,	84.800,	0.77744,	21
3	5.69700,	84.000,	0.80840,	6.61400,	83.300,	0.82904,	22
4	7.53800,	81.700,	0.80840,	7.95800,	80.500,	0.79292/	23
	DATA (PR(5,J),WAC(5,J),ETA(5,J),J=1,10)/						24
1	1.00000,	101.700,	0.57190,	2.51900,	101.700,	0.64242,	25
2	3.98200,	101.700,	0.72498,	5.27700,	101.700,	0.77744,	26
3	6.43800,	101.200,	0.80840,	7.20200,	101.000,	0.83936,	27
4	8.00000,	100.000,	0.86000,	8.56700,	99.500,	0.83936,	28
5	9.38500,	98.100,	0.80840,	9.59600,	97.400,	0.80582/	29
	DATA (PR(6,J),WAC(6,J),ETA(6,J),J=1,10)/						30
1	1.00000,	108.100,	0.57018,	2.85500,	108.100,	0.64242,	31
2	4.29700,	108.100,	0.72498,	5.61300,	108.100,	0.77744,	32
3	6.93600,	107.600,	0.80840,	7.62200,	107.100,	0.83936,	33
4	8.54500,	106.700,	0.86000,	9.13400,	106.000,	0.83936,	34
5	9.92500,	104.500,	0.80840,	10.21900,	104.000,	0.80410/	35
	DATA (PR(7,J),WAC(7,J),ETA(7,J),J=1,10)/						36
1	1.00000,	114.500,	0.55986,	3.26100,	114.500,	0.64242,	37
2	4.75900,	114.500,	0.72498,	6.11700,	114.500,	0.77744,	38
3	7.45400,	114.500,	0.80840,	8.30800,	114.300,	0.83936,	39
4	9.21800,	113.600,	0.84968,	9.63800,	113.300,	0.83936,	40
5	10.51300,	112.600,	0.80840,	10.99600,	112.400,	0.79808/	41
	DATA (PR(8,J),WAC(8,J),ETA(8,J),J=1,10)/						42
1	1.00000,	122.900,	0.53922,	1.68600,	122.900,	0.57018,	43
2	3.84900,	122.900,	0.64242,	5.46600,	122.900,	0.72498,	44
3	6.85500,	122.900,	0.77744,	8.37100,	122.900,	0.80840,	45
4	8.95500,	122.600,	0.82388,	9.88300,	122.100,	0.83936,	46
5	10.91200,	121.700,	0.80840,	11.81500,	120.700,	0.77744/	47

DATA (PR(9,J),WAC(9,J),ETA(9,J),J=1, 8)/	48
1 1.00000, 139.800, 0.47644, 4.35300, 139.800, 0.60114,	49
2 7.52200, 139.800, 0.72498, 10.21900, 139.800, 0.77744,	50
3 11.05900, 139.800, 0.78260, 11.89900, 139.500, 0.77744,	51
4 13.15900, 139.300, 0.72498, 13.65600, 139.000, 0.69918/	52
DATA (PR(10,J),WAC(10,J),ETA(10,J),J=1, 8)/	53
1 1.00000, 146.200, 0.46612, 3.76500, 146.200, 0.57018,	54
2 6.43100, 146.200, 0.64242, 9.17600, 146.200, 0.72498,	55
3 10.21900, 146.200, 0.75078, 11.47900, 146.200, 0.75078,	56
4 12.71100, 146.200, 0.72498, 14.41200, 146.200, 0.64242/	57
END	58

\$IBFTC CMBDAT DECK

BLOCK DATA	1
COMMON / COMB/PSI(15),DELT(15,15),ETA(15,15),N,NP(15)	2
DATA N,NP / 15,15*15 /	3
DATA PSI/4.9116,9.8232,14.735,19.646,24.558,29.470,34.381,	4
139.293,44.207,73.674,100.,200.,300.,400.,500./	5
DATA DELT/15*200.,15*300.,15*400.,15*500.,15*600.,15*700.,15*800.,	6
115*900.,15*1000.,15*1100.,15*1200.,15*1300.,15*1400.,15*1500.,	7
215*1600./	8
DATA ETA/	9
1.600.,.726.,.777.,.806.,.826.,.843.,.855.,.865,7*.870,	10
2.758.,.825.,.858.,.875.,.888.,.898.,.906.,.912.,.914,6*.915,	11
3.868.,.893.,.911.,.925.,.935.,.942.,.947.,.951,7*.953,	12
4.925.,.936.,.946.,.955.,.963.,.969.,.974.,.977.,.978,6*.979,	13
5.960.,.966.,.972.,.977.,.982.,.985.,.990.,.992.,.993,6*.995,	14
6.988.,.991.,.992.,.994.,.995.,.997.,.998,8*.999,	15
78*1.00,7*.999,120*1.00/	16
END	17

\$IBFTC HPTDAT DECK

BLOCK DATA	1
COMMON / HTURB/TFF(15),CN(15,15),DH(15,15),ETA(15,15),N,NP(15)	2
DATA N,NP/10,9*15,12,5*0/	3
DATA TFF / 39.670, 42.990, 47.460, 48.610, 49.175,	4
1 49.600, 50.000, 50.425, 50.920, 51.575, 5*0./	5
DATA (CN(1,J),DH(1,J),ETA(1,J),J=1,15)/	6
1 0.1872, 0.0032, 0.6219, 0.3372, 0.0057, 0.7078,	7
2 0.5156, 0.0084, 0.7868, 0.7128, 0.0108, 0.8090,	8
3 0.9382, 0.0133, 0.8090, 1.1442, 0.0152, 0.7963,	9
4 1.3138, 0.0164, 0.7779, 1.5382, 0.0174, 0.7422,	10
5 1.7264, 0.0179, 0.7078, 1.9324, 0.0176, 0.7635,	11
6 2.1500, 0.0167, 0.6068, 2.4058, 0.0144, 0.5309,	12
7 2.5892, 0.0120, 0.4773, 2.7862, 0.0082, 0.4045,	13
8 2.9460, 0.0034, 0.3034/	14
DATA (CN(2,J),DH(2,J),ETA(2,J),J=1,15)/	15
1 0.1872, 0.0038, 0.6068, 0.3942, 0.0080, 0.7078,	16
2 0.5814, 0.0113, 0.8090, 0.7128, 0.0136, 0.8292,	17
3 0.3442, 0.0156, 0.8363, 0.9804, 0.0176, 0.8393,	18
4 1.1068, 0.0192, 0.8368, 1.2754, 0.0212, 0.8302,	19
5 1.4450, 0.0228, 0.8254, 1.7068, 0.0248, 0.8090,	20

6	1.9596,	0.0260,	0.7696,	2.2706,	0.0261,	0.7078,	21
7	2.5970,	0.0241,	0.6068,	3.0960,	0.0188,	0.5056,	22
8	3.3774,	0.0128,	0.4197,				23
	DATA (CN(3,J),DH(3,J),ETA(3,J),J=1,15)/						24
1	0.1872,	0.0046,	0.5764,	0.4362,	0.0100,	0.7078,	25
2	0.5568,	0.0144,	0.8090,	0.8726,	0.0184,	0.8494,	26
3	1.0696,	0.0216,	0.8543,	1.2382,	0.0240,	0.8515,	27
4	1.4538,	0.0268,	0.8494,	1.6882,	0.0292,	0.8409,	28
5	1.9596,	0.0316,	0.8262,	2.2138,	0.0331,	0.8090,	29
6	2.5520,	0.0344,	0.7579,	2.8050,	0.0346,	0.7078,	30
7	3.0392,	0.0340,	0.6652,	3.2648,	0.0324,	0.6068,	31
8	3.3774,	0.0312,	0.5865,				32
	DATA (CN(4,J),DH(4,J),ETA(4,J),J=1,15)/						33
1	0.1872,	0.0052,	0.5643,	0.2550,	0.0068,	0.6068,	34
2	0.4784,	0.0120,	0.7078,	0.6942,	0.0164,	0.8090,	35
3	0.9148,	0.0204,	0.8494,	1.1442,	0.0244,	0.8596,	36
4	1.3882,	0.0280,	0.8596,	1.5618,	0.0304,	0.8575,	37
5	1.9010,	0.0336,	0.8535,	1.9794,	0.0356,	0.8494,	38
6	2.2794,	0.0388,	0.8363,	2.5138,	0.0412,	0.8262,	39
7	2.8334,	0.0441,	0.8090,	3.1422,	0.0472,	0.7797,	40
8	3.3774,	0.0494,	0.7584,				41
	DATA (CN(5,J),DH(5,J),ETA(5,J),J=1,15)/						42
1	0.1872,	0.0056,	0.5562,	0.3000,	0.0088,	0.6068,	43
2	0.5254,	0.0144,	0.7078,	0.7500,	0.0192,	0.8090,	44
3	0.9754,	0.0236,	0.8494,	1.2754,	0.0288,	0.8697,	45
4	1.4824,	0.0321,	0.8696,	1.7638,	0.0360,	0.8662,	46
5	2.0450,	0.0400,	0.8615,	2.3362,	0.0444,	0.8555,	47
6	2.5450,	0.0496,	0.8520,	2.8706,	0.0540,	0.8494,	48
7	3.0764,	0.0596,	0.8494,	3.1520,	0.0640,	0.8532,	49
8	3.1518,	0.0661,	0.8570,				50
	DATA (CN(6,J),DH(6,J),ETA(6,J),J=1,15)/						51
1	0.1872,	0.0068,	0.5309,	0.3568,	0.0120,	0.6068,	52
2	0.5196,	0.0192,	0.7078,	0.8628,	0.0252,	0.8090,	53
3	1.0932,	0.0300,	0.8494,	1.2852,	0.0340,	0.8697,	54
4	1.5010,	0.0384,	0.8819,	1.6882,	0.0421,	0.8899,	55
5	1.9138,	0.0472,	0.8940,	2.1246,	0.0524,	0.8969,	56
6	2.2706,	0.0564,	0.8975,	2.4226,	0.0612,	0.8976,	57
7	2.4950,	0.0640,	0.8968,	2.5372,	0.0668,	0.8937,	58
8	2.5558,	0.0698,	0.8896,				59
	DATA (CN(7,J),DH(7,J),ETA(7,J),J=1,15)/						60
1	0.1872,	0.0080,	0.5062,	0.4314,	0.0164,	0.6068,	61
2	0.5844,	0.0236,	0.7078,	0.9568,	0.0308,	0.8090,	62
3	1.2010,	0.0372,	0.8494,	1.3834,	0.0416,	0.8697,	63
4	1.5108,	0.0448,	0.8797,	1.6186,	0.0476,	0.8899,	64
5	1.7450,	0.0510,	0.8954,	1.8618,	0.0544,	0.9000,	65
6	1.9558,	0.0576,	0.9010,	2.0000,	0.0600,	0.9000,	66
7	2.0450,	0.0624,	0.8980,	2.0824,	0.0660,	0.8925,	67
8	2.1010,	0.0700,	0.8793,				68
	DATA (CN(8,J),DH(8,J),ETA(8,J),J=1,15)/						69
1	0.1872,	0.0088,	0.5051,	0.4834,	0.0196,	0.6068,	70
2	0.7314,	0.0272,	0.7078,	0.8814,	0.0316,	0.7665,	71
3	1.0226,	0.0356,	0.8090,	1.1442,	0.0392,	0.8292,	72
4	1.2804,	0.0432,	0.8494,	1.3696,	0.0460,	0.8596,	73
5	1.4538,	0.0488,	0.8697,	1.5950,	0.0528,	0.8808,	74
6	1.5746,	0.0560,	0.8848,	1.7450,	0.0596,	0.8848,	75
7	1.6010,	0.0640,	0.8788,	1.8156,	0.0664,	0.8697,	76
8	1.8196,	0.0693,	0.8590,				77
	DATA (CN(9,J),DH(9,J),ETA(9,J),J=1,15)/						78
1	0.1872,	0.0093,	0.4909,	0.3372,	0.0159,	0.5380,	79
2	0.5344,	0.0232,	0.6068,	0.6754,	0.0284,	0.5573,	80
3	0.8068,	0.0330,	0.7078,	0.9196,	0.0368,	0.7463,	81
4	1.0128,	0.0400,	0.7776,	1.1254,	0.0442,	0.8090,	82
5	1.2196,	0.0480,	0.8191,	1.3138,	0.0524,	0.8302,	83

6	1.3696,	0.0556,	0.8347,	1.4068,	0.0580,	0.8363,	84
7	1.4450,	0.0612,	0.8322,	1.4638,	0.0640,	0.8241,	85
8	1.4676,	0.0668,	0.8090/				86
	DATA (CN(10,J),DH(10,J),ETA(10,J),J=1,12)/						87
1	0.1872,	0.0132,	0.4257,	0.2814,	0.0180,	0.4747,	88
2	0.3804,	0.0228,	0.5056,	0.4686,	0.0268,	0.5359,	89
3	0.5528,	0.0314,	0.5683,	0.6382,	0.0352,	0.5941,	90
4	0.5392,	0.0380,	0.6068,	0.7362,	0.0412,	0.6178,	91
5	0.7696,	0.0440,	0.6240,	0.8068,	0.0476,	0.5310,	92
6	0.8254,	0.0504,	0.6265,	0.8304,	0.0530,	0.5118/	93
	END						94

\$IBFTC LPTDAT DECK

BLOCK DATA

COMMON / LTURB/TFF(15),CN(15,15),DH(15,15),ETA(15,15),N,NP(15)

DATA N,NP/11,9*15,12,9,4*0/

DATA TFF / 88.470, 102.795, 116.835, 129.330, 141.045,
1 145.725, 150.000, 153.345, 156.405, 159.780, 163.170,4*0./

DATA (CN(1,J),DH(1,J),ETA(1,J),J=1,15)/

1	0.3582,	0.0018,	0.7120,	0.5336,	0.0026,	0.7300,	7
2	0.7365,	0.0035,	0.7472,	0.9754,	0.0044,	0.7300,	8
3	1.2146,	0.0051,	0.7140,	1.4173,	0.0056,	0.7000,	9
4	1.5201,	0.0059,	0.6850,	1.7673,	0.0061,	0.6730,	10
5	2.0247,	0.0062,	0.6452,	2.2827,	0.0061,	0.6200,	11
6	2.4565,	0.0057,	0.6000,	2.6137,	0.0053,	0.5750,	12
7	2.3166,	0.0044,	0.5310,	2.9456,	0.0035,	0.5000,	13
8	3.3138,	0.0001,	0.3850/				14

DATA (CN(2,J),DH(2,J),ETA(2,J),J=1,15)/

1	0.3582,	0.0026,	0.8000,	0.5518,	0.0039,	0.8100,	15
2	0.7919,	0.0054,	0.8200,	1.0672,	0.0069,	0.8300,	16
3	1.2882,	0.0080,	0.8300,	1.4446,	0.0087,	0.8290,	17
4	1.5937,	0.0096,	0.8100,	1.8954,	0.0101,	0.8000,	18
5	2.0519,	0.0104,	0.7850,	2.2273,	0.0107,	0.7600,	19
6	2.3747,	0.0108,	0.7450,	2.6229,	0.0106,	0.7000,	20
7	2.8720,	0.0101,	0.6800,	3.0555,	0.0094,	0.6450,	21
8	3.3138,	0.0077,	0.5900/				22

DATA (CN(3,J),DH(3,J),ETA(3,J),J=1,15)/

1	0.3582,	0.0031,	0.8000,	0.5911,	0.0051,	0.8300,	23
2	0.8555,	0.0071,	0.8600,	1.0764,	0.0087,	0.8630,	24
3	1.2519,	0.0099,	0.8670,	1.4354,	0.0111,	0.8700,	25
4	1.5201,	0.0122,	0.8720,	1.8409,	0.0134,	0.8720,	26
5	2.0247,	0.0143,	0.8700,	2.2455,	0.0152,	0.8670,	27
6	2.4302,	0.0157,	0.8600,	2.5956,	0.0162,	0.8500,	28
7	2.7791,	0.0166,	0.8300,	3.0555,	0.0167,	0.8000,	29
8	3.3138,	0.0164,	0.7600/				30

DATA (CN(4,J),DH(4,J),ETA(4,J),J=1,15)/

1	0.3582,	0.0033,	0.7995,	0.4237,	0.0038,	0.8000,	31
2	0.5810,	0.0061,	0.8400,	0.8837,	0.0078,	0.8600,	32
3	1.1047,	0.0096,	0.8680,	1.2882,	0.0110,	0.8730,	33
4	1.5090,	0.0126,	0.8800,	1.7482,	0.0141,	0.8830,	34
5	2.0429,	0.0159,	0.8835,	2.2091,	0.0166,	0.8830,	35
6	2.3747,	0.0174,	0.8800,	2.6047,	0.0183,	0.8740,	36
7	2.8720,	0.0191,	0.8600,	3.1291,	0.0195,	0.8350,	37
8	3.3138,	0.0197,	0.8200/				38

DATA (CN(5,J),DH(5,J),ETA(5,J),J=1,15)/

1	0.3582,	0.0036,	0.7750,	0.5065,	0.0049,	0.8000,	39
2	0.7365,	0.0071,	0.8480,	0.9754,	0.0092,	0.8600,	40
3	1.2882,	0.0119,	0.8750,	1.5647,	0.0141,	0.8900,	41

4	1.7301,	0.0155,	0.8912,	1.9690,	0.0172,	0.8940,	46
5	2.0983,	0.0181,	0.8955,	2.2637,	0.0192,	0.8970,	47
6	2.4332,	0.0202,	0.8961,	2.6691,	0.0214,	0.8900,	48
7	2.9456,	0.0226,	0.8790,	3.1846,	0.0235,	0.8671,	49
8	3.3138,	0.0239,	0.8600/				50
DATA (CN(6,J),DH(6,J),ETA(6,J),J=1,15)/							51
1	0.3582,	0.0038,	0.7600,	0.6164,	0.0064,	0.8000,	52
2	0.8372,	0.0087,	0.8450,	1.1047,	0.0113,	0.8600,	53
3	1.2882,	0.0130,	0.8730,	1.5283,	0.0152,	0.8900,	54
4	1.7482,	0.0171,	0.8950,	1.9509,	0.0187,	0.9000,	55
5	2.2133,	0.0209,	0.9005,	2.4302,	0.0226,	0.9010,	56
6	2.5510,	0.0244,	0.9004,	2.8619,	0.0259,	0.9000,	57
7	3.1384,	0.0286,	0.8900,	3.2584,	0.0303,	0.8800,	58
8	3.3138,	0.0319,	0.8735/				59
DATA (CN(7,J),DH(7,J),ETA(7,J),J=1,15)/							60
1	0.3582,	0.0044,	0.7310,	0.7728,	0.0089,	0.8000,	61
2	1.0129,	0.0115,	0.8300,	1.2659,	0.0141,	0.8600,	62
3	1.4729,	0.0162,	0.8750,	1.6785,	0.0181,	0.8900,	63
4	1.8409,	0.0197,	0.8930,	2.0247,	0.0216,	0.8975,	64
5	2.1901,	0.0235,	0.8999,	2.3000,	0.0250,	0.9000,	65
6	2.3929,	0.0265,	0.8980,	2.5038,	0.0284,	0.8937,	66
7	2.5583,	0.0296,	0.8900,	2.6137,	0.0314,	0.8799,	67
8	2.5319,	0.0329,	0.8710/				68
DATA (CN(8,J),DH(8,J),ETA(8,J),J=1,15)/							69
1	0.3682,	0.0048,	0.7100,	0.6072,	0.0078,	0.7450,	70
2	0.7919,	0.0102,	0.7680,	0.9754,	0.0124,	0.8000,	71
3	1.2337,	0.0153,	0.8380,	1.4548,	0.0177,	0.8600,	72
4	1.5383,	0.0201,	0.8712,	1.8409,	0.0226,	0.8780,	73
5	1.9509,	0.0242,	0.8800,	2.0801,	0.0261,	0.8775,	74
6	2.1537,	0.0274,	0.8760,	2.2091,	0.0285,	0.8722,	75
7	2.2537,	0.0299,	0.8660,	2.3009,	0.0314,	0.8600,	76
8	2.3051,	0.0321,	0.8480/				77
DATA (CN(9,J),DH(9,J),ETA(9,J),J=1,15)/							78
1	0.3682,	0.0054,	0.6780,	0.5518,	0.0080,	0.7000,	79
2	0.5529,	0.0096,	0.7125,	0.8282,	0.0119,	0.7350,	80
3	1.0129,	0.0141,	0.7690,	1.1691,	0.0150,	0.8000,	81
4	1.2337,	0.0169,	0.8060,	1.3809,	0.0188,	0.8225,	82
5	1.5283,	0.0209,	0.8395,	1.6201,	0.0223,	0.8450,	83
6	1.7482,	0.0244,	0.8470,	1.8409,	0.0263,	0.8445,	84
7	1.9954,	0.0279,	0.8330,	1.9147,	0.0289,	0.8235,	85
8	1.9237,	0.0303,	0.8080/				86
DATA (CN(10,J),DH(10,J),ETA(10,J),J=1,12)/							87
1	0.3582,	0.0061,	0.6380,	0.4782,	0.0078,	0.6550,	88
2	0.5447,	0.0104,	0.6700,	0.7546,	0.0122,	0.6850,	89
3	0.8655,	0.0139,	0.7000,	0.9754,	0.0157,	0.7110,	90
4	1.1047,	0.0181,	0.7180,	1.2015,	0.0201,	0.7180,	91
5	1.2701,	0.0217,	0.7170,	1.3073,	0.0230,	0.7140,	92
6	1.3365,	0.0244,	0.7000,	1.3407,	0.0251,	0.6890/	93
DATA (CN(11,J),DH(11,J),ETA(11,J),J=1, 9)/							94
1	0.3682,	0.0069,	0.6000,	0.4418,	0.0086,	0.6000,	95
2	0.5518,	0.0106,	0.6120,	0.6447,	0.0123,	0.6170,	96
3	0.7365,	0.0141,	0.6210,	0.8282,	0.0159,	0.6258,	97
4	0.8837,	0.0172,	0.6250,	0.9391,	0.0186,	0.6230,	98
5	0.9715,	0.0201,	0.6009/				99
END							100

\$IBFTC	ETAAB3A	DECK	
	SUBROJTIME	ETAAB (FAR,EM6,P6,ETA,ETAADS,ETAASV,P6DS,P6DSAV,AM6DS,A	1
	1M6DSV,IDES,FAR7DS,FAR7SV)		2
	DIMENSION FART(25),ETABRT(25),EM6T(7),DELM6(7),P6T(14),DELP6(14)		3
	DIMENSION X(3),Y(3)		4
	DATA FART/.0390,.0585,.0732,.0878,.0976,.1171,.1268,.1463,.1619,		5
	1.1834,.1951,.2195,.2439,.2927,.3415,.4146,.4634,.5366,.5341,.7317,		6
	2.8293,.9268,1.000,1.0634,1.7/		7
	DATA ETABRT/.9400,.9887,1.0193,1.0306,1.0227,.9572,.9377,.9207,		8
	1.9354,.9626,.9773,1.0193,1.0532,1.077,1.0781,1.077,1.0747,1.0668,		9
	21.0573,1.0510,1.0374,1.0192,1.00,.9626,.9151/		10
	DATA EM6T/1.00,1.071,1.190,1.309,1.428,1.547,1.666/		11
	DATA DELM6/0.,.013,.041,.073,.110,.147,.187/		12
	DATA P6T/.220,.2267,.250,.300,.3333,.3767,.4167,.500,.5833,.6667,		13
	1.75,.8333,.9167,1.0/		14
	DATA DELP6/-.142,-.125,-.10,-.075,-.062,-.05,-.041,-.027,-.019,		15
	1-.013,-.008,-.004,-.0021,0./		15
	IF (IDES.NE.1) GO TO 5		17
	DO 1 K=1,25		18
1	ETABRT(K)=ETABRT(K)*ETAADS/ETAASV		19
	DO 2 K=1,25		20
2	FART(K)=FART(K)*FAR7DS/FAR7SV		21
	DO 3 K=1,7		22
3	EM6T(K)=EM6T(K)*AM6DS/AM6DSV		23
	DO 4 M=1,14		24
4	P6T(M)=P6T(M)*P6DS/P6DSAV		25
	ETAASV=ETAADS		26
	P6DSAV=P6DS		27
	FAR7SV=FAR7DS		28
	AM6DSV=AM6DS		29
	RETURN		30
5	CONTINUE		31
	N=0		32
	IF (FAR.GT.0.067) GO TO 8		33
	DO 6 J=1,25		34
6	IF (FAR.GE.FART(J)) N=J-1		35
	IF (N.EQ.0) N=1		36
	IF (N.GE.24) N=23		37
	DO 7 I=1,3		38
	NN=N-1+I		39
	X(I)=FART(NN)		40
7	Y(I)=ETABRT(NN)		41
	CALL PARABO (X,Y,FAR,ETA1)		42
	GO TO 9		43
8	ETA1=-2.*FAR+.1948		44
9	M=0		45
	DO 10 J=1,7		46
10	IF (EM6.GE.EM6T(J)) M=J-1		47
	IF (M.EQ.0) M=1		48
	IF (M.GE.6) M=5		49
	DO 11 I=1,3		50
	MM=M-1+I		51
	X(I)=EM6T(MM)		52
11	Y(I)=DELM6(MM)		53
	CALL PARABO (X,Y,EM6,COR1)		54
	L=0		55
	DO 12 J=1,14		56
12	IF (P6.GE.P6T(J)) L=J-1		57
	IF (L.EQ.0) L=1		58
	IF (L.GE.13) L=12		59
	DO 13 I=1,3		60
	LL=L-1+I		61
	X(I)=P6T(LL)		62

13	Y(I)=DELP6(LL)	63
	CALL PARABO (X,Y,P6,COR2)	64
	ETA=ETA1*(1.-COR1)*(1.+COR2)	65
	RETURN	66
	END	67

\$IBFTC	FRATIO	DECK	
	SUBROJTIME	FRASHO (PRATIO,ARATIO,CF)	1
	DIMENSION	PR(30),AR(10),CFR(30,10),XX(3),YY(3),ZZ(2)	2
	DATA	PR/1.0,1.25,1.5,1.75,2.0,2.25,2.5,2.75,3.0,3.25,3.5,3.75,4.0,	3
	DATA	14.25,4.5,4.75,5.,5.5,6.,6.5,7.,8.,9.,11.,13.,15.,17.,19.,21.,23./	4
	DATA	AR/ 1.0,1.05,1.092,1.228,1.318,1.423,1.8,1.9,1.97,2.15/	5
	DATA	(CFR(I,1),I=1,180)/	6
	1.971,.975,.978,.980,.981,.980,.9784,.9765,.9742,.9715,	7	
	2.9688,.9658,.9632,.960,.957,.9538,.9506,.9445,.938,.932,.927,.918,	8	
	3.910,.897,.8855,.875,.867,.859,.8515,.844,	9	
	4.963,.966,.970,.9743,.978,.9805,.982,.9805,.9786,.9762,.974,.9718,	10	
	5.9693,.967,.9645,.962,.9595,.954,.949,.944,.9395,.9308,.9238,	11	
	6.9117,.901,.892,.884,.8762,.869,.862,	12	
	7.956,.9605,.9638,.967,.970,.9731,.9754,.9775,.9795,.981,.980,.978,	13	
	8.9761,.9742,.9723,.9702,.968,.9637,.9594,.9554,.9517,.945,.9387,	14	
	9.9276,.9177,.9082,.8998,.891,.883,.8752,	15	
	\$.947,.949,.9517,.954,.957,.9601,.9632,.9668,.971,.9737,.976,.9777,	16	
	\$.978,.9772,.976,.9752,.974,.970,.9655,.9637,.9603,.955,.950,.9403,	17	
	\$.9317,.9232,.915,.907,.8996,.892,	18	
	\$.949,.9502,.9508,.9473,.9438,.941,.939,.940,.950,.956,.9605,.964,	19	
	\$.967,.968,.9692,.970,.9703,.9698,.9696,.9684,.966,.960,.9444,	20	
	\$.9438,.9347,.9265,.9193,.9128,.9067,.901,	21	
	\$.948,.9485,.9487,.9484,.947,.9435,.936,.9275,.9185,.9155,.9155,	22	
	\$.929,.941,.9455,.950,.953,.9558,.9591,.9608,.9609,.9601,.957,	23	
	\$.9531,.9448,.9362,.9279,.9188,.910,.901,.8921/	24	
	DATA (CFR(I,1),I=181,300)/	25	
	1.808,.815,.8245,.833,.8418,.852,.861,.871,.8811,.892,.903,.9142,	26	
	2.924,.935,.9428,.9505,.958,.968,.9748,.9784,.9808,.982,.9816,	27	
	3.9778,.974,.9695,.9646,.9598,.9543,.9487,	28	
	4.796,.804,.8121,.82,.8291,.838,.8465,.856,.8654,.8763,.8862,	29	
	5.8975,.9085,.922,.930,.939,.946,.9575,.9655,.971,.975,.9783,	30	
	6.9792,.977,.974,.970,.9665,.9623,.958,.9538,	31	
	7.792,.7981,.804,.812,.8201,.828,.8357,.846,.8545,.864,.874,	32	
	8.8837,.894,.905,.9146,.9247,.933,.95,.9613,.966,.970,.9755,.9768,	33	
	9.9782,.9770,.9742,.970,.966,.961,.956,	34	
	\$.774,.780,.7871,.794,.801,.8087,.8165,.8247,.833,.841,.850,.8583,	35	
	\$.868,.878,.887,.8962,.907,.9255,.944,.953,.9592,.966,.9705,.9725,	36	
	\$.9718,.97,.9685,.9664,.964,.9616/	37	
	Y(X1,X2,X,Y1,Y2)=(Y1-Y2)*(X-X1)/(X1-X2)+Y1	38	
	N=0	39	
	DO 1 J=1,30	40	
1	IF (PRATIO.GE.PR(J)) N=J-1	41	
	IF (N.EQ.0) N=1	42	
	IF (N.GE.29) N=28	43	
	M=0	44	
	DO 2 K=1,10	45	
2	IF (ARATIO.GE.AR(K)) M=K	46	
	IF (M.EQ.10.OR.M.EQ.0) GO TO 5	47	
	DO 4 L=1,2	48	
	LL=M+L-1	49	
	DO 3 I=1,3	50	

```

      NN=N+I-1
      XX(I)=PR(NN)
3     YY(I)=CFR(NN,LL)
      CALL PARABO (XX,YY,PRATIO,ANS)
4     ZZ(L)=ANS
      CF=Y(AR(M),AR(M+1),ARATIO,ZZ(1),ZZ(2))
      RETURN
5     WRITE (6,7) ARATIO,AR(10)
      DO 6 I=1,3
      NN=N+I-1
      XX(I)=PR(NN)
6     YY(I)=CFR(NN,10)
      CALL PARABO (XX,YY,PRATIO,CF)
      RETURN
C
C
7     FORMAT (8H0ARATIO=F7.3,4X,26HOUT OF RANGE,USE DATA FOR ,F7.3)
      END

```

```

$IBFTC INPUT  DECK
      SUBROUTINE INPUT(LIJNIT,LOUNIT,ID,D,IT)
      DIMENSION D(1), IT(1)
C
C          ' DEFINES ARGUMENTS AS ARRAYS
C          CONSTRUCTION OF THE TABLE  ITAB
C
C      BIT  STANDARD  RETURN  RETURN  RETURN  RETURN  RETURN
C      NUMBER  RETURN  1      2      3      4      5
C      0-2      $      A-Z      0-9( )  =      OTH
C      3-4      DETPC  RI OTH  NLSF
C      5-6      DER    TINL  OTH PS  CF
C      7-8      0-9.,+$)OTH , $A-Z=  -(*/
C      9        ,.)A-Z  OTH
C      9-10     .,      A-Z)      0-9+^  =$
C      10       OTH      $A-Z*)=/
C      10-11    0-9.,-OTH +^ (      $A-Z*  )=/
C      12       0-9 A-Z  OTH
C      12-13    A-Z      0-9      OTH
C      13-14    +-      OTH      0-9
C      15-16    +.      DE-      OTH
C      17-18    F        F        OTH
C      19-20    D        E        OTH
C      21       OTH      .
C      21-22    A-Z      OTH
C      23       '        OTH
C      24       $        OTH
C      25       =        OTH
C      26-27    (        A-Z      OTH
C      28-29    )        ,        OTH
C      30       (        OTH
C      31       )        OTH
C      ERROR CODE DESIGNATIONS
C      ROUTINE  TYPE
C      10 IXQTI  100 ILLEGAL CHARACTER
C      20 ITABLI 200 NAME TOO LONG
C      30 INMBRI 300 TABLE FULL OR BAD
C      40 INAMEI 400 SCALING ERROR
C      50 IARITI 500 NAME NOT IN TABLE
C      60 INAMEN 600 $DATA() INCOMPLETE

```

```

C      70 INPUT  700 FORMULA ILL FORMED 39
C      800 FUNCTION UNDEFINED 40
C      MCNVRT = TYPE OF LEFT HAND VARIABLE 41
C      KCNVRT = TYPE OF CURRENT VARIABLE 42
C      ITYPE  MEANING 43
C      1      REAL 44
C      2      INTEGER 45
C      3      DOUBLE PRECISION 46
C      4      TYPELESS OR NO CONVERSION 47
C      5      SUBROUTINE 48
C      6      FUNCTION 49
C      FORMAT OF TABLE 50
C      1ST WORD  012 3456 7-31 51
C      TYPE  NUMBER OF WORDS ADDRESS 52
C      NEXT 1 TO 15 WORDS  THE NAME, 4 CHARACTERS TO THE WORD 53
C      DIMENSION IFT( 31), IPTAB( 21), ITAB( 65) 54
C      DIMENSION ANAME(15) ,IMAGE(80) ,IMAGE1(81),IPARAM(9) 55
C      ,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27) 56
C 57
C      COMMON 58
C      ./ICOMVI/ VALUE ,ICOMP ,IFNTYP ,IMAGE1 ,IRADIX ,ISUB 59
C      ,KCH ,KCNVRT ,KCOUNT ,KDIF ,KFLD1 ,KFLD2 60
C      ,LCOMP ,LCNVRT ,LEVEL ,LFRT ,LOOK 61
C      ,MCNVRT ,MDIF ,MODALL ,MSTOR 62
C      ,NAME ,NERROR ,NONEW ,NOTARG 63
C      ,SMCHR ,TEST ,ERMARK 64
C      ./ICNSTI/ BLANK ,BLANKS ,DOLLAR ,EOS 65
C      ,ICOMMA ,IDOLAR ,IFT ,IPTAB ,ITAB 66
C      ,KAM10 ,KBPC ,KBPW ,KCPCD ,KERTYP 67
C      ,KZERO ,NOPRNT ,TAB1 68
C      ./IPARAM/ ABORT ,CIUNIT ,KOUNIT ,LIMALF ,LOCK ,LOCX 69
C      ,NOLIST ,NSTDIR ,TRACE 70
C      ./ISTAKI/ STACK ,ISTDIM ,KSTACK ,LEVLIM 71
C 72
C      INTEGER BLANK ,BLANKS ,EOS ,IDOLLAR ,TAB1 ,TRACE 73
C      ,CONTYP ,STORED 74
C      DOUBLE PRECISION STACK, VALUE 75
C      LOGICAL ABORT 76
C      * STOP FOR ERROR 77
C      LOGICAL DOLLAR 78
C      * TRUE IF A $T,$P, OR $C IS BEING PROCESSED 79
C      LOGICAL ERMARK 80
C      * .TRUE. IF ANY ERROR HAS OCCURED 81
C      LOGICAL LIMALF 82
C      * TERMINATE ALF CONSTANTS AT 15 WORDS. 83
C      LOGICAL LOCK 84
C      * DO NOT STORE ANY INPUT DATA 85
C      LOGICAL NOLIST 86
C      * DO NOT LIST INPUT CARDS 87
C      LOGICAL NONEW 88
C      * DO NOT READ A NEW CARD. SET IN INPJT 89
C      LOGICAL MODALL 90
C      * .TRUE. RETURN ALL CHARACTERS 91
C      * .FALSE. DELETE BLANKS AND SKIP FOR E 92
C      LOGICAL SHORT 93
C      * DO NOT STORE ALPHABETIC AND RADIX CO 94
C      DIRECTLY INTO D ARRAY 95
C      LOGICAL SMCHR 96
C      * SAME CHARACTER IS TO BE USED OVER BY 97
C      LOGICAL TEST 98
C      * STORE COMMA AT END OF CARD IF NEEDED 99
C      END ***** 100

```

	EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)	101
	EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)	102
	LOGICAL END2	103
C	* .TRUE. IF A \$END CARD HAS ALREADY BEEN FOUND	104
	KIUNIT=LIUNIT	105
	KOUNIT=LOUNIT	106
C	CALL DEBUGX	107
	LIMLF = .TRUE.	108
	DOLLAR = .FALSE.	109
	NOLIST = .FALSE.	110
	END2 = .FALSE.	111
	MODALL = .FALSE.	112
	ERMARK = .FALSE.	113
	LOCK = .FALSE.	114
	KERTYP = 0	115
	IFNTYP = -1	116
	LOCX=1	117
	L=LEVLIM+3	118
	DO 16 I=1,L	119
16	KSTACK(I)=0	120
	LEVEL=0	121
	LFRT=0	122
	NOTARG = ISTDIM	123
	LCNVRT = 3	124
C	* DOUBLE PRECISION, STANDARD	125
	MCNVRT = 1	126
	KDIF = 1	127
	MDIF = 1	128
1	NONEW = .FALSE.	129
	SMCHR = .TRUE.	130
	CALL ICHAR2(\$9470,24)	131
C	* ERROR IF \$ IS NOT 1ST CHARACTER READ	132
	LDOLLAR = KCOUNT	133
C	* CARD COLUMN OF LAST DOLLAR SIGN	134
	NONEW=.TRUE.	135
C	* \$DATA MUST BE ON A SINGLE CARD	136
C	D E OTH	137
	CALL ICHARI(\$3,\$9470,19)	138
C	(A-Z OTH	139
4	CALL ICHARI(\$4,\$9470,26)	140
	CALL ISUBI	141
	CALL ICHAR2(\$9470,31)	142
C	* ERROR IF NO)	143
	NONEW=.FALSE.	144
	IT(1)=ISUB	145
C	CALL DEBUG 2 (5HSETNO,IT(1))	146
C	CALL DEBUG 2(6HKERSIN,KERSIN)	147
	IF (ID.NE. IT(1)) GO TO 99	148
C	* RETURN BECAUSE WRONG DATA SET	149
	IF (IT(2).LT. 0)GO TO 9370	150
C	* TABLE FULL OR BAD	151
	GO TO 19	152
3	IF (END2) GO TO 99	153
	END2 = .TRUE.	154
	NONEW = .FALSE.	155
5	CALL ICHAR2(\$1,9)	156
C	* PASS LETTERS AND ,.)	157
	GO TO 5	158
9470	KERTYP = 470	159
	GO TO 999	160
99	CONTINUE	161

C	CALL DEBUG 2(6HSTATMT,99)	162
	KCH = IDOLAR	163
	KCOUNT = LDOLAR	164
C	*NEXT CALL BEGINS PROCESSING AT THIS CARD COLUMN	165
	IF(ERMARK) IT(1) = -IABS(IT(1))	166
C	* WARN PROGRAMER OF POSSIBLE ERRO	167
	IF(ERMARK.AND.ABORT) GO TO789	168
	RETURN	169
789	WRITE(6,787)	170
787	FORMAT(44H ERROR HAS OCCURED AND ABORT IS .TRUE., STOP)	171
	STOP	172
C	BEGIN TO INPUT THE DATA	173
C	STATEMENTS 19 AND 20 ARE THE SWITCHHOUSE	174
C	CONTROL COMES HERE FOR NEW DIRECTION.	175
C	LFRT = 0 INDICATES THAT THE PREVIOUS TASK WAS COMPLETED.	176
19	CONTINUE	177
C	CALL DEBUG2(6HSTATMT,19)	178
	TEST=.FALSE.	179
	GO TO 21	180
120	DOLLAR = .FALSE.	181
20	CONTINUE	182
C	CALL DEBUG 2(6HSTATMT,20)	183
	TEST=.TRUE.	184
C	ENTRY ICHAR4 LOADS LCOMP WITH TAB NO 10 AND RETURNS ON TAB NO 7	185
C	OTH , 'A-Z= -(*/	186
21	CALL ICHAR4(\$31, \$32, 7,10)	187
C NOW	WE TEST LCOMP ON EACH OF THE 3 POSSIBLE RETURNS. THIS IS A 12 WAY	188
C	0-9. + \$)	189
	GO TO (203,510, 64,430),LCOMP	190
C	, ' A-Z =	191
31	GO TO (460,202,530,450),LCOMP	192
C	- (* /	193
32	GO TO(520,420,440,441),LCOMP	194
64	IF(LFRT.NE.0) GO TO 9770	195
	LDOLAR = KCOUNT	196
C	* CARD COLUMN OF LAST DOLLAR SIGN	197
	DOLLAR = .TRUE.	198
C	CALL DEBUG 2 (6HSTATMT,64)	199
	TEST=.FALSE.	200
	CALL ICHAR4(\$9170,\$9170,3,5)	201
C	DE T P C	202
	GO TO(99,100,150,630),LCOMP	203
100	CALL ITABLI(IT)	204
	IF (KERTYP) 999,120,999	205
C		206
C		207
C	\$PARAMETER	208
C	(A-Z OTH	209
150	CALL ICHARI(\$150,\$9170,26)	210
151	CALL ICHAR2(\$152,0)	211
C	* GET FIRST CHAR OF NAME ..DON'T BRAN	212
152	CALL INAMEN	213
	CALL ILOOKI(\$153,IPTAB)	214
9570	KERTYP = 570	215
	GO TO999	216
153	CALL ICHAR2(\$9170,29)	217
C	* ERROR IF NO =	218
	CALL ICHAR2(\$9170,10)	219
C	* GET CHARACTER,SEPERATE PART OF	220
	CALL INMBRI	221
	KCNVRT=IFLD(0,3,IPTAB(LOOK))	222
	CALL ICNVTI(3,KCNVRT)	223

C		* CONVERT FROM DP TO TYPE OF VARIABLE	224
	IPT=IFLD(7,25,IPTAB(LOOK))		225
	IPARAM(IPT) = KVALUE		226
C	) , OTH		227
	CALL ICHAR1(\$151,\$9170,28)		228
C	IF(TRACE.GT.0) CALL DEBUGX		229
C		* IF TRACE TURNED ON PUT ON HEAD	230
	GO TO 120		231
C		* GO TO SWITCH HOUSE TEST FOR INSERTIO	232
C			233
C			234
C	CONSTANTS = LOGICAL,NUMERIC,ALPHAMERIC,RADIX.		235
201	ASSIGN 220 TO CONTYP		236
C		* RADIX	237
	IFNTYP = -1		238
	GO TO 210		239
202	ASSIGN 230 TO CONTYP		240
C		" "	241
	GO TO 210		242
203	ASSIGN 250 TO CONTYP		243
C		* NUMERIC,LOGICAL	244
C			245
C			246
C	ALL CONSTANTS 210		247
210	LOP = 0		248
	SHORT = .TRUE.		249
	MSTOR = 0		250
C	CALL DEBUG2(6HST CON,210)		251
	IF(LEVEL.EQ.0) LEVEL=3		252
	IF(KSTACK(LEVEL).EQ.1) GO TO 9770		253
C		* TWO CONSTANTS IN ROW E.G.	254
	GO TO CONTYP,(220,230,250)		255
C			256
C			257
C	RADIX CONSTANTS 220		258
C	(A-Z OTH		259
220	CALL ICHAR2(\$9170,30)		260
	CALL ISUBI		261
	NAME(2) = 0		262
221	CALL ICHAR2(\$2215,29)		263
C		* ERROR IF NO COMMA AFTER BASE NUMBER	264
	GO TO 9170		265
2215	CALL ICHAR2(\$9170,12)		266
C		* ERROR IF NO NUMBER	267
222	IRADIX = ISUB		268
	SMCHR = .TRUE.		269
223	CALL ISUBI		270
	MSTOR = MSTOR +1		271
	NAME(MSTOR) = ISUB		272
	IF (MSTOR.GE.15) GO TO 225		273
C	) , OTH		274
224	CALL ICHAR1(\$223,\$9170,28)		275
	IRADIX = 10		276
	GO TO 240		277
225	ASSIGN 224 TO NEXT		278
	GO TO 241		279
C			280
C			281
C	HOLLERITH CONSTANTS 230		282
230	MODALL = .TRUE.		283
	TEST = .FALSE.		284
	NAME(2) = BLANKS		285

231	CALL INAMEN	286
	IF(.NOT.MODALL)GO TO 240	287
C		288
	IF(LIMALF) GO TO 265	289
C	LONG CONSTANT GOES TO 234	290
234	ASSIGN 231 TO NEXT	291
	GO TO 241	292
C		293
C		294
C	STORE ALF + RADIX 240	295
240	ASSIGN 260 TO NEXT	296
	IF (MSTOR-2) 242,2405,241	297
2405	IF (LFRT.EQ.1) GO TO 242	298
	CALL ICHAR2(\$2406,29)	299
	SMCHR = .TRUE.	300
	GO TO 242	301
2406	SMCHR = .TRUE.	302
241	SHORT = .FALSE.	303
	IF (LFRT.NE.0) GO TO 265	304
C	CALL DEBUG2(6HST 241,0)	305
242	KVALUE = NAME(1)	306
	RVALUE(2) = ANAME(2)	307
	IF (NOTARG.GT.LEVEL) LCNVRT = 4	308
C	CALL DEBUG2(6HST 242,0)	309
C	CALL DEBUG2(6HSHORT ,SHORT)	310
C	CALL DEBUG2(6HLFRT ,LFRT)	311
243	IF (SHORT) GO TO 255	312
	DO 245 I= 1,MSTOR	313
	IF (LOCK) GO TO 245	314
	D(LOCK) = ANAME(I)	315
245	LOCK = LOCK + 1	316
246	GO TO NEXT, (224,231,260)	317
C	SEE NEXT AND MEANING	318
C	250 ' ALF OR) AND RADIX	319
C	224 MORE THAN 15 ELEMENTS IN RADIX FIELD	320
C	231 MORE THAN 15 ELEMENTS IN ALF FIELD	321
C		322
C		323
C	NUMERIC + LOGICAL 250	324
250	TEST = .TRUE.	325
	CALL INMBRI	326
255	CALL IARITI	327
	LFRT = 1	328
	IF (KERTYP) 999,20,999	329
C		330
C		331
C	RESET STACK BECAUSE IT WAS NOT USED	332
260	LEVEL = 0	333
	TEST = .TRUE.	334
	CALL ICHAR2(\$19,29)	335
C		336
	' SKIP COMMA	337
	KERTYP = 171	338
	GO TO 999	339
265	KERTYP = 270	340
	GO TO 999	341
C		342
C	TEST EMPTY PARENTHESES	343
400	IF ((KSTACK(LEVEL-1)+KSTACK(LEVEL-2)).NE.0) GO TO 997	344
C	CALL DEBUG 2(6HSTA (),400)	345
C	CALL STACKP	346
C		347
401	LEVEL=LEVEL-3	

C	EMPTY FUNCTION ARGUMENT IS NOT A CURRENT LEFT SIDE	348
	IF (KSTACK(LEVEL).LT.6) GO TO 404	349
405	VALUE=0.	350
	GO TO 403	351
404	RVALUE(1) = D (LOCK)	352
	RVALUE(2) = D(LOCK+1)	353
	CALL ICNVTI(MCNVRT,3)	354
403	CALL IARITI	355
	GO TO 20	356
C((((((((((((((((		357
420	IF (LEVEL.EQ.0) LEVEL=3	358
C	CALL DEBUG 2(6HSTAT (,420)	359
422	LOP=0	360
	LFRT = 1	361
	LEVEL=LEVEL+3	362
C	CALL STACKP	363
C	PRINT STACK	364
	IF (KSTACK(LEVEL-3)-1) 20, 997, 20	365
C))))))))))))))		366
430	IF (LEVEL.LT.6) GO TO 997	367
C	CALL DEBUG 2(6HSTAT),430)	368
C	CALL STACKP	369
C	PRINT STACK	370
	LFRT = 1	371
431	IF (KSTACK(LEVEL)-1) 400, 432, 997	372
432	DO 433 I=1,3	373
	VALUE=STACK(LEVEL)	374
	KSTACK(LEVEL)=0	375
	LEVEL=LEVEL-1	376
433	CALL IARITI	377
	IF (KERTYP) 999, 20, 999	378
C*****{KOP=2}		379
440	KOP=2	380
C	CALL DEBUG 2(6HSTAT *,440)	381
	GO TO 445	382
C/////////{KOP=3}		383
441	KOP = 3	384
C		385
C	CALL DEBUG 2(6HSTAT /,441)	386
445	IF (LJP.NE.0) GO TO 997	387
C	CALL DEBUG 2(6HSTATMT,445)	388
C	CALL STACKP	389
C	PRINT STACK	390
	LOP = 1	391
	LFRT = 1	392
	IF (LEVEL.EQ.0) LEVEL=3	393
	IF (KSTACK(LEVEL).NE.1) GO TO 997	394
444	VALUE=STACK(LEVEL)	395
	KSTACK(LEVEL)=0	396
	LEVEL=LEVEL-1	397
	CALL IARITI	398
	IF (KERTYP.NE.0) GO TO 999	399
446	KSTACK(LEVEL)=KOP	400
	LEVEL=LEVEL+1	401
C	CALL STACKP	402
C	PRINT STACK	403
	GO TO 19	404
C=====		405
450	IF (LEVEL.NE.0) GO TO 997	406
C	CALL DEBUG 2(6HSTAT =,450)	407
451	LOP=0	408
	LOCK = LOOK	409

MCNVRT = KCVRT	410
MDIF = KDIF	411
LCNVRT = 3	412
C	413
LEVEL=3	414
LFRT=0	415
KSTACK(3)=0	416
C CALL DEBUG2(6HLOCX ,LOCX)	417
C CALL DEBUG2(6MCNVRT,MCNVRT)	418
GO TO 19	419
C,,,,, ,,,,,, ,	420
460 CONTINUE	421
C CALL DEBUG 2(6HSTAT ,,460)	422
C CALL STACKP	423
C	424
IF (LEVEL - 3) 461,463, 600	425
461 IF (LEVEL.NE.0) GO TO 997	426
462 LEVEL=3	427
463 IF(LFRT.EQ.0)GO TO 480	428
C	429
LFRT WILL BE ZERO AFTER \$ EXPRESSION OR FOR CONSECUTIVE COMMAS	430
470 IF (KSTACK(3).NE.1) GO TO 997	431
471 DO 475 I = 1,2	432
VALUE=STACK(LEVEL)	433
KSTACK(LEVEL)=0	434
LEVEL=LEVEL-1	435
475 CALL IARITI	436
IF (KERTYP.NE.0) GO TO 999	437
476 KSTACK(1)=0	438
CALL ICNVTI (LCNVRT,MCNVRT)	439
474 IF(LOCK) GO TO 480	440
D(LOCK) = RVALUE(1)	441
IF (MCNVRT .EQ. 3) D(LOCK+1) = RVALUE(2)	442
480 LOP=0	443
C CALL STACKP	444
C	445
LFRT=0	446
LOCX = LOCX + MDIF	447
LEVEL=0	448
LCNVRT = 3	449
C	450
GO TO 19	451
C+++++ (KOP=4)	452
510 KOP=4	453
C CALL DEBUG 2(6HSTAT +,510)	454
GO TO 521	455
C----- (KOP=5)	456
520 KOP=5	457
C CALL DEBUG 2(6HSTAT -,520)	458
521 IF (LOP.NE.0) GO TO 997	459
522 LOP=1	460
LFRT = 1	461
IF (LEVEL.EQ.0) LEVEL=3	462
524 IF (KSTACK(LEVEL)-1) 525, 526, 997	463
525 STACK(LEVEL)=0. DO	464
C CALL DEBUG 2(6HSTATMT,525)	465
526 DO 528 I=1,2	466
C CALL DEBUG 2(6HSTATMT,526)	467
VALUE=STACK(LEVEL)	468
KSTACK(LEVEL)=0	469
LEVEL=LEVEL-1	470

528	CALL IARITI	471
	IF (KERTYP.NE.0) GO TO 999	472
527	KSTACK(LEVEL)=KOP	473
	LEVEL=LEVEL+2	474
	GO TO 19	475
CAB CDEFGHIJ KLMNOPQRST UVWXYZABCDEFGHIJ KLMNOPQRST UVWXYZABCDEFGHIJ KLMNOPQRS		476
530	LOP=0	477
C	CALL DEBUG 2(6HSTAT A,530)	478
	IF ((LEVEL.NE.0).AND.(KSTACK(LEVEL).EQ.1)) GO TO 997	479
535	TEST = .TRUE.	480
	CALL INAMEI(\$999,D,IT)	481
	IF (IFNTYP) 531,621,640	482
531	LFRT = 1	483
	IF(LEVEL.NE.0)GO TO 540	484
532	STACK(3)=VALUE	485
C	CALL DEBUG 2(6HSTATMT,532)	486
	KSTACK(3)=1	487
C	CALL STACKP	488
C	PRINT STACK	489
	GO TO 20	490
540	CONTINUE	491
C	CALL DEBUG 2(6HSTATMT,540)	492
	CALL IARITI	493
	IF(KERTYP) 999,20,999	494
C		495
C	PROCESS SUBROUTINES AND FUNCTIONS	496
C		497
C	COMMA SEPARATING FUNCTION ARGUMENTS	498
600	IF (KSTACK(LEVEL-3).LT.6) GO TO 997	499
C	CALL DEBUG 2(6HSTAT F,600)	500
601	IF (KSTACK(LEVEL)-1) 602, 603, 997	501
602	STACK(LEVEL)=0.	502
603	DO 610 I=1,2	503
	VALUE = STACK(LEVEL)	504
	KSTACK(LEVEL)=0	505
	LEVEL=LEVEL-1	506
610	CALL IARITI	507
	IF (KERTYP.NE.0) GO TO 999	508
611	KSTACK(LEVEL)=KSTACK(LEVEL-1)+1	509
	LEVEL=LEVEL+3	510
C	CALL STACKP	511
C	PRINT STACK	512
	GO TO 19	513
C	\$CALL	514
630	DO 631 I = 1,4	515
C	SKIP 'ALL' IN \$CALL	516
C	OTH OPERATORS	517
	CALL ICHAR2(\$9170,12)	518
631	CONTINUE	519
	DOLLAR = .FALSE.	520
	TEST = .TRUE.	521
	CALL INAMEI (\$999,D,IT)	522
	IF (IFNTYP .NE. 1) GO TO 640	523
	KVALUE = -KVALUE	524
C	* INDICATE THAT NO RESULT IS TO BE STORED FOR SJB	525
C	FUNCTION NAME	526
621	CONTINUE	527
C	* FUNCTIONS	528
C	CALL DEBUG2(6HST FNC,621)	529
C	CALL DEBUG2(6HIFNTYP,IFNTYP)	530

C	CALL DEBUG2(6HKVALUE,KVALUE)	531
	IFNTYP = -1	532
	IF (LEVEL.EQ.0) LEVEL = 3	533
	IF(NOTARG .GT. LEVEL) NOTARG = LEVEL	534
	LFRT = 1	535
	STACK(LEVEL)=KVALUE	536
	KSTACK(LEVEL)=6	537
	LEVEL = LEVEL + 3	538
C	CALL STACKP	539
C		540
C	(OTH	541
	CALL ICHAR2(\$622,30)	542
	GO TO 19	543
C	* IF THERE ARE ARGUMENTS	544
622	SMCHR = .TRUE.	545
	IF (KVALUE .LT. 0) LEVEL = LEVEL -3	546
	GO TO 405	547
640	IF (IFNTYP - 2) 9770,201,700	548
C	LOCX THE SUBROUTINE THAT STORES LOCX THE SUBSCRIPT OF D ARR	549
700	IF (IFNTYP .NE. 3) GO TO 9770	550
	IFNTYP = -1	551
	IF(LEVEL.NE.0) GO TO 9770	552
	LFRT = 0	553
	CALL ICHAR2(\$9170,30)	554
C	* ERROR IF NO (AFTER LOCX	555
	ASSIGN 704 TO NEXT	556
C	A-Z OTH	557
702	CALL ICHAR2(\$9170,9)	558
	CALL INAMEI(\$999,D,IT)	559
	IF (IFNTYP .GT. -1) GO TO 9770	560
	GO TO NEXT,(704,708)	561
704	LOOKX = LOOK	562
C	OTH	563
	CALL ICHAR2(\$706,29)	564
C	* ERROR IF NO COMMA	565
	GO TO 9170	566
706	ASSIGN 708 TO NEXT	567
	GO TO 702	568
708	KVALUE = LOCX - LOOKX + 1	569
	CALL ICNVTI(2,KCNVRT)	570
C	* CONVERT FROM INT TO TYPE OF 2ND ARG	571
	IF(LOCK) GO TO 710	572
	D(LOOK) = RVALUE(1)	573
710	CALL ICHAR2(\$9170,31)	574
C	* ERROR IF NO)	575
	TEST = .TRUE.	576
C	OTH	577
	CALL ICHAR2(\$19,29)	578
C	* SKIP THE COMMA THAT MUST FOLLOW	579
9170	KERTYP=170	580
	GO TO 999	581
9370	KERTYP=370	582
	GO TO 999	583
997	CONTINJE	584
9770	KERTYP = 770	585
	GO TO 999	586
999	CALL IERORI	587
	LFRT = 0	588
	LOP = 0	589
	NAME (1)=0	590
	GO TO 19	591
C	* NOW GO TO SWITCH HOUSE .. GOOD	592
	END	593

```

$IBFTC BLOCK   DECK
BLOCK DATA
C   ICOMVI      BLOCK DATA PROGRAM
    DIMENSION IFT( 31), IPTAB( 21), ITAB( 65)
    DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)
    .           ,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)
C
COMMON
./ICOMVI/ VALUE      ,ICOMP      ,IFNTYP      ,IMAGE1      ,IRADIX      ,ISUB
.           ,KCH        ,KCNVRT      ,KCOUNT      ,KDIF        ,KFLD1      ,KFLD2
.           ,LCOMP      ,LCNVRT      ,LEVEL        ,LFRT        ,LOOK
.           ,MCNVRT      ,MDIF        ,MODALL      ,MSTOR
.           ,NAME        ,NERROR      ,NONEW        ,NDTARG
.           ,SMCHR      ,TEST        ,ERMARK
./ICNSTI/ BLANK      ,BLANKS      ,DOLLAR      ,EOS
.           ,ICOMMA      ,IDOLAR      ,IFT          ,IPTAB      ,ITAB
.           ,KAM10      ,KBPC        ,KBPW        ,KCPCD      ,KERTYP
.           ,KZERO      ,NOPRNT      ,TAB1
./IPARAM/ ABORT      ,KIUNIT      ,KOUNIT      ,LIM Alf      ,LOCK      ,LOCX
.           ,NOLIST      ,NSTDIR      ,TRACE
./ISTACI/ STACK      ,ISTDIM      ,KSTACK      ,LEVLIM
C
INTEGER BLANK      ,BLANKS      ,EOS      ,IDOLAR      ,TAB1      ,TRACE
DOUBLE PRECISION STACK, VALUE
LOGICAL ABORT,DOLLAR,ERMARK,LIM Alf,LOCK,NOLIST,NONEW,MODALL,
.NSTDIR,SMCHR,TEST
DATA NERROR,TRACE/0,0/
DATA ITAB/ 65* 20572347824/
DATA ITAB( 49)/-11890146992/
., ITAB( 44)/ 11923699376/, ITAB( 61)/ 29355228304/
., ITAB( 29)/ 29053238816/, ITAB( 12)/ -3484760752/
., ITAB( 28)/-11821997744/, ITAB( 17)/-11903778480/
., ITAB( 33)/-12155961008/, ITAB( 45)/-12192136880/
., ITAB( 50)/-12208914096/, ITAB( 13)/-12041137840/
., ITAB( 60)/-11957255792/, ITAB( 11)/-11890146992/
., ITAB( 1)/ 29063724720/, ITAB( 2)/ 29063724720/
., ITAB( 3)/ 29063724720/, ITAB( 4)/ 29063724720/
., ITAB( 5)/ 29063724720/, ITAB( 6)/ 29063724720/
., ITAB( 7)/ 29063724720/, ITAB( 8)/ 29063724720/
., ITAB( 9)/ 29063724720/, ITAB( 10)/ 29063724720/
., ITAB( 18)/ 20572347824/, ITAB( 20)/ 18961735088/
., ITAB( 21)/ 17350532528/, ITAB( 22)/ 17350565296/
., ITAB( 23)/ 23256571312/, ITAB( 26)/ 20035476912/
., ITAB( 36)/ 22182960560/, ITAB( 38)/ 22182960560/
., ITAB( 40)/ 18424854176/, ITAB( 42)/ 19498606000/
., ITAB( 51)/ 22719831472/, ITAB( 52)/ 17887731120/
DATA BLANK,BLANKS,EOS,ICOMMA,IDOLAR,KAM10,KBPC,KBPW,KCH,KCPCD,
. KZERO,NOPRNT,TAB1,IMAGE1(2),IMAGE1(3)/
.           48,-17997958192, 10, 59, 43,
.           7, 6, 36, 43, 80,
.           0, 11555507248, 64, 11555507248, 11555507248/
DATA ABORT,IMAGE1(1),IMAGE1(81),IRADIX,ISTDIM,
. KCOUNT,KDIF,KSTACK,LEVLIM,NSTDIR,STACK/
. .FALSE.,1HE,1H ,10,27,0,1,27*0,24,.TRUE.,27*0.DO/
DATA IFT/
.           0, 31,-17716740112, 21051935792,-17716740128,
. 23481748528,-17716740144, -3864988720,-17716740160,-19756682288,
. -17715740176,-20009401392,-17716740192, 19113857200,-17716740208,
. -8210767088,-17716740224, -8428915760,-17716740240, -8433110064,
. -17715740256, 18568907824,-17716740272, 21907455024, -9126805536,
. -9954237936, -9126805552, -3863968816, -9126805696, -3863883248,
. 0/

```

DATA IPTAB/						62
.	0,	21,	-536870928,	18565733616,	-536870976,	63
.	-3650164950,	-536870992,	-3863882800,	-536871024,	-6015524019,	64
.	-535871040,	-6221022825,	17716740128,	-2580698739,	17716740144,	65
.	-2793802547,	17716740192,	-3863968816,	17716740240,	-21093496176,	66
.	0/					67
END						68

```

$IBFTC IARITI DECK
C ARITHMETIC OPERATIONS FOR INPUT R. U. A. S. 2 1
  SUBROUTINE IARITI 2
C CALLED FROM INPUT 3
  DIMENSION IFT( 31), IPTAB( 21), ITAB( 65) 4
  DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9) 5
  , KSTACK(27), NAME(15), RVALUE(2), STACK(27) 6
C 7
  COMMON 8
  ./ICOMVI/ VALUE , ICOMP , IFNTYP , IMAGE1 , IRADIX , ISUB 9
  . , KCH , KCNVRT , KCOUNT , KDIF , KFLD1 , KFLD2 10
  . , LCOMP , LCNVRT , LEVEL , LFRT , LOOK 11
  . , MCNVRT , MDIF , MODALL , MSTORE 12
  . , NAME , NERROR , NONNEW , NOTARG 13
  . , SMCHR , TEST , ERMARK 14
  ./ICNSTI/ BLANK , BLANKS , DOLLAR , EOS 15
  . , ICJMA , IDOLAR , IFT , IPTAB , ITAB 16
  . , KAMIO , KBPC , KBPW , KCPCD , KERTYP 17
  . , KZERO , NOPRNT , TAB1 18
  ./IPARAM/ ABORT , KIUNIT , KOUNIT , LIMAF , LOCK , LOCX 19
  . , NOLIST , NSTDIR , TRACE 20
  ./ISTACKI/ STACK , ISTDIM , KSTACK , LEVLIM 21
C 22
  INTEGER BLANK , BLANKS , EOS , IDOLAR , TAB1 , TRACE 23
  DOUBLE PRECISION STACK, VALUE 24
  LOGICAL ABORT, DOLLAR, ERMARK, LIMAF, LOCK, NOLIST, NONNEW, MODALL, 25
  NSTDIR, SMCHR, TEST 26
  EQUIVALENCE (STACK, ISTACK), (VALUE, KVALUE, RVALUE), (NAME, ANAME) 27
  EQUIVALENCE (ICOMNI, ISUB), (IMAGE, IMAGE1), (IPARAM, ABORT) 28
C CALL DEBUGC(6HIARITI) 29
C CALL STACKP 30
C 31
C PRINT STACK 32
C 33
  IF (KERTYP.NE.0) GO TO 100 34
  IF ( (LEVEL.LE.0).OR.(LEVEL.GT.LEVLIM) ) GO TO 120 35
C BRANCH ON KSTACK( LEVEL) = -, 0, 1, 2, 3, 4, 5, 6, 7, 8 AND UP 36
  2 K=MAX(1, KSTACK( LEVEL)+2) 37
  IF(K.3E.8) GO TO 60 38
C - 0 1 2 3 4 5 =KSTACK(LEVEL) 39
  3 GO TO (120, 90, 120, 20, 30, 40, 50), K 40
  20 VALUE=STACK( LEVEL)*VALUE 41
  GO TO 90 42
  30 VALUE=STACK( LEVEL)/VALUE 43
  GO TO 90 44
  40 VALUE=STACK( LEVEL)+VALUE 45
  GO TO 90 46
  50 VALUE=STACK( LEVEL)-VALUE 47
  GO TO 90 48
  60 LEVEL1= LEVEL-K+9 49
  DO 61 I=LEVEL1, LEVEL

```

61	KSTACK(I)=0	50
	LEVEL = LEVEL1-1	51
C	CALL DEBUG 2(5HLEVEL,LEVEL)	52
C	CALL DEBUG 2(5HVALUE,VALUE)	53
	IF (LOCK) GO TO 62	54
	CALL IXQTI(VALUE,STACK(LEVEL))	55
62	CONTINUE	56
C	CALL DEBUG 2(5HVALUE,VALUE)	57
	IF (LEVEL .LE. NOTARG) NOTARG = ISTDIM	58
	IF (STACK(LEVEL).LT.0.00) GO TO 110	59
	GO TO 90	60
90	STACK(LEVEL)=VALUE	61
	KSTACK(LEVEL)=1	62
100	CONTINUE	63
C	CALL STACKP	64
C		65
C	CALL DEBUGR	66
	RETURN	67
110	LFRT=0	68
C		69
C	CALL DEBUG2(6HSTATMT,110)	70
	KSTACK(LEVEL)=0	71
	LEVEL=0	72
C	OTH	73
	CALL ICHAR2(\$100,29)	74
	KERTYP = 150	75
	GO TO 100	76
120	KERTYP=750	77
	GO TO 100	78
	END	79

\$IBFTC	ICAR4	DECK	
	SUBROUTINE	ICAR4(*,*,LIST1,LIST2)	1
	DIMENSION	IFT(31), IPTAB(21), ITAB(65)	2
	DIMENSION	ANAME(15) , IMAGE(80) , IMAGE1(81),IPARAM(9)	3
		,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)	4
C			5
	COMMON		6
	./ICOMVI/	VALUE ,ICOMP ,IFNTYP ,IMAGE1 ,IRADIX ,ISUB	7
		,KCH ,KCNVRT ,KCOUNT ,KDIF ,KFLD1 ,KFLD2	8
		,LCOMP ,LCNVRT ,LEVEL ,LFRT ,LOOK	9
		,MCNVRT ,MDIF ,MODALL ,MSTOR	10
		,NAME ,NERROR ,NONEW ,NOTARG	11
		,SMCHR ,TEST ,ERMARK	12
	./ICNSTI/	BLANK ,BLANKS ,DOLLAR ,EOS	13
		,ICOMMA ,IDOLAR ,IFT ,IPTAB ,ITAB	14
		,KAM10 ,KBPC ,KBPW ,KCPCD ,KERTYP	15
		,KZERO ,NOPRNT ,TAB1	16
	./IPARAM/	ABORT ,KIUNIT ,KOUNIT ,LIMLF ,LOCK ,LOCX	17
		,NOLIST ,VSTDIR ,TRACE	18
	./ISTAKI/	STACK ,ISTDIM ,KSTACK ,LEVLIM	19
C			20
	INTEGER	BLANK ,BLANKS ,EOS ,IDOLAR ,TAB1 ,TRACE	21
	.,COMPR		22
	DOUBLE PRECISION	STACK, VALUE, IDEBUA, DEBGNA(2)	23
	DATA	DEBGNA/6HICAR ,5HITCHR/	24
	LOGICAL	ABORT,DOLLAR,ERMARK,LIMLF,LOCK,NOLIST,NONEW,MODALL,	25
		.NSTDIR,SMCHR,TEST	26

	EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE),(NAME,ANAME)	27
	EQUIVALENCE (ICOMNI,ISUB),(IMAGE,IMAGE1),(IPARAM,ABORT)	28
	LOGICAL GOTCD	29
C	* TRUE IF NEW CARD WAS READ	30
	LOGICAL SMCHR	31
C	* RE-PROCESS THE SAME CHARACTER AS LAS	32
	KFLD2=LIST2	33
	IBITS=2	34
	ASSIGN 36 TO COMPR	35
	IDEBUN = 4	35
	GO TO 10	37
	ENTRY ICHAR1(*,*,LIST1)	38
	IBITS = 2	39
	IDEBUN = 3	40
	GO TO 4	41
	ENTRY ICHAR2(*,LIST1)	42
	IBITS=1	43
	IDEBUN = 2	44
4	KFLD2 = -1	45
	ASSIGN 37 TO COMPR	46
	IF(MODALL)GO TO 40	47
10	ASSIGN 20 TO IGETR	48
12	KFLD1=LIST1	49
	IDEBUA = DEBGNA(1)	50
	ASSIGN 110 TO NEXT	51
	GOTCD=.FALSE.	52
	IF(SMCHR) GO TO 35	53
C	* SKIP GETTING NEW CHARACTER	54
	GO TO 200	55
C	* GET CHARACTER RETURN TO 20	56
20	IF(KCH.EQ.BLANK)GO TO 200	57
C	* BYPASS BLANKS	58
	IF (KCH.NE.EOS) GO TO 30	59
C	* EOS= END OF STATEMENT CHARACTER	60
	ASSIGN 200 TO INEWR	61
C	* PROVIDE TO GET FIRST CHARACTER FROM NEXT C	62
	GO TO 300	63
30	IF(TEST.AND.GOTCD)GO TO 100	64
35	ITEMP = ITAB(KCH+1)	65
	ICOMP=IFLD(LIST1,IBITS,ITEMP)	66
	GO TO COMPR,(36,37)	67
36	LCOMP=IFLD(LIST2,2,ITEMP)+1	68
37	CONTINUE	69
	IF(TRACE .LT.4) GO TO 38	70
C	CALL DEBUG2(IDEBUA,IDEBUN)	71
38	SMCHR = .FALSE.	72
	IF(ICOMP-1) 381,382,383	73
381	RETURN	74
382	RETURN1	75
383	RETURN2	76
40	ASSIGN 35 TO IGETR	77
	GO TO 12	78
C	- - - - - ITCR - - - - -	79
C	SPECIAL ROUTINE TO INSERT COMMA AT END OF CARD IF	80
C	THE NEXT CARD BEGINS WITH \$ OR A LEFT MEMBER	81
C	TO THIS ROUTINE, A LEFT MEMBER BEGINS WITH A-Z	82
C	FOLLOWED BY ANY OF 0-9()A-Z FOLLOWED BY =	83
100	IDEBUA = DEBGNA(2)	84
	ITEMP = ITAB(KCH+1)	85
	ICOMP=IFLD(0,3,ITEMP)	86
	GO TO NEXT,(110,120)	87

C		* 110 FOR 1ST CHR ON NEW CARD, 120 FOR FOLL	88
110	NONEW=.TRUE.		89
	ASSIGN 120 TO NEXT		90
C	\$ A-Z 0-9() = OTH		91
	GO TO(130,200,140,140,140),ICOMP		92
C			93
C	\$ A-Z 0-9() = OTH		94
120	GO TO(140,200,200,130,140),ICOMP		95
130	KCOUNT=0		96
	KCH=ICOMMA		97
	NONEW=.FALSE.		98
	GOTCD = .FALSE.		99
	GO TO 35		100
C	* COMMA IS CHARACTER RETURNED		101
140	KCOUNT=0		102
	NONEW=.FALSE.		103
	GOTCD = .FALSE.		104
C	GO TO 200		105
C	* FIRST NON-BLANK CHARACTER ON CARD IS RETURNED		106
C	- - - - -		107
C	ROUTINE TO GET NEXT CHARACTER		108
C			109
200	IF(KCOUNT.LT.KCPCD)GO TO 210		110
	ASSIGN 210 TO INEWR		111
	GO TO 300		112
210	KCOUNT=KCPCD+1		113
C	* CARD COLUMN OF NEW CHARACTER		114
	KCH=IFLD(0,KBPC,IMAGE(KCOUNT))		115
	GO TO IGETR,(20,35)		116
C		* 35 IS USED ONLY FOR MODALL .TRU	117
C	- - - - -		118
C	ROUTINE TO PRINT OLD CARD AND READ NEXT		119
C	IF NONEW IS TRUE, STORE TAB1 IN KCH AND RETURN		120
300	IF(NONEW)GO TO 310		121
	KCH=IMAGE(KCOUNT+1)		122
C	* GET CARRIAGE CONTROL (IMAGE(81) IS A BLANK)		123
	IF((KCH.EQ.NOPRNT).OR.NOLIST) GO TO 305		124
	WRITE(KOUNIT,398)KCH,IMAGE		125
398	FORMAT(1A1,5X,80A1)		126
305	READ(KIUNIT,399)IMAGE		127
399	FORMAT(80A1)		128
	GOTCD=.TRUE.		129
	KCOUNT=0		130
	GO TO INEWR,(200,210)		131
310	KCH=TAB1		132
	GO TO 30		133
C	* TAB1 IS RETURNED, INDICATES END OF C		134
	END		135

\$IBFTC	ICNVTI	DECK	
	SUBROUTINE	ICNVTI(IFROM,ITO)	1
	DIMENSION	IFT(31), IPTAB(21), ITAB(65)	2
	DIMENSION	ANAME(15) ,IMAGE(80) ,IMAGE1(81),IPARAM(9)	3
	.	,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)	4
C			5
	COMMON		6
	./ICOMVI/	VALUE ,ICOMP ,IFNTYP ,IMAGE1 ,IRADIX ,ISUB	7
	.	,KCH ,KCNVRT ,KCOUNT ,KDIF ,KFLD1 ,KFLD2	8

•	,LCOMP	,LCNVRT	,LEVEL	,LFRT	,LOOK	9
•	,MCNVRT	,MDIF	,MODALL	,MSTJR		10
•	,NAME	,NERROR	,NONEW	,NOTARG		11
•	,SMCHR	,TEST	,ERMARK			12
•/ICNSTI/	BLANK	,BLANKS	,DOLLAR	,EOS		13
•	,ICOMMA	,IDOLAR	,IFT	,IPTAB	,ITAB	14
•	,KAM10	,KBPC	,KBPW	,KCPCD	,KERTYP	15
•	,KZERO	,NDRNT	,TAB1			16
•/IPARAM/	ABORT	,KIUNIT	,KOUNIT	,LIMALF	,LOCK	,LOCKX
•	,NOLIST	,NSTDIR	,TRACE			18
•/ISTAKI/	STACK	,ISTDIM	,KSTACK	,LEVLIM		19
C						20
	INTEGER	BLANK	,BLANKS	,EOS	,IDOLAR	,TAB1
						,TRACE
	DOUBLE	PRECISION	STACK,	VALUE		
	LOGICAL	ABORT,DOLLAR,ERMARK,LIMALF,LOCK,NOLIST,NONEW,MODALL,				
	,NSTDIR,SMCHR,TEST					
	EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)					
	EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)					
	DIMENSION	NTYPE(4)				
	DOUBLE	PRECISION	NTYPE			
	DATA	NTYPE / 6HREAL ,6HINT ,6HDP ,6HNOCONV/				
	IF((IFROM.LE.0).OR.(ITO.LE.0))	GO TO 100				
	IF (IFROM - 4)	1,99,100				
1	IF (ITO - 4)	2,99,100				
2	ITOM2 = ITO - 2					
	IF (IFROM-2)	3,4,5				
3	IF (ITOM2)	99,10,20				
4	IF (ITOM2)	30,99,40				
5	IF (ITOM2)	50,60,99				
10	KVALUE = RVALUE(1)					
	GO TO 99					
20	VALUE = RVALUE(1)					
	GO TO 99					
30	RVALUE(1) = KVALUE					
	GO TO 99					
40	VALUE = KVALUE					
	GO TO 99					
50	RVALUE(1) = VALUE					
	GO TO 99					
60	KVALUE = VALUE					
99	CONTINJE					
C	CALL	DEBUG3 (6HICNVTI,0.00,3)				
C	CALL	DEBUG3(NTYPE(IFROM),NTYPE(ITO) ,3)				
	RETURN					
100	WRITE(KOUNIT,101)	IFROM,ITO				
101	FORMAT(35H ARGUMENTS OF ICNVTI BAD. IFROM = ,I13,8H, ITO = ,I13,					
	. 41H(1 TO 4 ALLOWABLE). CHECK IPTAB IN COMNJ)					
	GO TO 99					
	END					

\$IBFTC	IERORI	DECK	
	SUBROJTINE	IERORI	1
	DIMENSION	IFT(31), IPTAB(21), ITAB(65)	2
	DIMENSION	ANAME(15) ,IMAGE(80) ,IMAGE1(81),IPARAM(9)	3
•		,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)	4
C			5

	COMMON						5		
	./ICOMNI/	VALUE	,ICOMP	,IFNTYP	,IMAGE1	,IRADIX	,ISUB	7	
	.	,KCH	,KCNVRT	,KCOUNT	,KDIF	,KFLD1	,KFLD2	8	
	.	,LCOMP	,LCNVRT	,LEVEL	,LFRT	,LOOK		9	
	.	,MCNVRT	,MDIF	,MODALL	,MSTOR			10	
	.	,NAME	,NERROR	,NONEW	,NOTARG			11	
	.	,SMCHR	,TEST	,ERMARK				12	
	./ICNSTI/	BLANK	,BLANKS	,DOLLAR	,EJS			13	
	.	,ICOMMA	,IDOLAR	,IFT	,IPTAB	,ITAB		14	
	.	,KAM10	,KBPC	,KBPW	,KCPCD	,KERTYP		15	
	.	,KZERO	,NOPRNT	,TAB1				16	
	./IPARAM/	ABORT	,KIUNIT	,KOUNTIT	,LIMLF	,LOCK	,LOCX	17	
	.	,NOLIST	,NSTDIR	,TRACE				18	
	./ISTA<I/	STACK	,ISTDIM	,KSTACK	,LEVLIM			19	
C		INTEGER	BLANK	,BLANKS	,EJS	,IDOLAR	,TAB1	,TRACE	20
		DOUBLE	PRECISION	STACK, VALUE					21
		LOGICAL	ABORT,DOLLAR,ERMARK,LIMLF,LOCK,NOLIST,NONEW,MODALL,						22
		.NSTDIR,SMCHR,TEST							23
		EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)							24
		EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)							25
		INTEGER H(4,7)							26
		EQUIVALENCE (H(1,1),H1), (H(1,2),H2), (H(1,3),H3), (H(1,4),H4)							27
		., (H(1,5),H5), (H(1,6),H6), (H(1,7),H7)							28
		DATA KA/1H*/							29
		DATA <B/1H /							30
		DATA <C/1H,/							31
		DATA KD/1H*/							32
		INTEGER MSGTYP(2,2), H1(4), H2(4), H3(4), H4(4), H5(4),H6(4),H7(4)							33
		DATA MSGTYP(1,1)/24H*DIAGNOSTIC**ERROR**							34
		DATA +1(1) /24HINAPPROPRIATE CHARACTER							35
		DATA +2(1) /24HNAME TOO LONG							36
		DATA +3(1) /24HTABLE FULL OR DESTROYED							37
		DATA +4(1) /24HSD INCOMPLETE OR MISSING							38
		DATA +5(1) /24HNAME NOT IN TABLE							39
		DATA +6(1) /24HFUNCTION OR SUB NOT ABOARD							40
		DATA +7(1) /24HFORMULA ILL-FORMED							41
C		CALL DEBUGC(6HIERORI)							42
		NONEW = .FALSE.							43
		NERROR = NERROR + 1							44
C					* COUNT DIAGONISTICS				45
		ERMARK = .TRUE.							46
C					* WARN PROGRAMER OF POSSIBLE ERRO				47
		IOP = 0							48
C					* COUNT OPERATORS +-,*=				49
		MODE = 1							50
		IF(NOTARG.LT.LEVEL) MODE = 2							51
		KC1=K3							52
		IF(KCOUNT.EQ.0) KC1=KC							53
		L = KCOUNT + 1							54
		MODALL = .FALSE.							55
		TEST = .TRUE.							56
		M = IABS(KERTYP)							57
		J = M/100							58
		K = 2							59
		IF(DOLLAR) MODE = 2							60
		IF(KERTYP.GT.0) LOCK = .TRUE.							61
		IF(KERTYP.LT.0) K=1							62
11		WRITE(KOUNT,90)(MSGTYP(I,K),I=1,2),M,(H(I,J),I=1,4),KC1,							63
		.IMAGE,(KB,I=1,L),KA,(KB,I=L,81),NERROR,LOCK							64
90		FORMAT(1H ,2A6,2H (,I3,2H) ,4A6,2X,81A1/45X,83A1,							65
		. /14X,11HERROR COUNT,I4,13H LOCK = .,L1,1H.)							66
		IF(NERROR.GT.64)STOP							67
									68

```

      IF(KERTYP.LT.O) GO TO 99
      SMCHR = .TRUE.
30    CALL ICHAR4($31,$65,7,10)
C     0-9. + $ )
      GO TO (30,65,39,80),LCOMP
C     , ' A-Z =
31    GO TO (60,65,30,60),LCOMP
39    SMCHR = .TRUE.
40    L = KCOUNT + 1
      KCL = KB
      IF(KCOUNT.EQ.O)KCL=KC
      WRITE(KOUNIT,91)KCL,IMAGE,(KB,I=1,L),KD,(KB,I=L,81)
91    FORMAT(14X,43HSKIP AHEAD AND GUESS AT NEXT GOOD STATEMENT,
        /14X,32HBEGIN PROCESSING AT SYMBOL              ,81A1/45X,83A1
        /14X48HLOOK FOR DIAGNOSTICS BUT DON'T STORE ANYTHING./)
      IFNTYP = -1
      DO 1 I=1,ISTDIM
1       KSTACK(I) = O
          LEVEL = O
          NOTARG = ISTDIM
99      CONTINUE
          KERTYP = O
C       CALL DEBUGR
          RETURN
C     ,,,,,,,
60      IF(MODE.EQ.1) GO TO 40
C         ALL OPERATORS
65      IOP = IOP + 1
          IF(IOP.LE.16)   GO TO 30
C           IF YOU FIND 16 OPERATORS BEFORE )   GIVE UP
C
C     )))))))
C     80    MODE = 1
            DOLLAR = .FALSE.
            GO TO 30
C
C     MODE             ACTION
C
C         1           SKIP TO NEXT $ OR = OR , (UNCONDITIONAL)
C         2           SKIP TO NEXT $ (UNCONDITIONAL) OR TO = OR , AFTER )
END
```

```

$IBFTC ILOOKI DECK
      SUBROUTINE ILOOKI(*,IT)
C      CALLED FROM
C      ITABLI
C      INAMEI
      DIMENSION IT(1)
      DIMENSION IFT( 31), IPTAB( 21), ITAB( 65)
      DIMENSION ANAME(15) ,IMAGE(80) ,IMAGE1(81),IPARAM(9)
      .      ,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)
C
      COMMON
      ./ICOMVI/ VALUE      ,ICOMP      ,IFNTYP      ,IMAGE1      ,IRADIX      ,ISUB
      .      ,KCH          ,KCNVRT      ,KCOUNT      ,KDIF          ,KFLD1      ,KFLD2
      .      ,LCOMP        ,LCNVRT      ,LEVEL        ,LFRT          ,LOOK
      .      ,MCNVRT        ,MDIF        ,MODALL        ,MSTOR
      .      ,NAME          ,VERROR      ,NONEW          ,NOTARG

```

.	,SMCHR	,TEST	,ERMARK						16
./ICNSTI/	BLANK	,BLANKS	,DOLLAR	,EOS					17
.	,ICOMMA	,IDOLAR	,IFT	,IPTAB	,ITAB				18
.	,KAM10	,KBPC	,KBPW	,KCPCD	,KERTYP				19
.	,KZERO	,VOPRNT	,TAB1						20
./IPARAM/	ABORT	,KIUNIT	,KOUNIT	,LIMALF	,LOCK		,LOCX		21
.	,NOLIST	,NSTDIR	,TRACE						22
./ISTA<I/	STACK	,ISTDIM	,KSTACK	,LEVLIM					23
C									24
	INTEGER	BLANK	,BLANKS	,EOS	,IDOLAR	,TAB1	,TRACE		25
	DOUBLE PRECISION	STACK, VALUE							26
	LOGICAL	ABORT,DOLLAR,ERMARK,LIMALF,LOCK,NOLIST,NONEW,MODALL,							27
	,NSTDIR,SMCHR,TEST								28
	EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)								29
	EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)								30
	LOOK=3								31
C				* FIRST USABLE POSITION					32
1	IF(IT(LOOK) .EQ.0) GO TO 7								33
C				* RETURN WITH NEXT AVAILABLE ENTR					34
	J=IFLD(3,4,IT(LOOK))								35
C				* MAX POSSIBLE J IS 15					36
	IF (J .NE.MSTOR) GO TO 8								37
C				* 8 IF NAME IS WRONG LENGTH, TRY					38
	DO 4 <=1,J								39
C				* CHECK NAME FOR MATCH					40
	LOOKK = LOOK + K								41
	IF (NAME(K).NE.IT(LOOKK)) GO TO 8								42
4	CONTINUE								43
C	CALL DEBUG2(6H*LOOKF,LOOK)								44
C				* ENTRY WAS FOUND					45
	RETURN 1								46
7	CONTINUE								47
C	CALL DEBUG2(6H*LOOKN,LOOK)								48
C				* NO ENTRY WAS FOUND					49
	RETURN								50
8	LOOK = LOOK+J+1								51
									52
									53
C				* TRY NEXT ENTRY					54
	GO TO 1								55
	END								56

\$IBFTC	INAMEI	DECK							
	SUBROUTINE	INAMEI(*,D,IT)							1
C	CALL	FROM INPUT							2
C	CALL	DEBUGC(6HINAMEI)							3
C	SUBROUTINE	TO OBTAIN LOCATION, MODE, AND CONTENTS OF A NAMED CELL							4
C									5
	DIMENSION	IFT(31), IPTAB(21), ITAB(65)							6
	DIMENSION	ANAME(15) , IMAGE(80) , IMAGE1(81), IPARAM(9)							7
.		,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)							8
C									9
	COMMON								10
./ICOMNI/	VALUE	,ICOMP	,IFNTYP	,IMAGE1	,IRADIX	,ISUB			11
.	,KCH	,KCNVRT	,KCOUNT	,KDIF	,KFLD1	,KFLD2			12
.	,LCOMP	,LCNVRT	,LEVEL	,LFRT	,LOOK				13
.	,MCNVRT	,MDIF	,MODALL	,MSTOR					14
.	,NAME	,VERROR	,NONEW	,NOTARG					15
.	,SMCHR	,TEST	,ERMARK						16

./ICNSTI/	BLANK	,BLANKS	,DOLLAR	,EDS			17
.	,ICOMMA	,IDOLAR	,IFT	,IPTAB	,ITAB		18
.	,KAM10	,KBPC	,KBPW	,KCPCD	,KERTYP		19
.	,KZERO	,NOPRNT	,TAB1				20
./IPARAM/	ABORT	,<IUNIT	,KOUNIT	,LIMALF	,LOCK	,LOCKX	21
.	,NOLIST	,NSTDIR	,TRACE				22
./ISTAKI/	STACK	,ISTDIM	,KSTACK	,LEVLIM			23
C							24
	INTEGER	BLANK	,BLANKS	,EDS	,IDOLAR	,TAB1	25
	DOUBLE PRECISION	STACK, VALUE					25
	LOGICAL	ABORT,DOLLAR,ERMARK,LIMALF,LOCK,NOLIST,NONEW,MODALL,					27
	,NSTDIR,SMCHR,TEST						28
	EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)						29
	EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)						30
	DIMENSION IT(1)						31
C						* TABLE OF NAMES PROVIDED BY USER	32
	DIMENSION D(1)						33
C						* USERS VARIABLES ARE IN D ARRAY	34
	ISUB = 1						35
C						* UNDERSTOOD SUBSCRIPT	36
	KDIF = 1						37
	CALL INAMEN						38
	CALL ILOOKI(\$1,IT)						39
	CALL ILOOKI(\$64,IFT)						40
9540	KERTYP=540						41
C						* NAME NOT IN TABLE	42
	GO TO 99						43
1	CONTINJE						44
C	CALL DEBUG2(6HSTATMT,1)						45
	ITL = IT(LOOK)						46
	KCNVRT=IFLD(0,3,ITL)						47
C	CALL DEBUG 2 (6HKCNVRT,KCNVRT)						48
C	R I D T S F						49
	GO TO(12,12,10,12,62,61),KCNVRT						50
10	KDIF = 2						51
C	(DTH						52
12	CALL ICHAR2(\$90,30)						53
C						* GO TO 90 FOR NO SUBSCRIPT	54
C	A-Z 0-9 DTH						55
	CALL ICHARI(\$82,\$9140,12)						56
	CALL INAMEN						57
	CALL ILOOKI(\$83,IT)						58
	GO TO 9540						59
82	SMCHR = .TRUE.						60
	CALL ISUBI						61
	GO TO 84						62
83	ITYPE=IFLD(0,3,IT(LOOK))						63
	LOC=IFLD(7,25,IT(LOOK))						64
	RVALUE(1) = D(LOC)						65
	RVALUE(2) = D(LOC+1)						66
	CALL ICNVTI (ITYPE,2)						67
	ISUB = KVALUE						68
C	) DTH						69
84	CALL ICHAR2(\$9140,31)						70
C						* ERROR IF NO)	71
	GO TO 91						72
90	SMCHR = .TRUE.						73
91	LOC=IFLD(7,25,ITL)						74
C	CALL DEBUG 2 (5HLOC 1,LOC)						75
	LOC = LOC + (ISUB-1)*KDIF						76
50	RVALUE(1) = D(LOC)						77
	RVALUE(2) = D(LOC+1)						78

C	CALL DEBUG 3(5HIT(L),ITL,2)	79
	CALL ICNVTI (KCNVRT,3)	80
52	LOOK = LOC	81
98	CONTINUE	82
C	CALL DEBUG 3(5HVALUE,VALUE,4)	83
C	CALL DEBUG 2 (6HLOOK ,LOOK)	84
C	CALL DEBUG 2 (5HLOC 2,LOC)	85
C	CALL DEBUG 2(6HKCNVRT,KCNVRT)	86
C	CALL DEBUGR	87
	RETURN	88
61	IFNTYP = 0	89
C	* FUNCTION	90
	GO TO 63	91
62	IFNTYP = 1	92
C	* SUBROUTINE	93
63	KVALJE=IFLD(7,25,ITL)	94
	GO TO 98	95
64	IFNTYP = 0	96
C	* LIBRARY FUNCTION OR SUBROUTINE	97
	KVALJE=IFLD(7,25,IFT(LOOK))	98
C	* PROGRAM NUMBER (USED BY IXQTI)	99
	MSTOR = 0	100
	KCNVRT=IFLD(0,3,IFT(LOOK))	101
C	* 5 FOR SUBROUTINES, 6 FOR FUNCTIONS	102
	IF(KCNVRT.EQ.5)IFNTYP=KVALUE	103
C	* PROVIDE FOR EXECUTION OF INPUT SUBRO	104
C	CALL DEBUG2(6HLIBF ,KVALUE)	105
C	MEANING OF IFNTYP	106
C	VALUE NAME IS	107
C	-1(NORMAL) AN ORDINARY VARIABLE	108
C	0 FUNCTION (USER OR FORTRAN MATH)	109
C	1 USER SUBROUTINE	110
C	2 RADIX (INPUT FUNCTION)	111
C	3 LOCX (INPUT SUBROUTINE)	112
C	\$CALL LOCX(Y,I) CAUSES I TO BE SET SO THAT Y(I)REFERS TO THE CUR	113
C	LEFT SIDE.	114
	GO TO 98	115
9140	KERTYP =140	116
99	CONTINUE	117
C	CALL DEBUG2(6HRETURN,1)	118
C	CALL DEBUGR	119
C	* REGISTER RETURN	120
	RETURN 1	121
	END	122

\$IBFTC	INAMEN DECK	
	SUBROJTINE INAMEN	1
C	CALLED FROM	2
C	NAMEI	3
C	ITABLI	4
C	INPUT	5
C	CALL DEBUGC(6HINAMEN)	6
C		7
	DIMENSION IFT(31), IPTAB(21), ITAB(65)	8
	DIMENSION ANAME(15) ,IMAGE(80) ,IMAGE1(81),IPARAM(9)	9
	• ,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)	10
C		11

	COMMON						12	
	./ICOMVI/	VALUE	,ICOMP	,IFNTYP	,IMAGE1	,IRADIX	,ISUB	13
	.	,KCH	,KCNVRT	,KCOUNT	,KDIF	,KFLD1	,KFLD2	14
	.	,LCOMP	,LCNVRT	,LEVEL	,LFRT	,LOOK		15
	.	,MCNVRT	,MDIF	,MODALL	,MSTOR			16
	.	,NAME	,NERROR	,NONEW	,NOTARG			17
	.	,SMCHR	,TEST	,ERMARK				18
	./ICNSTI/	BLANK	,BLANKS	,DOLLAR	,EOS			19
	.	,ICOMMA	,IDOLAR	,IFT	,IPTAB	,ITAB		20
	.	,KAM10	,KBPC	,KBPW	,KPCPD	,KERTYP		21
	.	,KZERO	,NDPRNT	,TAB1				22
	./IPARAM/	ABORT	,KIUNIT	,KOUNIT	,LIMLF	,LOCK	,LOCX	23
	.	,NOLIST	,NSTDIR	,TRACE				24
	./ISTACKI/	STACK	,ISTDIM	,KSTACK	,LEVLIM			25
C								26
	INTEGER	BLANK	,BLANKS	,EOS	,IDOLAR	,TAB1	,TRACE	27
	DOUBLE PRECISION	STACK, VALUE, DNAME						28
	LOGICAL	ABORT, DOLLAR, ERMARK, LIMLF, LOCK, NOLIST, NONEW, MODALL,						29
	,NSTDIR, SMCHR, TEST							30
	EQUIVALENCE (STACK, ISTACK), (VALUE, <VALUE, RVALUE), (NAME, ANAME)							31
	EQUIVALENCE (ICOMNI, ISUB), (IMAGE, IMAGE1), (IPARAM, ABORT)							32
	EQUIVALENCE (NAME, DNAME)							33
C	COLLECTS NAME (UP TO 15 WORDS) TERMINATED BY ANY SPECIAL CHAR							34
	ASSIGN 6 TO NEXT							35
	IF (MODALL) ASSIGN 2 TO NEXT							35
	MSTOR = 0							37
	J = <BPW							38
	NAME(2) = BLANKS							39
1	GO TO NEXT, (2, 6, 7)							40
C	OTH							41
2	CALL ICHAR2(\$8, 23)							42
	TEST = .TRUE.							43
	MODALL = .FALSE.							44
	CALL ICHAR2(\$99, 23)							45
C								46
	TEST = .FALSE.							47
	MODALL = .TRUE.							48
	GO TO 8							49
6	ASSIGN 7 TO NEXT							50
	GO TO 8							51
C	A-Z-0-9 OTH							52
7	CALL ICHAR2(\$99, 12)							53
8	IF (J.LT.KBPW) GO TO 9							54
	IF (MSTOR.EQ.15) GO TO 10							55
	MSTOR = MSTOR + 1							56
	NAME(MSTOR) = BLANKS							57
	J = 0							58
9	NAME(MSTOR) = IFLD4(KCH, J, KBPC, NAME(MSTOR))							59
	J = J + KBPC							60
	GO TO 1							61
10	IF (MODALL) GO TO 99							62
	KERTYP = -260							63
	CALL IERORI							64
95	CALL ICHAR2(\$99, 12)							65
C								66
	GO TO 95							67
99	CONTINUE							68
	SMCHR = .TRUE.							69
C	CALL DEBUG3(6HNAME, DNAME, 3)							70
C	CALL DEBUG2(5HMSTOR, MSTOR)							71
C	CALL DEBUGR							72
	RETURN							73
	END							74

C	CALL DEBUG2(6HSTAT ,20)	63
	ASSIGN 15 TO NEXT	64
	SWITCH = .TRUE.	65
	GO TO 1	66
37	IPF=-1	67
	GO TO 36	68
30	CONTINUE	69
C	CALL DEBUG 2(6HSTAT E,30)	70
C	+ - OTH 0-9	71
	CALL ICHAR4(\$50,\$35, 13,15)	72
C	+ -	73
	GO TO(36,37),LCOMP	74
35	SMCHR=.TRUE.	75
36	CALL ISUBI	76
C	* TEST WILL BE TRUE	77
44	ICSC = ICSC+ISIGN(ISUB,IPF)	78
C	* RESOLVE SCALE FACTORS	79
50	CONTINUE	80
C	CALL DEBUG 2(6HSTAT ,50)	81
C	CALL DEBUG3 (5HDNBR1,DNBR ,4)	82
	H = 1.00	83
	IESC=IABS(ICSC)	84
	DO 63 I=1,4	85
61	IF (IESC.LT. LD(I)) GO TO 63	86
62	IESC=IESC-LD(I)	87
	H=H*FD(I)	88
	GO TO 61	89
63	CONTINUE	90
	IF (ICSC.LT.0) GO TO 65	91
64	DNBR = DNBR*H	92
	GO TO 98	93
C	T F OTH	94
70	CALL ICHAR1(\$73,\$19,17)	95
	LVALJE=.TRUE.	96
71	IF(NOTARG.GT.LEVEL)LCNVRT=4	97
C	* LOGICAL CONSTANTS NOT CONVERTED	98
	RVALUE(2)=0.	99
C	A-Z OTH .	100
72	CALL ICHAR1(\$9130,\$99,21)	101
C	*DISCARD REST OF WORD, MUST FIND	102
	GO TO 72	103
73	LVALUE=.FALSE.	104
	GO TO 71	105
65	DNBR = DNBR/H	106
98	VALUE = DNBR	107
	SMCHR = .TRUE.	108
99	CONTINUE	109
C	CALL DEBUG3(6HDNBR 2,DNBR,4)	110
C	CALL DEBUG3(5HVALUE,VALUE,4)	111
C	CALL DEBUG2(6HICSC ,ICSC)	112
C	CALL DEBUG2(6HIPF ,IPF)	113
C	CALL DEBUG 2(6HIESC ,IESC)	114
C	CALL DEBUGR	115
	RETURN	116
9130	KERTYP = 130	117
	GO TO 99	118
	END	119

```

$IBFTC ITABLI DECK
C SUBROUTINE TO CONSTRUCT TABLE ENTRIES
C SUBROUTINE ITABLI(IT)
C CALLED FROM INPUT
C CALL DEBUGC(6HITABLI)
C
C DIMENSION IFT( 31), IPTAB( 21), ITAB( 65)
C DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)
C , KSTACK(27), NAME(15), RVALUE(2), STACK(27)
C
C COMMON
C ./ICOMNI/ VALUE , ICOMP , IFNTYP , IMAGE1 , IRADIX , ISUB
C , KCH , KCNVRT , KCOUNT , KDIF , KFLD1 , KFLD2
C , LCOMP , LCNVRT , LEVEL , LFRT , LOOK
C , MCNVRT , MDIF , MODALL , MSTOR
C , NAME , NERROR , NONEW , NOTARG
C , SMCHR , TEST , ERMARK
C ./ICNSTI/ BLANK , BLANKS , DOLLAR , EOS
C , ICOMMA , IDOLAR , IFT , IPTAB , ITAB
C , KAM10 , KBPC , KBPW , KCPCD , KERTYP
C , KZERO , NOPRNT , TAB1
C ./IPARAM/ ABORT , KIUNIT , KOUNIT , LIMALF , LOCK , LOCX
C , NOLIST , NSTDIR , TRACE
C ./ISTACKI/ STACK , ISTDIM , KSTACK , LEVLIM
C
C INTEGER BLANK , BLANKS , EOS , IDOLAR , TAB1 , TRACE
C DOUBLE PRECISION STACK, VALUE
C LOGICAL ABORT, DOLLAR, ERMARK, LIMALF, LOCK, NOLIST, NONEW, MODALL,
C NSTDIR, SMCHR, TEST
C EQUIVALENCE (STACK, ISTACK), (VALUE, KVALUE, RVALUE), (NAME, ANAME)
C EQUIVALENCE (ICOMNI, ISUB), (IMAGE, IMAGE1), (IPARAM, ABORT)
C DIMENSION IT (1)
C
C KDIF = 1
C ISUBX = 1
C ITYPE = 1
C
C , ALWAYS INITIALIZE TO .REAL.
C ( A-Z OTH
1 CALL ICHAR1 ($1, $9120, 26)
C , SKIP 'ABLE' IN TABLE
3 CONTINUE
C CALL DEBUG2(6HSTATMT, 3)
C CALL ICHAR4($4, $9120, 7, 9)
C , 0-9+ OTH
C GO TO (30, 98, 10, 9120), LCOMP
C , A-Z OTH OTH
4 GO TO (3, 20, 9120, 9120), LCOMP
9120 KERTYP = 120
C , ILLEGAL CHARACTER
C GO TO 98
10 CONTINUE
C CALL DEBUG2(6HST 0-9, 10)
C SMCHR = .TRUE.
C CALL ISUB1
C ISUBX = ISUB
C CALL ICHAR2($9120, 25)
C , ERROR IF NO =
C GO TO 3
30 CONTINUE
C CALL DEBUG 2(6HSTATMT, 30)
31 KDIF = 1

```

C	SPLIT TYPES (INT,REAL,DP,NO CONVERSION,FUNCTION ,SUBROUTINE)	61
	CALL ICHAR4(\$32,\$33,3,5)	62
	IF(LCOMP.GT.1) GO TO 9120	63
	KDIF = 2	64
	GO TO 33	65
32	IF (LCOMP.EQ.3) GO TO 9120	66
	ITYPE = LCOMP	67
	GO TO 34	68
33	ITYPE = LCOMP+2	69
C	A-Z OTH .	70
34	CALL ICHARI(\$9120,\$3,21)	71
	GO TO 34	72
20	CONTINUE	73
C	CALL DEBUG2(6HSTATMT,20)	74
	CALL INAMEN	75
50	CONTINJE	75
C	CALL DEBUG2(6HSTATMT,50)	77
	ITBUFF=IFLD4(ITYPE,0,3,ITBUFF)	78
	ITBUFF=IFLD4(MSTOR,3,4,ITBUFF)	79
	ITBUFF=IFLD4(ISUBX,7,25,ITBUFF)	80
C	CALL DEBUG2 (5HITYPE,ITYPE)	81
C	CALL DEBUG2 (5HMSTOR,MSTOR)	82
C	CALL DEBUG2 (5HISUBX,ISUBX)	83
C	CALL DEBUG4 (6HITBUFF,ITBUFF,2)	84
	ISUBX = ISUBX + KDIF	85
	CALL ILOOKI(\$56,IT)	86
	IF((IT(2).NE.0).AND.((LOOK+MSTOR +2).GT.IT(2))) GO TO 9320	87
	IT(LOOK)=ITBUFF	88
	DO 55 K=1,MSTOR	89
	LOOK=LOOK+1	90
55	IT(LOOK)=NAME(K)	91
C	CALL DEBUG2(6HSTATMT,55)	92
	LOOK=LOOK+1	93
	IT(LOOK)=0	94
	GO TO 3	95
56	IT(LOOK)=ITBUFF	96
C	CALL DEBUG2(6HSTATMT,56)	97
	GO TO 3	98
98	CONTINJE	99
C	CALL DEBUGR	100
	RETURN	101
9320	KERTYP =-320	102
	CALL IERORI	103
	GO TO 3	104
	END	105

\$IBFTC	ISUBI DECK	
	SUBROJTINE ISUBI	1
C	ISUBI FINDS SUBSCRIPTS AND INTEGER CONSTANTS	2
C	CALLED FROM	3
C	INAMEI	4
C	INMBRI	5
C	INPUT	6
C	ITABLI	7
C	ISJB BEGINS PROCESSING WITH THE NEXT CHARACTER.	8
C	CALL DEBUGC(5HISUBI)	9
C	COLLECTS INTEGER OF BASE IRADIX TERMINATED BY A SPECIAL CHARACTER	10
	DIMENSION IFT(31), IPTAB(21), ITAB(65)	11

```

DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)
.      , KSTACK(27), NAME(15), RVALUE(2), STACK(27)
C
COMMON
./ICOMNI/ VALUE      , ICOMP      , IFNTYP      , IMAGE1      , IRADIX      , ISUB
.      , KCH          , KCNVRT      , KCOUNT      , KDIF          , KFLD1        , KFLD2
.      , LCOMP        , LCNVRT      , LEVEL        , LFRT          , LOOK
.      , MCNVRT        , MDIF          , MODALL        , MSTOR
.      , NAME          , NERROR        , NNEW          , NOTARG
.      , SMCHR         , TEST          , ERMARK
./ICNSTI/ BLANK      , BLANKS      , EOS
.      , ICOMMA        , IDOLAR        , IFT           , IPTAB         , ITAB
.      , KAM10         , KBPC          , KBPW          , KCPCD         , KERTYP
.      , KZERO         , NOPRNT        , TAB1
./IPARAM/ ABORT      , KIUNIT        , KOUNIT        , LIMAF         , LOCK         , LOCKX
.      , NOLIST        , NSTDIR        , TRACE
./ISTAKI/ STACK      , ISTDIM        , KSTACK        , LEVLIM
C
INTEGER BLANK      , BLANKS      , EOS           , IDOLAR        , TAB1         , TRACE
DOUBLE PRECISION STACK, VALUE
LOGICAL ABORT, DOLLAR, ERMARK, LIMAF, LOCK, NOLIST, NNEW, MODALL,
. NSTDIR, SMCHR, TEST
EQUIVALENCE (STACK, ISTACK), (VALUE, KVALUE, RVALUE), (NAME, ANAME)
EQUIVALENCE (ICOMNI, ISUB), (IMAGE, IMAGE1), (IPARAM, ABORT)
ISUB = 0
C      A-Z 0-9 OTH
80    CALL ICHARI($10,$99,12)
      IDIGIT=KCH-KAM10
C      * VALUE OF LETTER USED AS DIGIT
30    IF(IDIGIT .GE. IRADIX) GO TO 99
35    ISUB = ISUB * IRADIX + IDIGIT
C      * ACCUM TOTAL. IRADIX
GO TO 80
99    SMCHR = .TRUE.
C      * ALLOW SAME CHARACTER TO BE READ
CALL DEBUG2(6H*ISUBI, ISUB)
CALL DEBUGR
RETURN
10    IDIGIT=KCH-KZERO
C      * VALUE OF DIGIT
GO TO 30
END

```

```

$IBFTC IXQTI DECK
SUBROUTINE IXQTI (ARGL, ARGS)
C      USER MAY PUT HIS OWN COMMON STATEMENTS IN THIS ROUTINE AND
C      USE THEM TO SUPPLY ARGUMENTS TO HIS CALLS IF HE DESIRES
COMMON
./IPARAM/ ABORT , KIUNIT , KOUNIT, LIMAF , LOCK, LOCKX, NOLIST, NSTDIR
.      , TRACE
DOUBLE PRECISION ARG2(27), ARGL, ARG22
DIMENSION ARG2(2)
EQUIVALENCE (ARG2, ARG22)
C
M = DABS(ARGS(1))
IF (M.LT.1 .OR. M.GT.16) GO TO 99
GO TO (1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16), M

```

1	ARGL = COS(ARGL)	14
	GO TO 100	15
2	ARGL = EXP(ARGL)	15
	GO TO 100	17
3	ARGL = ALOG(ARGL)	18
	GO TO 100	19
4	ARGL = SIN(ARGL)	20
	GO TO 100	21
5	ARGL = SQRT(ARGL)	22
	GO TO 100	23
6	ARGL = ATAN2(ARGS(2),ARGL)	24
	GO TO 100	25
C	PRINT FUNCTION	26
7	ARG22 = ARGS(2)	27
	WRITE (KOUNIT,101) ARG2,ARGL	28
101	FORMAT(1H 2A6,3H = ,D26.17)	29
	GO TO 100	30
8	ARGL = SNGL(ARGS(2))*IFIX(SNGL(ARGL+DSIGN(.5D0,ARGL)))	31
	GO TO 100	32
9	ARGL = ABS(SNGL(ARGS(2)))*SNGL(ARGL)	33
	GO TO 100	34
10	ARGL = ABS(ARGL)	35
	GO TO 100	36
C	DISC(RIMINENT) FUNCTION	37
11	IF (ARGL) 102,103,104	38
102	ARGL = ARGS(2)	39
	GO TO 100	40
103	ARGL = ARGS(3)	41
	GO TO 100	42
104	ARGL = ARGS(4)	43
	GO TO 100	44
12	CALL LOCKX(ARGL)	45
	GO TO 100	46
13	CONTINUE	47
	GO TO 100	48
14	CONTINUE	49
	GO TO 100	50
15	CONTINUE	51
16	CONTINUE	52
100	CONTINUE	53
C	CALL DEBUGR	54
	RETURN	55
99	KERTYP =-610	56
	CALL IERORI	57
	GO TO 100	58
	END	59

\$IBFTC	LOCKX	DECK						1
	SUBROJTINE	LOCKX(J)						2
	COMMON							3
	./IPARAM/	ABORT	,KIUNIT	,KOUNIT	,LIMAF	,LOCK	,LOCKX	4
	.	,NOLIST	,NSTDIR	,TRACE				5
	LOGICAL	J,LOCK						6
	LOCK = J							7
	RETURN							8
	END							

```

$IBFTC DEBUGX DECK
SUBROUTINE DEBUGX
C
C          *INITIALIZATION
C          DIMENSION ISUBN(20)
C          TRACE = 0 NO PRINTING
C          TRACE = 1 PRINT DEBUG 2+3 CALLS ONLY
C          TRACE = 2 PRINT DEBUG 2+3 CALLS ONLY
C          TRACE = 3 PRINT DEBUG 2+3 AND STACK PRINT
C          TRACE = 4 PRINT DEBUG 2+3 AND STACK PRINT AND CALLS FROM C-AR
C          DIMENSION IFT(27), IPTAB(21), ITAB(65)
C          DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)
C          * ,KSTACK(27), NAME(15), RVALUE(2), STACK(27)
C
C          COMMON
C          ./ICOMNI/ VALUE ,ICOMP ,IFNTYP ,IMAGE1 ,IRADIX ,ISUB
C          * ,KCH ,KCNVRT ,KCOUNT ,KDIF ,KFLD1 ,KFLD2
C          * ,LCOMP ,LCNVRT ,LEVEL ,LFRT ,LOOK
C          * ,MCNVRT ,MODIF ,MODALL ,MSTOR
C          * ,NAME ,NERROR ,NONEW ,NOTARG
C          * ,SMCHR ,TEST ,ERMARK
C          ./ICNSTI/ BLANK ,BLANKS ,DOLLAR ,EOS
C          * ,ICOMMA ,IDOLAR ,IFT ,IPTAB ,ITAB
C          * ,KAMIO ,KBPC ,KBPW ,KCPCD ,KERTYP
C          * ,KZERO ,NOPRNT ,TAB1
C          ./IPARAM/ ABORT ,KIUNIT ,KOUNIT ,LIMLF ,LOCK ,LOCX
C          * ,NOLIST ,NSTDIR ,TRACE
C          ./ISTAKI/ STACK ,ISTDIM ,KSTACK ,LEVLIM
C
C          INTEGER BLANK ,BLANKS ,EOS ,IDOLAR ,TAB1 ,TRACE
C          DATA BADCAL/6HBADCAL/
C          DOUBLE PRECISION ALFARG, ISUBN, STACK, VALUE, DBLANK, DARG
C          LOGICAL ABORT, DOLLAR, ERMARK, LIMLF, LOCK, NOLIST, NONEW, MODALL,
C          * NSTDIR, SMCHR, TEST
C          EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)
C          EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)
C          * , (DBLANK,BLANK)
C          ISUBC = 0
C          DO 10 I = 1,10
C          ISUBN(I) = DBLANK
10 IF(TRACE.EQ.0) GO TO 99
WRITE (KOUNIT,410)
GO TO 99
ENTRY DEBUGC(ISUBNA)
DOUBLE PRECISION ISUBNA
C          * NEW SUBROUTINE CALLED
C          IF (ISUBC.GT.10) GO TO 98
C          ALFARG = DBLANK
C          NUMARG = -1
C          ISUBC = ISUBC+1
C          ISUBN(ISUBC) = ISUBNA
C          IF(TRACE .GE.4) GO TO 400
C          GO TO 99
C
C          ALFARG = BADCAL
C          NUMARG = ISUBC
C          GO TO 400
C
C          ENTRY DEBUGR

```

C	IF (ISUBC.LT.1) GO TO 98	59
	ALFARG = DBLANK	60
	NUMARG = -1	61
	ISUBN(ISUBC) = DBLANK	62
	ISUBC = ISUBC - 1	63
	IF(TRACE .GE.4) GO TO 400	64
	GO TO 99	65
C		66
	ENTRY DEBUG2(ALFAR,NUMAR)	67
	DOUBLE PRECISION ALFAR	68
	NUMARG = NUMAR	69
	ALFARG = ALFAR	70
20	IF (TRACE .LT.1) GO TO 99	71
400	WRITE (KUNIT,405) (ISUBN(I),I=1,4),ALFARG,NUMARG,	72
	.KCOUNT,SMCHR,KCH,KFLD1,ICOMP,KFLD2,LCOMP	73
99	RETURN	74
C		75
	ENTRY DEBUG3(ALFAR,DNUMAR,ITYPE)	76
	DOUBLE PRECISION DNUMAR	77
	ENTRY DEBUG4(ALFAR,DNUMAR,ITYPE)	78
	DARG = DNUMAR	79
	NUMARG=NUMAR	80
	ALFARG = ALFAR	81
	IF (TRACE .LT.1) GO TO 99	82
	GO TO (30,40,50,60), ITYPE	83
30	WRITE (KUNIT,406) (ISUBN(I),I=1,4),ALFARG,NUMARG,	84
	.KCOUNT,SMCHR,KCH,KFLD1,ICOMP,KFLD2,LCOMP	85
	GO TO 99	86
40	WRITE (KUNIT,407) (ISUBN(I),I=1,4),ALFARG,NUMARG,	87
	.KCOUNT,SMCHR,KCH,KFLD1,ICOMP,KFLD2,LCOMP	88
	GO TO 99	89
50	WRITE (KUNIT,408) (ISUBN(I),I=1,4),ALFARG,NUMARG,	90
	.KCOUNT,SMCHR,KCH,KFLD1,ICOMP,KFLD2,LCOMP	91
	GO TO 99	92
60	WRITE (KUNIT,409) (ISUBN(I),I=1,4),ALFARG,NUMARG,	93
	.KCOUNT,SMCHR,KCH,KFLD1,ICOMP,KFLD2,LCOMP	94
	GO TO 99	95
405	FORMAT(1H ,29X,5(A6,1X),I24,	96
	.1H(,I3,L2,1H),A6,2H (,I3,I2,1H),2H (,I3,I2,1H))	97
406	FORMAT(1H ,29X,5(A6,1X),E24.8,	98
	.1H(,I3,L2,1H),A6,2H (,I3,I2,1H),2H (,I3,I2,1H))	99
407	FORMAT(1H ,29X,5(A6,1X),11X,012,1X,	100
	.1H(,I3,L2,1H),A6,2H (,I3,I2,1H),2H (,I3,I2,1H))	101
408	FORMAT(1H ,29X,5(A6,1X),16X,A8,	102
	.1H(,I3,L2,1H),A6,2H (,I3,I2,1H),2H (,I3,I2,1H))	103
409	FORMAT(1H ,29X,5(A6,1X),D24.18,	104
	.1H(,I3,L2,1H),A6,2H (,I3,I2,1H),2H (,I3,I2,1H))	105
410	FORMAT(29X,11HDEBUG TRACE,52X,9HI--KCOUNT,5X,8HI--LIST1/	106
	.68X,94D ARG OR,15X,9HI I-SMCHR,5X,10HI I--ICOMP /	107
	.29X,19HSUBROUTINES CALLED,11X,7HALF ARG,7X,11HNUMERIC ARG,9X,	108
	.3HI I,1X,5HKCH-I,5X,16HI I I--LIST2 /	109
	.92X,34I I,5X,1HI,5X,18HI I I I--LCOMP)	110
	END	111
		112

```

$IBFTC STAC<P DECK
SUBROJTIME STACKP
DIMENSION IFT(27), IPTAB(21), ITAB(65)
DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)
.      ,KSTACK(27),NAME(15) ,RVALUE(2) ,STACK(27)
C
COMMON
./ICOMNI/ VALUE ,ICOMP ,IFNTYP ,IMAGE1 ,IRADIX ,ISUB
.      ,KCH ,KCNVRT ,KCOUNT ,KDIF ,KFLD1 ,KFLD2
.      ,LCOMP ,LCNVRT ,LEVEL ,LFRT ,LOOK
.      ,MCNVRT ,MDIF ,MODALL ,MSTOR
.      ,NAME ,NERROR ,NONEW ,NOTARG
.      ,SMCHR ,TEST ,ERMARK
./ICNSTI/ BLANK ,BLANKS ,DOLLAR ,EOS
.      ,ICOMMA ,IDOLAR ,IFT ,IPTAB ,ITAB
.      ,KAM10 ,KBPC ,KBPW ,KCPCD ,KERTYP
.      ,KZERO ,NOPRNT ,TAB1
./IPARAM/ ABORT ,KIUNIT ,KOUNIT ,LIMAF ,LOCK ,LOCX
.      ,NOLIST ,NSTDIR ,TRACE
./ISTACI/ STACK ,ISTDIM ,KSTACK ,LEVLIM
C
INTEGER BLANK ,BLANKS ,EOS ,IDOLAR ,TAB1 ,TRACE
DOUBLE PRECISION STACK, VALUE
LOGICAL ABORT,DOLLAR,ERMARK,LIMAF,LOCK,NOLIST,NONEW,MODALL,
.NSTDIR,SMCHR,TEST
EQUIVALENCE (STACK,ISTACK), (VALUE,KVALUE,RVALUE), (NAME,ANAME)
EQUIVALENCE (ICOMNI,ISUB), (IMAGE,IMAGE1), (IPARAM,ABORT)
IF(TRACE .LT. 3) GO TO 99
LEVELX = LEVEL + 3
WRITE(KOUNIT,85)LEVEL,VALUE,KVALUE,RVALUE,LOCX
WRITE(KOUNIT,89)(STACK(I),I=1,LEVELX)
WRITE(KOUNIT,86)(KSTACK(I),I=1,LEVELX)
89 FORMAT(11D12.4)
86 FORMAT(11I12)
85 FORMAT(7H LEVEL=I3,7H VALUE=D25.17,8H KVALUE=I3,3H ,0,2(1X,012),
.6H LOCX=I6)
99 RETURN
END

```

```

$IBMAP IFLD DECK
TTL IFLD AND IFLD4 PROGRAM
ENTRY IFLD
ENTRY IFLD4
* FUNCTION (I,N,DATA)
* START EXTRACTION WITH BIT I+1
* EXTRACT N BITS FROM DATA AND RETURN IN REGESTER 1
IFLD SAVE
CAL* 3,4 I
STA LGL1
CAL* 4,4 N
STA LGL2
LDQ* 5,4 DATA
LGL1 LGL ** **=I
ZAC

```

	SUBROUTINE STACKP	1
	DIMENSION IFT(27), IPTAB(21), ITAB(65)	2
	DIMENSION ANAME(15), IMAGE(80), IMAGE1(81), IPARAM(9)	3
	*, KSTACK(27), NAME(15), RVALUE(2), STACK(27)	4
C	COMMON	5
	./ICOMNI/ VALUE, ICOMP, IFNTYP, IMAGE1, IRADIX, ISUB	6
	*, KCH, KCNVRT, KCOUNT, KDIF, KFLD1, KFLD2	7
	*, LCOMP, LCNVRT, LEVEL, LFRT, LOOK	8
	*, MCNVRT, MDIF, MODALL, MSTOR	9
	*, NAME, NERROR, NONEW, NOTARG	10
	*, SMCHR, TEST, ERMARK	11
	./ICNSTI/ BLANK, BLANKS, DOLLAR, EOS	12
	*, ICOMMA, IDOLAR, IFT, IPTAB, ITAB	13
	*, KAM10, KBPC, KBPW, KCPCD, KERTYP	14
	*, KZERO, VOPRNT, TAB1	15
	./IPARAM/ ABORT, KIUNIT, KOUNIT, LIMALF, LOCK, LOCX	16
	*, NOLIST, NSTDIR, TRACE	17
	./ISTACKI/ STACK, ISTDIM, KSTACK, LEVLIM	18
C	INTEGER BLANK, BLANKS, EOS, IDOLAR, TAB1, TRACE	19
	DOUBLE PRECISION STACK, VALUE	20
	LOGICAL ABORT, DOLLAR, ERMARK, LIMALF, LOCK, NOLIST, NONEW, MODALL,	21
	NSTDIR, SMCHR, TEST	22
	EQUIVALENCE (STACK, ISTACK), (VALUE, KVALUE, RVALUE), (NAME, ANAME)	23
	EQUIVALENCE (ICOMNI, ISUB), (IMAGE, IMAGE1), (IPARAM, ABORT)	24
	IF(TRACE .LT. 3) GO TO 99	25
	LEVELX = LEVEL + 3	26
	WRITE(KOUNIT,85)LEVEL,VALUE,KVALUE,RVALUE,LOCX	27
	WRITE(KOUNIT,89)(STACK(I),I=1,LEVELX)	28
	WRITE(KOUNIT,86)(KSTACK(I),I=1,LEVELX)	29
89	FORMAT(11D12.4)	30
86	FORMAT(11I12)	31
85	FORMAT(7H LEVEL=I3,7H VALUE=D25.17,8H KVALUE=I3,3H ,0,2(1X,D12),	32
	6H LOCK=I6)	33
99	RETURN	34
	END	35

\$IBMAP	IFLD	DECK	1
	TTL	IFLD AND IFLD4 PROGRAM	2
	ENTRY	IFLD	3
	ENTRY	IFLD4	4
*	FUNCTION	(I,N,DATA)	5
*		START EXTRACTION WITH BIT I+1	6
*		EXTRACT N BITS FROM DATA AND RETURN IN REGISTER 1	7
IFLD	SAVE		8
	CAL*	3,4 I	9
	STA	LGL1	10
	CAL*	4,4 N	11
	STA	LGL2	12
	LDI*	5,4 DATA	13
LGL1	LGL	** **=I	14
	ZAC		

LGL2	LGL	**	**=N	15
	XCL			16
	XCA			17
	RETURN	IFLD		18
	SPACE	2		19
*	FUNCTION OR SUBROUTINE IFLD4(NUDATA,I,N,DATA)			20
*	PUT LOW ORDER N BITS OF NUDATA IN DATA			21
*	STARTING WITH THE ITH+1 BIT.			22
*	THE MODIFIED DATA IS LEFT IN REG 0 FOR FUNCTION TYPE USAGE			23
*	I.E. IX=IFLD4(NUDATA),I,N,DATA)			24
*	DATA IS UNCHANGED IX CONTAINS THE CHANGED DATA			25
IFLD4	SAVE			26
	CAL*	4,4	I	27
	STA	I1		28
	STA	I2		29
	CAL*	5,4	N	30
	STA	N1		31
	STA	N2		32
	ZAC			33
	LDQ*	6,4	DATA	34
I1	LGL	**	**=I	35
	SLW	TEMP		36
N1	LGL	**	**=N	37
	CAL*	3,4	Z	38
N2	LGR	**	**=N	39
	CAL	TEMP		40
I2	LGR	**	**=I	41
	XCA			42
	RETURN	IFLD4		43
TEMP	PZE			44
	END			45

HUFF INPUT ROUTINE

The first version of the Huff Input Routine was reported in reference 6. The Huff Input Routine provides more versatility in reading input data into the computer than the NAMELIST feature in FORTRAN. The Huff Input Routine has the ability to make simple arithmetic manipulations (such as conversion of units) during loading and to load alphanumeric data. While not an indispensable feature, it has been found to be quite convenient. The Huff Input Routine also allows for the automatic printout of data cards at execution time.

The following sections contain a general description of the Huff Input Routine and its usage.

Usage

The programmer transfers control to the INPUT routine with a standard FORTRAN IV call

CALL INPUT (5, 6, 1, X, ITABLE)

Argument 1 is the system input tape number (5 on the Lewis system). Argument 2 is the system output tape number (6 on the Lewis system). Argument 3 is the identifying number of a data group. This value is compared with an identification number occurring on the input card (\$DATA card). If the values agree, the data are processed until another \$DATA or end-of-data (\$END) card is encountered. If the values do not agree, no data are processed and control is returned to the calling program. Argument 4 is the Array X, which serves as a reference point for the storing of input data. Since all data are stored relative to X, the programmer must provide fixed relations between the location of X and other locations to be loaded (e. g., through the use of common blocks and/or equivalence statements). In this case, X is "WORD," the first name in the labeled common block "ALL." Common blocks ALL, DESIGN, FRONT, SIDE, BACK, and DUMMY are in all routines and hence are loaded sequentially so that the location of all variables is known. Argument 5 is the array ITABLE, which contains the names of the variables used on the cards and their subscript location relative to X. Sufficient space must be provided in the calling program for storing the table of names. This is done by a DIMENSION statement. The dimension of ITABLE(3) must initially be zero.

Types of Input Statements

\$DATA statement. - The \$DATA statement identifies a group of data with an identification number. It must be the first statement on a card. For example, \$DATA(1) or \$D(1) on the first card of a data group causes the value 1 to be compared with argument 3 in the calling sequence. If unequal, control is returned to the calling program. If equal, data are loaded until the next \$DATA or \$END statement is reached.

\$TABLE statement. - The \$TABLE statement makes a list of names needed for loading data. Consider for example that the real variable names VELOCITY, MASS, and RADIUS are to be assigned to memory locations X(1), X(2), and X(3), respectively. The card would be punched \$TABLE (.REAL., 1 = VELOCITY, 2 = MASS, 3 = RADIUS). These variables will be treated as real in any subsequent loading of data. A limit of 15 computer words is placed on the length of a name. Since .REAL. is what designates the mode of a name, a name may begin with any alphabetic letter. For example, the statement \$TABLE (.INTEGER., 20 = INDEX, SUBSCRIPT, I) will place these names in the table and any values subsequently loaded will be stored in X(20), X(21), and X(22), respectively, as integers. In a similar manner \$TABLE (.DOUBLE PRECISION., 10 = RADIUS DOUBLE, .LOGICAL., 12 = SWITCH1) causes the name RADIUS DOUBLE to be stored in the table as a double-precision variable equivalent to X(10) and X(11), and the logical variables SWITCH 1 and SWITCH 2 will be equivalent to X(12) and X(13).

Note that \$TABLE statements are loaded as Number = Name to avoid confusion with loading statements.

Loading Statement

The loading statement loads data by taking the name of a variable previously appearing in a \$TABLE statement and setting it equal to a value which may be of several forms.

Numeric values. - Standard FORTRAN language is used; for example, VELOCITY = 3.4, MASS = 32 (no decimal point is needed and MASS will have the REAL value 32), RADIUS = 4E21, and INDEX = 3. Data can be continued from one card to another; for example, SUBSCRIPT may appear at the end of one card and = 47 on the next card.

Subscripts may be used. Since 3 = RADIUS, RADIUS(2) = 6, 10, 12, , 14 will put real numbers in X(4), X(5), X(6), and X(8) and leave X(7) unchanged because of the double comma. If new values are assigned to a variable before the next \$D(1) card, the new value will override the previous one. For example, RADIUS(2) = 8 will override the RADIUS(2) = 6 card.

Internally addressed values. - An internally addressed value is one that refers to the contents of memory by name. RADIUS(7) = RADIUS(3); RADIUS(INDEX) causes RADIUS(7) to be replaced by the value of RADIUS(3) and RADIUS(8) to also be replaced

The statement $\text{RADIUS}(7) = \text{RADIUS}(\text{INDEX} + 1)$, however, is ILLEGAL.

Arithmetic expressions. - Provisions have been made to allow arithmetic operations to be performed on data at execution time. The operations + (addition), - (subtraction), *(multiplication), and /(division) and the functions, included among which are SQRT, EXP, SIN, COS, and $\text{PWR}(x, y) (= x^y)$, may be used with name or numbers (or any expression that has a value) to compute the value of an arithmetic expression. Parentheses may be used to indicate the order of performing the operations. The computations are analyzed from left to right and any intermediate results are stored in up to 24 locations in the core (the stack) which is sufficient for fairly complex expressions. All numeric operations are carried out in double-precision floating-point FORTRAN arithmetic. As an example, $\text{RADIUS}(2) = \text{RADIUS}(2) * \text{SQRT}(\text{RADIUS}(2))$ or $\text{RADIUS}(2) = \text{PWR}(\text{RADIUS}(2), 1.5)$ will set $\text{RADIUS}(2) = 8^{3/2}$.

Alphanumeric expressions. - Alphanumeric data may be entered by placing the variable name in the 'REAL' list and then setting the variable equal to the data by first enclosing in parentheses the length of the word to be read in. As an example,

$Q = (A39)\text{THIS IS AN EXAMPLE OF ALPHANUMERIC DATA}$

The (A39) gives the length of the data including imbedded blanks. Of course, since on the IBM 7094 there are six characters per word, Q must internally be dimensioned to at least 7.

Printing Input Cards

Each input card processed will normally be written on the tape specified by the second argument of the calling sequence. An end-of-statement symbol read on the card will cause interpretation of the card to stop at that point and permit comments to be placed on the remainder of the card to be printed with the output. In order to avoid printing the card at all, the nonprint character is placed in the next column following the end-of-statement character. The developers of the routine selected the sign $\ddagger$ for both characters. This is punched as a colon on an IBM Model 29 Key punch and corresponds to a 2-8 punch.

If the character following the end-of-statement symbol is other than a nonprint character, it is inserted as the printer control character in the first position of the output format before the card is written on the output tape. If no end-of-statement character occurs on the card, a blank printer control character is used. Comment cards having the end-of-statement character as the first nonblank character will be printed and may be placed anywhere except in a continued alphabetic field.

In summary, the end-of-statement character has the effect of moving the end of the card forward to the column ahead of the end-of-statement character. The column following it is printer control.

If the control parameter NOLIST is true, printing is suppressed for all cards.

SYMBOLS

General Symbols Internal to Program

Variables in program are formed by combining these symbols.

Station Numbers

See figures 1 to 4 for each type of engine.

Thermodynamic Properties

AM	Mach number
FAR	fuel-air ratio, f/a
H	enthalpy, Btu/lbm
P	total pressure, atm
PS	static pressure, atm
S	entropy, Btu/ $^{\circ}$ R/lbm
T	total temperature, $^{\circ}$ R
TS	static temperature, $^{\circ}$ R
V	velocity, ft/sec

Component Symbols

A,AFT	afterburner
B	burner
C	inner compressor
COM	combustor
D	fan duct
F	first or fan compressor

M	core nozzle
NOZ	nozzle
OB	overboard
T	total
THP	inner (high pressure) turbine
TLP	outer (low pressure) turbine

Engine Symbols

BL	bleed, lbm/sec
CN	ratio of corrected speed to design corrected speed
DHT	turbine delta enthalpy, Btu/lbm
DHTC	turbine delta enthalpy (temperature corrected), $(H_{in} - H_{out})/T_{in}$, Btu/ $^{\circ}$ R/lbm
DP	pressure drop, $\Delta P/P$
DT	temperature change, $^{\circ}$ R
ETA	efficiency
ETAR	ram recovery, P_2/P_1
HPEXT	horsepower extracted
PCBL	fractional bleed
PCN	percent of design shaft speed
PR	pressure ratio
TFF	turbine flow function, $\text{lbm} \sqrt{^{\circ}\text{R}} / (\text{psia})(\text{sec})$
WA	airflow, lbm/sec
WF	fuel flow, lbm/sec
WG	gas flow, lbm/sec
Z	ratio of pressure ratios

Miscellaneous Symbols

A	area, ft
ALTP	altitude, ft

BYPASS	bypass ratio (fan duct air/air entering intermediate compressor)
C	when following component symbol, signifies ''corrected''
CF	correction factor, when used following component symbol
CS	ambient speed of sound, ft/sec
CV	nozzle velocity coefficient
DEL	delta degradation coefficient
DS	design value
DUM	dummy value
FG	gross thrust, lbf
FGM	momentum thrust, lbf
FGP	pressure thrust, lbf
FN	net thrust, lbf
FRD	ram drag, lbf
GU	initial or guessed values
ITRYS	number of loops through engine before quitting
LOOP	variable counter
LOOPER	number of loops through engine counter
SFC	specific fuel consumption, lbm/lbf/hr
TOLALL	tolerance on convergence
VA	velocity of aircraft, ft/sec
VJ	jet velocity, ft/sec

Input Symbols

ALTP	altitude, ft
AM	Mach number of aircraft
AM6	design afterburner entrance Mach number
AM23	design duct-burner entrance Mach number

A6	area at afterburner entrance (calculated from AM6), ft ²
A8	main nozzle throat area (can be changed at off-design), ft ²
A28	fan duct nozzle throat area (see A8), ft ²
CNHPDS	design corrected speed - inner turbine
CNLPDS	design corrected speed - outer turbine
CVDNOZ	nozzle thrust coefficient (tabular lookup)
CVMNOZ	nozzle thrust coefficient (tabular lookup)
DELFG	gross-thrust delta degradation multiplier
DELFN	net-thrust delta degradation multiplier
DELSFC	specific-fuel-consumption delta degradation multiplier
DPAFDS	afterburner design pressure drop, $\Delta P/P$
DPCODS	combustor design pressure drop, $\Delta P/P$
DPDUDS	duct design pressure drop, $\Delta P/P$
DTCODS	combustor design temperature increase (automatically set to $T_4 - T_3$), °R
ETAA	afterburner efficiency (not required)
ETAADS	afterburner efficiency at design
ETABDS	combustor efficiency at design
ETACDS	inner compressor adiabatic efficiency at design
ETAD	duct-burner combustion efficiency
ETAFDS	front (outer) compressor adiabatic efficiency at design
ETAR	inlet pressure recovery (ram recovery)
ETHPDS	high-pressure-(inner) turbine design adiabatic efficiency
ETLPDS	low-pressure-(outer) turbine design adiabatic efficiency
FAN	logical control for fan and turbojet engines
HPEXT	horsepower extraction
IAFTBN	index on afterburning desired
IAMTP	index on ram or inlet operation desired
IDBURN	index on duct burning desired
IDCD	duct nozzle convergent-divergent when IDCD equals 1 (design or off-design)

atically

IDUMP	index for dumping of error matrix
IGASMIX	index for mixed-flow or non-mixed-flow turbofan
IMCD	main nozzle convergent-divergent when IMCD equals 1 (design or off-design)
ISPOOL	index for number of compressors for turbojet engines
ITRYS	index for maximum number of iterations
MODE	independent variable designator for engine operation
NOZFLT	index for floating main or duct nozzle
PCBLC	ratio of compressor bleed to turbines to compressor airflow
PCBLDU	ratio of compressor bleed leaked into fan duct to total compressor bleed flow
PCBLF	ratio of bleed from outer compressor to fan airflow dumped overboard (i. e., leakage)
PCBLHP	fraction of PCBLC used for high-pressure (inner) turbine (cooling)
PCBLLP	fraction of PCBLC used for low-pressure (outer) turbine (cooling)
PCBLOB	ratio of inner compressor bleed to compressor airflow (overboard for customer use)
PCNC	inner-compressor shaft speed as a percent of design shaft speed
PCNCDS	design inner-compressor shaft speed
PCNF	outer-compressor shaft speed as percent of design
PCNFDS	design outer-compressor shaft speed as percent of design shaft speed
PRCDS	design inner-compressor pressure ratio
PRFDS	design outer-compressor pressure ratio
PS55	static pressure at low-pressure-turbine exit
P2	compressor-face total pressure (for nonstandard days only), atm
TFHPDS	design inner-turbine flow function
TFLPDS	design outer-turbine flow function
TOLALL	tolerance on error matrix
T2	compressor-face total temperature (for nonstandard days only), T1 + T2
T24	duct-burner exit temperature, °R

T4	combustor exit temperature, $^{\circ}\text{R}$
T4DS	design combustor exit temperature, $^{\circ}\text{R}$
T7	afterburner exit temperature, $^{\circ}\text{R}$
T7DS	design afterburner exit temperature, $^{\circ}\text{R}$
WACCDS	design inner-compressor corrected airflow (turbofans only), lbm/sec
WAFCDs	design outer-compressor corrected airflow, lbm/sec
WFA	fuel flow rate to afterburner (IAFTBN = 2 only), lbm/sec
WFB	fuel flow rate to main burner (MODE = 2 only), lbm/sec
WFBDS	design fuel flow rate to main burner (MODE = 2 only), lbm/sec
ZCDS, ZFDS	design ratio of inner compressor and fan compressor pressure ratios, respectively; equals pressure ratio at design point on design speed line minus value of pressure ratio of lowest point on speed line divided by high (surge) value minus low value of pressure ratio on the design speed line

Output Symbols¹

A	area, ft^2
ALTP	altitude, ft
AM	Mach number
BLC	bleed flow out of compressor, lbm/sec
BLF	bleed flow out of fan (dumped overboard), lbm/sec
BLHP	bleed flow into high-pressure turbine, lbm/sec
BYPASS	ratio of airflow into fan duct to airflow into inner compressor
CNC	corrected shaft speed - inner compressor
CNF	corrected shaft speed - fan
CNHP	corrected shaft speed - high-pressure turbine, $\text{PCNC} / \sqrt{T_{\text{in}}}$
CNHPCF	corrected speed - high-pressure-turbine correction factor
CNLP	corrected speed - low-pressure turbine, $\text{PCNF} / \sqrt{T_{\text{in}}}$

¹Some symbols, such as T4, are followed by station numbers, see appropriate figure for each engine to determine station numbers.

CVDNOZ	velocity coefficient of fan nozzle
CVMNOZ	velocity coefficient of core nozzle
DHHPCF	high-pressure-turbine delta enthalpy correction factor
DHLPCF	low-pressure-turbine delta enthalpy correction factor
DHTC	work done by high-pressure turbine, Btu/lbm
DHTCHP	enthalpy change temperature corrected - high-pressure turbine, Btu/ ^o R/atm/lbm
DHTCLP	enthalpy change temperature corrected - low-pressure turbine, Btu/ ^o R/atm/lbm
DHTF	work done by low-pressure turbine, Btu/lbm
DPCOM	$(\Delta P/P)_{\text{combustor}}$
DPDUC	$(\Delta P/P)_{\text{fan duct}}$
DTCOCF	temperature-rise-across-combustor correction factor
ETAB	combustor efficiency
ETABCF	combustor efficiency correction factor
ETAC	inner-compressor adiabatic efficiency
ETACCF	inner-compressor efficiency correction factor
ETAD	duct-burner efficiency
ETAF	fan adiabatic efficiency
ETAFCF	fan efficiency correction factor
ETATHP	high-pressure-turbine adiabatic efficiency
ETATLP	low-pressure-turbine adiabatic efficiency
ETHPCF	high-pressure-turbine efficiency correction factor
ETLPCF	low-pressure-turbine efficiency correction factor
FAR	fuel-air ratio, f/a
FG	gross thrust, lbf
FGM	momentum thrust, lbf
FGP	pressure thrust, lbf
FN	net thrust, lbf

FRD	ram drag, lbf
HPEXT	horsepower extracted, hp
P	pressure, atm
PCBLC	fraction of compressor exit air bled for cooling or lost to cycle
PCBLDU	fraction of bled air out of compressor which leaks into fan duct
PCBLF	fraction of fan exit airflow lost overboard
PCBLHP	fraction of compressor bleed air put into high-pressure turbine
PCBLLP	fraction of compressor bleed air put into low-pressure turbine
PCNC	inner compressor shaft speed as fraction of design
PCNF	fan compressor shaft speed as fraction of design
PRC	pressure ratio of inner compressor
PRCCF	pressure-ratio-of-inner-compressor correction factor
PRF	pressure ratio of fan or outer compressor
PRFCF	pressure-ratio-of-fan-or-outer-compressor correction factor
PS	static pressure, atm
SFC	specific fuel consumption, lbm/(lbm/hr)
T	temperature, °R
TFFHP	high-pressure-turbine flow function, $(\text{lbm})\left(\sqrt{\text{°R}}\right)(\text{in.}^2)/(\text{sec})(\text{lbf})$
TFFLP	low-pressure-turbine flow function, $(\text{lbm})\left(\sqrt{\text{°R}}\right)(\text{in.}^2)/(\text{sec})(\text{lbf})$
TFHPCF	high-pressure-turbine flow function correction factor
TFLPCF	low-pressure-turbine flow function correction factor
T2DS	design exit temperature of fan, °R
T21DS	design exit temperature of inner compressor, °R
V	velocity, ft/sec
VA	velocity of aircraft, ft/sec
VJD	fan duct exhaust velocity, ft/sec
VJM	core exhaust velocity, ft/sec
WA	airflow, lbm/sec
WA3CDS	corrected airflow in combustor at design, lbm/sec
WAC	inner-compressor airflow, lbm/sec

WACCF	inner-compressor corrected airflow correction factor
WAD	fan duct airflow, lbm/sec
WAF	fan airflow, lbm/sec
W AFC	fan corrected airflow, lbm/sec
WAFCF	fan corrected airflow correction factor
WFA	fuel flow rate to afterburner, lbm/sec
WFB	fuel flow rate to combustor, lbm/sec
WFD	fuel flow rate to duct burner, lbm/sec
WFT	total fuel flow rate, lbm/sec
WG	gas flow rate, lbm/sec
WGT	total gas flow rate, lbm/sec
ZC	ratio of inner compressor pressure ratios
ZF	ratio of fan pressure ratios

REFERENCES

1. McKinney, John S.: Simulation of Turbofan Engine. Part I. Description of Method and Balancing Technique. Rep. AFAPL-TR-67-125, pt. 1, Air Force Systems Command, Nov. 1967. (Available from DDC as AD-825197.)
2. McKinney, John S.: Simulation of Turbofan Engine. Part II. User's Manual and Computer Program Listing. Rep. AFAPL-TR-67-125, pt. 2, Air Force Systems Command, Nov. 1967. (Available from DDC as AD-825198.)
3. Fishbach, Laurence H.; and Koenig, Robert W.: GENENG II - A Program for Calculating Design and Off-Design Performance of Two- and Three-Spool Turbofans with as Many as Three Nozzles. NASA TN D-6553, 1972.
4. Keenan, Joseph H.; and Kaye, Joseph: Gas Tables. John Wiley & Sons, Inc., 1948.
5. Anon.: U. S. Standard Atmosphere, 1962. Prepared under sponsorship of NASA, USAF, and USWB, 1962.
6. Turner, Don N.; and Huff, Vearl N.: An Input Routine Using Arithmetic Statements for the IBM 704 Digital Computer. NASA TN D-1092, 1961.

TABLE I. - VARIABLES AND ERRORS

	Two-spool turbofan	Mixed-flow turbofan	Two-spool turbojet	One-spool turbojet
Variable 1	ZF	ZF	ZF	ZF
Variable 2	PCNF	PCNF	PCNF	PCNF
Variable 3	ZC	ZC	ZC	TFFLP
Variable 4	PCNC	PCNC	PCNC	-----
Variable 5	TFFHP	TFFHP	TFFHP	-----
Variable 6	TFFLP	TFFLP	TFFLP	-----
Error 1	<u>TFHCAL - TFFHP</u> TFHCAL	<u>TFHCAL - TFFHP</u> TFHCAL	<u>TFHCAL - TFFHP</u> TFHCAL	<u>TFLCAL - TFFLP</u> TFLCAL
Error 2	<u>DHTCC - DHTCHP</u> DHTCC	<u>DHTCC - DHTCHP</u> DHTCC	<u>DHTCC - DHTCHP</u> DHTCC	<u>DHTCF - DHTCLP</u> DHTCF
Error 3	<u>TFLCAL - TFFLP</u> TFLCAL	<u>TFLCAL - TFFLP</u> TFLCAL	<u>TFLCAL - TFFLP</u> TFLCAL	<u>P7R - P7</u> P7R
Error 4	<u>DHTCF - DHTCLP</u> DHTCF	<u>DHTCF - DHTCLP</u> DHTCF	<u>DHTCF - DHTCLP</u> DHTCF	-----
Error 5	<u>P25R - P25</u> P25R	<u>PS25 - PS55</u> PS25	<u>WAF - WAC - BLF</u> WAC	-----
Error 6	<u>P7R - P7</u> P7R	<u>P7R - P7</u> P7R	<u>P7R - P7</u> P7R	-----
Matrix size	6 × 6	6 × 6	6 × 6	3 × 3

TABLE II. - INPUTS REQUIRED FOR BASIC CYCLES

Variable	Units	Definition	Two-spool turbofan	Mixed-flow turbofan	Two-spool turbojet	One-spool turbojet
PRFDS	-----	Fan pressure ratio	Yes	Yes	Yes	Yes
WAFCDs	lb/sec	Fan corrected airflow	↓	↓	↓	↓
ETAFDS	-----	Fan efficiency				
ZFDS	-----	Design Z of fan				
PCNFDS	-----	Corrected speed of fan				
PRCDS	-----	Compressor pressure ratio			↓	No
WACCDS	lb/sec	Compressor corrected airflow			No	↓
ETACDS	-----	Compressor efficiency			Yes	
ZCDS	-----	Design Z of compressor			↓	
PCNCDS	-----	Corrected speed of compressor				↓
ETABDS	-----	Combustor efficiency				Yes
DPCODS	-----	Combustor pressure drop, $\Delta P/P$				Yes
T4DS	$^{\circ}R$	Turbine inlet temperature				Yes
TFHPDS	$\frac{lb\sqrt{^{\circ}R}}{(sec)(psia)}$	High-pressure-turbine flow function				No
CNHPDS	-----	Corrected speed - high-pressure turbine				No
ETHPDS	-----	Efficiency - high-pressure turbine				No
TFLPDS	$\frac{lb\sqrt{^{\circ}R}}{(sec)(psia)}$	Low-pressure-turbine flow function				Yes
CNLPDS	-----	Corrected speed - low-pressure turbine				Yes
ETLPDS	-----	Efficiency - low-pressure turbine				Yes
DPDUDS	-----	$\Delta P/P$ of fan duct	↓	↓	No	No
DPAFDS	-----	$\Delta P/P$ of afterburner			Yes	Yes
FAN	-----	Logical variable	. TRUE.	. TRUE.	. FALSE.	. FALSE.
ISPOOL	-----	Number of spools	2	2	2	1

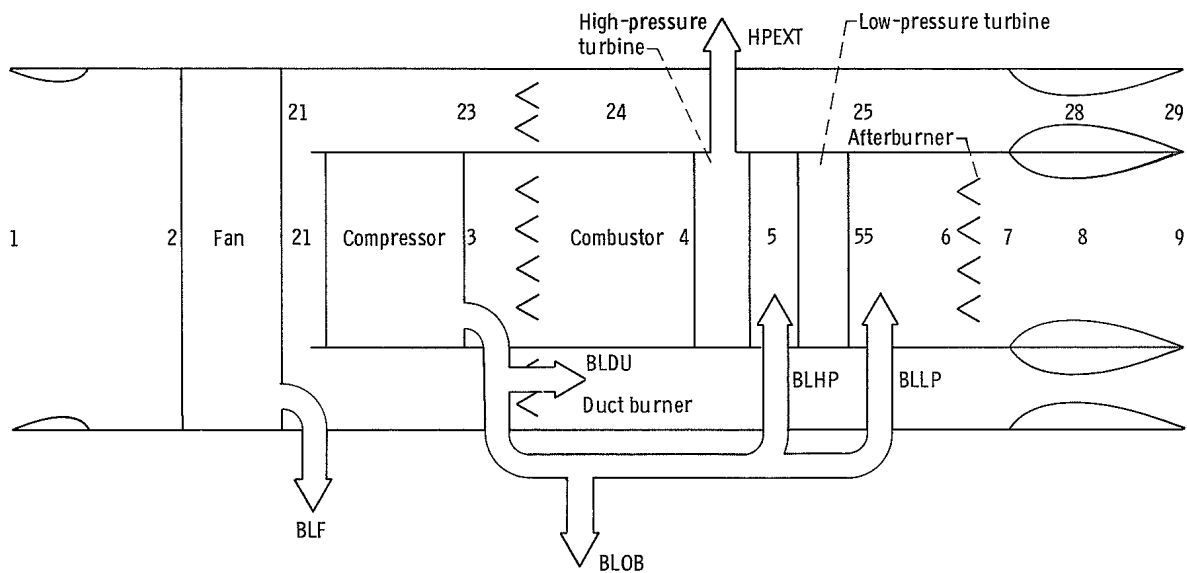


Figure 1. - Schematic of non-mixed-flow duct-burning and/or afterburning turbofan.

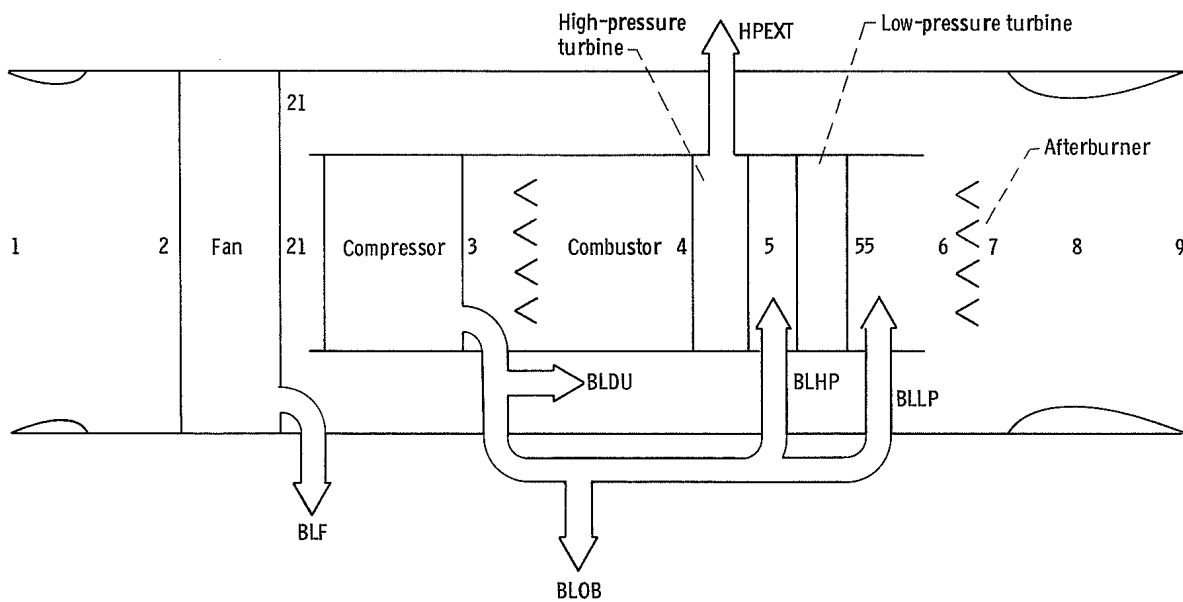


Figure 2. - Schematic of mixed-flow afterburning turbofan.

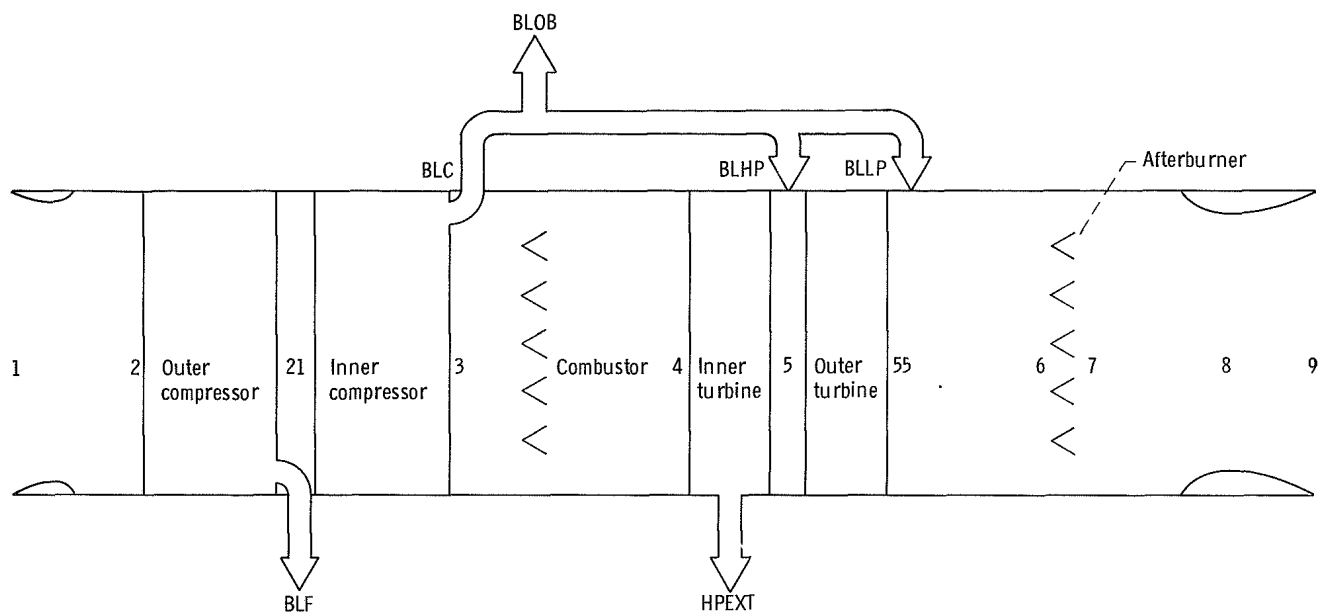


Figure 3. - Schematic of two-spool turbojet.

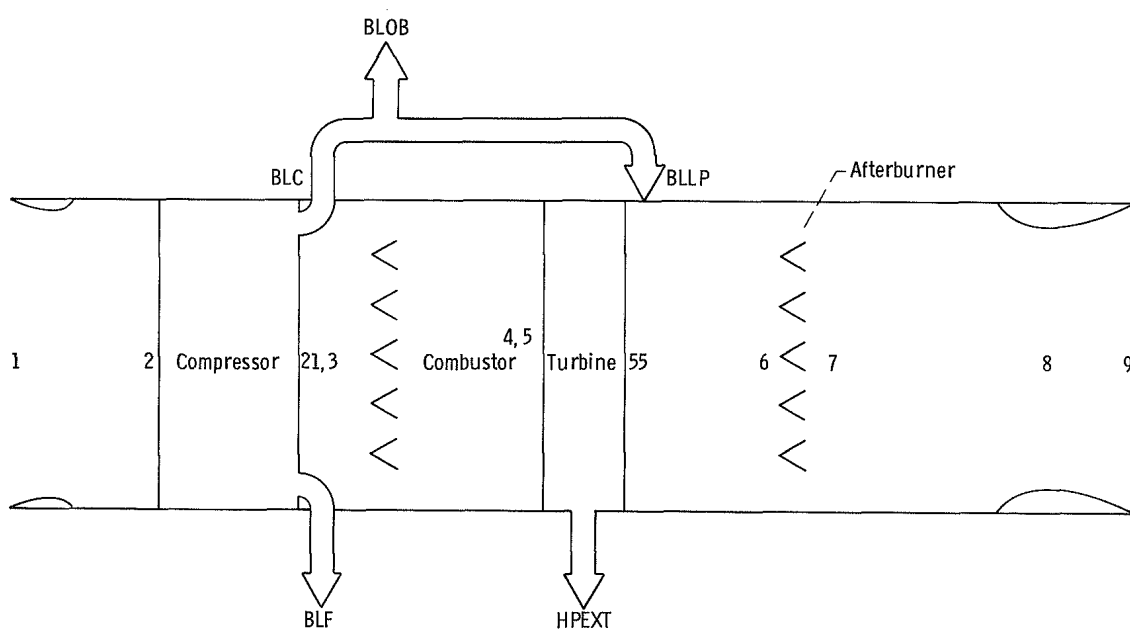


Figure 4. - Schematic of one-spool turbojet.

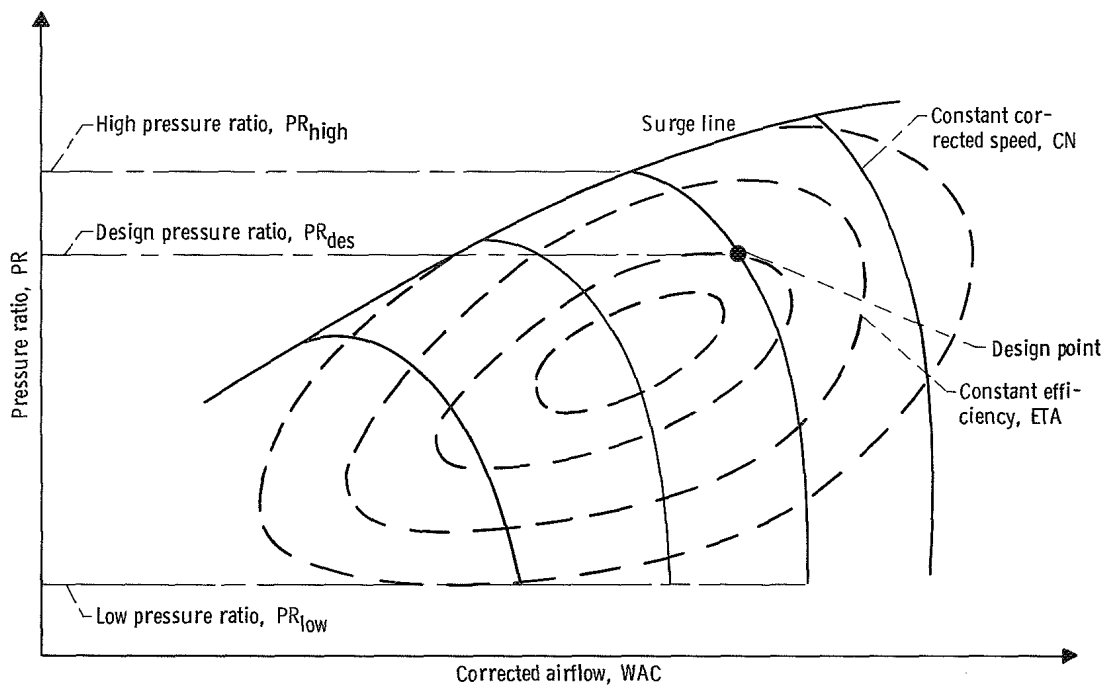


Figure 6. - Example of a specific fan-compressor map. $Z = (PR_x - PR_{low}) / (PR_{high} - PR_{low})$.

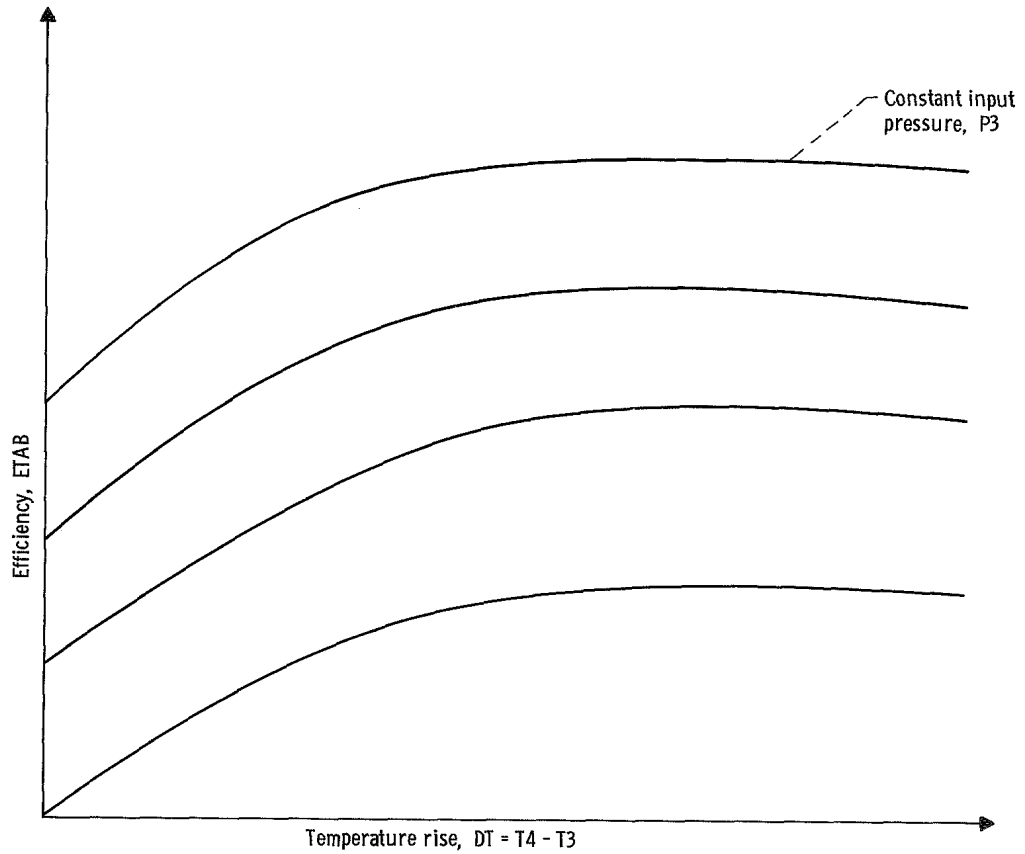


Figure 7. - Example of combustor map.

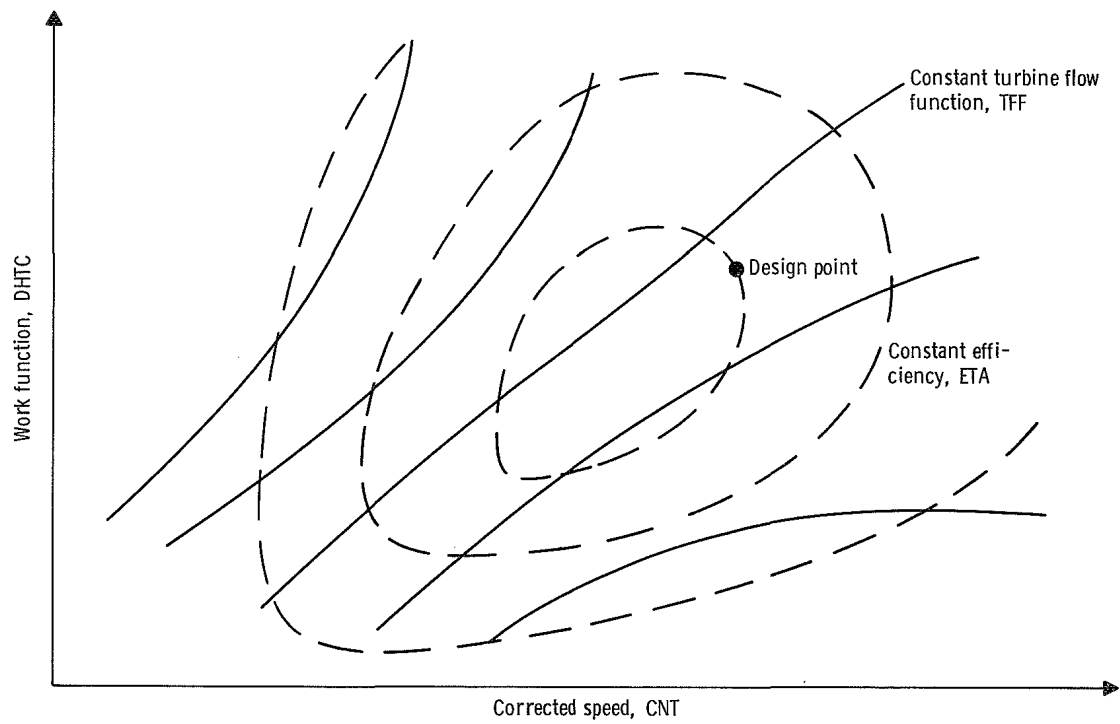
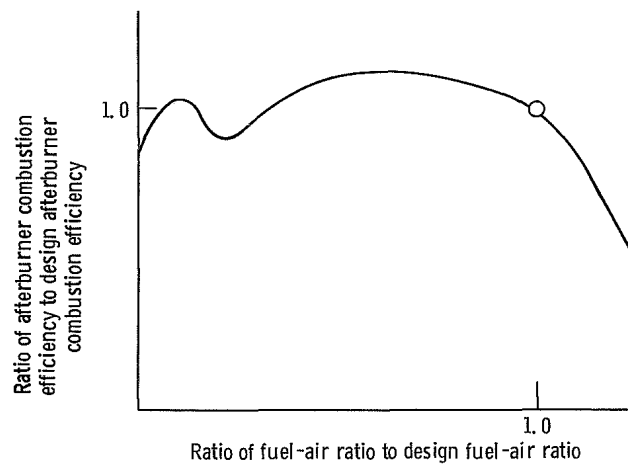
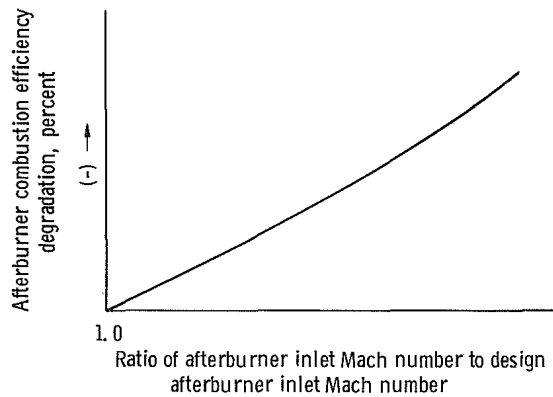


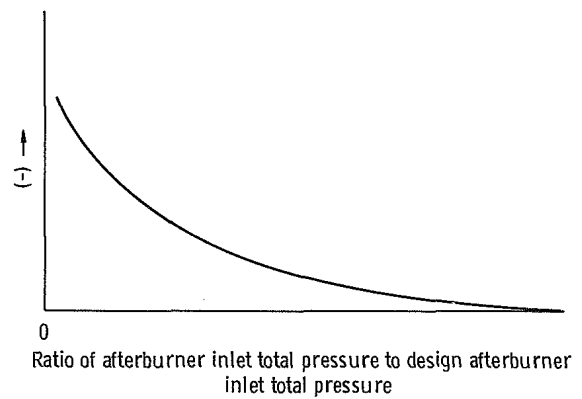
Figure 8. - Example of specific turbine map.



(a) Generalized afterburner combustion efficiency as function of fuel-air ratio.



(b) Efficiency correction factor as function of afterburner inlet Mach number.



(c) Efficiency correction factor as function of afterburner inlet total pressure ratio.

Figure 9. - Example of a generalized afterburner combustion efficiency performance map.

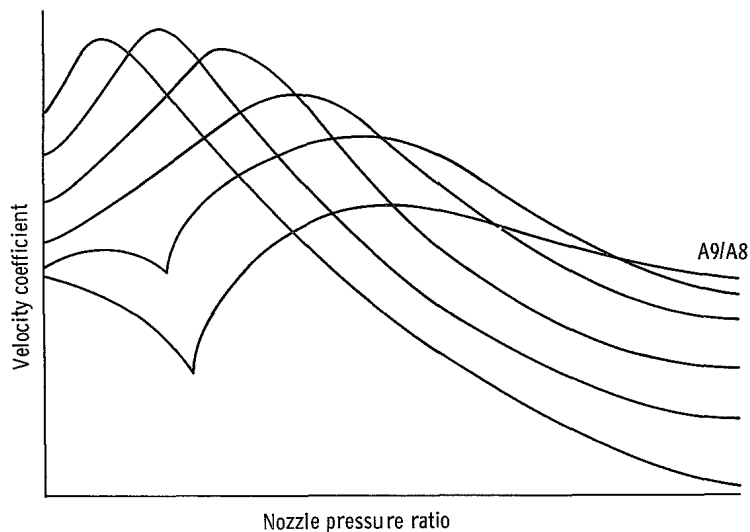


Figure 10. - Performance map for nozzle, giving velocity coefficient.

POSTMASTER: If Undeliverable (Section 158
Postal Manual) Do Not Return

"The aeronautical and space activities of the United States shall be conducted so as to contribute . . . to the expansion of human knowledge of phenomena in the atmosphere and space. The Administration shall provide for the widest practicable and appropriate dissemination of information concerning its activities and the results thereof."

— NATIONAL AERONAUTICS AND SPACE ACT OF 1958

NASA SCIENTIFIC AND TECHNICAL PUBLICATIONS

TECHNICAL REPORTS: Scientific and technical information considered important, complete, and a lasting contribution to existing knowledge.

TECHNICAL NOTES: Information less broad in scope but nevertheless of importance as a contribution to existing knowledge.

TECHNICAL MEMORANDUMS: Information receiving limited distribution because of preliminary data, security classification, or other reasons.

CONTRACTOR REPORTS: Scientific and technical information generated under a NASA contract or grant and considered an important contribution to existing knowledge.

TECHNICAL TRANSLATIONS: Information published in a foreign language considered to merit NASA distribution in English.

SPECIAL PUBLICATIONS: Information derived from or of value to NASA activities. Publications include conference proceedings, monographs, data compilations, handbooks, sourcebooks, and special bibliographies.

TECHNOLOGY UTILIZATION PUBLICATIONS: Information on technology used by NASA that may be of particular interest in commercial and other non-aerospace applications. Publications include Tech Briefs, Technology Utilization Reports and Technology Surveys.

Details on the availability of these publications may be obtained from:

**SCIENTIFIC AND TECHNICAL INFORMATION OFFICE
NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
Washington, D.C. 20546**